

CAPE COD NATIONAL SEASHORE PROVINCETOWN, MA

REPLACE AND RELOCATE SEPTIC SYSTEM FOR PROVINCE LANDS
VISITOR CENTER COMPLEX
CACO
PMIS NO. 299311

PROJECT SPECIFICATIONS



NATIONAL PARK SERVICE

2/6/2025
FINAL CONSTRUCTION DOCUMENTS

TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

01 11 00	Summary of Work
01 26 01	Contract Modification Procedures
01 27 00	Definition of Contract Line Items
01 31 00	Project Management & Coordination
01 32 16	Construction Schedule
01 32 33	Photographic Documentation
01 33 23	Submittal Procedures
01 35 23	Safety Requirements with Lock out/ Tag out Form
01 35 43	Environmental Protection
01 40 00	Quality Requirements
01 50 00	Temporary Facilities and Controls
01 51 29	Temporary Sanitary Facilities
01 67 00	Product Requirements
01 71 23	Survey and Record Drawings
01 73 29	Cutting and Patching
01 73 40	Execution
01 74 19	Cleaning Up-Civil
01 74 19	Construction Waste Management and Disposal
01 77 00	Closeout Procedures
01 78 23	Operation and Maintenance Data
01 79 00	Demonstration and Training

DIVISION 02 – EXISTING CONDITIONS

02 41 13	Selective Site Demolition
----------	---------------------------

DIVISION 03 - CONCRETE

03 20 00	Concrete Reinforcement
03 30 00	CIP Concrete

DIVISION 26 – ELECTRICAL

26 05 00	Common Work Results For Electrical
26 05 19	Low-Voltage Electrical Power Conductors and Cables
26 05 23	Control Voltage Electrical Power Cables
26 05 26	Grounding and Bonding for Electrical Systems
26 05 29	Hangers and Supports for Electrical Systems
26 05 33	Raceways and Boxes for Electrical Systems
26 05 43	Underground Ducts and Raceways for Electrical Systems
26 05 44	Sleeves and Sleeve Seals for Electrical Raceways and Cabling
26 05 53	Identification for Electrical Systems
26 24 16	Panelboards
26 43 13	Surge Protection

DIVISION 31 - EARTHWORK

31 10 00 Site Clearing
31 20 00 Earth Moving
31 25 00 Erosion Control
31 50 00 Excavation Support

DIVISION 32 – EXTERIOR IMPROVEMENTS

31 93 00 Plants

DIVISION 33 – UTILITIES

33 30 00 Sewage System

Attachment(s):

25-WP63-0002-APP Plan Approval
List of Submittals

SECTION 01 11 00 – SUMMARY OF WORK

PART 1 – GENERAL

1.1 SUMMARY

A. This Section includes the following:

1. Work covered by the Contract Documents.
2. Contractor use of site.
3. Public use of site.
4. Occupancy requirements for buildings.
5. Conduct of operations.
6. Utility fees.
7. Work restrictions.
8. Special construction requirements.
9. Soils investigation report.
10. Additional reports.

1.2 WORK COVERED BY CONTRACT DOCUMENTS

A. Project Location: Province Lands Visitor Center (PLVC), 171 Race Point Road, Provincetown, MA, within the Cape Cod National Seashore. Refer to contract drawings for exact construction limits and staging areas.

B. Scope of Work: The Work includes all supervision, labor, materials, equipment and plant necessary to:

1. Obtain all required permits, approvals and inspections from the Town of Provincetown, Massachusetts Department of Environmental Protection (MassDEP) under Permit 25-WP63-0002-APP, and the National Park Service. The Contractor shall be solely responsible for coordinating, applying for, paying all associated fees, and securing all required permits, inspections, and reports, including submitting of as-built documentation and certifications to the authority having jurisdiction.
2. Perform all inspections, testing, documentation, and as-built reports as required by MassDEP, Provincetown, and the Contracting Officer (CO).
3. Abandon and decommission the existing septic system in accordance with applicable federal, state, and local regulations. The Contractor shall pump, remove, and properly dispose of all sewage and effluent from existing septic tanks, seepage pits, and associated piping at an authorized treatment facility. All costs for pumping, transportation, and disposal are included in the Base Bid. No separate payment will be made. Contractor shall provide disposal documentation, including manifests or receipts, to the Contracting Officer's Representative (COR).
4. Furnish and install a new Title V septic system, including septic tanks, pumps, control panels, alarms, distribution boxes, leaching fields, and associated piping, wiring, raceways, and

terminations. Reuse of the existing force main is based on available information. No repair or replacement of the existing force main is included unless specifically identified in the Contract Line Items.

5. Coordinate with Eversource (public utility) for all required electrical service disconnections and reconnections required for electrical outages at the comfort station and PLVC main building. The Contractor shall prepare, submit, and manage all necessary work orders, pay all associated fees, stand-by costs, and any other charges required by Eversource, and schedule work in coordination with the COR. All costs associated with electrical service coordination, including work orders, fees, and required stand-by, are incidental to the Contract and are included in the turnkey scope. No separate payment will be made. The Contractor shall coordinate utility work sufficiently in advance to avoid impacts to the project schedule. Delays due to utility coordination that could have been reasonably anticipated shall not constitute grounds for additional compensation.
6. Restore all disturbed areas, including fine grading, replanting, and seeding in accordance with the drawings and specifications.
7. Provide all required coordination and scheduling with the NPS, MassDEP, and the Town of Provincetown for field inspections, pump tests, and system start-up verification prior to final acceptance.
8. Option 1- Submit and coordinate all required Eversource work orders for electrical disconnect/reconnect at the comfort station. Remove existing pump control panel, electrical panel, ejector pump, associated conductors, raceway system and disconnect. At existing ejector pump manhole, remove pumps and rails and prepare existing manhole to receive new rails, two (2) new duplex pumps (denoted as pump station-2), new floats and pump rail system. Pump and clean area to receive new rails, equipment and wiring. Replace existing comfort station electrical panel with new panel as indicated. Provide new free-standing pump control panel, new underground raceway, new conductors and new circuit breaker to service the new pump controller and ejector pumps. Prior to final inspection, provide a field test that demonstrates proper operation and performance of the ejector pump system and alarms to the COR.
9. Option 2- Provide plantings as shown on Drawings LP101 and LP510 and per specifications. Work includes soil preparation, planting, mulching, staking, and maintenance until final acceptance.
10. Option 3- Provide and maintain Temporary Sanitary Facilities: Provide and maintain the following temporary facilities during the duration of the septic system shutdown: one (1) ADA-accessible portable toilet and three (3) standard portable toilets. These facilities shall be provided for public and government use only. Work shall include pumping, restocking of toilet paper and sanitizer, and cleaning a minimum of three (3) times per week in accordance with Section 01 51 29, Temporary Sanitary Facilities. The provision of temporary sanitary facilities is supplemental and does not replace the requirements of Section 1.2.B.3, which governs the pumping, removal, and proper disposal of all sewage from the existing septic tanks, seepage pits, and associated piping. Duration of use shall be as required to support the Contractor's construction schedule and sequencing.
11. Option 4- Provide Jetting and Cleaning of Existing Force Main: Provide all labor, equipment, and materials necessary for Jetting and Cleaning of Existing Force Main between the lift station and the new septic tank (approximately 300LF) using high-pressure water jetting. Work includes the mobilization of jetting equipment suitable for sanitary force mains. Cleaning and removal of grease, sediment, debris and obstructions within the pipe. Flushing and clearing the full length of

the force main. Collection, handling and proper disposal of all removed debris in accordance with applicable federal, state and local regulations. Jetting and cleaning does not guarantee the suitability of the force main for continued use. No repair or replacement is included unless separately authorized.

12. Option 5- Provide closed-circuit television (CCTV) inspection: Provide all labor, equipment, and materials necessary to perform a closed-circuit television (CCTV) inspection of the existing force main between the lift station and septic tank (approximately 300 LF). Work includes Pre-inspection coordination and verification that the line is suitable for CCTV inspection (cleaning shall be completed prior to inspection if necessary). CCTV inspection of the full length of the pipe using equipment suitable for pressurized or previously pressurized sanitary lines. Continuous video recording of the inspection. Identification and documentation of defects, including but not limited to cracks, deformation, joint separation, root intrusion, blockages, and evidence of collapse. Measurement and logging of pipe length, locations of observed defects, and relative position from known structures. Delivery of a written inspection report summarizing findings and assessing overall condition of the pipe. Submission of video files in digital format (USB or electronic upload) to the CO/COR. CCTV inspection does not guarantee the suitability of the force main for continued use. No repair or replacement is included unless separately authorized.

13. Option 6 – Exterior Force Main Excavation and Check Valve Replacement: Provide all labor, equipment, and materials necessary to excavate in the vicinity of Pump Station No. 2, locate the existing force main, and remove and replace the existing check valve and valve box. Work includes field verification of the force main alignment and valve assembly; excavation; removal and disposal of the existing valve box, check valve, and associated piping; furnishing and installation of a new check valve and valve box; connection to the existing force main; backfill, compaction, and restoration of disturbed areas; and bypass pumping or temporary flow control as required to maintain system operation.

This work is based on the assumption that the existing force main and associated components remain in serviceable condition; however, the Government has identified uncertainty in the condition of the check valve and associated piping due to recent operational issues. This option provides a mechanism to perform localized repair if conditions encountered during construction indicate such work is necessary. No additional repair or replacement beyond that described herein is included unless separately authorized.

14. All work shall be performed under a single prime contract in accordance with the contract documents.

1.3 CONTRACTOR USE OF SITE

A. General: Contractor shall have limited use of the site for construction operations. Confine activities to areas indicated on the drawings. Do not disturb areas beyond those shown.

B. Limits:

1. Confine earthwork, clearing, grading, and staging to the minimum necessary.

2. Restrict site disturbance to 10 feet beyond walkways, patios, surface parking, and utilities <12 inches; 15 feet beyond primary roadway curbs and main utility trenches; 25 feet beyond constructed areas with permeable surfaces.
3. Follow City of Provincetown ordinances regarding noise, work hours, and public access.
4. Where construction impacts public roads, access roads and parking area, maintain traffic flow, provide signage, and employ flag personnel as needed.
5. The condition of the existing force main is not guaranteed. The Contractor shall verify suitability for reuse. Any cleaning, inspection, or verification required to confirm usability is incidental to the Base Bid. Repair or replacement, if required, will be addressed by modification unless specifically included in an exercised option.

C. Parking and Storage of Materials: Confine construction equipment and storage to designated staging areas. Protect from damage and environmental exposure.

D. Preservation of Natural Features:

1. Prevent damage to trees, plants, and other natural features. Restore damaged areas at no cost to the Government.
2. Provide temporary barriers and stakes to protect existing trees, plants, and root zones.
3. Do not fasten ropes, cables, or guys to trees.
4. Exercise caution during excavation, grading, filling, and demolition near vegetation.
5. Eastern Box Turtle: From April 1 to October 31, relocate any turtles found within the Limits of Disturbance (LOD) outside of the LOD. Conduct daily site sweeps before equipment use. Any injured or killed turtles must be reported to the Contracting Officer who will notify US Fish & Wildlife Service.
6. Trees ≥ 3 inches dbh shall be preserved where possible. Removal only between November 1 and March 31 to avoid bat and migratory bird impacts. Coordinate any additional removals with the CO and Park for bat clearance approval. Report any mortality to the CO and US Fish & Wildlife.

E. Driveways, Entrances & Accesses: Maintain clear access for Government employees, emergency vehicles, and the public. Schedule deliveries to minimize on-site congestion. Restore access roads, driveways, and paths to preconstruction or better condition.

F. Construction Camp: Not permitted within the park boundaries.

G. Hauling Restrictions: Comply with all legal load restrictions. Special permits do not relieve the contractor of liability for damages.

H. Parking Restrictions: Contractor parking for personal vehicles is available during work hours only. Do not park vehicles where they will restrict or inhibit public use of the Province Land Visitor Center. Confine construction equipment to the limit of disturbance (LOD) areas; minimize impact on neighboring buildings, public and government use of the site.

I. Archeological Monitoring: Required for all new excavation beyond prior disturbance. NPS archeologist may monitor ground-disturbing work. Limit excavation to the minimum required for removal and installation. Use flat-blade buckets and construction matting where equipment operates outside previously disturbed areas.

1.4 PUBLIC USE OF SITE

A. The Province Land Visitor Center building is open from April 1 to October 30. The public use the parking area, restrooms and adjacent trail system all year. Prohibit public access within the LOD. Provide barriers, signage, and caution tape as needed. Always maintain access for emergency vehicles, government personnel and the public. Unscheduled road closures are not permitted.

B. Coordinate visitor safety and closure notices with COR at least two (2) weeks before mobilization.

1.5 UTILITY FEES

A. The Contractor shall obtain and pay for all permits, fees, and inspections required for utility connections, including the septic system. Costs for such permits, inspections and fees are incidental to the Contract and shall not be paid separately.

B. There are no utility connections available at the site(s). The contractor shall provide its own utilities.

1.6 OCCUPANCY REQUIREMENTS FOR BUILDINGS

A. Government Occupancy: The Government will occupy premises during April 1 through October 30. The site is open to the public all year. Maintain safe access to all occupied facilities. Do not interfere with Government operations.

B. Coordinate access to restricted mechanical/electrical areas with the COR at least 48 hours in advance. Allow up to five (5) business days for key requests. Maintain all life-safety systems active and accessible in unaffected areas during construction.

C. Maintain safe access and egress for all occupied structures.

1.7 CONDUCT OF OPERATIONS

A. Comply with NPS rules, Secretary of the Interior regulations, and Park Superintendent directives.

B. Work on weekends, Federal holidays, or nights requires prior written approval from the CO. Submit requests at least 7 workdays in advance.

C. No signs or advertisements on site unless approved by the CO. Company signage limited to construction vehicles.

1.8 WORK RESTRICTIONS

Replace Septic PLVC

01 11 00-5
Summary of Work

A. On-Site Work Hours: 7:00 a.m. – 5:00 p.m., Monday – Friday, unless otherwise indicated.

1. Existing Utilities: The Contractor shall notify DIGSAFE, 888-344-7233, in advance of all excavations. In addition to DIGSAFE notification, the Contractor shall be responsible for locating and preventing damage to known utilities. Before starting any ground disturbance, the contractor shall perform a full ground penetrating radar (GPR) survey of all areas to be disturbed and impacted by construction. The GPR survey shall comply with ASTM D6432-Standard Guide for Using Ground Penetrating Radar for Subsurface Investigations. Clearly mark all detected and known utilities prior to the excavation. Where utilities are identified and suspected to cross the path of new work, utilize an excavation spotter to prevent damage. Any damage to utilities shall be repaired by the contractor at no additional cost to the Government. The GPR survey is intended to supplement, not replace, available utility information. The Contractor shall exercise due diligence. However, concealed or unknown utilities not reasonably detectable may be considered under the Differing Site Conditions clause.
2. Notify the CO and utilities in advance of all work affecting existing systems.
3. The transformer and primary electrical service are owned by Eversource. The Contractor shall coordinate all required disconnects/reconnects and pay associated costs.
4. If damage occurs to an unknown utility, repair utility and notify the Contracting Officer and COR as soon as possible. An equitable adjustment will be made in accordance with the Changes clause of the contract.
5. Submit a brief written report and site map indicating detected subsurface features and utility markings prior to excavation.
6. All utility coordination shall be documented in writing prior to commencement of excavation.
7. Whenever permanent restroom facilities are unavailable due to construction activities, utility shutdowns, or related work, the Contractor shall provide temporary sanitary facilities in accordance with Section 01 51 29 and Contract Line-Item 0004 (Option 3).

B. Utility Interruptions:

1. Notify the CO at least one week in advance of any planned utility shutdowns.
2. During building occupancy, utility outages shall be limited to a maximum of two (2) hours.
3. When the building is not occupied, utility outages shall be limited to a maximum of one workday.
4. Utilities shall not be left out of service overnight, including electricity, water, or other required services, without prior written approval from the CO. Requests shall include duration, justification, and proposed mitigation measures.
5. During replacement of the septic system, temporary sanitary facilities (Option 3) shall be provided to maintain sanitary services while the permanent system is offline.
6. All temporary interruptions of utilities must be coordinated, scheduled, and documented with the CO/COR.

C. Nonsmoking Area: Smoking, vaping, and tobacco use prohibited.

D. Best Management Practices: Include spill containment kits for fueling and comply with MassDEP regulations and the contract requirements.

1.9 SPECIAL CONSTRUCTION REQUIREMENTS

- A. **Project Website:** A project website administered by the contractor for document and communication management (per Section 01 31 00).
- B. The Contractor shall coordinate, schedule, and document all inspections, tests, and reports required by the Town of Provincetown, MassDEP, and NPS. No portion of the work shall be backfilled or concealed until inspected and approved by all authorities having jurisdiction. The Contractor shall coordinate inspection scheduling in advance. Delays due to failure to properly schedule or coordinate inspections shall not be grounds for additional time or compensation.
- C. Provide a minimum of five (5) business days' notice to the COR before each required inspection or test.
- D. **Seasonal Restrictions:**
 - 1.No construction activities shall occur from Memorial Day through September 30 to avoid impacts during the peak visitation season.
 - 2.No ground-disturbing work from April 15 to August 30 to avoid spadefoot toad breeding season and minimize box turtle impacts.
 - 3.Daily visual sweeps required; relocate turtles outside LOD.
 - 4.Tree removal and construction activities restricted June 1 – July 31 to avoid northern long-eared bat roosting and pupping. Coordinate with US Fish & Wildlife for exceptions.
- E. **Third-Party Inspection and Start-Up:** The Contractor shall engage a manufacturer-authorized service company to perform inspections of the duplex ejector pumps, pump controls, and free-standing pump control panel at the following milestones:
 - 1. Delivery
 - 2. Rough-in Installation
 - 3. Pre-Startup / System Commissioning: The Contractor is solely responsible for coordinating, scheduling, and paying for all third-party inspections. No backfill, concealment, or system start-up may occur until all required third-party inspections are complete and accepted by the CO/COR. The third-party inspector shall submit to the contracting officer (CO) and Contracting Officers Representative (COR) prior to system start up and final acceptance a complete written report for each inspection.
 - 4. All third-party inspections, testing, and operational verification of septic system components required under Section 333000, Sanitary Sewerage, shall be fully coordinated with the Contractor's construction schedule, preinstallation conferences, and project meetings as defined in Section 01 31 00. The Contractor shall provide notice of all third-party inspection dates at least five (5) business days in advance, include relevant subcontractors and manufacturer representatives in coordination meetings, and ensure all inspection reports and corrective actions are documented and submitted in accordance with 333000 requirements prior to system start-up or backfill.

1.10 SUBMITTALS (SUMMARY)

A. Submittals are required for materials, equipment, shop drawings, schedules, and other information necessary to demonstrate compliance with the Contract Documents.

B. Submit all required documentation in accordance with Section 01 33 00 – Submittal Procedures.

C. The following items are of particular importance to this project and shall be submitted for review and approval prior to the start of associated work:

1. Construction Schedule and Schedule of Values.
2. List of Subcontractors and Suppliers.
3. Product Data and Shop Drawings for septic tanks, pumps, alarm/control panels, electrical components, piping, and control wiring.
4. Excavation protection plan, trenching plan, and restoration details.
5. Permitting documentation, inspection reports, and coordination with the State of Massachusetts registered Sanitarian or Engineer and MassDEP.
6. Testing and start-up procedures for the septic systems, electrical controls, and alarms.
7. Record Drawings and O&M manuals at project closeout.
8. Submit final inspection and approval forms from the registered Sanitarian/Engineer, Town of Provincetown, and MassDEP as part of project closeout documentation prior to final payment and system acceptance.
9. Third-party test firm, inspector resume, and test/inspection reports: The Contractor is solely responsible for coordinating, scheduling, and paying for all supplemental and third-party inspections required for this project, including pre-delivery, rough-in, and pre-startup verification. All inspections, testing, and operational verification are part of the turnkey scope. No work shall be backfilled, concealed, or accepted until all required inspections and tests are complete and formally accepted by the Contracting Officer (CO) and Contracting Officer's Representative (COR). The Contractor shall submit all inspection reports, corrective actions, and related documentation to the CO/COR in a timely manner to support system acceptance. The Contractor remains fully responsible for compliance with Contract Documents regardless of third-party inspection or manufacturer involvement.

D. No material shall be fabricated, shipped, or installed until the corresponding submittal has been reviewed and accepted by the Contracting Officer.

E. Submittals shall be transmitted electronically through the project website in accordance with Section 01 31 00 – Project Administration.

1.11 SOILS INVESTIGATION REPORT

A. A copy of the Massachusetts Department of Environmental Protection (MassDEP) Form 11 – Soil Suitability Assessment for On-Site Sewage Disposal and Commonwealth of Massachusetts City of Provincetown Percolation Test (Mass DEP) Form 12 are provided in the solicitation drawings for reference. These documents are provided for reference only and illustrate the basis of design and field

data supporting the septic system design.

B. If conflicts occur between the report and contract documents, drawings and specifications govern.

C. The Contractor shall notify the CO immediately if subsurface conditions differ from those indicated in the report or impede design assumptions.

PART 2 – PRODUCTS

(Not Used)

PART 3 – EXECUTION

(Not Used)

END OF SECTION 01 11 00

SECTION 01 26 01 – CONTRACT MODIFICATION PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section consists of administrative and procedural requirements for contract modifications.

1.2 DEFINITIONS AND ALLOWANCES

- A. Home Office Overhead: Costs incurred in support of all of a contractor's projects and not attributable to a specific job. The cost for home office overhead is only allowed as a percentage of all direct work excluding profit. The following items represent allowable home office overhead costs identified in Part 31 of the Federal Acquisition Regulation (FAR):

1. Rent
2. Utilities
3. Furnishings
4. Office equipment
5. Executive and management staff not exclusively assigned to the project
6. Support, accounting, and administrative staff
7. Preparation of cost proposals, estimating, and schedule analyses connected with Modifications
8. Estimating and preconstruction services
9. Mortgage costs
10. Real estate and corporate taxes
11. Automobile maintenance and travel costs for home office personnel
12. Home office insurances i.e. structure, automotive, umbrella, flood, etc.
13. Depreciation of equipment and other assets
14. Home office supplies (paper, staples, etc.)
15. Legal services
16. Accounting and data processing
17. Professional fees/registration

- B. General Conditions (Field Office Overhead): Management and administrative costs incurred on site for the designated project. Costs associated with preparation of modifications will not be allowed. Costs for these items are to be included only in the general conditions of the modification estimate. Only in the case of a contract time extension are additional general conditions included in modifications. The following items, if applicable, are considered allowable costs for calculating General Conditions:

1. Project Manager (PM), Assistant Project Manager
2. Superintendent, Assistant Superintendent
3. Quality Control, Safety Officer, Environmental Manager, etc.
4. Engineers
5. Travel, lodging, and per diem (as established by Federal Travel Regulations)
6. Scheduling

7. Field Office Trailers and associated temporary utilities
8. Field office supplies
 - a. Mailing and couriers
 - b. Reproduction costs
 - c. Storage
 - d. Phones
 - e. Computers
 - f. Copiers
9. Personal vehicles i.e. Superintendent Pickup trucks

C. General Requirements: Costs directly associated with the project and are necessary to perform the actual work of the modification. These costs shall be shown as direct costs in the estimate. The following items, if applicable, are considered allowable costs for calculating General Requirements:

1. Hoisting
2. Material handling
3. Temporary fencing
4. Port-a-lets
5. Trash removal, dumpsters
6. Barricades
7. Small tools
8. Safety supplies
9. Scaffolding
10. Daily cleaning
11. Traffic control
12. Temporary signage
13. Temporary heating and power

D. Personnel Costs: Costs included in the modification must only be for General Conditions staff and workers actually present and working on project site. Modification costs for salaried workers are only allowed within the structure of a 40-hour week and no overtime or holiday pay will be allowed.

1. Worker Hourly Rates are costs directly associated with the individual worker and consist of the following:
 - a. Base Rate: The hourly rate paid directly to the worker
 - b. Labor Burden: Employer payments of all applicable burdens; includes insurance and taxes the business must pay on behalf of the worker to government entities and educational forums, such as:
 - 1) Social Security
 - 2) Medicare
 - 3) Workers Compensation – Policy and company calculation to be made available.
 - 4) Federal Unemployment Tax Act (FUTA) - Cap Rate and percentage to be proportionally allocated over one year.
 - 5) State Unemployment Tax Act (SUTA) - Cap Rate and percentage to be proportionally allocated over one year.

- 6) Union agreement costs - Other costs required under an enforceable collective bargaining agreement.
- c. Fringe Benefits: Various non-wage compensations provided to employees such as:
 - 1) Health Care Insurance Premiums
 - 2) Cell Phone
 - 3) Clothing
 - 4) 401K and Pensions
 - 5) Vehicle allowances
 - 6) Gas allowance
 - 7) Life insurance premiums
 - 8) Disability insurance
 - 9) Other Fringe Benefits required under an enforceable collective bargaining agreement
- E. Bonuses or Deferred Compensation: No Bonus or Deferred Compensation will be allowed within any components of pricing including Home Office Overhead, General Conditions, General Requirements, Hourly Worker Rates, or the direct costs of work.
- F. General Liability Insurance: An insurance policy that protects Contractor from claims resulting from bodily injury or property damage to a third party. Include as a separate line item within all modification proposals and provide a current insurance quote upon request.
- G. Performance and Payment Bonds: A performance bond is a surety bond issued by an insurance company or bank to guarantee satisfactory completion of a project. The Payment Bond guarantees the Contractor will pay the labor and material costs incurred. Banks and Insurance companies charge a premium for individual project based on a sliding scale related to the size of the project. Include as a separate line item in modification proposals and provide current company bonding rates upon request.
- H. Builder's Risk Insurance: Covers the contractor's loss due to fire, high winds, or other natural forces. Not reimbursed by the National Park Service (NPS) and shall not be included in modification proposals.

1.3 MODIFICATION PROPOSAL PRICING REQUIREMENTS

A. General:

- 1. Proposal be received in the format and within the time frame specified in the Request for Proposal (RFP) letter. Costs or delays resulting from failure of contractor to submit within the time frame specified will not be compensable.
- 2. Proposal shall be detailed with itemized lists of equipment, materials, labor, production rates, overhead, profit, and bond markup for each item. Labor costs must be itemized by craft and hourly rate, including Fringe Benefits and Labor Burden. If the costs of Fringe Benefits and Labor Burden are not itemized, it is assumed they are included in the hourly rate shown, or contractor is not requesting reimbursement. Contractor may utilize the government provided Contractor Estimate Form, or their own form, provided that it contains the same information and level of detail as the Government's form.

3. Requests for extensions of contract time as a result of change must be justified with a Time Impact Analysis (TIA). Refer to Section 01 32 16 "Construction Schedule", for time impact analysis requirements. TIA and associated costs shall be received with the proposal by the date shown within the Request for Proposal letter. Contractor's failure to submit within the specified time frame will be construed as the Contractor waiving right for additional time and no time extension will be allowed.
4. All supporting documentation used to justify the proposed modification will be made available to the Contracting Officer (CO) upon request.
5. Contractor shall review and approve all subcontractor/supplier pricing in detail for proper format, scope, production rates, and pricing prior to submission to NPS. All delay costs associated with not reviewing and approving subcontractor/supplier pricing will be borne by the Contractor.
6. All pricing and production rates within the estimate must be based on fair and reasonable pricing and cannot include built-in contingency.

B. Labor:

1. Contractor shall estimate cost of labor by itemizing each craft involved, indicating worker hourly rate (base rate + labor burden + fringe benefits) for each and itemizing hours required for each craft directly engaged in modification work. Any work proposed requiring overtime work or premium pay shall be itemized separately. Rates shall be in accordance with the Davis-Bacon Act as incorporated herein. Labor Burden may include payroll taxes, Social Security, unemployment insurances, workers compensation insurance, Federal Insurance Contributions Act (FICA), FUTA, and other direct costs resulting from Federal, State or local laws.
2. Itemize labor costs for equipment operators separate from equipment costs.
3. Labor cost for foremen shall only be costs for related work required for the modification.

C. Materials:

1. Estimated cost for materials shall include quotes from multiple sources. Material prices shall include applicable fees and credits, including but not limited to, sales tax, freight and delivery charges, and tax rebates.
2. No markup shall be applied to any material provided by NPS.

D. Equipment:

1. Equipment used for the project must be appropriately sized for work being performed.
2. Do not include costs for "miscellaneous tools and equipment", in your proposal for a replacement value of \$500 or less. Costs shown in excess of \$500 shall be broken out separately.
3. Regardless of ownership, rates to be used in determining equipment rental costs shall be the lowest cost from one of the following sources:
 - a. United States (U.S.) Army Corps of Engineers, Ownership and Operating Expense Schedule (use latest edition and applicable region)
 - b. Construction Blue Book
 - c. Local equipment rental rates, documented by actual invoice charges, or itemized vendor quotes.

4. Estimated equipment rates shall include operating costs of all fuel, oil, lubrication, supplies, small tools, necessary attachments, ground engaging components, tires and tracks, routine repairs and maintenance (cost of major repair and overhaul is not allowed per Federal Acquisition Regulation (FAR) 31.105(d)(2)), depreciation, storage, insurance, and all incidentals. Mobilization, if applicable, may be included for equipment solely used on the modification work but must be listed separately.
5. Estimate full rate for equipment only for duration that equipment will be utilized to accomplish work of the modification.
6. Standby unit rates used in accordance with paragraph 1.3, D, 2, above. If the U.S. Army Corp of Engineers is utilized then their standby rates prevail. If Bluebook or local equipment pricing is accepted, then 1/2 of equipment costs minus any operating costs, major repair and overhaul will be accepted.
7. If equipment is in standby mode due solely to a documented NPS delay, established standby rate shall apply from the first day of the delay.
8. Equipment not used and on job site for up to five consecutive days may be classified at standby rates, provided the equipment is or has been used solely to perform work on the modification and will be necessary to complete additional modification work. Equipment still on the jobsite but not in use after five consecutive days will not be considered in the modification pricing.
9. Requests for compensation for equipment stand by time must be justified, documented and itemized separately.
10. The estimated timeframe (daily, weekly, monthly) for use of the equipment must reflect the lowest cost to the Government.

E. Establishment and Application of Overhead and Profit Percentages:

1. Home Office Overhead and Profit (OH&P) shall be applied to direct costs only. Profit shall not be applied to overhead amounts; and overhead shall not be applied to profit. Home office overhead shall contain only allowable, allocable, and reasonable costs per the contract documents and FAR Part 31. Profit percentages are based on risk factors found in FAR Part 31 which have been applied to the specific type of work included in this project. Negotiated rates shall not exceed the following percentages for OH&P for contractor self-performed work:

Overhead.....7%

Profit.....5%
2. Total aggregate limit of markup (OH&P) for Contractor and Subcontractors on modification work shall not exceed 25%. The NPS will not be responsible for allocation of percentages between contractor and subcontractors at any tier.
3. If Contractors form a partnership, partnership may only receive home office overhead and profit in same amount as an individual Contractor (refer to paragraph 1.3,E,1 above). It is the responsibility of the partners to decide on division of revenue.
4. Combined Increases and Decreases: On proposals involving both increases and decreases in the Contract Price, overhead and profit mark-ups are required on net increases and deducted on net decreases.
5. At no time can profit be calculated on Overhead or itself, it must be calculated on direct costs of work only.

PART 2 - PRODUCTS

PART 3 - EXECUTION

END OF SECTION 012601

SECTION 01 27 00 – DEFINITION OF CONTRACT LINE ITEMS

PART 1 – GENERAL

1.1 SUMMARY

- A. This Section defines and clarifies the scope and limits of each Contract Line Items (CLINs) for the project.
- B. If specific line item is not identified for certain portions of work, include the associated costs under the most closely related CLIN.
- C. Each contract line item includes all work, materials, labor, equipment, permits, inspections, testing, documentation, restoration, and incidental activities necessary to deliver a complete and functional system as described in the Contract Documents.
- D. All work required to comply with Section 01 11 00, Summary of Work, including third-party inspections of pumps, pump controls, and if applicable, septic system components, is included in the appropriate CLIN and is not separately payable.
- E. No separate payment will be made for work required by the Contract Documents that is not explicitly listed as a separate CLIN. All such work shall be considered incidental to the appropriate CLIN.

1.2 COORDINATION WITH SUMMARY OF WORK

A. All contract line items include and are subject to the requirements described in Section 01 11 00 – Summary of Work, including but not limited to:

- 1. Environmental and seasonal restrictions (e.g., box turtle, bat, and migratory bird protections).
- 2. Utility coordination, locating, and protection, including DIGSAFE and GPR surveys.
- 3. Site access, public and emergency traffic maintenance, and safety measures.
- 4. Compliance with all applicable federal, state, and local regulations.
- 5. Preparation, submittal, and approval of as-built drawings and permit documentation.
- 6. All requirements described in Section 01 11 00 that are not specifically listed within a CLIN shall be considered incidental to that CLIN and included in the associated payment.
- 7. Refer to Section 01 11 00 for additional requirements applicable to all CLINs.

PART 2 – PRODUCTS

(Not Used)

PART 3 – EXECUTION

3.0 GENERAL

A. Reuse of the existing force main (approximately 300 LF) is assumed under the Base Bid. No cleaning, inspection, repair, or replacement is included unless specifically identified as a Contract Line Item.

B. The Government reserves the right to exercise any or all options.

3.1 LIST OF CONTRACT LINE ITEMS

A. Contract Line-Item Number 0001 – Base Bid – Province Land Visitor Center (PLVC) (171 Racepoint Road, Provincetown MA)

1. Provide all supervision, labor, materials, equipment, permits, inspections, testing, documentation, and incidental work necessary to furnish, install, and deliver a complete, operable septic system for the PLVC Property, including:
 - a. Obtaining all required permits and approvals from MassDEP, City of Provincetown, and the National Park Service.
 - b. Removing, pumping, and abandoning the existing septic system, including removal and proper disposal of all sewage and effluent from existing septic tanks, seepage pits, and associated piping at an authorized facility, and submission of disposal documentation. All requirements of Section 01 11 00, Paragraph 1.2.B.3 regarding pumping, removal, and disposal of sewage and effluent are included in this CLIN.
 - c. Excavation, trenching, backfilling with clean sand, compaction, grading, and surface restoration.
 - d. Installation of new septic tanks, distribution piping, leach field, low-pressure effluent pump assembly, control/alarm panel, quick disconnects, and all related appurtenances.
 - e. Coordinate all required electrical service disconnections/reconnections with Eversource, including preparation, submission, and management of all work orders; payment of all associated fees, standby costs, and utility charges; and full scheduling coordination. All such costs are incidental to this CLIN.
 - f. Electrical work from the existing panel LB-1 service, including conduit, wiring, NEMA 4X disconnects, breakers, and shutdowns in accordance with NEC and Massachusetts State Electrical Code.
 - g. Testing, inspection, and as-built documentation for system approval.
 - h. Compliance with all environmental restrictions, erosion control, and restoration requirements.
2. Payment Basis: Percentage of completion per approved schedule of values.
3. Includes all costs for supervision, labor, equipment, materials, permits, inspections, testing, and restoration. No separate payment will be made for incidental, administrative, or regulatory compliance activities.

B. Contract Line-Item Number 0002 – Option 1 – Replace SPVC Ejector Pump

1. Provide all work necessary to remove existing comfort station electrical panel, pump control panel, manhole-mounted ejector pumps and rails, existing wiring, and raceway. Furnish, install, and deliver a complete, operable duplex ejector pump system consisting of:
 - a. New freestanding pump control panel, duplex ejector pumps, pump rails, floats, and controls.
 - b. New raceway, direct-buried ducts, and conductors.
 - c. Obtaining required Eversource work order and approval for disconnect/reconnect.
 - d. Excavation, trenching, backfilling, grading, and surface restoration.
 - e. Testing, inspection, training, and as-built documentation for system approval.
 - f. Compliance with all seasonal and environmental restrictions.
 - g. All required third-party inspections of the duplex ejector pumps, pump controls, and free-standing pump control panels, as described in Section 01 11 00, are the sole responsibility of the Contractor. No backfill, concealment or system start-up may occur until inspections are complete and accepted by the CO/COR. Inspection reports shall be submitted to the CO/COR documenting deficiencies, required corrective actions, and verification of corrections completed.

2. Payment Basis: Percentage of completion per approved schedule of values.
3. Includes all costs as described above for the Base Bid, including incidental and regulatory-compliance activities. Each contract line item includes all work necessary to deliver a complete and functional system as described in the Contract Documents, including all incidental work required by Section 01 11 00 whether explicitly listed in the CLIN or not.

C. Contract Line-Item Number 0003 – Option 2 – New Planting Plan

1. Provide plantings as shown on Drawings LP101 and LP510 and per specifications. Work includes soil preparation, planting, mulching, staking, and maintenance until final acceptance.
2. Payment Basis: Percentage of completion per approved schedule of values.
3. The contract price for this item includes all supervision, labor, materials, equipment, permits, fees, inspections, testing, coordination, documentation, traffic control, and restoration. No separate payment will be made for incidental or regulatory-compliance activities.

D. Contract Line-Item Number 0004 – Option 3 – Temporary Sanitary Facilities

1. Provide and maintain Temporary Sanitary Facilities consisting of the following: one (1) ADA-accessible portable toilet and three (3) standard portable toilets. The temporary sanitary facilities shall be provided for public and government use during the period in which the permanent septic system is out of service due to construction activities. Work shall include pumping, restocking of toilet paper and sanitizer, and cleaning a minimum of three (3) times per week in accordance with Section 01 51 29, Temporary Sanitary Facilities. Duration of use shall be as required to support the Contractor's construction schedule and sequencing.
2. Payment Basis: Weekly, in arrears, upon successful completion of each week of required services. No partial or daily payments will be made.
3. Payment shall be made on a monthly construction requisition for the total number of full weeks of satisfactory service performed during the preceding month.
4. Includes all labor, materials, maintenance, pumping, cleaning, restocking, and coordination necessary to fulfill this requirement. No separate payment will be made for incidental or compliance activities.

E. Contract Line-Item Number 0005 – Option 4 – Jetting and Cleaning of Existing Force Main

1. This item is optional and will only be performed if authorized by the Contracting Officer. The existence of this CLIN does not imply that such work is required for successful system operation, nor does it relieve the Contractor of responsibility to deliver a fully functional system under other CLINs.
2. Provide all labor, equipment, and materials necessary to clean the existing force main between the lift station and septic tank (approximately 300 LF) using high-pressure water jetting. Work includes:
 - a. Mobilization of jetting equipment suitable for sanitary force mains
 - b. Cleaning and removal of grease, sediment, debris, and obstructions within the pipe
 - c. Flushing and clearing the full length of the force main
 - d. Collection, handling, and proper disposal of all removed debris in accordance with applicable federal, state, and local regulations
 - e. Coordination with the COR for access to lift station and downstream structures

3. Contractor shall take precautions to prevent damage to the existing pipe during cleaning operations.
4. This item is intended for cleaning and obstruction removal only. No repair, replacement, or reconstruction of piping is included under this CLIN. Cleaning and jetting does not constitute a condition assessment and does not guarantee the structural integrity or long-term suitability of the force main. No repair or replacement is included unless separately authorized.
5. Payment Basis: Lump sum.
6. Includes all supervision, labor, equipment, materials, disposal, and incidental work required to complete the cleaning operation. No separate payment will be made for mobilization, disposal, or ancillary activities. Claims for additional compensation based on omission of work from a specific CLIN will not be considered where such work is reasonably inferable from the Contract Documents.

F. Contract Line-Item Number 0006 – Option 5 – CCTV Inspection of Existing Force Main

1. This item is optional and will only be performed if authorized by the Contracting Officer. The existence of this CLIN does not imply that such work is required for successful system operation, nor does it relieve the Contractor of responsibility to deliver a fully functional system under other CLINs.
2. Provide all labor, equipment, and materials necessary to perform a closed-circuit television (CCTV) inspection of the existing force main between the lift station and septic tank (approximately 300 LF).
3. Work includes:
 - a. Pre-inspection coordination and verification that the line is suitable for CCTV inspection (cleaning shall be completed prior to inspection if necessary)
 - b. CCTV inspection of the full length of the pipe using equipment suitable for pressurized or previously pressurized sanitary lines
 - c. Continuous video recording of the inspection
 - d. Identification and documentation of defects, including but not limited to cracks, deformation, joint separation, root intrusion, blockages, and evidence of collapse
 - e. Measurement and logging of pipe length, locations of observed defects, and relative position from known structures
 - f. Delivery of a written inspection report summarizing findings and assessing overall condition of the pipe
 - g. Submission of video files in digital format (USB or electronic upload) to the CO/COR
4. The inspection report shall clearly indicate whether the pipe is suitable for continued use, requires repair, or shows evidence of structural failure.
5. This item is for inspection and documentation only. No repair, replacement, or corrective work is included under this CLIN. The CCTV inspection does not guarantee the suitability of the force main for continued use. No repair or replacement is included unless separately authorized.
6. Payment Basis: Lump sum.
7. Includes all supervision, labor, equipment, reporting, and incidental work required to complete the inspection and documentation. No separate payment will be made for mobilization or reporting.

G. Contract Line-Item Number 0007 – Option 6 – Exterior Force Main Excavation and Check Valve Replacement

1. This item is optional and will only be performed if authorized by the Contracting Officer. The Contract Documents are based on the assumption that the existing force main and associated components remain in serviceable condition. However, the Government has identified potential uncertainty in the condition of these components due to recent operational issues. This CLIN provides a mechanism to address localized excavation and replacement of the check valve and valve box if conditions encountered during construction indicate such work is necessary. The existence of this CLIN does not relieve the Contractor of responsibility to deliver a fully functional system under other CLINs.
2. Provide all labor, equipment, and materials necessary to excavate at the exterior of Pump Station no. 2, locate the force main, remove and replace the existing valve box and check valve.
3. Work includes:
 - a. Field verification and location of the force main and existing valve assembly.
 - b. Excavation adjacent to the lift station manhole, including removal of soil, walkway, fencing or other surface features as required.
 - c. Removal and disposal of the existing valve box, check valve, and associated piping and appurtenances.
 - d. Furnishing and installation of a new check valve.
 - e. Furnishing and installation of a new valve box with finish grade access.
 - f. Connection to existing force main, including fittings, couplings and restraints as required.
 - g. Backfill, compaction and surface restoration to match existing conditions.
 - h. Bypass pumping and temporary flow control, if required to maintain system operation
4. No condition assessment or system certification is included under this CLIN beyond verification of proper installation and operation of the replaced components.
5. Payment Basis: This item is a not to exceed allowance of \$30,000. Payment will be made based on actual work performed, as directed by the Contracting Officer, and supported by itemized cost documentation including labor, equipment, and materials. Work shall be performed on a time-and-materials basis consistent with contract provisions. The total amount paid shall not exceed \$30,000 without formal contract modification. Any unused portion shall be credited to the Government.
6. Includes all supervision, labor, equipment, materials, disposal, and incidental work required to complete the exaction at the exterior manhole containing pump station no. 2, the replacement of the check valve and valve box, back fill and restoration of the site. No separate payment will be made for mobilization, disposal, or ancillary activities. Claims for additional compensation based on omission of work from a specific CLIN will not be considered where such work is reasonably inferable from the Contract Documents.

END OF SECTION 01 27 00

SECTION 01 31 00 - PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following:

1. Definitions
2. Construction Coordination
3. Submittals
4. Coordination Drawings
5. Requests for Information (RFIs)
6. Project Meetings
7. Environmental Coordination
8. Permits

- B. Related Requirements:

1. Section 01 32 16 "Construction Schedule" for preparing and submitting Contractor's construction schedule.
2. Section 01 73 40 "Execution" for procedures for coordinating general installation and field-engineering services, including establishment of benchmarks and control points.
3. Section 01 77 00 "Closeout Procedures" for coordinating closeout of the Contract.

- C. Coordination with Summary of Work and Contract Line Items

All project management, coordination, submittal, and permitting requirements described in this section are included in the applicable contract line items defined in Section 01 27 00 and shall be performed in accordance with Section 01 11 00 – Summary of Work. The contractor shall perform all administrative, environmental, and regulatory coordination tasks necessary to comply with the requirements of the Summary of Work. No separate payment will be made for these activities unless specifically identified as a separate contract line item.

1.2 DEFINITIONS

- A. [Agency with Jurisdiction](#)
B. [Construction Permits – Contractor Provided](#)
C. [Government Furnished Permits](#)

1.3 CONSTRUCTION COORDINATION

- A. Coordination: Coordinate construction operations included in different Sections of Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations, included in different Sections, which depend on each other for proper installation, connection, and operation.

1. Schedule construction operations in sequence required to obtain best results where

installation of one part of the Work depends on installation of other components, before or after its own installation.

2. Coordinate installation of different components with other Contractors to ensure maximum accessibility for required maintenance, service, and repair.
 3. Make provisions to accommodate items scheduled for later installation.
 4. Where availability of space is limited, coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair of components, including mechanical and electrical.
 5. Properly plan construction operations to include permit requirements. Allow enough time to execute permit provisions to maintain work schedule, site visits, inspections, and reporting deadlines.
- B. Prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.
- C. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities and activities of other contractors to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to:
1. Preparation of Contractor's Construction Schedule
 2. Preparation of the Schedule of Values
 3. Installation and removal of temporary facilities and controls
 4. Delivery and processing of submittals
 5. Progress meetings
 6. Permit requirements
 7. Pre-installation conferences
 8. Project closeout activities
 9. Commissioning activities

1.4 SUBMITTALS

- A. Division 1 documents:
1. Letter designating Project Superintendent
 2. Construction Schedule
 3. A Comprehensive Schedule of Values
 4. Accident Prevention Plan
 5. A List of Subcontractors for this project
 6. Written statements from Subcontractors certifying compliance with applicable labor standard clauses.
 7. Certificates of Insurance and Standard Form SF1413 for Contractor and all Subcontractors
 8. Waste Management Plan
 9. Quality Control Plan
 10. Temporary Storm Water Pollution Prevention Plan (SWPP or UPPP)
- B. Provide items listed to Contracting Officer at least one week prior to Pre-Construction Conference, or as directed by Contracting Officer.
1. Failure to submit items may result in:
 - a. Construction conference may be cancelled, Notice to Proceed may not be issued, and Contracting Officer will consider other contractual remedies.

- b. Work shall not commence until written Notice to Proceed has been issued and Certificates of Insurance and SF1413 documents are received for Contractor and all Subcontractors.

1.5 REQUESTS FOR INFORMATION (RFIs)

- A. General: Immediately on discovery of the need for additional information or interpretation of Contract Documents, Contractor shall prepare and submit an RFI through email.
 1. Contracting Officer will not respond to RFIs submitted by other entities controlled by Contractor.
 2. Coordinate and submit RFIs in a prompt manner to avoid delays in the work.
- B. Content of RFI: Include detailed, legible description of item needing information or interpretation and the following:
 1. RFI number, numbered sequentially
 2. Date
 3. RFI subject
 4. Specification Section number and title and related paragraphs, as appropriate.
 5. Drawing number and detail references, as appropriate.
 6. Field dimensions and conditions, as appropriate.
 7. Contractor's suggested resolution: If Contractor's suggested resolution impacts Contract Time or Contract Sum, Contractor shall state impact in RFI.
 8. Contractor's signature
 9. Requested date for response
 10. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.
 - a. Include dimensions, thicknesses, structural grid references, and details of affected materials, assemblies, and attachments on attached sketches.

1.6 PROJECT MEETINGS

- A. Preconstruction Conference: Before start of construction, Contracting Officer will arrange an on-site meeting with Contractor. Meeting agenda will include the following as a minimum:
 1. Roles & Responsibilities / Lines of Authority
 2. Park rules and regulations
 3. Jobsite Safety
 4. Resolution of comments on required Division 1 documents
 5. Coordination of Subcontractors
 6. Labor law application
 7. Modifications
 8. Payments to Contractor
 9. Payroll reports
 10. Contract time
 11. Liquidated damages
 12. Contractor Performance Evaluation
 13. Display of Hotline posters

14. Notice to proceed
15. Correspondence procedures
16. Acceptance/rejection of work
17. Progress meetings
18. Submittal procedures
19. NPS Final Accessibility Inspection
20. Environmental requirements
21. Permit requirements
22. As-constructed drawings/operation and maintenance (O&M) manuals.
23. Saturday, Sunday, holiday and night work.
24. Reference materials
25. Value engineering
26. Schedule of Values

B. Progress Meetings: Contracting Officer will schedule weekly meetings with Contractor. All project meetings and sequencing must comply with access and public use requirements defined in Section 01 11 00.

1. Attendees: In addition to Government Representatives, each Contractor, Subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented. Participants at meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
2. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Meeting agenda will include:
 - a. Approval of minutes of previous meetings
 - b. Submittal status
 - c. Review of off-site fabrication and delivery schedules.
 - d. Requests for information (RFI) and other issues.
 - e. Modifications
 - f. Work in progress and projected.
 - 1) Status of required inspections (Special Inspections, Accessibility, etc.)
 - g. Inspections of work in progress and projected (Special inspections, Accessibility, etc.)
 - h. Construction Schedule update (provide updated Critical Path Method (CPM)).
 - i. Status of Project Record Drawings and O&M manuals.
 - j. Other business relating to work.
 - k. Permit requirements

C. Preinstallation Conferences: Conduct at Project site before each construction activity that requires coordination with other construction.

1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by installation and its coordination or integration with other materials and installations that have preceded or will follow, shall attend meeting. Advise Contracting Officer of scheduled meeting dates.
2. Agenda: Review progress of other construction activities and preparations for particular activity under consideration, including requirements for:
 - a. Contract Documents
 - b. Options

- c. Related RFIs
 - d. Related Change Orders
 - e. Purchases
 - f. Deliveries
 - g. Submittals
 - h. Review of mockups
 - i. Possible conflicts
 - j. Compatibility requirements
 - k. Time schedules
 - l. Weather limitations
 - m. Manufacturer's written instructions
 - n. Warranty requirements
 - o. Compatibility of materials
 - p. Acceptability of substrates
 - q. Temporary facilities and controls
 - r. Space and access limitations
 - s. Regulations of agency(ies) with jurisdiction
 - t. Testing and inspecting requirements
 - u. Installation procedures
 - v. Coordination with other work
 - w. Required performance results
 - x. Protection of adjacent work
 - y. Protection of construction and personnel
- 3. Record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.
 - 4. Reporting: Distribute minutes of meeting to each party present and to other parties requiring information.
 - 5. Do not proceed with installation if conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to performance of Work and reconvene conference at earliest feasible date.

1.7 ENVIRONMENTAL COORDINATION

- A. Contractor's Environmental Manager: Designate on-site party responsible for overseeing Contractor's conformance to environmental goals for project and implementing procedures for environmental protection.
 - 1. Qualifications: Minimum 3 years Construction experience on projects of similar size and scope; with environmental procedures similar to this project and t familiar with environmental regulations applicable to construction operations.
 - 2. Responsibilities: Responsibilities shall include:
 - a. Compliance with applicable Federal, State, and local environmental regulations, including maintaining required documentation.
 - b. Implementation of Waste Management Plan (WMP).
 - c. Present overview of environmental issues and summarize site specific procedures relating to management plans at Preconstruction conference.
 - d. Training for Contractor personnel in accordance with position requirements.
 - e. Monitoring and documentation of environmental procedures.
- B. Perform project quality control in accordance with requirements specified in Related Sections,

including:

1. Quality Requirements
2. Regulatory Requirements
3. Coordination for approvals with Mass DEP
4. Noise and Acoustics Management
5. Temporary Storm Water Pollution Prevention Environmental Management
6. Construction Waste Management

- C. Contractor's Environmental Training Program: Contractor shall provide environmental training for workers performing work on project site. Training shall include:

1. Overview of environmental issues related to Project.
2. Review of site-specific procedures and management plans:
 - a. Construction Waste Management
 - b. Coordination for approvals with Mass DEP
 - c. Noise and Acoustics Management
 - d. Temporary Storm Water Pollution Prevention
3. Compliance with environmental regulations: As specified in Regulatory Requirements. Submit Contractor 40 CFR (Code of Federal Regulations) employee training records upon request of Contracting Officer.

- D. Provide documentation for environmental procedures as specified herein and in accordance with approved Waste Management Plan.

1.8 PERMITS

- A. General:

1. Permits and Responsibilities: Contractor shall, without additional expense to the Government, be responsible for obtaining necessary licenses and permits, and for complying with Federal, State and municipal laws, codes, and regulations applicable to the performance of the work. Contractor shall also be responsible for damages to persons or property that occur as a result of Contractor's fault or negligence; and for materials delivered and work performed until completion and acceptance of the work.
2. For the purpose of this contract, Contractor will not be considered an agent of the Government. Contractor shall comply with appropriate Federal, State and local laws.

- B. Potential Permits: Permits listed below were identified during the design process as likely to be required based on typical means and methods of construction. The list is provided to assist Contractor in determining which permits will be required for contract's chosen means and

methods. The list shall not be considered complete; it is the Contractors' responsibility to determine means and methods and obtain required permits. Contractor shall obtain all permits and pay the associated fees required to legally conduct work as follows:

1. Construction trailer, Utility Company hookup permit
2. Sanitarian or Engineer Licensed in the Commonwealth of Massachusetts, Provincetown Board of Health, and Mass DEP for required inspections and approvals (See MassDEP System Approval Letter, 25-WR63-0002-APP)
3. Obtain Disposal System Construction Permit Provincetown Board of Health (Base).
4. Contractor is required to provide a site visit for Sanitarian or Engineer Licensed in the Commonwealth of Massachusetts for the following: Inspection and report for the bottom of bed excavation milestone to verify the elevation of the leaching field and inspection and report for six (6) inch crushed stone base for the septic tank and distribution box; inspection and report for installation of the soil absorption system; inspect and report prior to backfill of the distribution box and tank and an inspection and operational test of the entire system must be performed prior to the system being put on-line and prior to backfilling.
5. Pay required site visit and fees for Sanitarian or Engineer Licensed in the Commonwealth of Massachusetts to witness the clear water test.
6. Pay required site visits and fees for a Sanitarian or Engineer Licensed in the Commonwealth of Massachusetts to provide written certification that the system was constructed in accordance with the submitted design.
7. Within 30 days of the clear water test, provide as built plans to the Mass DEP.

C. Coordination with Agency(ies) with Jurisdiction Issuing Permits

1. Coordination: Contact the Agency(ies) with Jurisdiction as needed and sufficiently in advance to avoid delaying work: Coordinate meetings, reporting requirements, inspections, and other requirements.

D. Administrative Procedures:

1. Coordinate scheduling and timing of required administrative provisions of project permits with Agency(ies) with Jurisdiction, Construction Manager, and Park to avoid conflicts.
2. Supply needed information to Agency(ies) with Jurisdiction issuing permits, pay fees required and provide material needed to comply with permit's conditions and provisions.
3. Email permits to NPS Cape Cod National Seashore (CACO) management when permits are obtained.

1.9 Administrative and Permit Activities

All submittals, meetings, inspections, reporting, environmental coordination, and permit-related activities required under this section, or the Summary of Work, Section 01 11 00 are considered incidental to the relevant contract line items in Section 01 27 00. The contractor shall include all costs associated with these activities in the prices for the applicable line items. No separate payment will be made for administrative, permit, or coordination tasks unless specifically defined as a separate line item.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 31 00

SECTION 01 32 16 – CONSTRUCTION SCHEDULE

PART 1 - GENERAL

1.1 SUMMARY

- A. Section consists of Construction Schedule requirements including:
 - 1. Schedule of Values
 - 2. Construction Schedule Requirements.
 - 3. Construction Schedule Updates.
 - 4. Time Impact Analysis.
- B. Purpose: The Construction Schedule ensures adequate planning, coordination, scheduling, and reporting during execution of the work by the Construction Contractor. It shall assist the Construction Contractor and Contracting Officer (CO) in monitoring the progress of the work, evaluating proposed changes, and processing Construction Contractor's monthly progress payments. It shall include the dates in the contract, phases, milestones, occupancies, holidays, weather consideration, a critical path, and the requirements of this section.

1.2 DEFINITIONS

- A. Activity: A discrete part of a project identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources.
 - 1. Critical activities are activities on the critical path. They start and finish on the planned early start and finish times.
 - 2. Predecessor Activity: An activity that precedes another activity in the network.
 - 3. Successor Activity: An activity that follows another activity in the network.
- B. Cost Loading: Allocation of the Schedule of Values for completion of an activity as scheduled. The sum of costs for all activities must equal the total Contract Sum, unless otherwise approved by the Contracting Officer.
- C. Critical Path Method (CPM): Method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.
- D. Critical Path: Longest connected chain of interdependent activities through the network schedule that establishes minimum overall Project duration and contains no float.
- E. Float: Measure of leeway in starting and completing an activity.
 - 1. Float: Not for the exclusive use or benefit of the Government or Construction Contractor but is jointly owned.
 - 2. Free Float: Amount of time an activity can be delayed without adversely affecting the early start of the successor activity.

3. Total Float: Measure of leeway in starting or completing an activity without adversely affecting planned Project completion date.
- F. Resource Loading: Allocation of manpower and equipment necessary for completion of an activity as scheduled.
 - G. Fragnet: A partial or fragmentary network that breaks down activities into smaller activities for greater detail.

1.3 SUBMITTALS

- A. Electronic Copies: Schedules and reports shall be submitted in PDF (Portable Document Format) and posted to the Contractor's project management and communication platform. The Government's intent is to minimize printed reports, limiting hard copies to only those deemed essential by the project team.
- B. Schedule of Values: After contract award and before Pre-Construction conference, submit schedule of dollar values based on Contract Price Schedule.
- C. Construction Baseline Schedule: After contract award and before Pre-Construction conference, submit baseline schedule, large enough to show entire schedule for entire construction period. Utilize Schedule of Values in preparation of Construction Baseline Schedule.
- D. Critical Path Method (CPM) Reports: Concurrent with CPM schedule, submit computer-generated reports. For each activity, include activity number, activity description, cost and resource loading, original duration, remaining duration, early start date, early finish date, late start date, late finish date, and total float in calendar days.
 1. Activity Report: List of activities sorted by activity number and then early start date, or actual start date if known.
 2. Logic Report: List of predecessor and successor tasks for activities sorted in ascending order by activity number and then early start date, or actual start date if known.
 3. Total Float Report: List of activities sorted in ascending order of total float.
- E. Construction Schedule Updates: On or before 7th day preceding progress payment request date, submit estimates of percent completion of each schedule activity and necessary supporting data. Provide two paper copies.
- F. Construction Schedule Revisions and Time Impact Analysis: For each Construction Schedule revision, submit of a Time Impact Analysis. Incorporate a Fragmentary Network (Fragnet) into currently accepted Construction Schedule demonstrating proposal to incorporate a modification, change, delay, or Construction Contractor request.

1.4 QUALITY ASSURANCE

- A. Construction Contractor shall meet with Contracting Officer on day of the preconstruction conference to go over:
 1. Review software limitations, content and format for reports.

2. Verify availability of qualified personnel needed to develop and update schedule.
3. Discuss constraints, including phasing work stages area separations interim milestones and partial Government occupancy/substantial completions.
4. Review delivery dates for Government-furnished products.
5. Review schedule for work of separate Government contracts.
6. Review time required for review of submittals and re-submittals.
7. Review requirements for tests and inspections by independent testing and inspecting agencies.
8. Review time required for completion and startup procedures.
9. Review time required for obtaining and activating permits.
10. Review and finalize list of construction activities to be included in schedule.
11. Review baseline schedule comments, resolve issues and progress on incorporating them
12. Review procedures for updating schedule.
13. Discuss reporting requirements and establish protocol for naming and transmitting electronic schedules.

- B. Construction Contractor's Schedule Representative: Before the preconstruction conference, designate an authorized representative to be responsible for preparing and maintaining the Construction Schedule. Submit resume outlining qualifications of Scheduler to Contracting Officer for acceptance. Scheduler shall have prepared and maintained at least 5 previous schedules of similar size and complexity similar to this Contract, demonstrating proficiency of using scheduling software. Authorized representative will be responsible for preparing the Baseline Schedule, required updates, revisions, Time Impact Analyses, and reports.

1.5 COORDINATION

- A. Coordinate preparation and processing of schedules and reports with performance of construction activities and with scheduling and reporting of separate Construction Contractors.
- B. Coordinate Construction Baseline Schedule with Schedule of Values, list of subcontracts, Submittals Schedule, progress reports, payment requests, and other required schedules and reports.
 1. In developing Construction Baseline Schedule, ensure Subcontractor's work at all tiers, and prime Construction Contractor's work, is included and coordinated.
 2. Secure time commitments for performing critical elements of work from parties involved.
 3. Coordinate each construction activity in network with other activities and schedule in proper sequence.

PART 2 - PRODUCTS

2.1 SCHEDULE OF VALUES

- A. Breakdown each lump-sum item into component work activities used in the schedule for which progress payments may be requested. Work activities broken out within schedule of values shall be integrated into and made a logical part of the construction baseline schedule. Total costs for the component work activities shall equal contract price for that lump-sum item. Contracting Officer may request data to verify accuracy of dollar values. Include mobilization, general

condition costs, overhead and profit in the total dollar value of unit price items and in the component work activities for each lump-sum item. Do not include mobilization, general condition costs, overhead or profit as a separate item.

- B. Do not break down unit price items. Use only the contract price for unit price items.
- C. Total cost of all items shall equal the contract price. The Schedule of Values will form the basis for progress payments and the Construction Schedule.

2.2 CONSTRUCTION SCHEDULE REQUIREMENTS

- A. Construction Baseline Schedule: Prepare Construction Baseline Schedule using a computerized, cost and resource-based, time-scaled Critical Path Method network analysis diagram for the Work.
 - 1. Develop and finalize Construction Baseline Schedule so it can be accepted for use no later than **30** days after date established for the Notice of Award.
 - a. Failure to include any work item required for performance of this Contract shall not excuse Construction Contractor from completing work within applicable completion dates, regardless of Governments acceptance of schedule.
 - 2. Establish procedures for monitoring and updating Construction Baseline Schedule and for reporting progress. Coordinate procedures with progress meeting and payment request dates.
- B. Construction Baseline Schedule Preparation: Prepare a list of all activities required to complete the Work. Using preliminary Critical Path Method network diagram, prepare a skeleton network to identify probable critical paths.
 - 1. Activities: Indicate estimated duration, sequence requirements, and relationship of each activity in relation to other activities.
 - 2. Critical Path Activities: Identify critical path activities, including those for interim completion dates. Scheduled start and completion dates shall be consistent with Contract milestone dates.
 - 3. Processing: Process data to produce output data on a computer-drawn, time-scaled network. Revise data, reorganize activity sequences, and reproduce as often as necessary to produce the Critical Path Method schedule within the limitations of the Contract Time.
 - 4. Show sequence and interdependence of activities required for completion of work. Ensure work sequences are logical and Construction Baseline Schedule shows a coordinated plan of the work.
 - 5. Resource loading of each activity shall include personnel by labor category and equipment type and capacity proposed to complete the activity in duration shown.
 - 6. Consider seasonal weather conditions in planning and scheduling work influenced by high and low ambient temperatures, wind, or precipitation to ensure completion of work within contract time.
 - 7. Time Frame: Proposed duration assigned to each activity shall be Construction Contractor's best estimate of time required to complete activity considering the scope and resources planned for activity.

- a. An early finish date may be shown but the late finish date shall be same date as last day of contract period. An early completion schedule shall contain:
 - 1) Insert an activity titled "Project Float" as a successor to last activity in early project completion schedule network.
 - 2) Add a milestone titled "Contract End Date" as a successor to the activity "Project Float".
 - 3) Add duration to the activity "Project Float" as required so the milestone "Contract End Date" equals the last day of Contract Period.
- b. Contract completion date shall not be changed by submission of a schedule that shows an early completion date.
- c. Construction Contractor shall limit use of lead or lag duration's between schedule activities.
- d. Project Calendars: Develop and incorporate the following calendars:
 - 1) Administrative Calendar: Include calendar based on a 7-day week to be used on activities based on calendar days. Apply this calendar to administrative tasks or other tasks not affected by non-working days (Federal Holidays, weather, etc.).
 - 2) Project Calendar: Include calendar based on planned work week for the project. Include Federal Holidays, weekends, and non-workdays indicated in contract documents. Apply this calendar to activities not anticipated to be affected by weather. Be clear when identifying number of work days in work week.
 - 3) Weather Calendar: Utilize Project Calendar and show anticipated normal downtime related to weather as non-working time. Weather days shall be based on data for local area from a reliable source like the National Oceanic and Atmospheric Administration (NOAA), National Park Service records, or source acceptable to Contracting Officer. Apply this calendar to activities anticipated to be affected by weather.
- e. Activity Duration: Define so no activity is longer than **14** days, except for non-construction activities including mobilization, shop drawings and submittals, fabrication and delivery of materials and equipment.
- f. Procurement Activities: Include procurement process activities for long lead items and major items, requiring a cycle of more than 60 calendar days, as separate activities in the schedule. Procurement cycle activities can include submittals, approvals, purchasing, fabrication, and delivery.
 - 1) Review Time: Include review and re-submittal times indicated. Coordinate submittal review times in Construction Baseline Schedule.
- g. Startup and Testing Time: Include not less than 5 days for startup and testing-
- h. Substantial Completion: Allow time for Government administrative procedures necessary for certification of Substantial Completion. (For more information, refer to Specification 01 77 00 "Closeout Procedures.")
8. Constraints: Include constraints and work restrictions indicated in Contract Documents and as follows in schedule and show how the sequence of Work is affected.
 - a. Phasing: Arrange list of activities on schedule by phase.

- b. Work under More Than One Contract: Include a separate activity for each contract.
- c. Work Restrictions: Show effect of the following on the schedule:
 - 1) Coordination with existing construction
 - 2) Limitations of continued occupancies
 - 3) Uninterruptible services
 - 4) Partial occupancy before Substantial Completion
 - 5) Use of premises restrictions
 - 6) Provisions for future construction
 - 7) Seasonal variations
 - 8) Environmental control
 - 9) Permit provisions
- d. Work Stages: Indicate important stages of construction for each major portion of the Work.
 - 1) Subcontract awards
 - 2) Submittals
 - 3) Purchases
 - 4) Mockups
 - 5) Fabrication
 - 6) Sample testing
 - 7) Deliveries
 - 8) Installation
 - 9) Tests and inspections
 - 10) Adjusting
 - 11) Curing
 - 12) Building flush-out.
 - 13) Building commissioning activities.

C. Joint Review, Revision, and Acceptance:

- 1. Within seven calendar days of receiving Construction Contractor's proposed Construction Baseline Schedule, Contracting Officer shall review initial Construction Baseline Schedule.
- 2. Within seven calendar days after review, Construction Contractor shall revise and resubmit Construction Baseline Schedule in accordance with comments presented from review.
- 3. In the event the Construction Contractor fails to define any element of work, activity, or logic and the Contracting Officer review does not detect this omission or error, such omission or error, when discovered by Construction Contractor or Contracting Officer, shall be corrected by Construction Contractor within seven calendar days and shall not affect contract period.
- 4. Upon acceptance of the Construction Baseline Schedule, Contracting Officer saves schedule as a baseline and updates on a monthly basis. Construction schedule update will be used to evaluate Construction Contractor's monthly applications for payment based upon information developed at monthly Construction Schedule update meeting.

- D. Cost Correlation: In the heading of the schedule, provide a cost correlation line, indicating planned and actual costs. On the line, show dollar volume of Work performed as of dates used to prepare payment requests.

1. Construction Contractor shall assign cost to construction activities on Construction Baseline Schedule. Costs shall not be assigned to submittal activities unless specified otherwise but may, with Contracting Officer's approval, be assigned to fabrication and delivery activities. Costs shall be included for testing and commissioning activities, operation and maintenance manuals, punch list activities, Project Record Documents, and demonstration and training (if applicable).
 2. Each activity cost shall reflect an accurate value based on the Contract Price Schedule.
 3. Total cost assigned to activities shall equal total Contract Price.
- E. Recovery Schedule: When periodic schedule update indicates Work is **14** or more calendar days behind current accepted schedule, a separate recovery schedule indicating means by which Construction Contractor intends to regain compliance with the schedule shall also be submitted. Indicate changes to working hours, working days, crew sizes, and equipment required to achieve compliance, and date by which recovery shall be accomplished.
- F. Computer Software: Prepare schedules using a program developed specifically to manage construction schedules.
1. Use Microsoft Project

PART 3 - EXECUTION

3.1 CONSTRUCTION SCHEDULE UPDATES

- A. Progress Meeting Updates: Provide a **2** week look-ahead schedule, derived from the currently accepted schedule, before each weekly progress meeting. Utilize look-ahead schedule to facilitate and take notes on discussions held during progress meeting.
- B. Monthly Schedule Updates:
1. General: Update Construction Schedule on monthly basis to reflect construction progress and activities throughout entire contract period and until project substantial completion. The status date of each schedule update shall be the 7th day preceding the progress payment request date.
 2. Procedure: Construction Contractor shall meet with Contracting Officer each month at Construction Schedule update meeting to review progress made through the status date of the Construction Schedule update, including dates activities were started or completed and percentage of work completed on each activity started or completed.
 3. Reports: Concurrent schedule revisions, prepare tabulated reports showing:
 - a. Identification of activities that have changed
 - b. Changes in early and late start dates
 - c. Changes in early and late finish dates
 - d. Changes in activity durations in workdays
 - e. Changes in the critical path
 - f. Changes in total float or slack time
 - g. Changes in the Contract Time

4. Narrative: Report shall include a brief description of actual progress made during update period; actual and potential delaying activities; impediments to progress; issues related to inclement weather; progress toward established milestones and project float. Report shall include a brief description of work anticipated to be performed in the next month. Minor revisions to the schedule should be identified for evaluation and acceptance or rejection.
 5. As Work progresses, indicate Actual Completion percentage for each activity.
 6. If schedule update shows a late finish date after contract completion date, include:
 - a. Known delays
 - b. Actions to get back on schedule
 - c. Pending modifications
 - d. Impediments or constraints affecting progress
 7. Progress Payments: Monthly updating of the currently accepted Construction Schedule shall be an integral part of the process upon which progress payments will be made. If Construction Contractor fails to provide schedule updates or revisions, a portion of the monthly payment may be retained until corrections have been made.
- C. Distribution: Distribute copies of accepted schedule to Contracting Officer, Contracting Officers Representative, Construction Management Representative, Subcontractors, testing and inspecting agencies, and other parties identified by Construction Contractor with a need-to-know schedule responsibility.
1. Post copies in Project meeting rooms and temporary field offices.
 2. When revisions are made, distribute updated schedules to same parties and post in same locations. Delete parties from distribution when they have completed their assigned portion of the Work.
- D. Construction Schedule Revisions:
1. Required Revisions: If, as a result of the monthly schedule update, it appears the currently accepted Construction Schedule no longer represents actual prosecution and progress of the work, Contracting Officer will request, and Construction Contractor shall submit, a revision to the Construction Schedule. Construction Contractor may also request reasonable revisions to currently accepted Construction Schedule in event the Construction Contractor's planning for the work is revised. If Construction Contractor desires to make changes, Construction Contractor shall notify Contracting Officer in writing, stating reason for proposed revision. Accepted revisions shall be incorporated into currently accepted Construction Schedule for next monthly schedule update.
 2. Procedure: If revision to currently accepted Construction Schedule is contemplated, Construction Contractor or Contracting Officer shall advise the other in writing at least seven calendar days prior to next monthly schedule update meeting, describing revision and reasons for the revision. Government-requested revisions will be presented in writing to the Construction Contractor, who shall respond in writing within seven calendar days.
 3. Reports: Concurrent with making revisions to schedule, prepare tabulated reports showing:
 - a. Identification of activities changed
 - b. Changes in early and late start dates
 - c. Changes in early and late finish dates
 - d. Changes in activity durations in workdays
 - e. Changes in critical path

- f. Changes in total float or slack time

3.2 TIME IMPACT ANALYSIS FOR CONTRACT MODIFICATIONS CHANGES DELAYS AND CONSTRUCTION CONTRACTOR REQUESTS:

1. Requirements: When contract modifications or changes are initiated, delays experienced, or Construction Contractor desires to revise currently accepted Construction Schedule, Construction Contractor shall submit to Contracting Officer a written time impact analysis illustrating the influence of modification, change, delay, or Construction Contractor request on contract time.
2. Time Extensions: Activity delays, resulting in a late completion date projection, shall not automatically mean an extension of contract time is warranted or due to Construction Contractor. It is possible a modification, change, or delay will not affect existing critical path activities or cause non-critical activities to become critical. A modification, change, or delay may result in absorbing a part of available total float that may exist within an activity chain of the Schedule, not causing any effect on contract time. Time extensions will be granted in accordance with terms of contract.
3. Extension of contract time will be granted only to the extent the equitable time adjustments to activity or activities affected by modification, change, or delay exceeds total (positive or zero) float available on a particular activity.
4. Procedure: Each time impact analysis shall be submitted within time period stated in a request for proposal, or time period designated under the clauses entitled Changes or Default. In cases where Construction Contractor does not submit a written request for extension of time and a time impact analysis within the designated time, it is mutually agreed that the particular modification, change, delay, or Construction Contractor request does not require an extension of the contract time. Upon acceptance, time impact analysis shall be incorporated into currently accepted Construction Schedule at next monthly schedule update.
5. Contract Modifications: Prepare time-impact analysis using fragments to demonstrate effect of proposed change on overall Construction Schedule for each proposed contract modification concurrent with submission.

END OF SECTION 01 32 16

SECTION 01 32 33 – PHOTO DOCUMENTATION

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for:
 - 1. Existing condition images
 - 2. Periodic construction images
- B. See Section 01 77 00 "Closeout Procedures" for a complete listing of closeout documents.
- C. See Section 01 79 00 "Demonstration and Training" for submitting videotapes of demonstration of equipment and training of National Park Service (NPS) personnel.

1.2 SUBMITTALS

- A. Construction Images: Submit images electronically within seven days of taking the image. Include:
 - 1. Date, time and number (sequentially number all images) in filename.
 - 2. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction.
 - 3. Submit digital images exactly as originally recorded in digital camera, without alteration, or modifications using image-editing software.
- B. Closeout: Submit complete set of digital image electronic files as a Project Record Document. Submit on Digital Video Disc (DVD).
 - 1. Provide index as separate file on Disc. List each image as a file name with number, date, and time. Include description and or vantage point image was taken.
 - 2. Submit images that have the same aspect ratio as the sensor, un-cropped.

PART 2 - PRODUCTS

2.1 FORMAT REQUIREMENTS

- A. Media: DVD-R Archival Gold
- B. Media Labels: Archival DVD labeling markers, archival labels, or direct print.
- C. Images: Provide sRGB (standard Red Green Blue) color images in JPEG (Joint Photographic Experts Group) format. Minimum sensor size of 1200, and 300 dpi (dots per inch).

PART 3 - EXECUTION

3.1 CONSTRUCTION IMAGES

- A. General: Take digital images using the maximum range of depth of field, in-focus, to clearly show the Work. No blurry or out-of-focus areas accepted.
 - 1. Maintain index with each set of Construction images and identify the number, date, time, and description for each.
 - 2. Maintain one set of images accessible in field office at Project site available for reference.
- B. Existing Condition Images: Before commencement of demo or excavation. take color digital images of Project site and surrounding properties, including existing items to remain during construction, from different vantage points.
 - 1. Flag excavation areas construction limits before recording construction images.
 - 2. Take eight separate images to show existing conditions adjacent to property before starting Work.
 - 3. Take eight separate images of existing buildings either on or adjoining property to accurately record physical conditions at start of construction.
- C. Periodic Construction Images: Take 12 color, digital images weekly, with timing each month adjusted to coincide with cutoff date associated with each Application for Payment. Select vantage points to show status of construction and progress since last images were taken.
- D. Additional Images: Contracting Officer may issue requests for additional images.
 - 1. Three days advance, where feasible.
 - 2. In emergency situations, take additional images within 24 hours of request.
 - 3. Additional images include, but are not limited to:
 - a. Immediate follow-up when on-site events result in construction damage or losses.
 - b. Fabrication locations away from Project site.
 - c. Substantial Completion of a major phase or component of Work.
 - d. Extra record images at time of final acceptance.

END OF SECTION 01 32 33

SECTION 01 35 23 - SAFETY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes establishing an effective accident prevention program and providing a safe working environment for personnel and visitors.

1.2 CONDITIONS PRESENT FOR PROJECT

- A. Confined space work

1. The park and the local fire departments do not have resources to provide monitoring and stand by rescue for confined space. The contractor is required to provide third party confined space monitoring and standby rescue via a certified confined space standby rescue and monitoring firm.
2. The contractor shall include a site-specific confined space entry plan.

- B. Excavation in loose soils

3. Contractor to provided trench boxes to support excavation

- C. Electrical work

1. See requirement for Lock and Tag out (LOTO) section 2.4

1.3 SUBMITTALS

- A. Accident Prevention Plan (APP): Submit APP after contract award and before Pre-Construction conference. Contracting Officer (CO) will review proposed APP. If APP requires any revisions or corrections, Contractor shall resubmit Plan within 10 days. No progress payments will be made until the APP is accepted.
- B. LOTO acknowledgement

1.4 QUALITY ASSURANCE

- A. Comply with contract clauses "Accident Prevention" and "Permits and Responsibilities." In case of conflicts between Federal, State, and local safety and health requirements, the most stringent shall apply. Onsite equipment shall meet 29 CFR 1926 (Code of Federal Regulations) (Occupational Safety and Health Administration (OSHA)) requirements. Failure to comply with requirements of this section and related sections may result in suspension of work.

B. Site Safety Supervisor:

1. Designate authorized onsite representative for preparation and maintenance of the APP.
2. Shall be responsible for:
 - a. Implementation and enforcement of the APP
 - b. Daily safety inspections
 - c. Conducting and documenting weekly and monthly safety meetings
 - d. Review of safety requirements at progress meetings
 - e. Compilation and maintenance of Safety Data Sheets (SDS) and safety reference materials
 - f. Tracking and resolution of safety violations
 - g. Site personnel and visitor compliance with site safety and health requirements and APP
 - h. Investigation and reporting of accidents and injuries

C. Qualifications of Employees:

1. Physically and able to perform their assigned duties in a safe manner.
2. Do not allow employees whose ability or alertness is impaired because of prescription or illegal drug use, fatigue, illness, intoxication, or other conditions that may expose themselves or others to injury to perform work.
3. Provide operating instructions for equipment. Operators of vehicles, hoisting equipment, and hazardous plant equipment shall be able to understand signs, signals, operating instructions, and be fully capable of operating such equipment. Retain copies of operator licenses and certifications onsite.

1.5 ACCIDENT REPORTING

- A. Reportable Accidents: Defined as: death, occupational disease, and/or traumatic injury to employees or the public; fires; and/or property damage by accident in excess of \$100.
1. Notify Contracting Officer immediately in the event of a reportable accident.
 2. Fill out and forward an Accident/Property Damage Report Form (CM-22) to Contracting Officer within 7 days of a reportable accident. Obtain form from Contracting Officer.

1.6 RESOURCES

- A. COVID-19 (Coronavirus Disease 2019) information provided below is not intended to provide a complete analysis of requirements for Contractor and is provided as a courtesy.
1. [Coronoavirus.gov](https://www.cdc.gov/coronavirus)
 2. Occupational Safety and Health Administration (United States Department of Labor) - [COVID-19](#)
 3. Center for Disease Control (CDC)
 - a. [Get the Facts About Coronavirus](#)
 - b. [What Construction Workers Need to Know about COVID-19](#)
 4. Federal Emergency Management Agency (FEMA) - [Coronavirus \(COVID-19\) Response](#)
 5. National Park Service (NPS) - [NPS Public Health Update](#)

PART 2 - PRODUCTS

2.1 ACCIDENT PREVENTION PLAN (APP)

- A. APP shall be written to comply with OSHA and project requirements (generic plan is not acceptable) including but not limited to:
1. Name and qualifications of supervisor responsible to carry out program.
 2. Weekly and monthly safety meetings shall be documented with topics and attendees.
 3. First aid and rescue procedures.
 4. Job Hazard Analysis (JHA) for each major phase. List of hazards associated and methods proposed to provide for property protection and safety of the public, National Park Service personnel, and Contractor's employees. Include initial and continuing training.
 5. Planning for possible emergency situations, as detailed in Article 1.2. Such planning shall take nature of construction, site conditions, and degree of exposure of persons and property into consideration.
 6. Blasting Plan Requirements: See Section 01 35 25 "Explosives."
 7. Infectious Disease Preparedness:
 - a. Contractors are responsible for their employees' safety and the safety of job site visitors during the performance of this contract. We encourage Contractors to follow guidance from the Department of Labor (DOL), Occupational Safety and Health Administration (OSHA), the Centers for Disease Control and Prevention (CDC), and all other applicable local, city, and state mandates. We encourage Contractors to develop policies for infection prevention and an Infectious Disease Preparedness and Response Plan.
 - b. To the extent appropriate, Contractors should include the protective health and safety measures they intend to implement in any accident prevention or safety submittals required under this contract. These plans should contain preventive measures the Contractor intends to follow while performing work on government property as well as responsive and corrective actions to be taken if an employee exhibits symptoms or tests positive for contagion.
 - c. Upon contract award, Contractors should communicate with Contracting Officer regarding Contractor decisions and actions to protect the health and safety of workers for the duration of contract performance under which pandemic conditions exist.

2.2 FIRST AID FACILITIES

- A. Provide adequate facilities for number of employees and appropriate to construction hazards.

2.3 PERSONNEL PROTECTIVE EQUIPMENT (PPE)

- A. Selection shall conform to OSHA Subpart E.

2.4 Work on electrical systems

- A. The park has a lock out tag out policy. There is no work on energized equipment**
- B. "Lock-out Tag-out Acknowledge Form" must be filled out prior to proceeding with any type of electrical work. lock-out tag-out acknowledge form at the end of the spec section. Form included after this spec section.**

PART 3 - EXECUTION

3.1 DAILY SAFETY INSPECTIONS

- A. Conduct daily safety inspections and maintain daily safety reports which include:**
 - 1. Area/operation inspected**
 - 2. Date of inspection**
 - 3. Identified hazards**
 - 4. Corrective actions taken**

3.2 EMERGENCY INSTRUCTIONS

- A. Post telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and police in conspicuous locations at work site.**

3.3 FIRE AND LIFE SAFETY

- A. Comply with requirements of National Fire Protection Association (NFPA) 241 (Standard for Safeguarding Construction, Alteration, and Demolition Operations).**

3.4 HAZARDOUS MATERIALS

- A. Hazardous materials: Explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substances that could cause death or injury.**
- B. Store hazardous materials in accordance with manufacturer's and OSHA Subpart D requirements. Maintain Safety Data Sheets (SDS) for each chemical readily available on site.**
 - 1. Immediately report spills of hazardous materials to the Park.**
 - 2. Maintain a spill emergency response kit.**
 - 3. Train employees how to respond to a spill and use emergency response kit.**

3.5 PROTECTIVE EQUIPMENT

- A. Inspect personal protective equipment daily and maintain in a serviceable condition. Clean, sanitize, and repair personal items as appropriate before issuing to another individual.**

3.6 SAFETY MEETINGS

- A. As a minimum, conduct one weekly 15-minute "toolbox" safety meeting conducted by a foreman or supervisor and attended by construction personnel at worksite. Topics shall coincide with work scheduled for following week. Document and submit meeting minutes to Contracting Officer within one day after meeting.
- B. Conduct monthly safety meetings for personnel, contractors, and subcontractors performing work on the site. Notify Contracting Officer of meeting dates and times. Meetings shall be used to: review effectiveness of Contractor's safety effort; resolve current health and safety problems; provide a forum for planning safe construction activities, and for updating Accident Prevention Plan. Contracting Officers Representative will attend meetings and enter results of meetings into the daily log.

3.7 HARD HATS AND PROTECTIVE EQUIPMENT AREAS

- A. A hard hat use area shall be designated by Contractor. Hard hat area shall be posted by Contractor in a manner satisfactory to Contracting Officer.
- B. It is Contractor's responsibility to require persons working on or visiting site to wear hard hats and PPE in good repair at all times. As a minimum, maintain 6 hard hats and other APP required equipment.

3.8 TRAINING

- A. First Aid: Provide training to personnel to ensure prompt and efficient first aid.
- B. Hazardous Material: Train and instruct each employee exposed to hazardous material in safe and approved methods of handling and storage.

END OF SECTION 01 35 23

SECTION 01 33 23 - SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other submittals.

1.2 DEFINITIONS

- A. Action Submittals: Written, graphic information, and physical samples that require Government's responsive action.
- B. Informational Submittals: Written information that does not require Government's responsive action. Submittals may be rejected for not complying with requirements.
- C. File Transfer Protocol (FTP): Communications protocol that enables transfer of files to and from another computer over a network and that serves as the basis for standard Internet protocols. An FTP site is a portion of a network located outside of network firewalls within which internal and external users are able to access files.
- D. Portable Document Format (PDF): An open standard file format licensed by Adobe Systems used for representing documents in a device-independent and display resolution-independent fixed-layout document format.

1.3 GENERAL SUBMITTAL PROCEDURES

- A. General: Prepare and submit submittals required by individual Specification Sections and in some cases as requested in drawings. Types of submittals are indicated in individual specific sections.
 - 1. Contracting Officer (CO) reserves right to require submittals in addition to those called for in individual sections.
- B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities. Review for legibility, accuracy, completeness, and compliance with Contract Documents.
 - 1. Coordinate submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
 - 2. Coordinate transmittal of different types of submittals for related parts of Work so processing will not be delayed because of need for concurrent review coordination.
 - a. Contracting Officer reserves right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.

- C. Submittal List: Submittal list is attached to the end of this Specification Section. The intent is to provide an overall summary of submittal requirements. The requirements of individual Specification Sections and terms and conditions of the Contract still apply regardless of what is shown on submittal list.
- D. Processing Time: Allow time for submittal review, including time for re-submittals, as follows. Time for review shall commence when e-mail notification is received by Contracting Officer (or designee) indicating submittal has been posted on contractor's management software website and is ready for review. When Contracting Officer has completed review, e-mail notification will be sent to Contractor indicating submittal has been processed. No extension of Contract Time will be authorized because of failure to transmit submittals in advance of Work to permit processing, including re-submittals.
 - 1. Action Submittals
 - a. Initial Review: Allow 14 days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required.
 - b. Re-submittal Review: Allow 14 days for review of each re-submittal.
 - 2. Informational submittals
 - a. Review: Allow 14 days for review of each submittal.
- E. Approved Equals:
 - 1. For each item proposed as an "approved equal," submit supporting data, including:
 - a. Drawings and samples as appropriate.
 - b. Comparison of the characteristics of the proposed item with that specified.
 - c. Changes required in other elements of the work because of the substitution.
 - d. Name, address, and telephone number of vendor.
 - e. Manufacturer's literature regarding installation, operation, and maintenance, including schematics for electrical and hydraulic systems, lubrication requirements, and parts lists. Describe availability of maintenance service, and state source of replacement materials.
 - 2. A request for approval constitutes a representation that Contractor:
 - a. Has investigated the proposed item and determined that it is equal or superior in all respects to that specified.
 - b. Will provide the same warranties for the proposed item as for the item specified.
 - c. Has determined that the proposed item is compatible with interfacing items.
 - d. Will coordinate installation of an approved item and make changes required in other elements of the work because of the substitution.
 - e. Waives claims for additional expenses that may be incurred as a result of the substitution.
- F. Electronic Submittals: Identify and incorporate information in each electronic submittal file as follows:

1. Transmittal Form (CM-16): All submittals shall be transmitted using National Park Service Transmittal Form (CM-16). The form can be downloaded from the DSC Workflows website's [Submittal Review](#) page. No action will be taken on a submittal item unless accompanied by this Transmittal Form.
 - a. Complete the general information at the top of form.
 - b. Provide all required information based on submittal type
 - c. Attach all related documents.
 - d. Sign the Contractor section at bottom of the Transmittal Form (CM-16).
 2. Physical samples: Complete Transmittal Form (CM-16) Deliver physical sample to the Contracting Officer (or designee) on site for processing. All comments and actions will be documented on the Transmittal Form (CM-16) and sent to NPS/DSC management.
- G. Identification: Submittal number or other unique identifier, including revision identifier.
1. Submittal number shall use a sequential number (e.g. .001). Re-submittals shall include alphabetic suffix after another decimal point (e.g. .001.A).
- H. Re-submittals: Make re-submittals using same process used with initial submittal.
1. Note date and content of previous submittal.
 2. Note date and content of revision in the title block on the Transmittal Form (CM-16) and clearly indicate extent of revision.
 3. Re-submit submittals until they are marked "Approved" or "Approved with notations".
- I. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, and others as necessary for performance of construction activities.
- J. Use for Construction: Use only final submittals with mark indicating "Approved" or "Approved with notations". Ensure notations have been incorporated and, at a minimum, keep one copy of final approved submittal on site for use during construction.
- 1.4 CONTRACTOR'S USE OF CAD/BIM FILES
- A. General: At Contractor's written request, copies of pdf files will be provided to Contractor for Contractor's use in connection with Project, subject to:
1. Files provided as is; no format or other changes to files or changes to objects in the drawing will be done by the Government

PART 2 - PRODUCTS

2.1 ACTION SUBMITTALS

- A. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.

1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
 2. Mark each submittal to show which products and options are applicable.
 3. As applicable, include:
 - a. Manufacturer's product specifications.
 - b. Manufacturer's installation instructions: When Contract Documents require compliance with manufacturer's printed instructions, provide one complete set of instructions to Contracting Officer and keep another complete set of instructions at the project site until substantial completion.
 - c. Manufacturer's catalog cuts: Submit only pertinent pages; mark each page of standard printed data to identify specific products proposed for use.
 - d. Wiring diagrams showing factory-installed wiring.
 - e. Printed performance curves.
 - f. Operational range diagrams.
 - g. Compliance with specified referenced standards.
 - h. Testing by recognized testing agency.
 4. Submit product data in PDF (portable document format) file format before or concurrent with samples.
- B. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data
1. Preparation: Fully illustrate requirements in Contract Documents. As applicable, include:
 - a. Dimensions
 - b. Identification of products
 - c. Fabrication and installation drawings
 - d. Roughing-in and setting diagrams
 - e. Wiring diagrams showing field-installed wiring, including power, signal, and control wiring
 - f. Shopwork manufacturing instructions
 - g. Templates and patterns
 - h. Schedules
 - i. Notation of coordination requirements
 - j. Notation of dimensions established by field measurement
 - k. Relationship to adjoining construction clearly indicated
 - l. Seal and signature of professional engineer if specified
 - m. Wiring Diagrams: Differentiate between manufacturer-installed and field-installed wiring
 2. Submit shop drawings as PDF electronic file
- C. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other elements and for a comparison of these characteristics between submittal and actual component as delivered and installed.
1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.
 2. Complete and email the Transmittal Form (CM-16) to the NPS/DSC management for processing and documentation of action on submitted samples.

3. Identification: Attach label on unexposed side of Samples that includes:
 - a. Generic description of Sample
 - b. Product name and name of manufacturer
 - c. Sample source
 - d. Submittal Number and title of appropriate Specification Section
 4. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
 5. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing full range of colors, textures, and patterns available.
 - a. Number of Samples: Submit two full set(s) of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Contracting Officer will return with options selected.
 6. Samples for Verification: Submit full-size units or samples of size indicated, prepared from same material to be used for the Work, cured and finished in manner specified, and physically identical with material or product proposed for use, and that show full range of color and texture variations expected. Samples include: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.
 - a. Number of Samples: Submit four sets of Samples. Contracting Officer will retain three Sample sets; remainder will be returned to the jobsite.
- D. Construction Materials: Contractor is encouraged to submit products made out of recycled or environmentally responsible material. Every effort will be made by National Park Service to approve these materials.

2.2 INFORMATIONAL SUBMITTALS

- A. General: Prepare and submit Informational Submittals required by individual Specification Sections.
1. Post informational submittals as PDF electronic files directly to the NPS management software website.
 2. Certificates and Certifications: Provide a notarized statement with signature of entity responsible for preparing certification. Certificates and certifications shall be signed by officer or other individual authorized to sign documents on behalf of that entity.
 3. Informational submittals that do not comply with requirements specified in Contract Documents will be rejected and one copy will be returned.
- B. Coordination Drawings: Comply with requirements specified in Section 01 31 00 "Project Management and Coordination."
- C. Contractors Construction Schedule: Comply with requirements specified in Section 01 32 16 "Construction Progress Schedule."

- D. Accident Prevention Plan: Comply with requirements specified in Section 01 35 23 "Safety Requirements."
- E. Schedule of Values: Comply with requirements specified in Section 01 32 16 "Construction Progress Schedule."
- F. Waste Recycling Plan: Comply with requirements specified in Section 01 74 19 "Construction Waste Management and Disposal."
- G. Quality Control Plan: Comply with requirements specified in Section 01 40 00 "Quality Requirements."
- H. Storm Water Pollution Prevention Plan: Comply with requirements specified in Section 01 57 23 "Temporary Storm Water Pollution Prevention" and storm water permit requirements identified in Section 01 31 00 "Project Management and Coordination."
- I. Qualification Data: Prepare written information demonstrating capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.
- J. Welding Certificates: Prepare written certification that welding procedures and personnel comply with Contract Documents. Submit record of Welding Procedure Specification (WPS) and Procedure Qualification Record (PQR) on American Welding Society (AWS) forms. Include names of firms and personnel certified.
- K. Installer Certificates: Prepare written statements on manufacturer's letterhead certifying Installer complies with Contract Documents and, where required, is authorized by manufacturer for this specific Project.
- L. Manufacturer Certificates: Prepare written statements on manufacturer's letterhead certifying manufacturer complies with Contract Documents. Include evidence of manufacturing experience where required.
- M. Product Certificates: Prepare written statements on manufacturer's letterhead certifying product complies with Contract Documents.
- N. Material Certificates: Prepare written statements on manufacturer's letterhead certifying material complies with Contract Documents.
- O. Material Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in Contract Documents.
- P. Product Test Reports: Prepare written reports indicating current product produced by manufacturer complies with requirements in Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.
- Q. Research/Evaluation Reports: Prepare written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project.

- R. Preconstruction Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in Contract Documents.
- S. Compatibility Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.
- T. Field Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in Contract Documents.
- U. Design Data: Prepare written and graphic information, including: performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.
- V. Manufacturer's Instructions: Prepare written or published information documenting manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer.
- W. Manufacturer's Field Reports: Prepare written information documenting factory-authorized service representative's tests and inspections. As applicable, include:
 - 1. Statement on condition of substrates and their acceptability for installation of product.
 - 2. Summary of installation procedures being followed, compliance with requirements and, if not, what corrective action was taken.
 - 3. Results of operational and other tests and a statement of whether observed performance complies with the requirements.
- X. Permit Compliance Products: Prepare required information for compliance with permit provisions. Products include written notification of project startup, suspension, and completion of work; photo documentation of site conditions; reports; and drawings.

PART 3 - EXECUTION

3.1 CONTRACTOR'S REVIEW

- A. Review each submittal and check for coordination with other Work of Contract and for compliance with Contract Documents. Note corrections and field dimensions.

3.2 CONTRACTING OFFICER'S ACTION

- A. General: Submittals will be disapproved without technical review if identification information is missing, not filled in, or if placed on back of submittal; an incorrect format of submittals is

provided; transmittal form is incorrectly filled out; submittals are not coordinated; or submittals do not show evidence of Contractor's approval.

1. Any work done or orders for materials or services placed before approval shall be at Contractor's own risk.
- B. Action Submittals: Contracting Officer will review each submittal, generate comments on corrections or modifications required, and indicate appropriate action on the Transmittal Form (CM-16). Submittal will be marked as defined below:
1. APPROVED: Acceptable with no corrections.
 2. APPROVED WITH NOTATIONS: Minor corrections or clarifications required. Comments are clear and no further review is required. Contractor shall address review comments when proceeding with the work.
 3. DISAPPROVED - RESUBMIT: Rejected as not in accordance with the contract or as requiring major corrections or clarifications. Contracting Officer will identify reasons for disapproval. Contractor shall revise and resubmit with changes clearly identified.
- C. Informational Submittals: Contracting Officer will review each submittal and will either accept or reject it.
- D. Partial submittals are not acceptable, will be considered non-responsive, and will be returned without review.

END OF SECTION 01 33 23

CONTRACTOR ACKNOWLEDGEMENT OF NPS LO/TO POLICY AND PRACTICE

Pursuant to Occupational Safety and Health Administration Regulation 29 CFR 1910.247 (f) (2) (i):

“Whenever outside servicing personnel are to be engaged in activities covered by the scope and application of this standard, the on-site employer and the outside employer shall inform each other of their respective lockout or tagout procedures.”

And Cape Cod National Seashore “SOP#S006 Lockout Tagout” Lockout and Tagout (LOTO) Program”, Section 3:

3.1. All CACO employees and outside contractors working on park property are required (through this program and the OSHA regulations) to comply with the restrictions and limitations imposed on them during the use of lockout or tagout procedures. All employees, upon observing a machine or piece of equipment that has been "locked out" to perform servicing or maintenance shall not attempt to start, energize or use that machine or equipment. Contractors hired to install, repair or perform servicing activities for the Medical Center’s machinery, systems or equipment shall be notified of the facility’s lockout procedure by the representative coordinating the project, and given copies of any pertinent policies or procedures.

Undersigned acknowledges that the contractor, it’s SSHO and it’s on-site competent person have reviewed the Cape Cod National Seashore LOTO program and provided any and all policies, procedures, instructions, and supplementary documentation and/or materials in accordance with the above and as required by construction activities to be performed under

Contract Number: _____

Contractor	Date
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COR	Date
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SECTION 013543 – ENVIROMENTAL PROTECTION

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 SUMMARY

- A. Furnish all labor, materials, equipment, and perform all work required for the prevention of environmental pollution in conformance with applicable laws and regulations, during and, as the result, of construction operation under this Contract. For the purpose of this Section, environmental pollution is defined as the presence of chemical, physical or biological elements, or agents, which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humans; or degrade the utility of the environment for aesthetic and/or recreational purposes.
- B. The control of environmental pollution requires consideration of air, water, and land, and involves management of runoff, dust, noise, and solid waste, as well as other pollutants. Work will include installing, maintaining, and removing sedimentation and erosion control components within the Limits of Work.
- C. This Section does not address erosion and sedimentation control requirements, which are addressed in Section 31 2500 of these Specifications. The Contractor is responsible for completing a Stormwater Pollution Prevention Plan (SWPPP) and will submit to the Engineer prior to construction commencement.

1.3 SECTION INCLUDES

- A. Applicable Regulations
- B. Notifications
- C. Protection of Groundwater
- D. Protection of Streams And Wetlands
- E. Protection of Land Resources
- F. Protection of Air Quality
- G. Maintenance of Pollution Control Facilities During Construction
- H. Noise Control
- I. Diesel Equipment Emission Controls
- J. Spill And Discharge Control

1.4 RELATED SECTIONS

- A. Section 01 5000 - TEMPORARY FACILITIES AND CONTROLS:
- B. Section 02 4113 – SELECTIVE SITE DEMOLITION:
- C. Section 31 1000 – SITE CLEARING:
- D. Section 31 2000 – EARTH MOVING:
- E. Section 31 2500 – EROSION CONTROL:
- F. Section 31 5000 – EXCAVATION SUPPORT:

1.5 APPLICABLE REGULATIONS

- A. The General Contractor will comply with all applicable Federal, State, and local laws and regulations concerning environmental pollution control and abatement.
- B. Fines and related costs resulting from failure to provide adequate protection against any environmentally objectionable acts and corrective action to be taken are the obligations of the General Contractor.

1.6 NOTIFICATIONS

- A. Engineer may notify the General Contractor, in writing, of any non-compliance with the foregoing provisions or of any environmentally objectionable acts and corrective action to be taken. State or local agencies responsible for verification of certain aspects of the environmental protection requirements may notify the General Contractor, in writing, through the Engineer, of any non-compliance with State or local requirements. After receipt of such notice from the Engineer, the General Contractor will immediately take corrective action. Such notice, when delivered to the General Contractor their authorized representative at the site of the Work, will be deemed sufficient for the purpose. No part of the time lost, due to any such stop orders, will be made the subject of a claim for extension of time or for excess costs or damages by the General Contractor, unless it is later determined that the General Contractor was in compliance.

PART 2 - MATERIALS

2.1 WATER

- A. Water used for dust control and equipment washes will be clean and free of salt, oil, and other injurious materials. The General Contractor will provide all necessary water.
- B. ONSITE SPILL KIT
 - 1. The General Contractor will provide the following minimum equipment to be kept onsite, at all times, during site work activities for any unexpected spills or discharges:
 - a. Sand, clean fill, and absorbent pillows;
 - b. Four (4) drum drums (55 gallon, U.S. DOT 17-E or 17-H);
 - c. Shovels; and
 - d. Steam cleaner for decontamination of tools and equipment.

PART 3 - EXECUTION

3.1 PROTECTION OF GROUNDWATER

3.2 PROTECTION OF STREAMS AND WETLANDS

- A. Care will be taken to prevent, or reduce to a minimum, any damage to any wetland from pollution by debris, sediment, or other material. Manipulation of equipment and/or materials in delineated wetland areas is prohibited. Water that has been used for washing or processing, or that contains oils or sediments that will reduce the quality of the water in downstream waters of the state will not be discharged from the Site. Such waters will be collected and disposed of by the General Contractor, in accordance with all applicable Federal, State and local regulations.

3.3 PROTECTION OF LAND RESOURCES

- A. Land resources, within the project boundaries and outside the limits of permanent work, will be restored to a condition, after completion of remediation activities that will appear to be natural and not detract from the appearance of the project. Confine all construction activities to Limits of Work areas shown on the Drawing.
- B. Outside of the Limits of Work as shown on the Drawing, do not deface, injure, or destroy trees or shrubs, nor remove or cut them without prior approval. Snow fence or other approved equal will be erected at the "fall line" of the tree canopy, and no vehicles or storage will be permitted within, to prevent damage to trees.
- C. The locations of storage and other facilities, required in the performance of the Work, will not be within wetlands or resource areas.

3.4 PROTECTION OF AIR QUALITY

- A. Burning – The use of burning at the project site for the disposal of refuse and debris will not be permitted.
- B. Dust Control – Maintain all demolition excavations, stockpiles, waste areas and all other work areas within or without the project boundaries free from dust, which could cause the standards for air pollution to be exceeded (MADEP 310 CMR 7.09.-7.10) and, which would cause a hazard or nuisance to others.
- C. The General Contractor will provide adequate means for the purpose of preventing dust and odor caused by construction operations throughout the period of the construction contract. If the Engineer indicates that the level of dust or odors is unacceptable, the General Contractor will employ measures necessary to reduce dust or odors to an acceptable level.
- D. The General Contractor will implement engineering controls (e.g. watering, misting) to control dust.

3.5 MAINTENANCE OF POLLUTION CONTROL FACILITIES DURING CONSTRUCTION

- A. During the life of this Contract, maintain all facilities constructed for pollution, erosion and sedimentation control as long as the operations creating the particular pollutant area being carried out.

3.6 NOISE CONTROL

- A. The General Contractor will develop and maintain a noise-abatement program and enforce strict discipline over all personnel to keep noise to a minimum. Local noise ordinances will govern.
- B. The General Contractor will execute construction work by methods and by use of equipment, which will reduce excess noise.
- C. Equipment will be equipped with silencers or mufflers designed to operate with the least possible noise in compliance with Federal and State regulations.
- D. The General Contractor will manage vehicular traffic and scheduling to reduce noise.

3.7 DIESEL EQUIPMENT EMISSION CONTROLS

- A. All motor vehicles and construction equipment will comply with all pertinent local, state, and federal regulations covering exhaust emission controls and safety.
- B. All General Contractor and Subcontractor diesel-powered, non-road construction equipment with engine horsepower (HP) ratings of 50HP and above, which are used on the Project Site, for a period in excess of 30 calendar days over the course of the construction period on the Project Site, will be retrofitted with Emission Control Devices to reduce diesel emissions.
- C. The reduction of emissions of volatile organic compounds (VOCs), carbon monoxide (CO), and particulate matter (PM) from diesel-powered equipment will be accomplished by installing Retrofit Emission Control Devices.

- D. The General Contractor will submit monthly reports, updating the same information stated in Paragraph D above, including the quantity of Clean Fuel utilized. The addition, or deletion, of non-road diesel equipment will be indicated in the report.
- E. The General Contractor will use methods to control nuisance odors associated with diesel emissions from construction equipment including, but not limited to, the following: (1) turning off diesel combustion engines on construction equipment not in active use and on trucks that are idling for five (5) minutes or more; and (2) locating diesel equipment away from the general public and sensitive receptors.
- F. All costs associated with implementation of the diesel equipment emissions control will be borne by the respective General Contractor and included in their cost for performing the work of the Contract.

3.8 SPILL AND DISCHARGE CONTROL

- A. The General Contractor will provide equipment and personnel to perform emergency measures required to contain any spillage and to remove spilled materials and soils or liquids that become contaminated due to spillage. The collected spill material will be properly disposed of at the General Contractor's expense.
- B. Costs to provide the above spill and discharge control materials will be included in the contract base bid price.

END OF SECTION 013543

SECTION 01 40 00 - QUALITY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for quality assurance and quality control.
- B. Testing and inspecting services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with Contract Document requirements. Quality of work shall be responsibility of the Contractor.
 - 1. Specified tests, inspections, and related actions do not limit Contractor's other quality assurance and control procedures that facilitate compliance with Contract Document requirements.
- C. See Divisions 2 through 49 Sections for specific test and inspection requirements.

1.2 DEFINITIONS

- A. Quality Assurance Services: Activities, actions, and procedures performed before and during execution of the work to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.
- B. Quality Control Services: Tests, inspections, procedures, and related actions during and after execution of the work to evaluate actual products incorporated into the work and completed construction comply with requirements.
- C. Preconstruction Testing: Tests and inspections performed specifically for project before products and materials are incorporated into work to verify performance or compliance with specified criteria.
- D. Product Testing: Tests and inspections performed by a Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or a testing agency qualified to conduct product testing, to establish product performance and compliance with industry standards.
- E. Source Quality Control Testing: Tests and inspections performed at the source, i.e., plant, mill, factory, or shop.
- F. Field Quality Control Testing: Tests and inspections performed on-site for installation of work and for completed work.
- G. Testing Agency or Laboratory: Entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.

- H. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, Subcontractor, or Sub-subcontractor, to perform a particular construction operation, including installation, erection, application, and similar operations.
 - 1. Using a term such as "carpentry" does not imply certain construction activities must be performed by accredited or unionized individuals of a corresponding generic name, such as "carpenter." It also does not imply that requirements specified apply exclusively to trades people of corresponding generic name.

1.3 CONFLICTING REQUIREMENTS

- A. Reference Standards: If compliance with two or more standards is specified and standards establish different or conflicting requirements for minimum quality levels, comply with most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Contracting Officer before proceeding.
- B. Minimum Quality Levels: Quality level shown or specified shall be minimum provided or performed. Actual installation may comply exactly with minimum quality specified, or it may exceed minimum within reasonable limits. To comply with requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Contracting Officer before proceeding.

1.4 SUBMITTALS

- A. Quality Control Plan:
 - 1. After contract award and before Pre-Construction conference, submit a written Contractor Quality Control (CQC) plan.
 - 2. If plan requires revisions or corrections, Contractor shall resubmit plan within 10 days.
 - 3. Government reserves the right to require changes in plan during contract period as necessary to obtain the quality specified.
 - 4. No change in the approved plan may be made without written concurrence by Contracting Officer.
 - 5. Include provisions for coordinating and documenting third-party inspections and manufacturer's field services.
 - 6. Include provisions for monitoring and controlling quality of temporary facilities, including toilets, site enclosures, and utilities, as required by Section 01 50 00.
- B. Qualification Data: For testing agencies specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include proof of qualifications in form of a recent report on inspection of testing agency by a recognized authority.
- C. Contractor Quality Control (CQC) Daily Reports: Submit showing inspections and tests on first workday following date covered by report. Quality Control Supervisor shall utilize [DSC Forms](#).
 - 1. Review Construction Management Representative (CMR) Daily report if applicable and reconcile any differences prior to posting.
- D. Test Reports

1. Test reports shall be completed by person performing test.
 2. Submit Daily Test Information Sheets with Quality Control Daily Reports.
 3. Submit failing test results and proposed remedial actions within four hours of noted deficiency.
 4. Submit three copies of complete test results no later than one calendar day after test was performed.
- E. Off-Site Inspection Reports: Submit prior to shipment.
- F. If Contractor Quality Control plan and Quality Control Daily Reports are not submitted as specified, Contracting Officer may retain payments until such time plan(s) is/are accepted and implemented, or may retain payments for work completed on days with no Quality Control Daily Reports.
- G. Permits, Licenses, and Certificates: For National Park Service (NPS) records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of work.

1.5 QUALITY ASSURANCE

- A. General: Qualifications paragraphs in this Article establish minimum qualification levels required; individual Specification Sections specify additional requirements.
- B. Contractors Quality Control Staff:
1. Contractor's Quality Control Supervisor may also perform other duties.
 2. Contractor's Quality Control Supervisor shall be assigned no other duties.
 3. Contractor's designated Quality Control Supervisor shall be on the project site whenever contract work is in progress.
 4. Contractor's job supervisory staff may be used to assist Quality Control Supervisor supplemented, as necessary, by additional certified testing technicians.
- C. Installer Qualifications: Firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent indicated for Project, whose work has resulted in construction with a record of successful in-service performance.
- D. Manufacturer Qualifications: Firm experienced in manufacturing products or systems similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- E. Fabricator Qualifications: Firm experienced in producing products similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- F. Professional Engineer Qualifications: Professional engineer legally qualified to practice in jurisdiction where Project is located and is experienced in providing engineering services of kind indicated (including Structural Tests and Special Inspections (STSI)). Engineering services are defined as those performed for installations of system, assembly, or products similar to those indicated for Project in material, design, and extent.

- G. Specialists: Certain sections of the Specifications require that specific construction activities shall be performed by entities who are recognized experts in those operations. Specialists shall satisfy qualification requirements indicated and shall be engaged for activities indicated.
 - 1. Requirement for specialists shall not supersede building codes and regulations governing Work.
- H. Testing Agency Qualifications: A Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or an independent agency with experience and capability to conduct testing and inspecting indicated, according to ASTM E 329; and with additional qualifications specified in individual Sections; and where required by Contract, is acceptable to Contracting Officer.
 - 1. Nationally Recognized Testing Laboratory (NRTL): Nationally recognized testing laboratory according to 29 CFR 1910.7 (Code of Federal Regulations).
 - 2. National Voluntary Laboratory Accreditation Program (NVLAP): Testing agency accredited according to National Institute of Standards and Technology's (NIST) National Voluntary Laboratory Accreditation Program.
 - 3. Measuring devices, laboratory equipment, and instruments shall be calibrated at established intervals against certified standards in accordance with NIST requirements. Measuring and testing devices shall be made available for use by Government for verification tests.
- I. Factory-Authorized Service Representative Qualifications: Authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of manufacturer's products similar in material, design, and extent to those indicated for Project.

1.6 QUALITY CONTROL

- A. Contractor is responsible for testing and inspections, including any third-party inspections identified in Section 01 11 00 and Section 01 27 00. Inspect and test work as needed to ensure quality of materials, workmanship, construction, finish, and functional performance are in compliance with applicable specifications, drawings, and those required by the Building Code.
 - 1. Engage qualified testing agency to perform quality-control services.
 - 2. Submit appropriate report for each quality-control service.
 - 3. Testing and inspecting requested by Contractor and not required by Contract Documents are Contractor's responsibility.
 - 4. Contracting Officer may designate test locations.
- B. Manufacturer's Field Services: Where indicated, engage factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing.
- C. Re-testing/Re-inspecting: Regardless of whether original tests or inspections were Contractor's responsibility, provide quality-control services, including retesting and re-inspecting, for construction of replaced work that failed to comply with Contract Documents.
- D. Testing Agency Responsibilities: Cooperate with NPS and Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.

1. Notify Contracting Officer and Contractor promptly of irregularities or deficiencies observed in work during performance of services.
 2. Determine location from which test samples will be taken and in which in-situ tests are conducted.
 3. Conduct and interpret tests and inspections, State in each report whether tested and inspected work complies with or deviates from requirements.
 4. Submit 3 copies of certified written report of each test, inspection, and similar quality-control service through Contractor.
 5. Do not release, revoke, alter, or increase Contract Document requirements or approve or accept any portion of Work.
- E. Associated Services: Cooperate with agencies performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide:
1. Access to Work.
 2. Incidental labor and facilities necessary to facilitate tests and inspections.
 3. Adequate quantities of representative samples of materials that require testing and inspecting. Assist agency in obtaining samples.
 4. Facilities for storage and field curing of test samples.
 5. Delivery of samples to testing agencies.
 6. Preliminary design mix proposed for material mixes that require control by testing agency.
 7. Security and protection for samples and testing and inspecting equipment at Project site.
- F. Coordination: Coordinate sequence of activities to accommodate required quality assurance and control services with minimum delay and to avoid removing and replacing construction to accommodate testing and inspecting.
1. Schedule times for tests, inspections, obtaining samples, and similar activities.
- G. This Party Inspections: Where specified in Section 01 11 00, Summary of Work, Section 01 50 00, Temporary Facilities and Controls or in individual specification sections (e.g., ejector pumps, pump controls, and septic system components), the Contractor shall coordinate, schedule, and pay for all required third-party inspections by manufacturer-authorized service companies or independent testing agencies. No concealment, backfill, or system start-up may occur until such inspections are complete and accepted by the CO/COR. Reports shall document deficiencies, corrective actions, and confirmation of corrections.

PART 2 - PRODUCTS

2.1 QUALITY CONTROL PLAN

- A. Quality Control Plan shall include:
1. List of personnel responsible for quality control and assigned duties. Include each person's qualifications. Include alternate(s) and qualifications.
 2. Copy of letter of direction to Contractor's Quality Control Supervisor(s) outlining assigned duties and authorities designated by principal or owner.
 3. Names, qualifications / accreditations, and descriptions of laboratories to perform sampling and testing, and samples of proposed report forms from laboratories.

4. Methods of performing, documenting, and enforcing quality control of work including Contractor report forms and acknowledgment of NPS forms.
5. Methods of monitoring and controlling environmental pollution and contamination as required by regulations and laws.
6. Specific discussion regarding mockups, off-site visits, receiving inspections, manufacturers representation, startup requirements, and other aspects of performance specific to Project.
7. Provisions for substantial completion(s) and final inspection(s) per Contract.

PART 3 - EXECUTION

3.1 OFF-SITE CONTROL

- A. Items fabricated or assembled off-site shall be inspected for quality control at the place of fabrication, if required by the Contract Documents or manufacturer. The Contractor shall be responsible for scheduling, performing, and documenting such inspections.

3.2 ON-SITE CONTROL

- A. Notification:

1. Notify Contracting Officer at least 48 hours in advance of preparatory phase meeting.
2. Notify Contracting Officer at least 24 hours in advance of initial and follow-up phases.

- B. Preparatory Phase: Perform before beginning each feature of work.

1. Review control submittal requirements with personnel directly responsible for quality assurance and quantity control of the work. As a minimum, Contractor's Quality Control Supervisor and foreman responsible for feature of work shall be in attendance.
2. Review applicable specifications sections and drawings related to feature of work.
3. Review requirements for temporary facilities (toilets, storage sheds, site enclosures, fencing) and ensure installation and maintenance comply with approved submittals and contract requirements.
4. Ensure copies of referenced standards related to sampling, testing, and execution for feature of work are available on site.
5. Ensure provisions have been made for field control testing.
6. Examine work area to ensure preliminary work has been completed.
7. Verify field dimensions and advise Contracting Officer of discrepancies with contract documents.
8. Ensure necessary equipment and materials are at project site and they comply with approved shop drawings and submittals.
9. Document preparatory phase activities and discussions on Contractor's Quality Control Daily Report.

- C. Initial Phase:

1. As soon as work begins, inspect and test representative portion of particular feature of work for quality of workmanship.
2. Review control testing procedures to ensure compliance with contract requirements.

3. Document initial phase activities and discussions on Contractor's Quality Control Daily Report. Exact location of initial phase shall be indicated for future reference and comparison with follow-up phases.
- D. Follow-Up Phase: Inspect and test as work progresses to ensure compliance with contract requirements until completion of work.
- E. Additional Preparatory and Initial Phases: Additional preparatory and initial phases may be required on same feature of work for following reasons:
 1. Quality of on-going work is unacceptable.
 2. Changes in quality control staff, on-site production supervision, or work crew.
 3. Work on particular feature of work is resumed after substantial period of inactivity.

3.3 DOCUMENTATION

- A. Maintain Quality Control Daily Reports, Daily Test Report Information Sheets, and Accessibility Inspection Reports of quality control activities and tests. (Download from DSC Workflows website > Forms/Templates/Samples/Guidelines page > [Construction Forms](#) section.)
- B. Quality Control Daily Reports shall not be substituted for other written reports required under clauses of contract, such as Disputes, Differing Site Conditions, or Changes.
- C. Quality Control Daily Reports shall document inspections, testing, and maintenance of temporary facilities, including toilets, site enclosures, barriers, and utilities, in addition to permanent construction features.
- D. Quality Control Daily Reports shall accurately portray all work and materials. The materials shall match the materials as approved from the Construction Documents (CD) requirements. Combustible/flammable materials shall be documented as well.

3.4 ENFORCEMENT

- A. Contractor shall stop work on any item or feature pending satisfactory correction of deficiency noted by quality control staff or Contracting Officer.

3.5 REPAIR AND PROTECTION

- A. General: On completion of testing, inspecting, sample taking, and similar services, repair damaged construction and restore substrates and finishes.
 1. Provide materials and comply with installation requirements specified in other Specification Sections. Restore patched areas and extend restoration into adjoining areas with durable seams as invisible as possible.
 2. Comply with Contract Document requirements for Section 01 73 29 "Cutting and Patching."
- B. Protect construction exposed by or for quality-control service activities.

- C. Repair and protection are Contractor's responsibility, regardless of the assignment of responsibility for quality-control services.

END OF SECTION 01 40 00

SECTION 01 50 00 - TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes requirements for temporary utilities, support facilities, and security and protection facilities and Porta Johns.

DEFINITIONS

- B. Permanent Enclosure: As determined by Contracting Officer (CO), permanent or temporary roofing is complete, insulated, and weathertight; exterior walls are insulated and weathertight; and openings are closed with permanent construction or substantial temporary closures.

1.2 USE CHARGES

- A. General: Cost or use charges for temporary facilities shall be included in Contract Sum as required.
- B. Water Service: Potable Water from existing water system is available for use with metering and with payment of use charges. Provide connections and extensions of services as required for construction operations with any additional permit costs.
- C. Electric Power Service: Electric power from existing system is available for use with metering and with payment of use charges. Provide connections and extensions of services as required for construction operations with any additional permit costs.
- D. Temporary Toilets: All labor, materials, equipment, maintenance, cleaning, pumping, delivery, and incidental activities required for temporary facilities are included in the associated CLIN price (CLIN 0004) and are not separately payable.

1.3 QUALITY ASSURANCE

- A. Electric Service: Comply with National Electrical Contractors Association (NECA), National Electrical Manufacturers Association (NEMA), and Underwriter Laboratories (UL) standards and regulations for temporary electric service. Install service to comply with National Fire Protection Association (NFPA) 70.
- B. Environmental Protection: Provide environmental protection as required by agency(ies) with jurisdiction and as indicated in Contract Documents. Coordinate with requirements of the following:
 - 1. Regulatory Requirements
 - 2. Environmental Management
 - 3. Construction Waste Management

- C. Accessible Temporary Egress: Comply with applicable provisions in the United States (U.S.) Architectural & Transportation Barriers Compliance Board's Architectural Barriers Act Accessibility Standard (ABAAS) Accessibility Guidelines.

1.4 PROJECT CONDITIONS

- A. Temporary Use of Permanent Facilities: Installer of each permanent service shall assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before NPS acceptance, regardless of previously assigned responsibilities.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Temporary materials may be new or used, but must be adequate in capacity for required usage, must not create unsafe conditions, and must not violate requirements of applicable codes and standards.
- B. Portable Chain-Link Fencing: Minimum 2 inch, 9-gage, galvanized steel, chain-link fabric fencing; minimum 6 feet high with galvanized steel pipe posts; minimum 2-3/8 inch OD line posts and 2-7/8 inch OD corner and pull posts, with 1-5/8 inch OD top and bottom rails.
- C. Safety Barrier Fence: Orange plastic fence, minimum height, 4 feet.
- D. Barrier Tape: Yellow tape Imprinted with "CAUTION: CONSTRUCTION AREA," manufactured by Reef Industries, Inc., Houston, Texas, or approved equal.

2.2 TEMPORARY FACILITIES

- A. Field Offices, General: Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.
- B. Storage and Fabrication Sheds: Temporary weather tight sheds or other covered facilities for storage of materials subject to weather damage. Number and size of structures shall be subject to Contracting Officer's approval.
- C. Toilets: Sufficiently lighted and ventilated toilet facilities in weatherproof, sight proof, handicap accessible, sturdy enclosures with privacy locks.
 - 1. Provide separate toilet facilities for men and women.
 - 2. Statement of Work
 - 3. Porta John Rental and Pumping Service, see below

This Contract is to rent, pump, recharge, replace hand sanitizer, replace toilet paper and clean three(3) days per week (Monday, Wednesday and Saturday) from day after award for a duration of 70 days (10

weeks), for six (6) Porta-Johns located in the National Seashore at Race Point Beach, in Provincetown MA.

Location and quantities of portable toilets owned by Contractor *

4 Standard Porta Johns, Race Point Beach Provincetown, MA.

2 ADA compliant Porta-John, Race Point Beach Provincetown, MA.

* All Contractor owned portable toilets to include urinals, hand antiseptic dispenser and toilet paper dispenser

Delivery

Call Contracting Officer Representative (John DeFoe) at (508) 957-0707, or the assigned Contracting Officer, 48 hrs. prior to delivery of toilets for exact location information. Toilets to be delivered as per below schedule.

Description of Work

1. Vendor transports, delivers and offloads porta-Johns to the site.
2. Pump portable toilets three days per week before 9 am from day after award for a duration of 70 days (10 weeks, 180 pumps).
3. Pump out holding tank. Recharge with 5 gal. biodegradable deodorant and water solution. Contractor to provide deodorant.
4. Clean and restock each portable toilet with each pump. Provide large roll of toilet paper and cleaning supplies. Toilet paper is to be 100% recyclable, free of dyes and perfumes
 - a. Re-stock toilet paper. There must be a minimum of two (2) ½ rolls at the time toilets are inspected.
 - b. Provide and re-stock hand antiseptic with ethanol based green product.
 - c. Scrub unit interior, seating and urinal, seat and seat panel to be towel dry.
 - d. Wash door inside and floor, squeegee floor dry, no chemicals, liquids or solids to be left on surface.
5. Make minor repairs to units at no additional charge to the Government.
6. The government reserves the right to cancel services anytime due to the Race Point Beach Bath-house repair being completed and open for public use.
7. Remove, Load and transport Porta-Johns back to vendor.

2.3 EQUIPMENT

- A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.

1. Locate facilities to limit site disturbance and as directed by the Contracting Officer.
- B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.2 TEMPORARY UTILITY INSTALLATION

- A. General: Install temporary service or connect to existing service.
 1. Arrange with utility company, NPS, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services. Acquire necessary permits.
- B. Storm Sewers and Drainage: Provide temporary utilities to remove effluent lawfully.
- C. Non-potable water for construction is not available within the park boundaries. Contractor shall furnish non-potable water from a source outside park boundary.
- D. Potable water is available on site. Make connections to existing facilities as needed. Facilities must be cleaned and maintained in a condition acceptable to NPS. At Substantial Completion, restore these facilities to condition existing before initial use.
- E. Potable water is not available on site. Furnish cool, potable water for construction personnel in locations convenient to work stations.
- F. Sanitary Facilities: Provide temporary toilets, and wash facilities for use by construction personnel.
 1. Place in approved locations secluded from public observation and convenient to work stations. Relocate as work progress requires.
 2. Maintain and clean toilet facilities at least weekly.
 3. Completely remove sanitary facilities on completion of work.
 4. Toilets: Use of existing toilet facilities will be permitted, as long as facilities are cleaned and maintained in a condition acceptable to NPS. At Substantial Completion, restore facilities to condition existing before initial use.
- G. Ventilation and Humidity Control: Provide temporary ventilation required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed. Coordinate ventilation requirements to produce ambient condition required and minimize energy consumption.
- H. Electric Power Service: Use of existing electric power service will be permitted, as long as equipment is maintained in a condition acceptable to NPS.
 1. When temporary connections are removed, restore existing utility services to original condition.
- I. Electric Power Service: Provide electric power service and distribution system of sufficient size, capacity, and power characteristics required for construction operations.

1. Install electric power service overhead, unless otherwise indicated.
- J. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, inspections, and traffic conditions.
 1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.
- K. Telephone Service: No telephone service is available on site for Contractor's use. Make arrangements with Telephone Company and pay costs.

3.3 SUPPORT FACILITIES INSTALLATION

- A. General: Comply with the following:
- B. Waste Disposal Facilities: Provide waste-collection containers in sizes adequate to handle waste from construction operations. Comply with requirements of agency(ies) with jurisdiction.
- C. Lifts and Hoists: Provide facilities necessary for hoisting materials and personnel.
 1. Truck cranes and similar devices used for hoisting materials are considered "tools and equipment" and not temporary facilities.

3.4 SECURITY AND PROTECTION FACILITIES INSTALLATION

- A. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects. Contractor shall be responsible for any inspections or manufacturer-authorized verifications required for temporary facilities, whether on-site or off-site, as needed to meet contract, regulatory, or permit requirements.
- B. Cleaning of Equipment: Contractor shall ensure prior to moving on to Project Area, equipment, is free of soil, seeds, vegetative matter, or other debris that could contain or hold seeds. Ensure equipment has been pressure washed and is free of exotic species. Equipment shall be considered free of soil, seeds, and other debris when visual inspection does not disclose such material. Disassembly of equipment components or specialized inspection tools are not required.
- C. Temporary Erosion and Sedimentation Control: Refer to Section 01 57 23 "Temporary Storm Water Pollution Prevention".
- D. Tree and Plant Protection: Refer to Section 01 11 00 "Summary of Work".
- E. Pest Control: Follow NPS requirements to minimize attraction and harboring of rodents, roaches, and other pests and perform extermination and control procedures at regular intervals so Project will be free of pests and residues at Substantial Completion. Perform control operations lawfully, using environmentally safe materials.
- F. Site Enclosure Fence: Before construction operations begin, furnish and install chain link fencing to prevent people and animals from easily entering site except by entrance gates.

1. Extent of Fence: As required to enclose entire Project site or portion determined sufficient to accommodate construction operations As indicated on Drawings and in accordance with site protection requirements in Section 01 11 00.
 2. Locate vehicular gates to avoid interference with traffic on public thoroughfares.
 3. Locate pedestrian entrance gates as required to provide controlled personnel entry.
 4. Maintain security by limiting number of keys and restricting distribution to authorized personnel. Provide Contracting Officer with one set of keys.
- G. Security Enclosure and Lockup: Install substantial temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security.
- H. Barricades, Warning Signs, and Lights: Comply with requirements of Manual on Uniform Traffic Control Devices (MUTCD), Part 6, latest edition for erecting structurally adequate barricades, including warning signs and lighting.
- I. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.
- J. Temporary Fire Protection: Install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241.
1. Responsible Person: Capable and qualified person shall be placed in charge of fire protection. Responsibilities shall include locating and maintaining fire protective equipment and establishing and maintaining safe torch cutting and welding procedures.
 2. Tobacco Use, Smoking, and Vaping: Smoking within buildings or temporary storage sheds is prohibited.
 3. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition according to requirements of NPS Agency(ies) with Jurisdiction. Check with Park; many require "burn permits" for welding.
 4. Develop and supervise overall fire-prevention and -protection program for personnel at Project Site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
 5. Provide temporary standpipes and hoses for fire protection. Hang hoses with warning sign stating hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.
 6. Hazard Control: Take necessary precautions to prevent fire during construction. Do not store flammable or combustible liquids in existing buildings. Provide adequate ventilation during use of volatile or noxious substances. Ensure cleanup procedures and storage requirements are followed at close of every work session.
 7. Spark Arresters: Equip gasoline or diesel powered equipment used during periods of potential fire hazards or in potential forest and grass fire locations with spark arresters approved by United States Department of Agriculture (USDA) Forest Service.
 - a. Written determinations of periods and areas of potential fire hazard will be issued by Contracting Officer.
 8. Buildings: Furnish a minimum of one extinguisher for each 1,500 square feet of area or major fraction thereof.

- a. Travel distance from any work station to nearest extinguisher shall not exceed 75 feet.
- 9. Vehicles and Equipment: Provide one extinguisher on each vehicle or piece of equipment.
- 10. Service and Refueling Areas: Locate areas a minimum of 50 feet from buildings. Shut down equipment before refueling.

3.5 OPERATION, TERMINATION, AND REMOVAL

- A. Supervision: Enforce strict discipline in use of temporary facilities. Minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.
- B. Maintenance: Maintain facilities in good operating condition until removal.
 - 1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on 24-hour basis where required to achieve indicated results and avoid possibility of damage.
 - 2. All activities related to temporary facilities, including delivery, pumping, cleaning, and maintenance, shall be documented in the Contractor Quality Control Daily Reports in accordance with Section 01 40 00.
- C. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.
- D. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.
 - 1. Materials and facilities that constitute temporary facilities are property of Contractor. NPS reserves right to take possession of Project identification signs.
 - 2. At Substantial Completion, clean and renovate permanent facilities used during construction period.

END OF SECTION 01 50 00

SECTION 01 51 29 – TEMPORARY SANITARY FACILITIES

PART 1 – GENERAL

1.1 SUMMARY

A. This section includes requirements for furnishing, maintaining, and servicing temporary sanitary facilities during the project. Contractor shall provide temporary sanitary facilities for its own personnel. Temporary sanitary facilities for public and Government personnel are option 3, and shall be provided only if the related bid option is awarded.

B. The intent of this section is to establish minimum standards for contractor-provided portable toilet units for project personnel and visitors during construction activities at the Province Lands Visitor Center (PLVC), Cape Cod National Seashore.

1.2 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.3 DESCRIPTION OF WORK

A. Work includes providing, maintaining, servicing, and removing temporary sanitary facilities as specified herein. Facilities for contractor personnel are mandatory and the responsibility of the contractor. Facilities for public and Government personnel are optional and only required if awarded as part of a bid option

B. If bid option is awarded, contractor shall provide the following public/Government portable toilet units:

1. One (1) ADA-compliant portable toilet unit.
2. Three (3) standard portable toilet units.

C. Services shall include rental, delivery, pumping, cleaning, restocking, and removal upon completion or when directed by the Contracting Officer's Representative (COR).

1.4 QUALITY ASSURANCE

A. Portable toilets shall comply with ADA requirements where designated. Contractor shall ensure that its own personnel facilities are clean and operable; these facilities are not subject to COR inspection. Public and Government facilities provided under the bid option shall meet all requirements specified herein and be documented in quality control reports.

B. Units shall be clean, in good working condition, free of damage, and equipped with privacy door lock, hand sanitizer, toilet paper, and deodorizing solution.

C. Service personnel shall be trained and experienced in cleaning and maintaining portable toilets to sanitary standards.

1.5 COORDINATION

- A. Coordinate delivery, placement, and removal of facilities with the COR.
- B. Contact the COR at least 48 hours before delivery for location confirmation.

PART 2 – PRODUCTS

2.1 PORTABLE TOILET UNITS

- A. Contractor-provided portable toilets shall include:
 - 1. Urinal, seat, and toilet paper dispenser.
 - 2. Hand antiseptic dispenser.
 - 3. Adequate ventilation, door latch and privacy door lock.
- B. Toilet paper shall be 100% recyclable, free of dyes and perfumes.
- C. Hand sanitizer shall be an ethanol-based, environmentally friendly product.

PART 3 – EXECUTION

3.1 DELIVERY AND INSTALLATION

- A. Deliver and install units as directed by the COR.
- B. Locate units to provide convenient access for workers while minimizing visibility from public areas.
- C. Secure units against movement or tipping.

3.2 MAINTENANCE AND SERVICING

- A. Contractor's employee facilities shall be maintained at the contractor's discretion. The following maintenance schedule applies only to public and Government portable toilets provided under the bid option 3. These units shall be serviced three (3) times per week (Monday, Wednesday, and Saturday) before 9:00 a.m.
- B. Each service visit shall include:
 - 1. Pumping waste from holding tank.
 - 2. Recharging tank with 5 gallons biodegradable deodorant and water solution.
 - 3. Cleaning and sanitizing interior surfaces, including seat, urinal, door, and floor.
 - 4. Drying interior surfaces after cleaning.
 - 5. Replacing and restocking supplies:
 - i. Minimum of two (2) rolls of toilet paper per unit.
 - ii. Refill hand sanitizer dispensers.
- C. Make minor repairs to units at no additional cost to the Government.
- D. Maintain facilities in clean and operable condition throughout the rental period.

3.3 REMOVAL

A. Upon completion of the project or when directed by the COR, remove units, clean area of placement, and restore ground surface to pre-installation condition. Contractor shall remove its own personnel facilities at project completion. Public and Government facilities shall be removed only if bid option 3 was awarded

B. The Government reserves the right to cancel services early upon reopening of permanent bathhouse facilities.

END OF SECTION 01 51 29

SECTION 01 67 00 - PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for selection of products for use in Project; product delivery, storage, and handling; manufacturers' standard warranties on products; special warranties; and environmental requirements.

1.2 DEFINITIONS

- A. Products: Items purchased for incorporating into Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes "material," "equipment," "system," and terms of similar intent.
 - 1. Named Products: Items identified by manufacturer's product name, including make or model number and other designation shown or listed in manufacturer's published product literature, current as of date of Contract Documents.
 - 2. New Products: Items that have not previously been incorporated into another project or facility, except that products consisting of recycled-content materials are allowed, unless explicitly stated otherwise. Products salvaged or recycled from other projects are not considered new products.
 - 3. Approved Equals: Product demonstrated and approved through submittal process, or where indicated as a product substitution, to have indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product.
- B. Definitions pertaining to sustainable development: As defined in ASTM E2114.
- C. Biobased Materials: As defined in Farm Security and Rural Investment Act, for purposes of Federal procurement of biobased products, "biobased" means a "commercial or industrial product (other than food or feed) composed, in whole or in significant part, of biological products or renewable domestic agricultural materials (including plant, animal, and marine materials) or forestry materials." Biobased materials also include fuels, chemicals, building materials, or electric power or heat produced from biomass as defined by The Biomass Research and Development Act of 2000.
 - 1. Biobased content: Amount of biobased carbon in material or product as a percentage of weight (mass) of total organic carbon in material or product.
- D. Chain-of-Custody: Process whereby a product or material is maintained under physical possession or control during its entire life cycle.
- E. Environmentally preferable products: Products and services with lesser or reduced effect on the environment in comparison to conventional products and services. Refer to Environmental

Protection Agency's (EPA) Final Guidance on [Environmentally Preferable Purchasing](#) for more information.

- F. Stewardship: Responsible use and management of resources in support of sustainability.
- G. Sustainability: Maintenance of ecosystem components and functions for future generations.
 - 1. Recycled Content Materials: Products containing pre-consumer or post-consumer materials as all or part of their feedstock. Recycled content claim shall be consistent with International Organization for Standardization (ISO) 140001 Standard for the Use of Environmental Marketing Claims.
 - 2. Rapidly Renewable Material: Material made from plants typically harvested within a ten-year cycle.
 - 3. Regional Materials: Materials manufactured and extracted, harvested, or recovered within a radius of 500 miles from Project location.

1.3 SUBMITTALS

- A. Record Submittals as specified in – Sustainable Design Close-Out Documentation, submit:
 - 1. Affirmative Procurement Reporting Form. Submit on form in Appendix A of this Section, or similar form as approved by Contracting Officer (CO).
 - 2. Submit environmental data in accordance with Table 1 of ASTM E2129 for these products:
 - a. Masonry
 - b. Finish Carpentry
 - c. Plastic Fabrications
 - d. Building Insulation
 - e. Roofing
 - f. Joint Sealers
 - g. Wood & Plastic Doors
 - h. Windows
 - i. Skylights
 - j. Glazed Curtain Wall
 - k. Gypsum Board
 - l. Tile
 - m. Acoustical Ceilings
 - n. Resilient Flooring
 - o. Carpet
 - p. Wall Coverings
 - q. Paints & Coatings
 - r. Toilet Compartments
 - s. Loading Dock Equipment
 - t. Office Equipment
 - u. Furnishings & Accessories
 - v. Renewable Energy Equipment
 - w. Elevators
 - x. Plumbing fixtures and equipment.
 - y. HVAC equipment
 - z. Lighting equipment

.Material Safety Data Sheets (MSDS): For each product required by OSHA to have a MSDS, submit an MSDS. MSDS shall be prepared within the previous five years. Include information for MSDS Sections 1 to 16 in accordance with ANSI Z400.1 and as follows:

- a. Section 1: Chemical Product and Company Identification
 - b. Section 2: Composition/Information on Ingredients
 - c. Section 3: Hazards Identification
 - d. Section 4: First Aid Measures
 - e. Section 5: Fire Fighting Measures
 - f. Section 6: Accidental Release Measures
 - g. Section 7: Handling and Storage
 - h. Section 8: Exposure Controls/Person Protection
 - i. Section 9: Physical and Chemical Properties
 - j. Section 10: Stability and Reactivity Data
 - k. Section 11: Toxicological Information. Include data used to determine the hazards cited in Section 3. Identify acute data, carcinogenicity, reproductive effects, and target organ effects. Provide written description of the process used in evaluating chemical hazards relative to preparation of the MSDS.
 - l. Section 12: Ecological Information. Include data regarding environmental impacts during raw materials acquisition, manufacture, and use. Include data regarding environmental impacts in event of accidental release.
 - m. Section 13: Disposal Considerations. Include data regarding proper disposal of the chemical. Include information regarding recycling and reuse. Indicate whether or not product is considered to be "hazardous waste" according to United States EPA Hazardous Waste Regulations 40 CFR 261 (Code of Federal Regulations).
 - n. Section 14: Transportation Information. Identify hazard class for shipping.
 - o. Section 15: Regulatory Information. Identify federal, state, and local regulations applicable to the material.
 - p. Section 16: Other Information. Include additional information relative to recycled content, biobased content, and other information regarding environmental and health impacts. Identify the date MSDS was prepared.
4. Chain of Custody: Submit chain-of-custody documentation for sustainable forestry for these products:
- a. Rough Carpentry
 - b. Finish Carpentry
 - c. Wood Doors
 - d. Windows
 - e. Wood Flooring
 - f. Furnishings & Accessories

1.4 QUALITY ASSURANCE

- A. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, product selected shall be compatible with products previously selected, even if previously selected products were also options.

1.5 PRODUCT DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, and handle products using means and methods that prevent damage, deterioration, and loss, including theft. Comply with manufacturer's written instructions.
- B. Delivery and Handling:
 - 1. Schedule delivery to minimize long-term storage at Project site and prevent overcrowding of construction spaces.
 - 2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
 - 3. Deliver products to Project site in undamaged condition; in manufacturer's original sealed container or other packaging system; complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
 - 4. Inspect products on delivery to ensure compliance with Contract Documents. Ensure products are undamaged and properly protected.
 - 5. Obtain materials in biodegradable or recyclable/reusable packaging which uses minimum amount of packaging possible.
- C. Storage:
 - 1. Allow for inspection and measurement of quantity or counting of units.
 - 2. Store materials in manner to not endanger Project structure.
 - 3. Store products subject to damage by the elements, under cover in weather tight enclosure above ground, with ventilation adequate to prevent condensation.
 - 4. Store cementitious products and materials on elevated platforms.
 - 5. Store foam plastic from exposure to sunlight, except to extent necessary for period of installation and concealment.
 - 6. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
 - 7. Protect stored products from damage and liquids from freezing.
 - 8. Store loose granular materials in well-drained area on solid surfaces to prevent mixing with foreign matter.

1.6 PACKAGING

- A. Where Contractor has option to provide one of listed products or equal, preference shall be given to products with minimal packaging and easily recyclable packaging as defined in ASTM D5834.
- B. Maximize use of source reduction and recycling procedures outlined in ASTM D5834.

1.7 ENVIRONMENTALLY PREFERABLE PRODUCTS

- A. Provide environmentally preferable products to greatest extent possible.
 - 1. To greatest extent possible, provide products and materials with a lesser or reduced effect on the environment considering raw materials acquisition, production, manufacturing, packaging, distribution, reuse, operation, maintenance, and/or disposal of the product.

2. Eliminate use of ozone depleting compounds during and after construction where alternative environmentally preferable products are available, consistent with either Montreal Protocol and Title VI or Clean Air Act Amendments of 1990, or equivalent overall air quality benefits that take into account life cycle impacts.
3. Use products meeting or exceeding EPA's recycled content recommendations for EPA-designated products. Use materials with recycled content such that the sum of post-consumer recycled content plus one-half of the pre-consumer content constitutes at least 10% (based on cost) of total value of the materials in project.

1.8 PRODUCT WARRANTIES

- A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of Contract Documents.
 1. Manufacturer's Warranty: Preprinted written warranty published by individual manufacturer for product specifically endorsed by manufacturer to Owner.
 2. Special Warranty: Written warranty required by or incorporated into Contract Documents, either to extend time limit provided by manufacturer's warranty or to provide more rights for Owner.
- B. Special Warranties: Prepare written document containing appropriate terms and identification, ready for execution. Submit draft for approval before final execution.
 1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.
 2. Specified Form: When specified forms are included with Specifications, prepare written document using appropriate form properly executed.
 3. Refer to Divisions 2 through 49 Sections for specific content requirements and particular requirements for submitting special warranties.
- C. Submittal Time: Comply with requirements in Section 01 77 00 "Closeout Procedures."

PART 2 - PRODUCTS

2.1 PRODUCT SELECTION PROCEDURES

- A. General Product Requirements: Provide products to comply with Contract Documents, undamaged and, unless otherwise indicated, new at time of installation.
 1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
 2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types produced and used successfully in similar situations on other projects.
 3. Government reserves right to limit selection to products with warranties not in conflict with requirements of Contract Documents.

4. Where products are accompanied by term "as selected," Contracting Officer will make selection.
5. Where products are accompanied by term "match sample," sample to be matched is Governments.
6. Descriptive, performance, and reference standard requirements in Specifications establish "salient characteristics" of products.

B. Product Selection Procedures:

1. Product: Where Specifications name single product and manufacturer, provide named product that complies with requirements or approved equal.
2. Manufacturer/Source: Where Specifications name single manufacturer or source, provide product by named manufacturer or source that complies with requirements or approved equal.
3. Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed that complies with requirements or approved equal.
4. Manufacturers: Where Specifications include list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements or approved equal.
5. Available Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed, or an unnamed product, that complies with requirements. Comply with provisions in Part 2 "Products" Articles for consideration of an unnamed product.
6. Available Manufacturers: Where Specifications include list of manufacturers, provide a product by one of the manufacturers listed, or an unnamed manufacturer, that complies with requirements. Comply with provisions in Part 2 "Products" Articles for consideration of an unnamed product.
7. Product Options: Where Specifications indicate sizes, profiles, and dimensional requirements on Drawings are based on a specific product or system, provide specified product, system, or approved equal.
8. Visual Matching Specification: Where Specifications require matching an established Sample, select product that complies with requirements and matches Architect's sample. Contracting Officers decision will be final on whether a proposed product matches.
 - a. If no product available within specified category matches and complies with other specified requirements, comply with provisions in Part 2 "Product Substitutions" Article for proposal of product.
9. Visual Selection Specification: Where Specifications include phrase "as selected from manufacturer's colors, patterns, textures" or a similar phrase, select a product that complies with other specified requirements.
 - a. Standard Range: Where Specifications include phrase "standard range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that does not include premium items.
 - b. Full Range: Where Specifications include phrase "full range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

2.2 APPROVED EQUAL PRODUCTS

- A. Conditions: Contracting Officer will consider Contractor's request when the following conditions are satisfied. If following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:
1. Evidence proposed product does not require revisions to Contract Documents, that it is consistent with Contract Documents and will produce indicated results and is compatible with other portions of Work.
 2. Detailed comparison of significant qualities of proposed product with those named in Specifications. Significant qualities include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
 3. Evidence proposed product provides specified warranty.
 4. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners, if requested.
 5. Samples, if requested.

PART 3 - EXECUTION

3.1 PROTECTION AFTER INSTALLATION

- A. Provide adequate coverings as necessary to protect installed materials from damage resulting from natural elements, traffic, and subsequent construction. Remove when no longer needed.

END OF SECTION 01 67 00

SECTION 017123 – SURVEYS AND RECORD DRAWINGS

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 RECORD DRAWINGS

- A. Prior to final payment, the Contractor will complete an “on-the ground” detailed survey and provide an as-built plan of all facilities within the limit of work. An electronic version of the as-built plan in AutoCAD format shall be provided. As-Built drawings that consist of the Engineer’s electronic design file will not be accepted.
- B. Record Drawings will consist of all the Contract Drawings with mark-ups made during construction.
- C. The Contractor shall keep a record of all as-built conditions, noting the exact location of their work as actually installed. This will include the location and dimensions of underground and concealed Work and any variations from the Contract Drawings. All changes, including those issued by Addendum, Change Order, or instructions by the Engineer will be recorded. Marked-up record drawings will be prepared for the entire project and include all Work, including, but not limited to:
 - 1. The location of all underground utilities and appurtenances referenced to permanent surface improvements, both horizontally and vertically, at ten-foot (10’) intervals and at all changes of direction.
- D. The government may periodically inspect the marked-up record drawings at the site. The proper and current maintenance of the information required on these drawings will be a condition precedent to approval of the monthly applications for payment.
- E. At Substantial Completion, the Contractor will submit the complete set of marked-up as-built drawings to the government. The Contractor will check all marked-up record drawings prepared by subcontractors and certify, in writing, on the title sheet of the drawings, that they are complete and correct prior to submission to the Engineer.
- F. The Government will review the marked-up record drawings. The Contractor will incorporate any and all changes into the as-built drawings.
- G. The Contractor may make a written request for copies of the completed Record Drawings. The Contractor will reimburse the government directly for the cost of printing of any requested Record Drawings.

- H. Submission of accurate marked-up record drawings, as-built drawings, and their approval by the government will be a condition precedent to final payment.
- I. Regulatory Submittals: Contractor is responsible for submitting all as-built drawings and related information required by the Massachusetts Department of Environmental Protection (MassDEP) for permit closeout. Submission of as-builts to the Government as specified herein does not relieve the Contractor of responsibility to provide all required as-built documentation to MassDEP or any other regulatory agency. The format and certification requirements specified in this section apply solely to Government submittals.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 017123

SECTION 01 73 40 - EXECUTION

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes general procedural requirements governing execution of Work including:

1. Coordination with utility service providers
2. Construction layout
3. Field engineering and surveying
4. General installation of products
5. Progress cleaning
6. Starting and adjusting
7. Protection of installed construction
8. Correction of the Work

1.2 SUBMITTALS

- A. Certificates: Submit certificate signed by professional engineer certifying location and elevation of improvements comply with requirements.
- B. Landfill Receipts: Submit copy of receipts issued by landfill facility, licensed to accept hazardous materials, for hazardous waste disposal.
- C. Certified Surveys: Submit two copies signed by professional engineer.
- D. Quantity Surveys: Submit 2 copies showing quantities of work performed and actual construction completed in place.

1.3 QUALITY ASSURANCE

- A. Land Surveyor Qualifications: Professional engineer legally qualified to practice in jurisdiction where Project is located and—is experienced in providing land-surveying services of kind indicated.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Existing Conditions: Existence and location of site improvements and other construction indicated as existing are not guaranteed.

1. Before beginning work, investigate and verify the existence and location of mechanical and electrical systems and other construction affecting the Work.
 2. Before construction, verify the location and points of connection of utility services.
- B. Existing Utilities: Existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Before beginning sitework, investigate and verify existence and location of underground utilities and other construction affecting Work.
1. Before construction, verify location and invert elevation at points of connection of sanitary sewer, storm sewer, and water-service piping, and underground electrical services.
 2. Furnish location data for work related to Project that must be performed by public utilities serving Project site.
- C. Acceptance of Conditions: Examine substrates, areas, and conditions, with Installer or Applicator present where indicated, for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.
1. Verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.
 2. Examine roughing-in for mechanical and electrical systems to verify actual locations of connections before equipment and fixture installation.
 3. Examine walls, floors, and roofs for suitable conditions where products and systems are to be installed.
 4. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

3.2 COORDINATION WITH UTILITY SERVICE PROVIDERS

- A. Coordination with Utility Service Providers: Contact following Utility Service providers, sufficiently in advance to avoid delaying the work, to coordinate Contractor's portion of Work, testing requirements, inspections, etc.
1. Electrical: Service Contact: Contact to coordinate Electrical service requirements.
 - a. Construction Contractor Responsibilities: Contractor is responsible for, see electrical drawings and specifications.
 2. Data Service Contact: Contact NPS provider to coordinate Telephone/Data service requirements.
 - a. Construction Contractor Responsibilities: You are responsible for see drawings and specs.

3.3 PREPARATION

- A. Field Measurements: Take field measurements as required to fit Work properly. Recheck measurements before installing each product. Where portions of Work are indicated fit to other

construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying Work.

- B. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.
- C. Review of Contract Documents and Field Conditions: Immediately on discovery of need for clarification of the Contract Documents caused by differing field conditions outside control of Contractor, submit request for information to Contracting Officer in accordance with Section 01 31 00 "Project Management and Coordination."

3.4 CONSTRUCTION LAYOUT

- A. Verification: Verify layout information shown on Drawings, in relation to the existing benchmarks before proceeding to lay out Work. Notify Contracting Officer promptly if discrepancies are discovered.
- B. General: Engage a professional engineer to lay out Work using accepted surveying practices.
 - 1. Establish benchmarks and control points to set lines of construction and elsewhere as needed to locate each element of Project.
 - 2. Establish dimensions within tolerances indicated. Do not scale Drawings to obtain required dimensions.
 - 3. Inform installers of lines and levels to which they must comply.
 - 4. Check location, level and plumb, of every major element as Work progresses.
 - 5. Notify Contracting Officer when deviations from required lines and levels exceed allowable tolerances.
 - 6. Close site surveys with an error of closure equal to or less than the established standard.
- C. Site Improvements: Locate and lay out site improvements, including pavements, grading, fill and topsoil placement, utility slopes, and invert elevations.
- D. Building Lines and Levels: Locate and lay out control lines and levels for structures, building foundations, column grids, and floor levels, including those required for mechanical and electrical work. Transfer survey markings and elevations for use with control lines and levels. Level foundations and piers from two or more locations.
- E. Record Log: Maintain log of layout control work. Record deviations from required lines and levels. Include beginning and ending dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make log available for review by National Park Service (NPS).

3.5 FIELD ENGINEERING

- A. Reference Points: Locate existing permanent benchmarks, control points, and similar reference points before beginning Work. Preserve and protect permanent benchmarks and control points during construction operations. Controls destroyed by Contractor will be replaced by Contractor at their expense.

1. Existing Monuments: All benchmarks, land corners, and triangulation points, established by other surveys, existing within construction area shall be preserved. If existing monuments interfere with Work, secure written permission before removing them.
- B. Benchmarks: Establish and maintain a minimum of two permanent benchmarks on Project site, referenced to data established by survey control points. Comply with NPS requirements for type and size of benchmark.
 1. Record benchmark locations, with horizontal and vertical data, on Project Record Documents.
- C. Certified Survey: On completion of foundation walls, major site improvements, and other work requiring field-engineering services, prepare a certified survey showing dimensions, locations, angles, and elevations of construction and sitework.

3.6 INSTALLATION

- A. General: Locate Work and components of Work accurately in correct alignment and elevation, as indicated.
 1. Make vertical work plumb and make horizontal work level.
 2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
 3. Conceal pipes, ducts, and wiring in finished areas, unless otherwise indicated.
- B. Comply with manufacturer's written instructions and recommendations for installing products in applications indicated.
- C. Install products at the time and under conditions for best possible results. Maintain conditions required for product performance until Substantial Completion.
- D. Conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.
- E. Tools and Equipment: Do not use tools or equipment that produce harmful noise levels.
- F. Templates: Obtain and distribute to parties involved templates for work specified to be factory prepared and field installed. Check Shop Drawings of other work to confirm adequate provisions are made for locating and installing products to comply with indicated requirements.
- G. Anchors and Fasteners: Provide anchors and fasteners as required to anchor each component securely in place, accurately located and aligned with other portions of Work.
 1. Mounting Heights: Where mounting heights are not indicated, mount components at heights directed by Contracting Officer.
 2. Allow for building movement, thermal expansion, and contraction.
 3. Coordinate installation of anchorages. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors to be embedded in concrete or masonry. Deliver to Project site in time for installation.

- H. Joints: Make joints of uniform width. Where joint locations in exposed work are not indicated, arrange joints for best visual effect. Fit exposed connections together to form hairline joints.
- I. Hazardous Materials: Use products, cleaners, and installation materials not considered hazardous.
- J. Quantity surveys: Shall be conducted, and data derived from these surveys shall be used in computing quantities of work performed and actual construction completed and in place.
 - 1. Contractor shall conduct original and final surveys and surveys for any periods for which progress payments are requested. These surveys shall be conducted under direction of a representative of the Contracting Officer, unless Contracting Officer waives requirement in a specific instance. Government shall make such computations as are necessary to determine quantities of work performed or finally in place. Contractor shall make computations based on surveys for any periods for which progress payments are requested.
 - 2. Promptly upon completing a survey, Contractor shall furnish originals of field notes and other records relating to survey or layout of Work to Contracting Officer. Contractor shall retain copies of all such material furnished to Contracting Officer.

3.7 PROGRESS CLEANING

- A. General: Clean Project site, work areas, and common areas daily. Coordinate progress cleaning for joint-use areas where more than one Installer has worked. Enforce requirements strictly. Dispose of materials lawfully.
 - 1. Comply with requirements in National Fire Protection Association (NFPA) 241 for removal of combustible waste materials and debris at the end of every work session.
 - 2. Do not hold materials more than 7 days during normal weather or 3 days if the temperature is expected to rise above 80 degrees Fahrenheit (27 degrees Celsius).
 - 3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, according to regulations.
- B. Site: Maintain Project site free of waste materials and debris.
- C. Work Areas: Clean areas where work is in progress to level of cleanliness necessary for proper execution of Work.
 - 1. Remove liquid spills promptly.
 - 2. Where dust would impair proper execution of Work, broom-clean or vacuum entire work area, as appropriate.
 - 3. Contractor shall provide progress cleaning that minimizes sources of food, water, and harborage available to pests.
- D. Installed Work: Keep installed work clean. Clean installed surfaces according to written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials not hazardous to health or property and will not damage exposed surfaces.
 - 1. Utilize non-toxic cleaning materials and methods.

- a. Comply with Green Seal Standard (GS) 37 for general purpose cleaning and bathroom cleaning.
- b. Use natural cleaning materials where feasible. Natural cleaning materials include:
 - 1) Abrasive cleaners: substitute 1/2 lemon dipped in borax.
 - 2) Ammonia: substitute vinegar, salt and water mixture, or baking soda and water.
 - 3) Disinfectants: substitute 1/2 cup borax in gallon water.
 - 4) Drain cleaners: substitute 1/4 cup baking soda and 1/4 cup vinegar in boiling water.
 - 5) Upholstery cleaners: substitute dry cornstarch.
- E. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.
- F. Exposed Surfaces in Finished Areas: Clean exposed surfaces and protect from damage and deterioration at time of Substantial Completion.
- G. Waste Disposal: Do not bury or burn waste materials on-site. Do not wash waste materials down sewers or into waterways.
- H. Clean and protect construction in progress and adjoining materials already in place during handling and installation. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.
- I. Clean and provide maintenance on completed construction as frequently as necessary through remainder of the construction period. Adjust and lubricate operable components to ensure operability without damaging effects.
- J. Limiting Exposures: Supervise construction operations so that no part of construction completed or in progress, is subject to harmful, dangerous, damaging, or deleterious exposure during construction period.
- K. Final Cleaning: At completion of Work, remove remaining waste materials, rubbish, tools, equipment, machinery and surplus materials. Clean exposed surfaces and leave Project clean and ready for occupancy.
 - 1. Provide final cleaning in accordance with ASTM E1971.

3.8 STARTING AND ADJUSTING

- A. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.
- B. Adjust operating components for proper operation without binding. Adjust equipment for proper operation.
- C. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

- D. Manufacturer's Field Service: If factory-authorized service representative is required to inspect field-assembled components and equipment installation, comply with qualification requirements in Section 01 40 00 "Quality Requirements."

3.9 PROTECTION OF INSTALLED CONSTRUCTION

- A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.
- B. Comply with manufacturer's written instructions for temperature and relative humidity.

3.10 CORRECTION OF THE WORK

- A. Repair or remove and replace defective construction. Restore damaged substrates and finishes. Comply with requirements in Section 01 73 29 "Cutting and Patching."
 - 1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.
- B. Restore permanent facilities used during construction to specified condition.
- C. Remove and replace damaged surfaces exposed to view if surfaces cannot be repaired without visible evidence of repair.
- D. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.
- E. Remove and replace chipped, scratched, and broken glass or reflective surfaces.

END OF SECTION 01 73 40

SECTION 01 73 29 - CUTTING AND PATCHING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes procedural requirements for cutting and patching on buildings that do not contain Historic Fabric.

1.2 SUBMITTALS

- A. Cutting and Patching Plan: Submit Plan describing procedures at least 10 days before cutting and patching will be performed, requesting approval to proceed. Include:
 - 1. Extent: Describe cutting and patching, show how performed, and indicate why they cannot be avoided.
 - 2. Changes to In-Place Construction: Describe anticipated results. Include changes to structural elements and operating components as well as changes in building's appearance and other significant visual elements.
 - 3. Products: List products to be used and firms or entities that will perform Work.
 - 4. Dates: Indicate when cutting and patching will be performed.
 - 5. Utility Services and Mechanical/Electrical Systems: List services/systems that cutting and patching procedures will disturb or affect. List services/systems that will be relocated and those that will be temporarily out of service. Indicate how long services/systems will be disrupted.
 - 6. Structural Elements: Where cutting and patching involve adding reinforcement to structural elements, submit details and engineering calculations showing integration of reinforcement with original structure. Do not cut and patch structural elements in a manner that could change their load carrying capacity or increase deflection.
 - 7. Contracting Officer's (CO) Approval: Obtain approval of cutting and patching plan before cutting and patching. Approval does not waive right to later require removal and replacement of unsatisfactory work.

1.3 QUALITY ASSURANCE

- A. Miscellaneous Elements: Do not cut and patch miscellaneous elements or related components in a manner that could change their load-carrying capacity, that results in reducing their capacity to perform as intended, or that results in increased maintenance or decreased operational life or safety.
 - 1.
- B. Visual Requirements: Do not cut and patch construction in a manner that results in visual evidence of cutting and patching. Do not cut and patch construction exposed on exterior or in occupied spaces in a manner that would, in Contracting Officer's opinion, reduce the building's aesthetic

qualities. Remove and replace construction that has been cut and patched in a visually unsatisfactory manner.

1.4 WARRANTY

- A. Existing Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during cutting and patching operations, by methods and with materials so as not to void existing warranties.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. General: Comply with requirements specified in other Sections.
- B. In-Place Materials: Use materials identical to in-place materials. For exposed surfaces, use materials that visually match in-place adjacent surfaces to the fullest extent possible.
 - 1. If identical materials are unavailable or cannot be used, use materials that will match the visual and functional performance of in-place materials when installed.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine surfaces to be cut and patched and conditions under which cutting and patching are to be performed.
 - 1. Compatibility: Before patching, verify compatibility with and suitability of substrates, including compatibility with in-place finishes or primers.
 - 2. Proceed with installation only after unsafe or unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Temporary Support: Provide temporary support of Work to be cut.
- B. Protection: Protect in-place construction during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portions of Project that might be exposed during cutting and patching operations.
- C. Adjoining Areas: Avoid interference with use of adjoining areas or interruption of free passage to adjoining areas.

- D. Existing Utility Services and Mechanical/Electrical Systems: Where existing services/systems are required to be removed, relocated, or abandoned, bypass such services/systems before cutting to prevent interruption to occupied areas.

3.3 PERFORMANCE

- A. General: Employ skilled workers to perform cutting and patching. Proceed with cutting and patching at earliest feasible time. Complete without delay.
 - 1. Cut in-place construction to provide for installation of other components or performance of other construction subsequently. Patch as required to restore surfaces to original condition.
- B. Cutting: Cut in-place construction by sawing, drilling, breaking, chipping, grinding, and similar operations, including excavation, using methods least likely to damage elements retained or adjoining construction. If possible, review proposed procedures with original Installer and original Installer's written recommendations.
 - 1. In general, use hand or small power tools designed for sawing and grinding, not hammering and chopping. Cut holes and slots as small as possible, neatly to size required, and with minimum disturbance of adjacent surfaces. Temporarily cover openings when not in use.
 - 2. Finished Surfaces: Cut or drill from exposed or finished side into concealed surfaces.
 - 3. Concrete: Cut using an abrasive saw or a diamond-core drill.
 - 4. Excavating and Backfilling: Comply with requirements in applicable Division 31 Sections where required by cutting and patching operations.
 - 5. Mechanical and Electrical Services: Cut off pipe or conduit in walls or partitions to be removed. cap, valve, or plug and seal remaining portion of pipe or conduit to prevent entrance of moisture or other foreign matter after cutting.
 - 6. Proceed with patching after construction operations cutting are complete.
- C. Patching: Patch construction by filling, repairing, refinishing, closing up, and similar operations following performance of other Work. Patch with durable seams as invisible as possible. Provide materials and comply with installation requirements specified in other Sections.
 - 1. Inspection: Where feasible, test and inspect patched areas after completion to demonstrate integrity of installation.
 - 2. Exposed Finishes: Restore exposed finishes of patched areas and extend finish restoration into retained adjoining construction in a manner that will eliminate evidence of patching and refinishing.
 - 3. Floors and Walls: Where walls or partitions that are removed extend one finished area into another. Patch and repair floor and wall surfaces in new space. Provide an even surface of uniform finish, color, texture, and appearance. Remove in-place floor and wall coverings and replace with new materials, if necessary, to achieve uniform color and appearance.
 - 4. Ceilings: Patch, repair, or re-hang in-place ceilings as necessary to provide an even-plane surface of uniform appearance.
 - 5. Exterior Building Enclosure: Patch components in a manner that restores enclosure to a weather tight condition.

- D. Cleaning: Clean areas and spaces where cutting and patching are performed. Remove paint, mortar, oils, putty, and similar materials.

END OF SECTION 01 73 29

SECTION 017419 – CLEANING UP-CIVIL

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 RELATED DOCUMENTS

- A. Consult the individual sections of the specifications for cleaning of Work installed under those sections.

1.3 CLEANING DURING CONSTRUCTION

- A. Conduct cleaning and disposal operations to comply with local ordinances, anti-pollution laws, and the National Park Service (NPS).
- B. Do not burn or bury rubbish and waste materials on the site.
- C. Do not dispose of volatile wastes such as mineral spirits, oil, or paint thinner in storm or sanitary drains.
- D. Do not dispose of wastes into streams or waterways.
- E. Wet down dry materials and rubbish to lay dust and prevent blowing dust.
- F. Maintain the site free from accumulations of waste, debris, and rubbish.
- G. Provide on-site containers for collection of waste materials and rubbish.
- H. At the end of each day, remove and legally dispose of waste materials and rubbish from site.
- I. Schedule cleaning operations so that dust and other contaminants, resulting from cleaning process, will not fall on wet, newly applied surfaces.
- J. Disposal of materials will be in compliance with all applicable laws, ordinances, codes, and by-laws.

1.4 FINAL CLEANING

- A. Prior to submitting a request to the Engineer to certify Substantial Completion of the Work, the Contractor will inspect all spaces and verify that all waste materials, rubbish, tools, equipment, machinery, and surplus materials have been removed, and that all exposed surfaces are clean. Leave the Project clean and ready for occupancy.
- B. Unless otherwise specified under other sections of the Specifications, the Contractor will perform final cleaning operations as herein specified prior to final inspection.
- C. Cleaning will include all surfaces which Contractor has had access to, whether new or existing.
- D. Employ experienced workmen or professional cleaners for final cleaning.
- E. Use only cleaning materials recommended by the manufacturer of the surface to be cleaned.
- F. Use cleaning materials which will not create a hazard to health or property and will not damage surfaces.
- G. Remove grease, mastic, adhesive, dust, dirt, stains, labels, fingerprints, and other foreign materials from sight-exposed surfaces. This includes cleaning of the Work of all finishing trades where needed, whether or not cleaning by such trades is included in their respective specifications.
- H. Repair, patch and touch up marred surfaces to the specified finish, to match adjacent surfaces.
- I. In cleaning items with manufacturer's finish, or items previously finished by a Subcontractor, care will be taken not to damage such manufacturer's or Subcontractor's finish. Any damage to finishes caused by cleaning operations will be repaired at the Contractor's expense.
- J. Broom clean exposed concrete surfaces and paved surfaces. Rake clean other surfaces of grounds.
- K. The NPS responsibility for cleaning commences at Substantial Completion and transfer of occupancy from the Contractor to NPS.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 017419

SECTION 01 74 19 - CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for:
 - 1. Recycling nonhazardous demolition and construction waste.
 - 2. Disposing of nonhazardous demolition and construction waste.

1.2 DEFINITIONS

- A. Construction Waste: Building and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.
- B. Demolition Waste: Building and site improvement materials resulting from demolition or selective demolition operations.
- C. Solid Waste: Garbage, debris, sludge, or other discharged material (except hazardous waste) including solid, liquid, semisolid, or contained gaseous materials resulting from domestic, industrial, commercial, mining, or agricultural operations.
- D. Debris: Non-hazardous solid waste generated during construction, demolition, or renovation of a structure which exceeds 2.5 inch (60 millimeter) particle size that is: a manufactured object; plant or animal matter; or natural geologic material (e.g. cobbles and boulders). A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if mixture is comprised primarily of debris by volume, based on visual inspection.
- E. Disposal: Removal off-site of demolition and construction waste and subsequent sale, recycling, reuse, or deposit in landfill or incinerator acceptable to authorities having jurisdiction.
- F. Environmental Pollution and Damage: Presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade utility of environment for aesthetic, cultural, or historical purposes.
- G. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.
- H. Hazardous Materials: Material regulated as a hazardous material in accordance with 49 CFR 173 (Code of Federal Regulations), requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets or has components which meet or have potential to meet the definition of Hazardous Waste in accordance with 40 CFR 261.
- I. Recycle: Recovery of demolition or construction waste for subsequent processing in preparation for reuse.

- J. Salvage: Recovery of demolition or construction waste and subsequent sale or reuse in another facility.
- K. Single-use plastic products: Plastic items intended to be disposed of immediately after use, including plastic and polystyrene food and beverage containers, bottles, straws, cups, cutlery, and disposable plastic bags.

1.3 PERFORMANCE REQUIREMENTS

- A. General: Project shall minimize creation of construction, deconstruction, and demolition waste to protect and restore natural habitat and resources. Minimize factors contributing to waste such as over packaging, improper storage, ordering error, poor planning, breakage, mishandling, single-use plastic products, and contamination. A Waste Management Plan shall be developed to ensure that existing site and building materials are reused, salvaged, or recycled. Minimize waste disposal in landfills.
- B. Salvage /Recycle Requirements: Develop waste management plan resulting in end-of-Project rates for salvage/recycling of 75 percent by weight of total waste generated by the Work. The following waste categories, at a minimum, shall be diverted from a landfill:
 - 1. Land clearing debris (chipped debris can be used on site for mulch or erosion control)
 - 2. Clean dimensional wood, palettes
 - 3. Plywood, OSB (oriented strand board), and particle board
 - 4. Concrete (can be ground and used for fill on site)
 - 5. Asphaltic concrete (can be ground and used for fill on site)
 - 6. Cardboard, paper, packaging, newsprint
 - 7. Metals (from banding, stud trim, piping, rebar, roofing, other trim, steel, iron, galvanized sheet steel, stainless steel, aluminum, copper, zinc, lead, brass, and bronze)
 - 8. Gypsum drywall - unpainted
 - 9. Non-hazardous paint and paint cans
 - 10. Beverage containers: Aluminum, glass, and plastic containers
 - 11. Insulation
 - 12. Ceiling grid and tiles
 - 13. Ductwork
 - 14. Wiring
 - 15. Other mixed construction and demolition waste as appropriate.
- C. If waste materials encountered during deconstruction/demolition or construction phase are found to contain lead, asbestos, polychlorinated biphenyls (PCBs), (such as fluorescent lamp ballasts), or other harmful substances, they are to be handled and removed in accordance with local, state, and federal laws and requirements concerning hazardous waste.
- D. Existing items and material to be removed during deconstruction/demolition phase shall be reused in construction phase of the Project. Items that cannot be reused shall be recycled. Items considered for reuse must be in refurbishable condition and must meet quality standards set forth in these specifications. Contractor shall ensure quality of the item(s) in question will meet or exceed accepted industry or trade standards for first quality commercial grade application. During construction, deconstruction, or demolition Contracting Officer (CO) may designate other objects or materials for reuse.

- E. Salvage/Recycle Requirements: Government goal is to salvage and recycle as much nonhazardous demolition and construction waste as possible. Government has established minimum goals for the following materials:

- 1. **Packaging material plastic, cardboard and wood**

1.4 SUBMITTALS

- A. Waste Management Plan: After award of contract and prior to scheduled Pre-Construction Conference, Contractor shall submit a draft Waste Management Plan to Contracting Officer for approval. Submit [3] copies of plan. Revise and resubmit Plan as required by Contracting Officer. Approval of Contractor's Plan will not relieve Contractor of responsibility for compliance with applicable environmental regulations.
- B. Progress Documentation: Supplemental to Waste Management Plan, document solid waste disposal, diversion, and cost/revenue analysis and submit completed worksheet on a monthly basis. See Project Waste Management Plan Worksheet Sample, attached to the end of the Division 1 Specifications, and report totals to date for column headings.
- C. Recycling and Processing Facility Records: Indicate receipt and acceptance of recyclable waste by recycling and processing facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.
- D. Landfill and Incinerator Disposal Records: Indicate receipt and acceptance of waste by landfills and incinerator facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.
- E. Progress payment requirements:
 - 1. With each Application for payment, submit an updated Project Waste Management Plan worksheet for solid waste disposal and diversion.
 - 2. With each Application for Payment, submit manifests, weight tickets, receipts, and invoices specifically identifying the Project and waste material.
- F. Closeout Submittals
 - 1. With Closeout Submittals, submit a summary of a Project Waste Management Plan worksheet for solid waste disposal and diversion.

1.5 QUALITY ASSURANCE

- A. Regulatory Requirements: Comply with hauling and disposal regulations of authorities having jurisdiction.
- B. Waste Management Meeting: Conduct separate meeting or cover in Pre-Construction Conference and comply with requirements in Section 01 31 00 "Project Management and Coordination." Review methods and procedures related to waste management including:
 - 1. Review and discuss waste management plan including responsibilities of Waste Management Coordinator.

2. Review requirements for documenting quantities of each type of waste and its disposition.
3. Review and finalize procedures for materials separation and verify availability of containers and bins needed to avoid delays.
4. Review procedures for periodic waste collection and transportation to recycling and disposal facilities.
5. Review procedures for reduction of single-use plastic products on site.
6. Review waste management requirements for each trade.

PART 2 - PRODUCTS

2.1 WASTE MANAGEMENT PLAN

- A. General: Develop plan consisting of waste identification and waste reduction work plan. Include separate sections in plan for demolition and construction waste. Indicate quantities by weight or volume but use same units of measure throughout waste management plan.
- B. Waste Identification: Indicate anticipated types and quantities of demolition, site-clearing and construction waste generated by the Work. Include estimated quantities and assumptions for estimates.
- C. Waste Reduction Work Plan: List each type of waste and whether it will be salvaged, recycled, or disposed of in landfill or incinerator. Include points of waste generation, total quantity of each type of waste, quantity for each means of recovery, and handling and transportation procedures.
 1. Salvaged Materials for Reuse: For materials salvaged and reused in Project, describe methods for preparing salvaged materials before incorporation into the Work.
 2. Salvaged Materials for Sale: For materials sold to individuals and organizations, include list of names, addresses, and telephone numbers.
 3. Salvaged Materials for Donation: For materials donated to individuals and organizations, include list of names, addresses, and telephone numbers.
 4. Recycled Materials: Include list of local receivers and processors and type of recycled materials each will accept. Include names, addresses, and telephone numbers.
 5. Disposed Materials: Indicate how and where materials will be disposed of. Include name, address, and telephone number of each landfill and incinerator facility.
 6. Handling and Transportation Procedures: Include method used for separating recyclable waste including sizes of containers, container labeling, and designated location on Project site where materials separation will be located.
 7. Reduction of single-use plastic products: Include methods for water distribution to include bottle refilling stations if available for personnel.
- D. Cost/Revenue Analysis: Indicate total cost of waste disposal as if there was no waste management plan and net additional cost or net savings resulting from implementing waste management plan. Include:
 1. Landfill tip fees per ton.
 2. If diverted, tip fee savings from landfill diversion.
 3. Costs of recycling, salvage, or reuse.
 4. Revenue from recycling, salvage, or reuse.
 5. Total cost or savings from diversion. (Calculate by using tip fee savings and subtracting costs of recycling or adding revenue from recycling.)

PART 3 - EXECUTION

3.1 PLAN IMPLEMENTATION

- A. General: Implement waste management plan as approved by Contracting Officer. Provide handling, containers, storage, signage, transportation, and other items as required to implement waste management plan during entire duration of Contract.
- B. Waste Management Coordinator: Engage waste management coordinator responsible for implementing, monitoring, and reporting status of waste management work plan. Coordinator shall be present at Project site full time for duration of Project.
- C. Contractor shall establish contacts with local recycling and reuse companies to set up lines of responsibility. Contractor shall be responsible for coordination in terms of identifying materials, pickup schedules, and standard quality for recycled materials.
- D. Training: Train workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work occurring at Project site.
 - 1. Distribute waste management plan to everyone concerned within three days of submittal return.
 - 2. Distribute waste management plan to entities when they begin work on-site. Review plan procedures and locations established for salvage, recycling, and disposal.
- E. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.
- F. Separation facilities:
 - 1. Contractor shall designate and Contracting Officer shall approve specific area or areas to facilitate separation of materials for potential reuse, salvage, recycling, and return.
 - 2. Place waste and recycling bins near each other, and close to point of waste generation but out of traffic pattern.
 - 3. Keep recycling and waste bin areas neat, clean, and clearly marked in order to avoid comingling of materials.
 - 4. Protect bins during non-working hours from off-site contamination.
 - 5. Check garbage dumpsters periodically for recyclables being thrown away and undocumented materials that could be recycled.
- G. Materials handling procedures: Material to be recycled shall be protected from contamination and shall be handled, stored, and transported in a manner that meets requirements set by designated facilities for acceptance. Establish defined area for operations of each trade, especially woodcutting so off-cuts are kept in one area and can be sorted by dimension for future reuse.

3.2 SALVAGING DEMOLITION WASTE

- A. Salvaged Items for Reuse in the Work:
 - 1. Clean salvaged items.
 - 2. Pack or crate items after cleaning. Identify contents of containers.
 - 3. Store items in secure area until installation.

4. Protect items from damage during transport and storage.
 5. Install salvaged items to comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make items functional for use indicated.
- B. Salvaged Items for Sale and Donation: Permitted Not permitted on Project site.
- C. Salvaged Items for Governments Use:
1. Clean salvaged items.
 2. Pack or crate items after cleaning. Identify contents of containers.
 3. Allow for inspection if necessary.
 4. Store items in secure area until delivery to Government.
 5. Transport items to storage area [**on-site**] [**off-site**] [**designated by Government**].
 6. Protect items from damage during transport and storage.

3.3 RECYCLING DEMOLITION AND CONSTRUCTION WASTE, GENERAL

- A. General: Recycle paper and beverage containers used by on-site workers.
- B. Recycling Incentives: Revenues, savings, rebates, tax credits, and other incentives received for recycling waste materials shall accrue to Contractor.
- C. Procedures: Separate recyclable waste from other waste materials, trash, and debris. Separate recyclable waste by type at Project site to maximum extent practical.
1. Provide appropriately marked containers or bins for controlling recyclable waste until removed from Project site. Include list of acceptable and unacceptable materials at each container and bin.
 - a. Inspect containers and bins for contamination and remove contaminated materials if found.
 2. Stockpile processed materials on-site without intermixing with other materials. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
 3. Stockpile materials away from construction area. Do not store within drip line of remaining trees.
 4. Store components off ground and protect from weather.
 5. Remove recyclable waste off Government property and transport to recycling receiver or processor.

3.4 RECYCLING DEMOLITION WASTE

- A. Asphaltic Concrete Paving: Grind asphalt to maximum 4 inch size.
- B. Asphaltic Concrete Paving: Break up and transport paving to asphalt-recycling facility.
1. Crush asphaltic concrete paving and screen to comply with requirements in Division 31 Section "Earth Moving".
- C. Concrete: Remove reinforcement and other metals from concrete and sort with other metals.

1. Pulverize concrete to maximum 4 inch size.
- D. Masonry: Remove metal reinforcement, anchors, and ties from masonry and sort with other metals.
1. Pulverize masonry to maximum 4 inch size.
 2. Clean and stack undamaged, whole masonry units on wood pallets.
- E. Wood Materials: Sort and stack members according to size, type, and length. Separate lumber, engineered wood products, panel products, and treated wood materials.
- F. Metals: Separate metals by type.
1. Structural Steel: Stack members according to size, type of member, and length.
 2. Remove and dispose of bolts, nuts, washers, and other rough hardware.
- G. Asphalt Shingle Roofing: Separate organic and glass-fiber asphalt shingles and felts. Remove and dispose of nails, staples, and accessories.
- H. Gypsum Board: Stack large clean pieces on wood pallets and store in a dry location. Remove edge trim and sort with other metals. Remove and dispose of fasteners.
- I. Acoustical Ceiling Panels and Tile: Stack large clean pieces on wood pallets and store in dry location.
1. Separate suspension system, trim, and other metals from panels and tile and sort with other metals.
- J. Carpet and Pad: Roll large pieces tightly after removing debris, trash, adhesive, and tack strips.
1. Store clean, dry carpet and pad in closed container or trailer provided by Carpet Reclamation Agency or carpet recycler.
- K. Equipment: Drain tanks, piping, and fixtures. Seal openings with caps or plugs. Protect equipment from exposure to weather.
- L. Plumbing Fixtures: Separate by type and size.
- M. Piping: Reduce piping to straight lengths and store by type and size. Separate supports, hangers, valves, sprinklers, and other components by type and size.
- N. Lighting Fixtures: Separate lamps by type and protect from breakage.
- O. Electrical Devices: Separate switches, receptacles, switchgear, transformers, meters, panelboards, circuit breakers, and other devices by type.
- P. Conduit: Reduce conduit to straight lengths and store by type and size.
- Q. Electronic Products: Ensure non-usable electronic products are reused, donated, sold, or recycled using environmentally sound management practices at end of life.

3.5 RECYCLING CONSTRUCTION WASTE

A. Packaging:

1. Cardboard and Boxes: Break down packaging into flat sheets. Bundle and store in dry location.
2. Polystyrene Packaging: Separate and bag materials.
3. Pallets: As much as possible, require deliveries using pallets to remove pallets from Project site. For pallets that remain on-site, break down pallets into component wood pieces and comply with requirements for recycling wood.
4. Crates: Break down crates into component wood pieces and comply with requirements for recycling wood.

B. Site-Clearing Wastes: Chip brush, branches, and trees at landfill facility.

C. Wood Materials:

1. Clean Cut-Offs of Lumber: Grind or chip into small pieces.
2. Clean Sawdust: Bag sawdust not containing painted or treated wood.

D. Gypsum Board: Stack large clean pieces on wood pallets and store in dry location.

1. Clean Gypsum Board: Grind scraps of clean gypsum board using small mobile chipper or hammer mill. Screen out paper after grinding.

3.6 DISPOSAL OF WASTE

A. General: Except for items or materials to be salvaged, recycled, or otherwise reused, remove waste materials from Project site and legally dispose in landfill or incinerator acceptable to authorities having jurisdiction.

1. Except as otherwise specified, do not allow waste materials to accumulate on-site.
2. Remove and transport debris in manner preventing spillage on adjacent surfaces and areas.

B. Burning: Do not burn waste materials.

C. Burning: Burning of waste materials allowed only at designated areas on Government property, provided required permits are obtained. Provide full-time monitoring for burning materials until fires are extinguished.

D. Disposal: Transport waste materials off Government property and legally dispose of them.

END OF SECTION 01 74 19

SECTION 01 77 00 - CLOSEOUT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for contract closeout, including:
 - 1. Project Record Drawings
 - 2. Closeout Submittals
 - 3. Substantial Completion and Final Inspection
 - 4. Permit Closure and Transfer
 - 5. Final Acceptance of the Work
 - 6. Warranties

1.2 PROJECT RECORD DRAWINGS

- A. Maintain one complete full-size set of contract drawings and one full-size set of vendor-supplied drawings. Clearly mark changes, deletions, and additions using National Park Service (NPS) drafting standards to show actual construction conditions. Show additions in red, deletions in green and special instructions in blue.
- B. Keep record drawings current. Make record drawings available to Contracting Officer (CO) for inspection at the time of monthly progress payment requests. If project record drawings are not current, Contracting Officer may retain an appropriate amount of progress payment.
- C. Submit complete record drawings on completion of total project. Include shop drawings, sketches, and additional drawings to be included in final set, with clear instructions showing the location of these drawings.

1.3 CLOSEOUT SUBMITTALS

- A. A list of closeout requirements has been attached at the end of the Division 1 Specifications for your convenience. The intent is to provide an overall summary of requirements and not a comprehensive list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list. Submit the following before requesting final inspection:
 - 1. Specific warranties, guarantees, workmanship bonds, final certifications, and similar documents.
 - 2. NPS required forms for occupancy, Fire Sprinkler/Alarm acceptance, and other similar forms or certificates.
 - 3. Project Record Documents, operation and maintenance manuals, final completion construction digital images recorded on CD-R (compact disc-recordable) or DVD-R (digital video disc-recordable) with index and descriptions, and similar final record information.

4. Posted Operating Instructions: As specified in individual sections. Furnish operating instructions attached to or posted adjacent to equipment. Include wiring diagrams, control diagrams, control sequence, start-up, adjustment, operation, lubrication, shut-down, safety precautions, procedures in the event of equipment failure, and other items of instruction recommended by manufacturer.
5. Deliver tools, spare parts, extra materials, and similar items to location designated by Contracting Officer. Label with manufacturer's name and model number where applicable.
 - a. Special Tools: One set of special tools required to operate, adjust, dismantle, or repair equipment. Special tools are those not normally found in possession of mechanics or maintenance personnel.
6. Keys and Keying Schedule: Submit keys including duplicates. Wire keys for each lock securely together. Tag and plainly mark with lock number, equipment identification, or panel or switch number, and indicate location, building, and room name or number.
7. Make final changeover of permanent locks and deliver keys to Contracting Officer. Advise Park personnel of changeover in security provisions.
8. Approved pre-functional checklists and functional performance testing reports from commissioning documentation.
 - a. Equipment start-up requires coordination with commissioning process. Refer to drawings and specs. Equipment shall not be "temporarily" started for commissioning.
9. Test and balance report.
10. Terminate and remove temporary facilities, mockups, construction tools, and similar elements from Project site, complete final cleaning requirements, including touchup painting.
11. Touch up and repair and restore marred exposed finishes to eliminate visual defects.
12. Instruct NPS personnel in operation, adjustment, and maintenance of products, equipment, and systems. Submit demonstration and training videos.

1.4 FINAL INSPECTION, SUBSTANTIAL COMPLETION AND ACCEPTANCE PROCEDURES

- A. Request final inspection in writing when project or designated portion of project is substantially complete. Contracting Officer will proceed with inspection within 10 days of receipt of written request or will advise Contractor of items that prevent project from being substantially complete.
- B. If work is determined substantially complete, following final inspection, Contracting Officer will prepare Punch List and issue a Letter of Substantial Completion.
- C. If work is not determined substantially complete following final inspection, Contracting Officer will notify Contractor in writing. Contractor shall request new final inspection after completing work. Re-inspection costs may be charged against Contractor in accordance with Inspection of Construction contract clause.
- D. Contractor shall complete Punch List within 30 calendar days, documented weather permitting.
 1. Prior to requesting final inspection:

- a. Complete commissioning requirements of drawings and specs unless approved in writing by Contracting Officer.
- E. If Contractor completes items of work on Punch List and contractually required items, Contracting Officer will issue Letter of final acceptance of work.
- F. If Contractor fails to complete work within the time frame, Contracting Officer may correct work with an appropriate reduction in contract price or charge for re-inspection costs in accordance with Inspection of Construction contract clause.

1.5 PERMIT CLOSURE AND TRANSFER

- A. When work covered by the permits is complete, create list of tasks required to close or transfer permits to Park. Submit to Contracting Officer for approval.
- B. After substantial completion and Punch List completion, permits shall be closed and documented by Agency(ies) with Jurisdiction for the permit.
- C. If responsibility for permits is to be transferred to Park, Park shall be informed of permit provisions completed and responsibilities transferring to Park staff.

1.6 WARRANTIES

- A. Submittal Time: Submit written warranties on request of Contracting Officer for designated portions of Work where commencement of warranties other than date of Substantial Completion is indicated.
- B. Organize warranty documents into an orderly sequence based on table of contents of Project Manual.
 - 1. Bind warranties and bonds in heavy-duty, 3-ring, vinyl-covered, loose-leaf binders, thickness as necessary to accommodate contents, and sized to receive 8-1/2 by 11 inch paper.
 - 2. Provide heavy paper dividers with plastic-covered tabs for each separate warranty. Mark tab to identify product or installation. Provide typed description of product or installation, including name of product and name, address, and telephone number of Installer.
 - 3. Identify each binder on front and spine with typed or printed title "WARRANTIES," Project name, and name of Contractor.
 - 4. Warranty Electronic File: Scan warranties and bonds and assemble complete warranty and bond submittal package into a single indexed electronic PDF (portable document format) file with links enabling navigation to each item. Provide bookmarked table of contents at beginning of document.
- C. Provide additional copies of each warranty in operation and maintenance manuals.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. See Division 1 Specification Section "Execution" for information on cleaning agents.

PART 3 - EXECUTION

3.1 FINAL CLEANING

- A. General: Conduct final cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.
- B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.
 - 1. Complete following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a portion of Project:
 - a. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
 - b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
 - c. Rake grounds that are neither planted nor paved to smooth, even-textured surface.
 - d. Remove tools, construction equipment, machinery, and surplus material from Project site.
 - e. Remove snow and ice to provide safe access to building.
 - f. Clean exposed exterior and interior hard-surfaced finishes to dirt-free condition, free of stains, films, and foreign substances. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to original condition.
 - g. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.
 - h. Sweep concrete floors broom clean in unoccupied spaces.
 - i. Vacuum carpet and similar soft surfaces, removing debris and excess nap; shampoo, soil or stains remain.
 - j. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and vision-obscuring materials. Replace chipped or broken glass and transparent materials. Polish mirrors and glass.
 - k. Remove labels that are not permanent.
 - l. Touch up, repair, and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.
 - 1) Do not paint over "UL" (Underwriters Laboratories) and similar labels, including mechanical and electrical nameplates.

- m. Wipe surfaces of mechanical and electrical equipment[,and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.
 - n. Replace parts subject to unusual operating conditions.
 - o. Clean plumbing fixtures to sanitary condition, free of stains, including stains resulting from water exposure.
 - p. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of diffusers, registers, and grills.
 - q. Clean light fixtures, lamps, globes, and reflectors to function with full efficiency. Replace burned-out and noticeably dimmed bulbs, and defective or noisy starters in fluorescent and mercury vapor fixtures to comply with requirements for new fixtures.
 - r. Leave Project clean and ready for occupancy.
- C. Pest Control: Engage experienced, licensed exterminator to make a final inspection and rid project of rodents, insects, and other pests. Provide Government with report.
- D. Waste Disposal: Comply with requirements of Section 01 74 19 "Construction Waste Management and Disposal."

END OF SECTION 01 77 00

SECTION 01 78 23 - OPERATION AND MAINTENANCE DATA

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for preparing operation and maintenance manuals, including:
 - 1. Manuals, general
 - 2. Emergency manuals
 - 3. Operation manuals for systems, subsystems, and equipment
 - 4. Maintenance manuals for care and maintenance of systems and equipment
- B. See Divisions 2 through 49 Sections for additional operation and maintenance manual requirements for Work in those Sections.

1.2 SUBMITTALS

- A. Manual: Submit two copies of each manual in draft form or one electronic copy at least 15 days before final inspection. Contracting Officer (CO) will return copy or edit version with comments within 15 days of receipt.
- B. Format: Submit operations and maintenance manuals in following format:
 - 1. PDF (portable document format) electronic file. Assemble each manual into composite electronically indexed file. Submit on digital media acceptable to Contracting Officer.
 - a. Name each indexed document file in composite electronic index with applicable item name. Include a complete electronically linked operation and maintenance directory.
 - b. Enable inserted reviewer comments on draft submittals.
 - 2. Hard copy manual: In accordance with Part 2 of this Section.
 - 3. Correct or modify each manual to comply with Contracting Officers comments. Submit 2 copies of each corrected manual within 15 days of receipt of Contracting Officers comments.

1.3 QUALITY ASSURANCE

- A. Coordinate with Section 01 91 14 "Total Building Commissioning." Commissioning Agent shall review Operation and Maintenance Manuals for commissioned systems.

PART 2 - PRODUCTS

2.1 MANUALS, GENERAL

- A. Organization: Unless otherwise indicated, organize manual into separate sections for each system and subsystem, and separate sections for each piece of equipment not part of a system. Manual shall contain title page, table of contents, and manual contents.
- B. Title Page: Enclose title page in transparent plastic sleeve. Include:
 - 1. Project Title
 - 2. Location
 - 3. Park
 - 4. Contract Number
 - 5. Prime Contractors Name and Address
 - 6. Date of Substantial Completion
 - 7. Binder Volume Number
- C. Table of Contents: List each product included in manual, identified by product name, indexed to content of the volume, and cross-referenced to Specification Section number in Project Manual.
- D. Manual Contents: Organize into sets of manageable size. Arrange contents alphabetically by system, subsystem, and equipment. Assemble instructions for subsystems, equipment, and components of one system into a single binder if needed.
 - 1. Binders: White, commercial quality, hard back, 3-ring, vinyl-covered, loose-leaf binders, in thickness necessary to accommodate contents, sized to hold 8-1/2 by 11 inch paper; with clear plastic window sleeve on front and spine to hold label describing contents and pockets inside covers to hold folded oversize sheets.
 - a. Cover Sheet: Identify binders on front and spine, with project title, location, park, contract number, prime contractor's name and address, date of substantial completion, and binder volume number. Insert cover sheet into clear plastic view pocket on front of binder. Insert sheet into clear plastic view pocket on spine with title "OPERATION AND MAINTENANCE MANUAL," Project title or name, and subject matter of contents. Indicate volume number for multiple-volume sets.
 - 2. Data: Fill binders to no more than 75 percent of capacity. Punch holes shall not obscure any data. When contents of a single tabbed section cover more than one item, provide colored paper sheets to separate the data for each item.
 - a. Manufacturers' Data: Provide originals for color or copyrighted data. Black and white data may be originals or clean, good quality reproductions. No copies produced by facsimile transmission and sheets with stamps, such as submittal approval stamps. Include only sheets that apply to items installed; cross out inapplicable data.
 - b. Vendor Furnished As-Built Drawings: Maximum 24 inch by 36 inch sheets with minimum character or lettering size of 1/8 inch. Reduced-size reproductions may be provided instead of full-size drawings if reproductions are clear and legible. If

- reduced-size drawings are used, identify as "REDUCED SIZE" and provide graphic scales, if applicable.
 - c. Custom Data: Data supplemented by drawings and schematics necessary to describe systems adequately.
 - d. Equipment Data Sheet: Data, using form at end of this section.
 - e. Schedules: Schedules reflecting final, as-installed conditions.
 - f. Poorly reproduced or illegible data will be rejected.
3. Dividers: Divider sheets with Mylar reinforced edges and pre-printed numbered tabs aligned with numbers and title lines on index sheet. Include typed list of products and major components of equipment included in section on each divider, cross-referenced to Specification Section number and title of Project Manual.
 4. Protective Plastic Sleeves: Transparent plastic sleeves designed to enclose diagnostic software diskettes for computerized electronic equipment.
 5. Drawings: Attach reinforced, punched binder tabs on drawings and bind with text.
 - a. If oversize drawings are necessary, fold drawings to same size as text pages and use as foldouts.
 - b. If drawings are too large to be used as foldouts, fold and place drawings in labeled envelopes and bind envelopes in rear of manual. At appropriate locations in manual, insert typewritten pages indicating drawing titles, descriptions of contents, and drawing locations.

2.2 EMERGENCY MANUALS

- A. Content: Organize manual into separate section for type of emergency, emergency instructions, and emergency procedures.
- B. Type of Emergency: Where applicable for each type of emergency indicated below, include instructions and procedures for each system, subsystem, piece of equipment, and component for equipment failure.
- C. Emergency Instructions: Describe and explain warnings, trouble indications, error messages, and similar codes and signals. Include responsibilities of National Park Service (NPS) operating personnel for notification of Installer, supplier, and manufacturer to maintain warranties.
- D. Emergency Procedures: Include instructions on stopping, shutdown instructions for each type of emergency, operating instructions for conditions outside normal operating limits, and required sequences for electric or electronic systems.

2.3 OPERATION AND MAINTENANCE MANUALS

- A. Operation Requirements
 1. Content: In addition to requirements in Section, include operation data required in individual Specification Sections and equipment descriptions, operating standards, operating procedures, operating logs, wiring and control diagrams, and license requirements.
 2. Descriptions: Include:

- a. Product name and model number
 - b. Manufacturer's name
 - c. Equipment identification with serial number of each component
 - d. Equipment function
 - e. Operating characteristics
 - f. Limiting conditions
 - g. Performance curves
 - h. Engineering data and tests
 - i. Complete nomenclature and number of replacement parts
3. Operating Procedures: Include start-up, break-in, and control procedures; stopping and normal shutdown instructions; routine, normal, seasonal, and weekend operating instructions; and required sequences for electric or electronic systems.
 4. Systems and Equipment Controls: Describe sequence of operation, and diagram controls as installed.
 5. Piped Systems: Diagram piping as installed and identify color-coding where required for identification.

B. Maintenance Requirements for Systems and Equipment

1. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranty and bond information, and equipment data sheets as described below.
2. Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.
3. Manufacturers' Maintenance Documentation: Manufacturers' maintenance documentation including maintenance instructions, drawings and diagrams for maintenance, nomenclature of parts and components, and recommended spare parts for each component part or piece of equipment:
4. Maintenance Procedures: Test and inspection instructions, troubleshooting guide, disassembly instructions, and adjusting instructions, detailing essential maintenance and environmental procedures.
5. Maintenance and Service Schedules: Service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.
6. Spare Parts List and Source Information: Lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.
7. Warranties and Bonds: Copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

2.4 PRODUCT MAINTENANCE MANUAL

- A. Content: Organize manual into separate section for each product, material, and finish. Include source information, product information, maintenance procedures, repair materials and sources, and warranties and bonds, as described below.
- B. Source Information: List each product included in manual identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.
- C. Product Information: Include:
 - 1. Product name and model number
 - 2. Manufacturer's name
 - 3. Color, pattern, and texture
 - 4. Material and chemical composition
 - 5. Reordering information for specially manufactured products
- D. Environmental Requirements
 - 1. Identify environmentally preferable products incorporated into Project. Include: product model; manufacturer's name, address, phone, and website; and local technical representative.
 - a. Verify plastic products to be incorporated into Project are labeled in accordance with ASTM D1972. Where products are not labeled, provide product data indicating polymeric information in Operation and Maintenance Manual.
 - 1) Type 1: Polyethylene Terephthalate (PET, PETE)
 - 2) Type 2: High Density Polyethylene (HDPE)
 - 3) Type 3: Vinyl (Polyvinyl Chloride or PVC)
 - 4) Type 4: Low Density Polyethylene (LDPE)
 - 5) Type 5: Polypropylene (PP)
 - 6) Type 6: Polystyrene (PS)
 - 7) Type 7: Other. Use of this code indicates that package in question is made with a resin other than the six listed above or is made of more than one resin listed above and used in a multi-layer combination.
 - b. Describe maintenance procedures associated with environmentally preferable materials and systems. Provide cleaning recommendations in accordance with ASTM E1971 and approved Integrated Pest Management (IPM) plan.
 - 1) Include potential environmental impacts of recommended maintenance procedures and materials.
 - 2) Include potential indoor air quality impacts of recommended maintenance procedures and materials.
 - 3) Where proposed maintenance procedures incorporate composting of plastics, assess potential effect of each type of plastic to be included in composting process in accordance with ASTM D5509 or ASTM D6002

- c. Material Safety Data Sheets (MSDS): Include MSDSs as specified.
- 2. Develop environmental management programs for facility as follows:
 - a. Waste management program: Develop in accordance with ASTM E1609. Maximize use of source reduction and recycling procedures outlined in ASTM D5834.
 - b. Indoor Air Quality (IAQ) management program: Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of VOCs (volatile organic compounds) in indoor air in accordance with ASTM D6345.
 - c. Water management program: Develop water monitoring program for surface and ground water on project site in accordance with ASTM D5851 and consistent with water management program utilized during construction operations.
- E. Maintenance Procedures: Include manufacturer's written recommendations and inspection procedures, types of cleaning agents, methods of cleaning, schedule for cleaning and maintenance, and repair instructions.
- F. Repair Materials and Sources: Include lists of materials and local sources of materials and related services.
- G. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

2.5 SYSTEMS AND EQUIPMENT MAINTENANCE MANUALS

- A. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranty and bond information, as described below.
- B. Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual and drawing or schedule designation or identifier where applicable.
- C. Manufacturers' Maintenance Documentation: Manufacturers' maintenance documentation including the following information for each component part or piece of equipment:
 - 1. Standard maintenance instructions and bulletins.
 - 2. Drawings, diagrams, and instructions required for maintenance, including disassembly and component removal, replacement, and assembly.
 - 3. Identification and nomenclature of parts and components.
 - 4. List of items recommended to be stocked as spare parts.
- D. Maintenance Procedures: Include the following and items detailing essential maintenance procedures:

1. Test and inspection instructions
 2. Troubleshooting guide
 3. Precautions against improper maintenance
 4. Disassembly; component removal, repair, and replacement; and reassembly instructions
 5. Aligning, adjusting, and checking instructions
 6. Demonstration and training video recording
- E. Maintenance and Service Schedules: Include service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.
1. Scheduled Maintenance and Service: Tabulate actions for daily, weekly, monthly, quarterly, semiannual, and annual frequencies.
 2. Maintenance and Service Record: Include manufacturers' forms for recording maintenance.
- F. Spare Parts List and Source Information: Include lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.
- G. Maintenance Service Contracts: Include copies of maintenance agreements with name and telephone number of service agent.
- H. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.
1. Include procedures and required notifications for warranty claims.

PART 3 - EXECUTION

3.1 GENERAL

- A. At start of project, begin accumulating operation and maintenance data and initiate index. Install and index data in binders within 30 days after delivery of items. As custom written data and test results are produced, add to operation and maintenance data file.
- B. List of Operation and Maintenance requirements has been attached at end of the Division 1 Specifications for your convenience. Intent is to provide an overall summary of requirements and not a comprehensive list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list.
- C. Keep operation and maintenance data current. Make operation and maintenance binders available to Contracting Officer for inspection at time of monthly progress payment requests. If operation and maintenance binders are not current, Contracting Officer may retain an appropriate amount of the progress payment.

3.2 MANUAL PREPARATION

- A. Manual Types

1. Emergency Manual: Assemble complete set of emergency information indicating procedures for use by emergency personnel and by NPS operating personnel for types of emergencies indicated.
2. Product Maintenance Manual: Assemble complete set of maintenance data indicating care and maintenance of each product, material, and finish incorporated into Work.
3. Operation and Maintenance Manuals: Assemble complete set of operation and maintenance data indicating operation and maintenance of each system, subsystem, and piece of equipment not part of a system.

B. Manual Contents: Including:

1. Manufacturers' Data: Where manuals contain manufacturers' standard printed data, include only sheets pertinent to product or component installed. Mark sheet to identify product or component incorporated into Work. If data include more than one item in a tabular format, identify each item using appropriate references from Contract Documents. Identify data applicable to Work and delete references to information not applicable.
2. Custom Written Data: For data not in manufacturer's standard literature, provide text, drawings, and schematics specifically applicable to installed systems. Include step-by-step descriptions of operating procedures; identification of individual components and their functions; descriptions of how system components relate to one another and operate together to accomplish a common process or function; and sequence of operation for system control circuits. For seasonally operated systems, provide start-up and shutdown instructions.
3. Equipment Data Sheets: For each item of equipment included in operation and maintenance data, provide Equipment Data Sheet using form at the end of this section. For equipment consisting of a driven machine and a driver (for example, a pump and a motor), equipment data shall cover both the driven machine and the driver. For similar type equipment (for example, multiple exhaust fans of the same model and type), provide a single equipment data sheet with an attached schedule listing individual equipment items.
4. Vendor Furnished As-Built Drawings: Provide for each electrical and each mechanical control system.
 - a. For each control system, provide control circuit schematic drawings. Identify each wire and terminal block number. Show terminal numbers on control devices. Show control wires and devices remote from control panel.
 - b. For each control panel, provide general arrangement drawing showing location of each control component and terminal block on the panel front and interior. Include materials list of panel-mounted control components as well as field-installed control components remote from the panel, identifying components, manufacturer, model number, and initial set points or sensing ranges of devices where applicable.
 - c. For packaged equipment systems, provide general arrangement drawings showing interrelationships of the various items of equipment and components.
 - d. In addition to control wiring schematic, provide power wiring schematic drawing showing power flow to each motor. Identify each power conductor. Show over-current protection and motor starting devices.

C. Comply with Section 01 77 00 "Closeout Procedures" for schedule for submitting operation and maintenance documentation.

END OF SECTION 01 78 23

EQUIPMENT DATA SHEET	
Equipment Item: _____ Designation: _____ Function: _____ Location: _____ Project: _____ Model Number: _____ Serial Number: _____	
Manufacturer Address and Phone: _____ _____ _____	Supplier Address and Phone: _____ _____ _____
Preventive Maintenance Tasks: _____ _____ _____ _____ _____	
Nameplate Data: _____ _____ _____ _____ _____	
Spare Parts Furnished and Other Information: _____ _____ _____ _____ _____	

SECTION 01 79 00 - DEMONSTRATION AND TRAINING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes administrative and procedural requirements for instructing National Park Service (NPS) personnel, including:
 - 1. Demonstration of operation of systems, subsystems, and equipment.
 - 2. Training in operation and maintenance of systems, subsystems, and equipment, including environmental considerations.
 - 3. Demonstration and training video.
- B. See Divisions 2 through 49 Sections for specific requirements for demonstration and training for products in those Sections.

1.2 GENERAL REQUIREMENTS

- A. List of System Demonstration and Training requirements has been attached at end of Division 1 Specifications for your convenience. Intent is to provide overall summary of requirements and not a comprehensive list. Terms and conditions of contract still require satisfaction of requirements of individual specification sections regardless of what is shown on list.

1.3 SUBMITTALS

- A. Instruction Program: Submit outline of instructional program for demonstration and training, including schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Include learning objective and outline for each training module.
 - 1. For each training session, Contractor shall submit for approval a proposed outline of subjects to be covered. Training shall not be conducted until outline is approved.

1.4 QUALITY ASSURANCE

- A. Facilitator Qualifications: Firm or individual experienced in training or educating maintenance personnel in training program similar in content and extent for this Project, and whose work has resulted in training or education with a record of successful learning performance.
- B. Instructor Qualifications: Factory-authorized service representative, complying with requirements in Section 01 40 00 "Quality Requirements," experienced in operation and maintenance procedures and training.
- C. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals. Do not submit instruction program until operation and maintenance data has been reviewed and approved by Contracting Officer (CO).

PART 2 - PRODUCTS

2.1 INSTRUCTION PROGRAM

- A. Program Structure: Develop instruction program that includes individual training modules for each system and equipment not part of a system, as required by individual Specification Sections, and:

1. Septic Pumps. Controls, Alarms and BAS connection

2. Environmental Topics

- a. Overview of environmental issues related to building industry.
- b. Overview of environmental issues related to Project.
- c. Review of site-specific procedures and management plans implemented during construction:
 - 1) Regulatory Requirements
 - 2) Indoor Air Quality (IAQ) Management
 - 3) Noise and Acoustics Management
 - 4) Environmental Management
 - 5) Construction Waste Management
- d. Review of site-specific procedures and management plans to be implemented during operation and maintenance.
 - 1) Include review of environmentally related aspects of Operations and Maintenance Manual.
 - 2) Integrated Pest Management (IPM)

- B. Training Modules: Develop a learning objective and teaching outline for each module. Include description of specific skills and knowledge that participant is expected to master. For each module, include instruction for:

- 1. Basis of System Design, Operational Requirements, and Criteria: Include system and equipment descriptions, operating standards, regulatory requirements, equipment function, operating characteristics, limiting conditions, and performance curves.
- 2. Documentation: Review emergency, operations, and maintenance manuals; Project Record Documents; identification systems; warranties and bonds; and maintenance service agreements.
- 3. Emergencies: Instructions on stopping; shutdown instructions; operating instructions for conditions outside normal operating limits; instructions on meaning of warnings, trouble indications, and error messages; and required sequences for electric or electronic systems.
- 4. Operations: Startup, break-in, control, and safety procedures; stopping and normal shutdown instructions; routine, normal, seasonal, and weekend operating instructions; operating procedures for emergencies and equipment failure; and required sequences for electric or electronic systems.
- 5. Adjustments: Alignments and checking, noise, vibration, economy, and efficiency adjustments.
- 6. Troubleshooting: Diagnostic instructions and test and inspection procedures.
- 7. Maintenance: Inspection procedures, types of cleaning agents, methods of cleaning, procedures for preventive and routine maintenance, and instruction on use of special tools.

8. Repairs: Diagnosis, repair, and disassembly instructions; instructions for identifying parts; and review of spare parts needed for operation and maintenance.

PART 3 - EXECUTION

3.1 INSTRUCTION

- A. Facilitator: Engage qualified facilitator to prepare instruction program and training modules, to coordinate instructors, and to coordinate between Contractor and Contracting Officer for number of participants, instruction times, and location.
- B. Engage qualified instructors to instruct NPS personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.
- C. Scheduling: Provide instruction at mutually agreed on times. For equipment that requires seasonal operation, provide similar instruction at start of each season.
 1. Schedule training with NPS through Contracting Officer with at least 7 days' advance notice.
 2. Conduct training sessions after equipment or system has been accepted and turned over to Government. Coordinate with commissioning requirements.
 3. Coordinate with Integrated pest management requirements. Refer to specifications section and approved IPM plan.
 4. Individual sections specify duration of training required. If no duration is listed, provide training of sufficient duration to adequately cover subjects.
 - 5.

END OF SECTION 01 79 00

SECTION 024113 – SELECTIVE SITE DEMOLITION

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 SUMMARY

- A. Work to include the demolition and abandonment of existing beehive leaching pits, a portion of the existing gravity sanitary line, and any auxiliary features of the existing leaching field, and selective demolition of topsoil and groundcover.
- B. Except for items or materials indicated to be reused, salvaged, reinstalled or otherwise indicated to remain the Owner's property, demolished materials shall become the Contractor's property and shall be removed from the site with further disposition at the Contractor's option and in full compliance with all applicable disposal regulations.

1.3 DESCRIPTION OF WORK

- A. Work Included:
 - 1. Demolition and removal of selected site elements as required for new work. Refer to the Drawings for additional requirements.
 - 2. Salvage of existing items to be reused or turned over to the owner.
 - 3. Removal and legal disposal of demolished materials off site. Except those items specifically designated to be relocated, reused, or turned over to the owner, all existing removed materials, items, trash, unsuitable soils, stumps, and debris shall become property of the Contractor and shall be completely removed from the site and legally disposed of at the Contractor's expense. Salvage value belongs to the Contractor. On-site sale of materials is not permitted.
 - 4. Demolition and removal work shall properly prepare for alteration work and new construction to be provided under the Contract.
 - 5. Scheduling and sequencing operations without interrupting utilities serving occupied areas. If interruption is required, obtain written permission from the utility company. Provide temporary services as necessary to serve occupied and usable facilities when permanent utilities must be interrupted, and schedule interruption when the least amount of inconvenience will result.

- B. Related Work: The following items are not included in this Section and will be performed under the designated Sections:

- 1. Section 01 5000 - TEMPORARY FACILITIES: Maintenance of access, cleaning during construction, dust, and noise control.

1.4 DEFINITIONS

- A. Remove: Detach items from existing construction and legally dispose of them off-site, unless indicated to be removed and salvaged or removed and reinstalled.
- B. Remove and Salvage: Detach items from existing construction and deliver them to the Owner ready for reuse, at a location designated by the Owner. Protect from weather until accepted by Owner.
- C. Remove and Reinstall: Detach items from existing construction, prepare them for reuse, and reinstall them where indicated. Protect from weather until reinstallation.
- D. Existing to Remain: Existing items of construction that are not to be removed and that are not otherwise indicated to be removed, removed and salvaged, or removed and reinstalled.

1.5 MATERIALS OWNERSHIP

- A. Historic items, relics, and similar objects that may be encountered during demolition shall remain property of the Owner as applicable. Carefully remove each item or object in a manner to prevent damage and deliver promptly to a location acceptable to the Owner.
- B. Excess topsoil: Refer to Site Clearing 31 1000.
- C. Except for materials indicated to be stockpiled, reused, or to remain as the Owner's property, cleared materials shall become the Contractor's property and shall be removed from the site.

1.6 SUBMITTALS

- A. Schedule of Selective Demolition Activities: Indicate the following:
 - 1. Detailed sequence of selective demolition and removal work, with early and late starting and finishing dates for each activity. Ensure Owner's on-site operations are uninterrupted if applicable.
 - 2. Coordination of Owner's continuing occupancy of portions of existing site.
 - 3. Means of protection for items to remain and items in path of removal.
- B. Submit photographs or videotape, sufficiently detailed, of existing conditions of adjoining construction and site improvements that might be misconstrued as damage caused by selective demolition operations. Submit photographs of existing items to be removed and reinstalled to record original condition of objects to be retained.

1.7 RECORD DRAWINGS

- A. Record drawings at Project Closeout shall be in accordance with Division 01.
- B. Identify and accurately locate capped utilities and other subsurface conditions.

1.8 REGULATORY REQUIREMENTS

- A. Comply with governing local, State and EPA notification regulations, before starting selective demolition. Comply with the hauling and disposal regulations of any authorities having jurisdiction.
- B. The Owner will occupy portions of the facilities immediately adjacent to selective demolition areas. Conduct selective demolition so that the Owner's operations will not be disrupted. Provide not less than 72 hours of notice to the Owner of activities (if any) that may affect the Owner's operations.
- C. The Owner assumes no responsibility for the actual condition of facilities or items to be selectively demolished or removed and reused.
- D. Storage or sale of removed items or materials on-site will not be permitted without the Owner's permission.

1.9 QUALITY ASSURANCE

- A. Examination of Existing Conditions: The Contractor shall examine the Contract Drawings for demolition and removal requirements and provisions for new work. Verify all existing conditions and dimensions before commencing work. The Contractor shall visit the site and examine the existing conditions as he finds them and shall inform themselves of the character, extent, and type of demolition and removal work to be performed. Submit any questions regarding the extent and character of the demolition and removal work in the manner and within the time period established for receipt of such questions during the bidding period.

1.10 WARRANTY

- A. Existing Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during selective demolition, by methods and with materials so as not to void existing warranties.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 EXECUTION

- A. Prior to commencing any excavation or demolition, the Contractor shall take all actions necessary to fully protect the existing facilities from damage. The Contractor shall take all actions required to repair any damage and return the fields to their existing conditions.
- B. Survey the condition of the site to determine whether removing any element might result in the undesirable damage of any portion of the adjacent facilities during selective demolition.

- C. Perform surveys as the Work progresses to detect hazards resulting from selective demolition activities.
- D. Conduct demolition operations and remove debris to ensure minimum interference with roads, parking lots, streets, walks, and other adjacent occupied and utilized facilities.
- E. Conduct demolition operations to prevent injury to people and damage to adjacent buildings, facilities, and site improvements to remain. Ensure safe passage of people around selective demolition areas.
- F. Use water mist and other suitable methods, as necessary, to limit the spread of dust and dirt. Refer to ENVIRONMENTAL PROTECTION 01 3543.
- G. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
- H. Clean adjacent structures and improvements of dust, dirt, and debris caused by selective demolition operations. Return adjacent areas to conditions existing before the start of selective demolition. Refer to CLEANING UP-CIVIL 01 7419.
- I. Demolish and remove existing construction only to the extent required by new construction and as indicated. The Contractor is to be responsible for any cutting and patching that is required.
- J. Promptly patch and repair holes and damaged surfaces caused to adjacent construction by selective demolition operations.
- K. Where repairs to existing surfaces are required, patch to produce surfaces suitable for new materials.
- L. Restore exposed finishes of patched areas and extend finish restoration into adjoining construction to remain in a manner that eliminates evidence of patching and refinishing.
- M. Disposal: Promptly dispose of demolished materials. Do not allow demolished materials to accumulate on-site.
- N. Do not burn demolished materials.
- O. Transport demolished materials off of the Owner's property and legally dispose of them, if they are not designated for salvage by the Owner or reuse.
- P. In areas where bituminous concrete is to be removed, the edge of any bituminous concrete to remain must be a sawcut edge.
- Q. Items to be removed and reset may be stored on site, at a location approved by the Owner.
- R. Maintain existing utilities indicated to remain in service and protect them against damage during selective demolition operations.

3.2 PREPARATION

- A. Site Access and Temporary Controls: Conduct selective demolition and debris-removal operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.
 - 1. Comply with requirements for access and protection specified in Section 01 5000 - TEMPORARY FACILITIES AND CONTROLS.
- B. Conduct demolition operations to prevent injury to people and damage to adjacent buildings and facilities to remain. Ensure safe passage of people around demolition area(s).
 - 1. Erect temporary protection, such as walks, fences, railings, canopies, and covered passageways, where required by authorities having jurisdiction. Provide temporary barricades as required to limit access to demolition areas.
 - 2. Protect existing site improvements, appurtenances, and landscaping to remain.

3.3 DISCOVERY OF HAZARDOUS MATERIALS

- A. If hazardous materials, such as chemicals, asbestos-containing materials, or other hazardous materials are discovered during the course of the work, cease work in affected area only and immediately notify the Engineer of such discovery. Do not proceed with work in such areas until instructions are issued by the Engineer. Continue work in other areas.
- B. If unmarked containers are discovered during the course of the work, cease work in the affected area only and immediately notify the Engineer of such discovery. Do not proceed with work in such areas until instructions are issued by the Engineer. Take immediate precautions to prohibit endangering the containers integrity. Continue work in other areas.

3.4 CUTTING

- A. Provide a flush saw cut edge where pavement, curb, and concrete removals abut new construction work or existing surfaces to remain undisturbed.

3.5 DISPOSAL OF DEMOLISHED MATERIALS

- A. General: Comply with requirements of Section 01 7419 - CLEANING UP-CIVIL and the following.
 - 1. Do not allow demolished materials to accumulate on-site.
 - 2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
 - 3. Remove debris from elevated portions of building by chute, hoist, or other device that will convey debris to grade level in a controlled descent.
- B. Burning: Do not burn demolished materials.

3.6 CLEANING

- A. Clean adjacent structures and improvements of dust, dirt, and debris caused by selective demolition operations. Premises shall be left in a clean condition and ready to accept alteration work and new construction.

END OF SECTION 024113

SECTION 032000 – CONCRETE REINFORCEMENT

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 RELATED WORK

- A. Examine Contract Documents for requirements that affect work of this Section. Other Specification Sections that directly relate to work of this Section include, but are not limited to:

- 1. Section 03 3000 – Cast-in-Place Concrete

1.3 WORK INCLUDED

- A. In general, the Contractor shall supply all labor, materials, equipment, temporary protection, tools, and appliances necessary for the proper completion of the work in this section, as required in the specifications and in accordance with good construction practice. The work under this section generally includes the following:
 - 1. Furnish and install reinforcing bars, tie wires, and supports, as required.
 - 2. All reinforcing must be grounded in accordance with the National Electric Code and MA State Building Code. Coordinate with electrical engineer and contractor.
 - 3. Clean all areas affected by the work.

1.4 REFERENCES

- A. Comply with applicable requirements of the following standards (latest edition). Where these standards conflict with other requirements, the most restrictive requirements shall govern.
 - 1. American Concrete Institute Standards (ACI), Latest Edition.
 - a. 117- Specifications for Tolerances for Concrete Construction and Materials and Commentary
 - b. 301- Specifications for Structural Concrete
 - c. 318 - Building Code Requirements for Structural Concrete
 - d. 315 - Details and Detailing of Concrete Reinforcement
 - e. 315R Manual of Engineering and Placing Drawings for Reinforced Concrete Structures
 - 2. American Society for Testing and Materials (ASTM)
 - a. A615 Deformed and Plain Billet Steel Bars for Concrete Reinforcement
 - 3. Concrete Reinforcing Steel Institute (CRSI)
 - a. Manual of Standard Practice, Latest Edition
 - b. Placing Reinforcing Bars, Latest Edition

1.5 SUBMITTALS

- A. The following shall be submitted in accordance with Section 01 3323 SUBMITTAL PROCEDURES:
- B. Shop Drawings:

Show sizes and dimensions for fabrication and placing of reinforcing steel and bar supports. Indicate bar schedules, stirrup spacing, and diagrams of bent bars, including all accessories. Detail reinforcing in accordance with ACI 215.
- C. Certificates:
 - 1. Mill test certificates; identifying the chemical and physical analysis of each load of reinforcing steel delivered.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver reinforcement to project site, in bundles, marked with metal tags indicating bar size and length.
- B. Handle and store materials to prevent contamination and contact with the ground. Store reinforcing steel under cover and protect from rusting, oil, grease and/or distortion.
- C. Remove from storage only those materials needed for immediate use.

1.7 QUALITY CONTROL

- A. Reinforcing steel shall be fabricated to conform to the required shapes, dimensions and tolerances specified in the CRSI Manual.
- B. Allowable Tolerances:
 - 1. Fabrication Tolerances: Sheared Length: Plus or minus 1 inch for bends. Stirrups, ties, and spirals: Plus or minus 1/2 inch.
 - 2. Placement Tolerances:
 - a. When member depth (or thickness) is 4 in. or less: Plus or minus 1/4 inch.
 - b. When member depth (or thickness) is over 4 in. and not over 12 in.: Plus or minus 3/8 inch.
 - c. When member depth (or thickness) is over 12 in.: Plus or minus 1/2 in.
 - 3. Minimum spacing between bars: The greater of the bar diameter or 1 in. for unbundled bars.
 - 4. Top bars in slabs and beams: Members 8 inches deep or less: Plus or minus 1/4in.; Members between 8 inches and 2 feet: Plus or minus 1/2 inch; Members 2 feet deep or greater: Plus or minus 1 inch.
 - 5. Crosswise of members: Spaced evenly within 2 inches.
 - 6. Lengthwise of members: Plus or minus 1 inch.
 - 7. Maximum bar movement, to avoid interference with other reinforcing steel, conduits, or embedded items: 2 bar diameters.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Reinforcing Bars; Ties and Stirrups:
 - 1. Deformed billet steel: ASTM A615, Grade 60.
 - 2. Bend test: Meet 90° bend test at 60° F minimum temperature, around a 10 bar diameter bend, without cracking.
- B. Tie Wire:
 - 1. Annealed Steel - Federal Specification QQ-W-461, 16 gage minimum.
- C. Bar Supports; Accessories:
 - 1. Conform to "Bar Support Specifications", CRSI Manual of Standard
 - 2. Practice, Class B – Pre-galvanized Cold-Drawn Wire.

3. All necessary spacers, ties, chains, bolsters and other devices required to properly support and fasten reinforcing shall be galvanized or plastic, in accordance with ACI 315. Legs or other parts in contact with forms of exposed surfaces shall be plastic coated.

2.2 FABRICATION

- A. In accordance with CRSI Manual of Standard Practice.
- B. Fabricate and mark reinforcing bars, in accordance with ACI 315 and ACI 318.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Remove all mud, oil, loose rust, or mill scale and other foreign materials that may reduce bond prior to placing concrete. "Tight" rust or mill scale will be permissible without cleaning or brushing, provided weights and dimensions are not less than the minimum required by referenced specifications.
- B. Form oils shall be placed prior to erecting around reinforcement; or, if applied in the vicinity of reinforcement, adequate masking shall be temporarily placed to avoid coating the reinforcing steel with form release agents.
- C. All bars shall be cold bent to the required shapes before they are placed in the forms. Reinforcement shall not be straightened or re-bent in a manner that will injure, damage or weaken the material.
- D. Bars with kinks or bends not required shall not be used.

3.2 INSTALLATION

- A. Reinforcing Bar Placement:
 1. Conform to CRSI-WCRSI, "Placing Reinforcing Steel".
 2. Position bars in accordance with above tolerances and secure in place.
 3. All rods shall be securely wired together at all intersections.
 4. Where continuous bars are called for, they shall be run continuously around corners and lapped at necessary splices or booked at discontinuous ends. Laps shall be forty (40) bar diameters, unless otherwise shown.
- B. Notify the Engineer for inspection of the completed installation of reinforcement at least forty-eight (48) hours prior to the scheduled placement of concrete.

C. Bar Supports:

1. Provide minimum number of supports, as required by ACI 315.
2. Do not use pebbles, pieces of broken stone, brick, concrete, metal pipe, or wood blocks to support reinforcement. Do not use bar supports as support for runways for concrete buggies or similar loads. Do not place bars more than 2 inches beyond the last leg at the end of a run of continuous supports.
3. All reinforcement shall be securely held in place with approved supporting, spacing and tying devices. Concrete supports require the Engineer's approval.
4. Maximum space between bar supports shall be 24" in each direction.

D. Concrete Cover:

1. Except as otherwise indicated on the construction drawings; provide the minimum clearance for concrete cover, in accordance with ACI 318.

E. Reinforcing Adjustment:

1. Move only within allowable tolerances to avoid interference with other reinforcing steel, conduits, or embedded items. Do not move bars beyond allowable tolerance without approval of the Engineer.
2. Do not heat, bend or cut bars without the approval of the Engineer.

F. Splices:

1. Do not splice bars, except at locations on the details, without the approval of the Engineer.
2. Minimum lap distance is as shown on the details and as specified in ACI 318. Tie splices securely with wire to prevent displacement during the placing of concrete.

END OF SECTION 032000

SECTION 26 00 50 - COMMON WORK RESULTS FOR ELECTRICAL

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Electrical equipment coordination and installation.
 - 2. Common electrical installation requirements.

1.2 DESCRIPTION OF WORK

- A. Requirements of this Section are applicable to work in Division 26.
- B. Contract Documents
 - 1. Contract drawings for electrical work are diagrammatic, intended to convey scope and general arrangement.
 - 2. Refer questions involving document interpretation or discrepancies to Contracting Officer for review and direction.
 - 3. Correct faulty work due to resolving discrepancies without proper approval.
 - 4. Specifications establish quality of materials, equipment, workmanship and methods of construction.
 - 5. Follow drawings and specifications in laying out work. Consult other applicable contract drawings and specifications. Become familiar with conditions affecting work.
- C. Scope
 - 1. Furnish and install the electrical work complete and ready for satisfactory service.
- D. Definitions: The following are definitions of terms and expressions used in Divisions 26 and 27.
 - 1. "Accessible" – Capable of being removed or exposed without damaging the building or structure or finish or not permanently closed in by other equipment or by the structure or finish of the building.
 - 2. "Approve" - To permit use of material, equipment or methods conditional upon compliance with contract document requirements.
 - 3. "Concealed" - Hidden from normal sight; includes work in crawl spaces, above ceilings, and in building shafts.
 - 4. "Directed" - directed by Contracting Officer.
 - 5. "Equal, equivalent" - possessing the same performance qualities and characteristics and fulfilling the same utilitarian function.
 - 6. "Exposed" - not concealed.
 - 7. "Furnish" - Supply and deliver to project site, ready for unloading, unpacking, assembly, installation, and similar operations.
 - 8. "Indicated" - indicated in Contract Documents.
 - 9. "Install" - Operations at project site including unloading, temporarily storing, unpacking, assembling, erecting, placing, anchoring, applying, working to dimensions, finishing, curing, protecting, cleaning and similar operations.
 - 10. "Provide" - furnish and install, complete and ready for the intended use.
 - 11. "Removable" - detachable from the structure or system without physical alteration of materials or equipment and without disturbance to other construction.

12. "Review" - limited observation or checking to ascertain general conformance with design concept of the work and with information given in contract documents. Such action does not constitute a waiver or alteration of the contract requirements.

1.3 QUALITY ASSURANCE

- A. Regulations: Comply with regulations of NFPA, state, county, and municipal building ordinances, and other applicable codes and regulations.
- B. Provide UL label on electric powered equipment or certification that equipment has been tested by a testing agency approved by the local authority as equivalent in safety to UL labeled equipment.
- C. Material and Equipment Requirements
 1. All materials and equipment shall be new and free from defects.
 2. Use products of one manufacturer where two or more items of same kind of equipment are required.
 3. For certain items of equipment, the specification and the project design are based upon the specified manufacturer's product. Other manufacturers' names are listed. Contractor may purchase, conditional upon meeting project requirements, equipment from the listed manufacturers.
 4. Only the manufacturer's equipment upon which the specification and the project design has been based, has been checked for this project. Check allocated space and structure for suitability of equipment of other listed manufacturers, including parts replacement and servicing.
- D. Workmanship
 1. Remove and replace, at no extra cost, work not in conformance with contract requirements.
 2. Coordinate with Other Trades
 - a. Coordinate work and cooperate with other trades to facilitate execution of work.
 - b. Give full cooperation and coordination with other trades and furnish information necessary to permit the work of all trades to be installed satisfactorily with the least possible interference or delay.
 - c. Furnish to other trades, as required, necessary templates, patterns, setting plans and shop details for the proper installation of the work and for the purpose of coordinating adjacent work.
 3. Accessible Equipment and Systems: Consider all materials and equipment installations and coordinate with the work of other trades to ensure equipment or systems are accessible for operations, maintenance, repairs, and replacement. Install materials and equipment, including but not limited to, supports and electrical conduit, to permit complete unobstructed access to panelboards, transformers, and other items requiring access for inspection, maintenance, and operations. The installation of new equipment or materials which renders new or existing equipment inaccessible will be disapproved by the Engineer and shall be corrected by the Contractor.

1.4 COORDINATION

- A. Coordinate arrangement, mounting, and support of electrical equipment as follows:

1. To allow maximum possible headroom unless specific mounting heights that reduce headroom are indicated.
 2. To provide for ease of disconnecting the equipment with minimum interference to other installations.
 3. To allow right of way for piping and conduit installed at required slope.
 4. So that connecting raceways, cables, wireways, cable trays, and busways will be clear of obstructions and of the working and access space of other equipment.
- B. Coordinate installation of required supporting devices and set sleeves in cast-in-place concrete, masonry walls, and other structural components as they are constructed.
- C. Coordinate location of access panels and doors for electrical items that are behind finished surfaces or otherwise concealed.
- D. Coordinate sleeve selection and application with selection and application of firestopping specified in Division 7 Section "Penetration Firestopping."

1.5 SHOP DRAWINGS AND SUBMITTALS

- A. Refer to Division 01 for complete requirements.
- B. Submit all products for a single specification section as a complete submittal. All products specified within a division shall be included, otherwise submittal will be returned as incomplete.
- C. Clearly mark submittals to indicate actual intended products to be utilized. Marks may include highlighting, circling, boxing, checking, etc. Do not provide submittal data which lists multiple product's options and features without clearly indicating which data applies to the products intended to be used on project.
- D. Coordinate drawings and data before submitting and certify that provisions of the contract documents have been met.
- E. Call attention, in writing, to deviations from contract requirements.
- F. Do not fabricate, deliver to site, or install items requiring shop drawing review, until the review has been completed by the Engineer and the shop drawing has been marked to indicate "No Exception Taken" or "Make Corrections Noted."
- G. Use only final or corrected drawings and data for construction. This includes all Addendums, Architectural Supplemental Information (ASIs), and Change Bulletins.
- H. The Engineer's review of submittals shall not be construed as a complete check, but will indicate only that the general method of construction, materials, detailing and other information are satisfactory. Approval will not relieve the Contractor of the responsibility for any error which may exist, as the Contractor under the requirements of this contract is responsible for dimensions, the design of adequate connections and details, and the satisfactory construction of all work.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 COMMON REQUIREMENTS FOR ELECTRICAL INSTALLATION

- A. Comply with NECA 1.
- B. Measure indicated mounting heights to bottom of unit for suspended items and to center of unit for wall-mounted items.
- C. Headroom Maintenance: If mounting heights or other location criteria are not indicated, arrange and install components and equipment to provide maximum possible headroom consistent with these requirements.
- D. Equipment: Install to facilitate service, maintenance, and repair or replacement of components of both electrical equipment and other nearby installations. Connect in such a way as to facilitate future disconnecting with minimum interference with other items in the vicinity.
- E. Right of Way: Give to piping systems installed at a required slope.

END OF SECTION 26 00 50

SECTION 033000 – CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 RELATED REQUIREMENTS

- A. Section 03 20 00 – Concrete Reinforcement

1.3 SCOPE OF WORK

- A. In general, the Contractor shall supply all labor, materials, equipment, temporary protection, tools, formwork, and appliances necessary for the proper completion of the work in this section, as required in the specifications and in accordance with good construction practice. The work under this Section includes cast-in-place concrete as shown on the contract documents.
- B. Clean all areas affected by the work to the satisfaction of the Owner.

1.4 JOB CONDITIONS

- A. The Contractor shall provide all protection, barriers, and guards necessary to segregate his work area and the areas below, from pedestrian and vehicular traffic. Also protect existing buildings, landscaping, and paved areas from damage.
- B. The Contractor shall be responsible for securing and protecting their equipment, materials and tools (as well as partially completed construction) from wind blow-off and vandalism or abuse.
- C. Environmental Requirements: Do not place concrete during rain, sleet, or snow unless adequate protection is provided and the Engineer's approval is obtained. Do not allow rainwater to increase the mixing water or damage the surface finish.

D. Cold Weather Concreting:

1. Conform to ACI 306 latest edition, "Recommended Practice for Cold Weather Concreting."
2. Temperature of concrete when placed shall not be less than the following:

Minimum Concrete Temperature °F
Sections with Least Dimension

Air Temp (°F)	Under 12"	12" and Over
30 to 45	60	50
0 to 30	65	55
Below 0	70	60

3. When placed, heated concrete shall not be warmer than 80° F.
4. Prior to placing concrete, all ice, snow, and surface and subsurface frost shall be removed, and the temperature of the surfaces to be in contact with the new concrete shall be raised to the temperature specified above for placing.
5. Protect the concrete from freezing for four (4) days after placement.
6. Heated enclosures shall be strong and windproof to ensure adequate protection of corners, edges, and thin sections. Do not permit heating units to locally heat or dry the concrete. Do not use combustion heaters during the first 24 hours unless the concrete is protected from exposure to exhaust gases which contain carbon dioxide.
7. When air temperature gets below 25 degrees F, two (2) additional ASTM C39 cylinders shall be made and located at the site in a location and under conditions which will match the placement that they represent. After seven (7) days of site conditions, the cylinders shall be placed in a steam room for twenty-one (21) days.

E. Hot Weather Concreting:

1. Conform to ACI 305 latest edition, "Recommended Practice for Hot Weather Concreting." Take precautions when the ambient air temperature is 90° or above. Temperature of the concrete when placed shall not exceed 80° F. Cool forms and reinforcing to a maximum of 90° F by spraying with water prior to placing concrete. Do not use cement that has reached temperatures in excess of 170° F.

F. Prevent plastic shrinkage cracking due to rapid evaporation of moisture. Do not place concrete when the evaporation rate (actual or anticipated) equals or exceeds 0.20 pounds per square foot per hour, as determined by Figure 2.1.4 of ACI 305.

1. Set-retarding admixtures may be used with Engineer's approval when the ambient air temperature is 90° F or above to off-set the accelerating effects of high temperatures.

1.5 QUALITY ASSURANCE

- A. Reference Standards: Except as modified or supplemented herein, all concrete materials, placing, furnishing, curing, and all other appurtenant work shall meet the requirements of the latest edition of the following Standard Specifications. Pertinent portions of the reference standards are included herein. Refer to the standards for detailed requirements.
 - 1. AMERICAN CONCRETE INSTITUTE STANDARDS (ACI)
 - a. 301 - Standard Specifications for Structural Concrete for Buildings.
 - b. 304 - Recommended Practice for Measuring, Mixing, Transporting, and Placing Concrete.
 - c. 316 - Building Code Requirements for Reinforced Concrete
 - 2. AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
 - a. ATMS C109 "Test Method for Compressive Strength of Hydraulic Cement Mortars"

1.6 SUBMITTALS

- A. Refer to Section 01 3323 – Submittal Procedures. Supplement with the following:
- B. Test Reports: Perform and submit test reports for the following products in accordance with above general reference standards and specific standards set forth hereafter.
- C. Proposed Mix Design:
 - 1. Prior to commencing concrete work submit and obtain Engineer's approval of certified test report describing proposed concrete mix design, including:
 - a. Fine Aggregates - Source, type, gradation, deleterious substances and saturated surface dry specific gravity (ASTM C128).
 - b. Coarse Aggregates - Source, type, gradation, deleterious substances and saturated surface dry specific gravity (ASTM C127); soundness (ASTM C88).
 - c. Ratio of fine to total aggregates.
 - d. Weight (surface dry) of each aggregate per cubic yard.
 - e. Total water content (gallons) per cubic yard, water/cementitious materials ratio and proposed source.
 - f. Slump on which design is based, ASTM C143.
 - g. Brand, type and quantity of cement.
 - h. 7-day and 28-day compressive strength results from each of two sets of test cylinders for each proposed mix.
 - i. Air Content, ASTM C231 or ASTM C173.
 - j. Certifications of Chloride Content of admixtures.
 - k. Water soluble chloride ion content of concrete, ASTM G1218.
 - l. Proportions of all ingredients including all admixtures added either at time of batching or at job site.

D. Cylinder Compression Test Reports:

1. Submit two copies of certified test reports to Engineer indicating results of tests required in Part 3 hereof.

E. Ready-Mix Delivery Tickets:

1. Submit one copy to the Engineer of ready-mix delivery ticket for each load delivered.
2. Include identification and quantity of concrete supplied.
3. Include time loaded and time unloaded.
4. Reading of revolution counter at times initial water added, supplemental water added, and unloading completed.
5. Amounts of initial and supplemental water added, and name of individual authorizing supplementing water.

1.7 DELIVERY, STORAGE, AND HANDLING

- A. Store cement in watertight enclosures and protect against dampness, contamination and warehouse set.
- B. Stock pile aggregates to prevent segregation, or contamination with other materials or other sizes of aggregates. Use only one supply source for each aggregate stockpile.
- C. Store admixtures to prevent contamination, evaporation or damage. Protect liquid admixtures from freezing or harmful temperature ranges. Agitate emulsions prior to use.
- D. Store rubber and plastic materials in a cool place away from direct sunlight.

1.8 INSPECTION AND TESTING

- A. The Contractor agrees to accept as final the results of tests, inspection, and reports as may be made by the testing laboratory.
- B. Inspection
 1. During the progress of the work, the General Contractor shall provide free and safe access to the work at all times to the Engineer and the Owner's representative. He/she shall cooperate with the Engineer to obtain proper inspection of all work and shall furnish any required samples of concrete for testing.

C. Laboratory Inspection and Testing

1. During the progress of the work, a testing laboratory paid for by the Contractor, contacted and coordinated by the Contractor, and approved by the Engineer, shall conduct necessary field tests and make compensation for any variation in water content of the aggregate; and shall further direct that all batches shall be as nearly uniform as possible by the use of selected materials which are accurately measured, thoroughly mixed, and maintained at a constant water-cement ratio and consistency.
2. Provide the Owner and Engineer with necessary reports covering all of the above.
3. The payment for laboratory inspection and testing will be the responsibility of the Contractor.
4. Coordination and scheduling of tests by the testing lab shall be the responsibility of the Contractor.
5. Testing required because of changes requested by the Owner in materials, sources of materials, or mix proportions; and extra testing of concrete or materials because of failure to meet the Specification requirements are to be paid for by the Owner.

D. Required Testing During Construction:

The following minimum testing shall be performed, and field/ lab- results submitted to the structural Engineer for approval:

1. Air entrainment at placement – ASTM C231
2. Slump – ASTM C143
3. Compressive strength – ASTM C39

Concrete cylinder samples shall be obtained from each concrete delivery truck for compressive strength testing. Five (5) cylinders shall be made from each sample. Each cylinder shall be standard 6" diameter by 12" tall. One (1) cylinder will be tested at 7-day cure, and three (3) cylinders will be tested at 28-day cure to determine compressive strength of the concrete in accordance with ASTM C39. Air entrainment and slump will be tested at each sample as well. Retain the fifth cylinder sample for potential 56 day compressive testing and/ or petrographic examination. Test results which are determined by the Engineer to be deficient or questionable will require that the contractor pay for additional testing and coring of the in-place concrete, including petrographic examination with report as directed by the Engineer. Concrete determined by the Engineer to be deficient after final testing shall be entirely removed and replaced at no additional cost.

1.9 GUARANTEES

Upon completion of the work and prior to final payment, the Contractor shall submit a guarantee of his work as free from defect in materials and workmanship. The guarantee shall be for a period of three (3) years. The guarantee shall be signed by an officer of the Contractor's firm and sealed if a corporation.

PART 2 - PRODUCTS

2.1 CONCRETE MATERIALS

- A. Portland Cement: ASTM C 150, Type I or II.
- B. Aggregate: ASTM C 33, uniformly graded, from a single source. Maximum aggregate size = $\frac{3}{4}$ " at slabs.
- C. Water: ASTM C 94.
- D. Air-Entraining Admixture: ASTM C 260.
- E. Water-Reducing Admixture: ASTM C 494, Type A.
- F. High-Range, Water-Reducing Admixture: ASTM C 494, Type F.
- G. Water-Reducing and Accelerating Admixture: ASTM C 494, Type E.
- H. Water-Reducing and Retarding Admixture: ASTM C 494, Type D.

2.2 CONCRETE PRODUCTION

- A. Concrete Mixes, General - Prepare design mixes, proportioned according to ACI 211.1 and ACI 301-05.

Refer to the Contract Drawings for additional information.

- B. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete according to ASTM C 94 and ASTM C 1116, and furnish batch ticket information.
- C. Concrete shall have a minimum compressive strength of 4500 psi for foundations and 4500 psi for slabs at 28 days with a slump of no more than 4" and air entrainment of $4\frac{1}{2}$ to $7\frac{1}{2}\%$.
- D. Proportioning: Proportion ingredients to produce a well-graded mix of high density and maximum workability consistent with approved mix design and subject to the characteristics as specified in the Contract Drawings.
- E. Mixing:
 - 1. Central Mixed Concrete - 1 minute for mixer capacities one cubic yard or less plus 15 seconds for each cubic yard or fraction thereof of additional capacity.
 - 2. Truck Mixed Concrete - 100 revolutions after the introduction of all ingredients.

F. Tempering and Control of Mixing Water:

1. Mix concrete only in quantities for immediate use. Do not use concrete which has stiffened due to initial set or concrete which cannot be discharged within 1-1/2 hours or 300 revolutions of the mixer drum after the introduction of the mixing water.
2. Water may be added to concrete arriving at the site, only if neither the maximum slump nor the maximum water cement ratio is exceeded. Provide additional cement if required by the addition of water to maintain water cement ratio within specified limits. Obtain Engineer's approval prior to adding water or cement.
3. Incorporate any added water or cement by additional mixing equal to half the total mixing required.

2.3 CURING MATERIALS

- A. Impervious-sheet materials shall conform to ASTM C 171, type optional, except that polyethylene sheet shall not be used.
- B. Burlap and cotton mat used for curing shall conform to AASHTO M 182, Class 2.

2.4 WATER

- A. Water for mixing and curing shall be fresh, clean, potable, and free of injurious amounts of oil, acid, salt, or alkali, except that non-potable water may be used if it meets the requirements of ASTM C94.

2.5 EMBEDDED ITEMS

- A. Embedded items shall be of the size and type indicated or as needed for the application.
- B. All other embedded items shall also be securely anchored and protected from damage or displacement.

2.6 JOINT MATERIALS

- A. Expansion joint fillers shall be preformed materials conforming to ASTM D 1751.
- B. Sawable type contraction joint inserts shall conform to COE CRD-C 540. Nonsawable joint inserts shall have sufficient stiffness to permit placement in plastic concrete without undue deviation from a straight line and shall conform to the physical requirements of COE CRD-C 540, with the exception of Section 3.4 "Resistance to Sawing". Plastic inserts shall be polyvinyl chloride conforming to the materials requirements of COE CRD-C 572.

- C. Expansion joint fillers shall be a closed-cell, non-absorbent, synthetic foam, and as recommended by the sealant manufacturer. Filler shall be totally compatible with sealant, primer, and substrates. Backers shall conform to the requirements of ASTM C 962, Type A, such as Ceramar as manufactured by W.R. Meadows, Expansion Joint Filler as manufactured by BASF-Sonneborn, or approved equal.

PART 3 - EXECUTION

3.1 PREPARATION FOR PLACING

- A. Before commencing concrete placement, the following shall be performed:
- B. Surface Preparation:
 - 1. Surfaces to receive concrete shall be clean and free from frost, ice, mud, and water.
 - 2. Earth (subgrade, base, or subbase courses) surfaces upon which concrete is to be placed shall be clean, damp, and free from debris, frost, ice, and standing or running water. The foundation shall be well drained and shall be satisfactorily graded and uniformly compacted.
- C. Equipment:
 - 1. Transporting and conveying equipment shall be in-place, ready for use, clean, and free of hardened concrete and foreign material.
 - 2. Equipment for consolidating concrete shall be at the placing site and in proper working order.
 - 3. Equipment and material for curing and for protecting concrete from weather or mechanical damage shall be at the placing site, in proper working condition, and in sufficient amount for the entire placement.
- D. When hot, windy conditions during concreting appear probable, equipment and material shall be at the placing site to provide windbreaks, shading, fogging, or other action to prevent plastic shrinkage, cracking, or other damaging drying of the concrete.
- E. Before placement of concrete, care shall be taken to determine that all embedded items are firmly and securely fastened in place as indicated on the drawings, or required. Conduit and other embedded items shall be clean and free of oil and other foreign matter such as loose coatings or rust, paint, and scale. The embedding of wood in concrete will be permitted only when specifically authorized or directed. Voids in sleeves, inserts, and anchor slots shall be filled temporarily with readily removable materials to prevent the entry of concrete into voids. Welding shall not be performed on embedded metals within 2 feet of the surface of the concrete. Tack welding shall not be performed on or to embedded items.
- F. Forms shall be in place, cleaned, coated, and adequately supported. Reinforcing steel shall be in place, cleaned, tied, and adequately supported, in accordance with Section 03 2000, CONCRETE REINFORCEMENT.

3.2 INSTALLATION

A. Conveying:

1. Convey concrete from mixer to final position as rapidly as practical without segregation or loss of material.
2. Use only metal or metal lined chutes with maximum length of 20 feet, maximum slope 1 vertical to 2 horizontal and minimum slope 1 vertical to 3 horizontal.
3. Provide a hopper at the end of long belt conveyors and chutes not meeting the above requirements.
4. Conveying by pumping methods shall conform to ACI 304. Maximum loss of slump, 2 inches. Do not use pipe made of aluminum or aluminum alloy to convey concrete. Should pumping be required for this project, all costs for pumping shall be borne by the Contractor. No additional compensation will be considered for any pumping costs.

B. Depositing:

1. Deposit concrete in a continuous operation until the section is completed. Regulate rate of placement so concrete remains plastic and flows into position.
2. Maximum height of concrete free fall is 4 feet.
3. All concrete shall be placed within 2 hours of batching. All concrete on site more than 2 hours from batching time shall be rejected and sent back to the plant.

C. Consolidation:

1. Use mechanical vibrating, rodding, or spading for consolidation. Conform to 309-72, "Recommended Practice For Consolidation of Concrete."
2. Do not use vibrators to transport concrete in forms.
3. Minimum vibrator speed 8000 rpm.
4. Vertically invert vibrators at points 18 inches apart to a depth sufficient to penetrate 6 inches into the preceding layer. Vibrate each location for a length of time to obtain adequate consolidation (generally 5 to 15 seconds).

D. Embedments:

1. Accurately position and securely fasten all anchor bolts, castings, steel shapes, conduit, sleeves, and other materials to be embedded in the concrete.
2. Embedments shall be clean when installed. Remove concrete spatter from all surfaces not in contact with concrete.

E. Wash-out:

1. The Contractor shall remove residue from concrete mixing wash-out from all landscape, walkways, curbs, driveways, and similar surfaces to the satisfaction of the Owner.

3.3 CURING

A. Normal Conditions

1. All concrete shall be prevented from drying for at least the first 7 days after placing. All slabs shall be cured by spraying on the specified curing compound as per the manufacturer's printed instructions. Concrete walls shall be cured as carefully as the slabs. However, instead of covering the sides with the curing compound, it would be satisfactory if the forms were "loosened after the concrete had hardened" and the wall sprinkled with water frequently for at least five (5) days allowing the water to flow down the sides between the forms and the concrete. After the five-day wetting the forms may be removed. Curing compounds that discolor the concrete are not permitted.

B. Cold Weather Conditions

1. Whenever the temperature of the surrounding air is below 40 degrees F, all concrete shall be maintained at a temperature of not less than 50 degrees F for at least 72 hours and shall be protected from freezing for at least another 72 hours, or for as much time as is necessary to insure proper curing of the concrete. The housing, covering, or other protection used in connection with the curing shall remain in place and intact for at least 24 hours after the artificial heating is discontinued. No dependence shall be placed on salt or other chemicals for the prevention of freezing. The approved practice for Winter Concreting are those outlined in ACI 306.

C. Alternates

1. Methods of curing other than those specified above shall be approved by the Engineer before being used.

3.4 FINISHING CONCRETE

A. Defective Concrete:

1. Any concrete which is not formed as shown on the plans or for any reason is out of alignment or level, or shows a defective surface, shall be corrected or replaced as directed by the Engineer.
2. Repair all surface defects and tie holes immediately after form removal.
3. Remove honeycombed or otherwise defective concrete to sound concrete with square cut edges to avoid feathering.

B. Patching:

1. Immediately after removing the forms, all concrete surfaces shall be inspected and any poor joints, voids, stone pockets, other defective areas, and all tie holes shall at once be patched before the concrete is thoroughly dry. The patching shall be done in such a manner that it shall form a homogeneous part, in appearance, and action of the main concrete. Fins shall be removed and patched as required where concrete is exposed.

C. Exposed Concrete:

1. All exposed concrete finish shall be as produced through the use of new smooth plywood or metal forms.

D. Trowel and Fine-Broom Finish:

1. Apply a partial trowel finish, stopping after second troweling, to surfaces indicated and to surfaces where ceramic or quarry tile is to be installed by either thickset or thin-set method. Immediately after second troweling, and when concrete is still plastic, slightly scarify surface with a fine broom.

3.5 FIELD QUALITY CONTROL

A. Concrete Tests: Conduct the following minimum tests in accordance with the requirements of ACI 301, Section 16.3.

1. Strength Test:
 - a. Mold and cure five (5) cylinders from each sample. Test one at 7 days for information and three (3) at 28 days for acceptance. Retain one (1) cylinder for potential 56-day compressive testing and/ or petrographic examination.
2. Slump Test: Conduct test for each strength test sample and whenever consistency of concrete appears to vary.
3. Air Content: Conduct test from one of first three batches mixed each day and for each strength test sample.

B. Acceptance of Concrete:

1. The strength level of concrete will be considered satisfactory so long as the average of all sets of three consecutive strength test results equals or exceeds the specified 28-day strength and no individual strength test result falls below the specified strength by more than 200 psi.
2. Upon failure of test cylinder results, the Owner may require the Contractor, at their expense, to obtain and test at least three 2-inch diameter core samples from the area in question. Conform to ASTM C42. Concrete will be considered adequate if the average of the three cores is at least 85% of, and if no single core is less than 75% of the specified 28-day strength.
3. Upon failure of core test results, the Owner may require the Contractor, at their expense, to perform load tests as specified in ACI 318, Chapter 20. Should load tests fail to prove the concrete has reached the required strength; the Contractor shall remove and replace all defective concrete at no additional cost to the Owner. No contract extension will be considered for the time required to remove and replace defective concrete.
4. Fill all core holes as specified for repairing defective concrete.

END OF SECTION 033000

SECTION 26 05 19 - LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the following:
 - 1. Wires and cables rated 600 V and less.
 - 2. Connectors, splices, and terminations rated 600 V and less.

1.2 DEFINITIONS

- A. EPDM: Ethylene-propylene-diene terpolymer rubber.
- B. NBR: Acrylonitrile-butadiene rubber.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

1.4 INFORMATIONAL SUBMITTALS

- A. Field quality-control test reports.

1.5 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.
- B. Comply with NFPA 70.

PART 2 - PRODUCTS

2.1 CONDUCTORS AND CABLES

- A. Description: Flexible, insulated and uninsulated, drawn copper current-carrying conductor with an overall insulation layer or jacket, or both, rated 600 V or less.
- B. Standards:
 - 1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
 - 2. RoHS compliant.
 - 3. Conductor and Cable Marking: Comply with wire and cable marking according to UL's "Wire and Cable Marking and Application Guide."
- C. Copper Conductors: Comply with NEMA WC 70.
- D. Conductor Insulation: Comply with NEMA WC 70 for Types THW, THWN, and XHHW.

2.2 CONNECTORS AND SPLICES

- A. Description: Factory-fabricated connectors and splices of size, ampacity rating, material, type, and class for application and service indicated; listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.

PART 3 - EXECUTION

3.1 CONDUCTOR MATERIAL APPLICATIONS

- A. Feeders: Copper. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.
- B. Branch Circuits: Copper. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.

3.2 CONDUCTOR INSULATION APPLICATIONS AND WIRING METHODS

- A. Service Entrance: Type XHHW, single conductors in raceway.
- B. Exposed Feeders: Type THWN, single conductors in raceway.
- C. Class 1 Control Circuits: Type THWN, in raceway.
- D. Class 2 Control Circuits: Type THWN, in raceway.

3.3 INSTALLATION OF CONDUCTORS AND CABLES

- A. Install wiring on this project, including, but not limited to, line voltage, low voltage, fiber optics, etc., in raceway and junction box systems to prevent potential impacts from rodents. All penetrations through boxes, cabinets, enclosures, tank walls, and domes, etc. shall be sealed to prevent potential rodent migration.
- B. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.
- C. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips, that will not damage cables or raceway.
- D. Support cables according to Section 26 05 29 "Hangers and Supports for Electrical Systems."
- E. Identify and color-code conductors and cables according to Section 26 05 53 "Identification for Electrical Systems."

3.4 CONNECTIONS

- A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A and UL 486B.

- B. Make splices and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.
- C. Wiring at Outlets: Install conductor at each outlet, with at least 6 inches of slack.

3.5 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

- A. Install sleeves and sleeve seals at penetrations of exterior wall assemblies. Comply with requirements in Section 26 05 33 "Raceways and Boxes for Electrical Systems."
- B. Seal all penetrations through tank walls and domes to prevent potential rodent migration into the structure.

3.6 FIELD QUALITY CONTROL

- A. Perform tests and inspections and prepare test reports.
- B. Tests and Inspections:
 - 1. After installing conductors and cables and before electrical circuitry has been energized, test service entrance, feeder conductors, and branch circuit conductors for compliance with requirements.
 - 2. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
- C. Test Reports: Prepare a written report to record the following:
 - 1. Test procedures used.
 - 2. Test results that comply with requirements.
 - 3. Test results that do not comply with requirements and corrective action taken to achieve compliance with requirements.
- D. Remove and replace malfunctioning units and retest as specified above.

END OF SECTION 26 05 19

SECTION 26 05 23 - CONTROL-VOLTAGE ELECTRICAL POWER CABLES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Low-voltage control cabling.
2. Control-circuit conductors.
3. Identification products.

1.2 DEFINITIONS

- A. IDC: Insulation displacement connector.
- B. Low Voltage: As defined in NFPA 70 for circuits and equipment operating at less than 50 V or for remote-control and signaling power-limited circuits.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

1.4 INFORMATIONAL SUBMITTALS

- A. Field quality-control reports.

1.5 QUALITY ASSURANCE

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

PART 2 - PRODUCTS

2.1 PATHWAYS

- A. Conduit and Boxes: Comply with requirements in Section 26 05 33 "Raceways and Boxes for Electrical Systems."
 1. Outlet boxes shall be no smaller than 4 inches square, and 2-1/2 inches deep.

2.2 LOW-VOLTAGE CONTROL CABLE

- A. Paired Cable: NFPA 70, Type CMG.
 1. One pair, twisted, No. 16 AWG, stranded (19x29) tinned-copper conductors.
 2. PVC insulation.
 3. Unshielded.
 4. PVC jacket.
 5. Flame Resistance: Comply with UL 1581.

B. Paired Cable: NFPA 70, Type CMG.

1. One pair, twisted, No. 18 AWG, stranded (19x30) tinned-copper conductors.
2. PVC insulation.
3. Unshielded.
4. PVC jacket.
5. Flame Resistance: Comply with UL 1581.

2.3 CONTROL-CIRCUIT CONDUCTORS

- A. Class 1 Control Circuits: Stranded copper, Type THWN, Type XHHN, in raceway, complying with UL 83.
- B. Class 2 Control Circuits: Stranded copper, Type THWN, in raceway, Type XHHN, in raceway, complying with UL 83.

2.4 IDENTIFICATION PRODUCTS

- A. Comply with UL 969 for a system of labeling materials, including label stocks, laminating adhesives, and inks used by label printers.
- B. Comply with requirements in Section 26 05 53 "Identification for Electrical Systems."

2.5 SOURCE QUALITY CONTROL

- A. Cable will be considered defective if it does not pass tests and inspections.
- B. Prepare test and inspection reports.

PART 3 - EXECUTION

3.1 INSTALLATION OF PATHWAYS

- A. Comply with requirements in Section 26 05 33 "Raceways and Boxes for Electrical Systems" for installation of conduits and wireways.
- B. Install manufactured conduit sweeps and long-radius elbows if possible.

3.2 INSTALLATION OF CONDUCTORS AND CABLES

- A. Comply with NECA 1.
- B. General Requirements for Cabling:
1. Cold-Weather Installation: Bring cable to room temperature before de-reeling. Heat lamps shall not be used for heating.
- C. Installation of Control-Circuit Conductors:
1. Install wiring in raceways. Comply with requirements specified in Section 26 05 33 "Raceways and Boxes for Electrical Systems."

3.3 CONTROL-CIRCUIT CONDUCTORS

A. Minimum Conductor Sizes:

1. Class 1 remote-control and signal circuits, No 12 AWG.
2. Class 2 low-energy, remote-control, and signal circuits, No. 12 AWG.

3.4 GROUNDING AND BONDING

- #### A.
- For low-voltage wiring and cabling, comply with requirements in Section 26 05 26 "Grounding and Bonding for Electrical Systems."

3.5 IDENTIFICATION

- #### A.
- Comply with requirements for identification specified in Section 26 05 53 "Identification for Electrical Systems."

3.6 FIELD QUALITY CONTROL

- #### A.
- Perform tests and inspections.
- #### B.
- End-to-end cabling will be considered defective if it does not pass tests and inspections.
- #### C.
- Prepare test and inspection reports.

END OF SECTION 26 05 23

SECTION 26 05 26 - GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes grounding and bonding systems and equipment.

1.2 ACTION SUBMITTALS

- A. Product Data: For each type of product.

1.3 INFORMATIONAL SUBMITTALS

- A. Field quality-control reports.

PART 2 - PRODUCTS

2.1 SYSTEM DESCRIPTION

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. Comply with UL 467 for grounding and bonding materials and equipment.

2.2 CONDUCTORS

- A. Insulated Conductors: Copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction.
- B. Bare Copper Conductors:
 - 1. Solid Conductors: ASTM B 3.
 - 2. Stranded Conductors: ASTM B 8.
 - 3. Bonding Cable: 28 kcmil, 14 strands of No. 17 AWG conductor, 1/4 inch (6 mm) in diameter.
 - 4. Bonding Conductor: No. 4 or No. 6 AWG, stranded conductor.
 - 5. Bonding Jumper: Copper tape, braided conductors terminated with copper ferrules; 1-5/8 inches (41 mm) wide and 1/16 inch (1.6 mm) thick.

2.3 CONNECTORS

- A. Listed and labeled by an NRTL acceptable to authorities having jurisdiction for applications in which used and for specific types, sizes, and combinations of conductors and other items connected.
- B. Welded Connectors: Exothermic-welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.
- C. Bus-Bar Connectors: Mechanical type, cast silicon bronze, solderless compression-type wire terminals, and long-barrel, two-bolt connection to ground bus bar.
- D. Cable-to-Cable Connectors: Compression type, copper or copper alloy.
- E. Conduit Hubs: Mechanical type, terminal with threaded hub.
- F. Ground Rod Clamps: Mechanical type, copper or copper alloy, terminal with hex head bolt.
- G. Lay-in Lug Connector: Mechanical type, copper rated for direct burial terminal with set screw.
- H. U-Bolt Clamps: Mechanical type, copper or copper alloy, terminal listed for direct burial.
- I. Water Pipe Clamps:
 - 1. Mechanical type, two pieces with zinc-plated bolts.
 - a. Material: Tin-plated aluminum.
 - b. Listed for direct burial.
 - 2. U-bolt type with malleable-iron clamp and copper ground connector.

2.4 GROUNDING ELECTRODES

- A. Ground Rods: Copper-clad steel; 3/4 inch by 10 feet.

PART 3 - EXECUTION

3.1 APPLICATIONS

- A. Conductors: Install solid conductor for No. 8 AWG and smaller, and stranded conductors for No. 6 AWG and larger unless otherwise indicated.
- B. Underground Grounding Electrode Conductors: Install bare copper conductor, No. 4 AWG minimum.
 - 1. Bury at least 24 inches below grade.
- C. Conductor Terminations and Connections:

1. Pipe and Equipment Grounding Conductor Terminations: Bolted connectors.
2. Underground Connections: Welded connectors.
3. Connections to Ground Rods: Exothermic welds.

3.2 GROUNDING AND BONDING AT THE SERVICE

- A. Equipment grounding conductors and grounding electrode conductors shall be connected to the ground bus. Install a main bonding jumper between the neutral and ground buses.

3.3 GROUNDING DISTRIBUTION SYSTEM COMPONENTS

- A. Maintain connections to existing underground components.
- B. Install new grounding electrode systems where indicated.
- C. Comply with IEEE C2 grounding requirements.

3.4 EQUIPMENT GROUNDING AND BONDING

- A. Install insulated equipment grounding conductor with all feeder and branch circuit conductors.

3.5 INSTALLATION

- A. Grounding Conductors: Route along shortest and straightest paths possible unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.
- B. Ground Rods: Drive rods until tops are 2 inches below final grade unless otherwise indicated.
 1. Interconnect ground rods with grounding electrode conductor below grade and as otherwise indicated. Make connections without exposing steel or damaging coating if any.
 2. Use exothermic welds for all below-grade connections.
 3. Bonding For grounding electrode system, install at least three rods spaced at least one-rod length from each other and located at least the same distance from other grounding electrodes, and connect to the service grounding electrode conductor.
- C. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance except where routed through short lengths of conduit.
 1. Bonding to Structure: Bond straps directly to basic structure, taking care not to penetrate any adjacent parts.
 2. Bonding to Equipment Mounted on Vibration Isolation Hangers and Supports: Install bonding so vibration is not transmitted to rigidly mounted equipment.
 3. Use exothermic-welded connectors for outdoor locations; if a disconnect-type connection is required, use a bolted clamp.
- D. Grounding and Bonding for Piping:

1. Metal Water Service Pipe: Install insulated copper grounding conductors, in conduit, from building's main service equipment, or grounding bus, to main metal water service entrances to building. Connect grounding conductors to main metal water service pipes; use a bolted clamp connector or bolt a lug-type connector to a pipe flange by using one of the lug bolts of the flange. Where a dielectric main water fitting is installed, connect grounding conductor on street side of fitting. Bond metal grounding conductor conduit or sleeve to conductor at each end.
 2. Water Meter Piping: Use braided-type bonding jumpers to electrically bypass water meters. Connect to pipe with a bolted connector.
- E. Connections: Make connections so possibility of galvanic action or electrolysis is minimized. Select connectors, connection hardware, conductors, and connection methods so metals in direct contact are galvanically compatible.
1. Use electroplated or hot-tin-coated materials to ensure high conductivity and to make contact points closer in order of galvanic series.
 2. Make connections with clean, bare metal at points of contact.
 3. Make aluminum-to-steel connections with stainless-steel separators and mechanical clamps.
 4. Make aluminum-to-galvanized-steel connections with tin-plated copper jumpers and mechanical clamps.
 5. Coat and seal connections having dissimilar metals with inert material to prevent future penetration of moisture to contact surfaces.

3.6 FIELD QUALITY CONTROL

- A. Perform tests and inspections.
- B. Tests and Inspections:
1. After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements.
 2. Inspect physical and mechanical condition. Verify tightness of accessible, bolted, electrical connections with a calibrated torque wrench according to manufacturer's written instructions.
- C. Grounding system will be considered defective if it does not pass tests and inspections.
- D. Prepare test and inspection reports.
- E. Excessive Ground Resistance: If resistance to ground exceeds 10 ohms, notify Contracting Officer promptly and include recommendations to reduce ground resistance.

END OF SECTION 26 05 26

SECTION 26 05 29 - HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Hangers and supports for electrical equipment and systems.

PART 2 - PRODUCTS

2.1 SUPPORT, ANCHORAGE, AND ATTACHMENT COMPONENTS

- A. Steel Slotted Support Systems: Comply with MFMA-4 factory-fabricated components for field assembly.
 - 1. Material: Galvanized steel.
 - 2. Channel Width: 1-5/8 inches.
 - 3. Metallic Coatings: Hot-dip galvanized after fabrication and applied according to MFMA-4.
 - 4. Protect finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.
 - 5. Channel Dimensions: Selected for applicable load criteria.
- B. Conduit and Cable Support Devices: Steel hangers, clamps, and associated fittings, designed for types and sizes of raceway or cable to be supported.
- C. Support for Conductors in Vertical Conduit: Factory-fabricated assembly consisting of threaded body and insulating wedging plug or plugs for nonarmored electrical conductors or cables in riser conduits. Plugs shall have number, size, and shape of conductor gripping pieces as required to suit individual conductors or cables supported. Body shall be made of malleable iron.
- D. Structural Steel for Fabricated Supports and Restraints: ASTM A 36/A 36M steel plates, shapes, and bars; black and galvanized.
- E. Mounting, Anchoring, and Attachment Components: Items for fastening electrical items or their supports to building surfaces include the following:
 - 1. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete, steel, or wood, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
 - 2. Mechanical-Expansion Anchors: Insert-wedge-type, stainless steel, for use in hardened portland cement concrete, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
 - 3. Concrete Inserts: Steel or malleable-iron, slotted support system units are similar to MSS Type 18 units and comply with MFMA-4 or MSS SP-58.
 - 4. Hanger Rods: Threaded steel.

PART 3 - EXECUTION

3.1 APPLICATION

- A. Comply with NECA 1 and NECA 101 for application of hangers and supports for electrical equipment and systems unless requirements in this Section are stricter.
- B. Comply with requirements for raceways and boxes specified in Section 26 05 33 "Raceways and Boxes for Electrical Systems."
- C. Maximum Support Spacing and Minimum Hanger Rod Size for Raceway: Space supports for EMTs, IMCs, and RMCs as scheduled in NECA 1, where its Table 1 lists maximum spacing that are less than those stated in NFPA 70. Minimum rod size shall be 1/4 inch in diameter.
- D. Multiple Raceways or Cables: Install trapeze-type supports fabricated with steel slotted or other support system, sized so capacity can be increased by at least 25 percent in future without exceeding specified design load limits.
 - 1. Secure raceways and cables to these supports with two-bolt conduit clamps.
- E. Spring-steel clamps designed for supporting single conduits without bolts may be used for 1-1/2-inch and smaller raceways serving branch circuits fastening raceways to trapeze supports.

3.2 SUPPORT INSTALLATION

- A. Comply with NECA 1 and NECA 101 for installation requirements except as specified in this article.
- B. Raceway Support Methods: In addition to methods described in NECA 1, EMTs may be supported by openings through structure members, according to NFPA 70.
- C. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb.
- D. Mounting and Anchorage of Surface-Mounted Equipment and Components: Anchor and fasten electrical items and their supports to building structural elements by the following methods unless otherwise indicated by code:
 - 1. To Wood: Fasten with lag screws or through bolts.
 - 2. To Masonry: Approved toggle-type bolts on hollow masonry units and expansion anchor fasteners on solid masonry units.
 - 3. To Existing Concrete: Expansion anchor fasteners.
 - 4. Items Mounted on Hollow Walls and Nonstructural Building Surfaces: Mount cabinets, panelboards, disconnect switches, control enclosures, pull and junction boxes, transformers, and other devices on slotted-channel racks attached to substrate.
- E. Drill holes for expansion anchors in concrete at locations and to depths that avoid the need for reinforcing bars.

3.3 INSTALLATION OF FABRICATED METAL SUPPORTS

- A. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing metal fabrications. Set metal fabrications accurately in location, alignment, and elevation to support electrical materials and equipment.
- B. Finish exposed surfaces smooth and blended so no roughness shows.
- C. Set metal fabrications with edges and surfaces level, plumb, true, and free of rack; and measured from established lines and levels.
- D. Field Welding: Comply with AWS D1.1/D1.1M.
 - 1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
 - 2. Obtain fusion without undercut or overlap.
 - 3. Remove welding flux immediately.
 - 4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.
- E. Fastening to In-Place Construction: Provide anchorage devices and fasteners where metal fabrications are required to be fastened to in-place construction. Provide threaded fasteners for use with concrete and masonry inserts, toggle bolts, through bolts, lag screws, wood screws, and other connectors.

3.4 PAINTING

- A. Touchup: Comply with requirements in Section 09 91 00 "Painting and Finishing" for cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal.
- B. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

END OF SECTION 26 05 29

SECTION 26 05 33 - RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Metal conduits, tubing, and fittings.
 - 2. Nonmetal conduits, tubing, and fittings.
 - 3. Metal wireways and auxiliary gutters.
 - 4. Boxes, enclosures, and cabinets.
- B. Related Requirements:
 - 1. Section 26 05 43 "Underground Ducts and Raceways for Electrical Systems" for exterior duct banks and underground utility construction.

1.2 DEFINITIONS

- A. GRC: Galvanized rigid steel conduit.
- B. IMC: Intermediate metal conduit.

1.3 ACTION SUBMITTALS

- A. Product Data: For surface raceways, wireways and fittings, floor boxes, hinged-cover enclosures, and cabinets.
- B. Shop Drawings: For custom enclosures and cabinets. Include plans, elevations, sections, and attachment details.

PART 2 - PRODUCTS

2.1 METAL CONDUITS, TUBING, AND FITTINGS

- A. Listing and Labeling: Metal conduits, tubing, and fittings shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. GRC: Comply with ANSI C80.1 and UL 6.
- C. IMC: Comply with ANSI C80.6 and UL 1242.
- D. EMT: Comply with ANSI C80.3 and UL 797.
- E. FMC: Comply with UL 1; zinc-coated steel or aluminum.

- F. LFMC: Flexible steel conduit with PVC jacket and complying with UL 360.
- G. Fittings for Metal Conduit: Comply with NEMA FB 1 and UL 514B.
 - 1. Conduit Fittings for Hazardous (Classified) Locations: Comply with UL 1203 and NFPA 70.
 - 2. Fittings for EMT:
 - a. Material: Steel or die cast.
 - b. Type: Setscrew or compression.
 - 3. Expansion Fittings: PVC or steel to match conduit type, complying with UL 651, rated for environmental conditions where installed, and including flexible external bonding jumper.
 - 4. Coating for Fittings for PVC-Coated Conduit: Minimum thickness of 0.040 inch, with overlapping sleeves protecting threaded joints.
- H. Joint Compound for IMC, GRC, or ARC: Approved, as defined in NFPA 70, by authorities having jurisdiction for use in conduit assemblies, and compounded for use to lubricate and protect threaded conduit joints from corrosion and to enhance their conductivity.

2.2 NONMETALLIC CONDUITS, TUBING, AND FITTINGS

- A. Listing and Labeling: Nonmetallic conduits, tubing, and fittings shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. ENT: Comply with NEMA TC 13 and UL 1653.
- C. RNC: Type EPC-40-PVC, complying with NEMA TC 2 and UL 651 unless otherwise indicated.
- D. LFNC: Comply with UL 1660.
- E. Rigid HDPE: Comply with UL 651A.
- F. Continuous HDPE: Comply with UL 651B.
- G. Coilable HDPE: Preassembled with conductors or cables, and complying with ASTM D 3485.
- H. RTRC: Comply with UL 1684A and NEMA TC 14.
- I. Fittings for ENT and RNC: Comply with NEMA TC 3; match to conduit or tubing type and material.
- J. Fittings for LFNC: Comply with UL 514B.
- K. Solvents and Adhesives: As recommended by conduit manufacturer.

2.3 METAL WIREWAYS AND AUXILIARY GUTTERS

- A. Description: Sheet metal, complying with UL 870 and NEMA 250, Type 1 or Type 3R unless otherwise indicated, and sized according to NFPA 70.

1. Metal wireways installed outdoors shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. Fittings and Accessories: Include covers, couplings, offsets, elbows, expansion joints, adapters, hold-down straps, end caps, and other fittings to match and mate with wireways as required for complete system.
- C. Wireway Covers: Hinged type unless otherwise indicated.
- D. Finish: Manufacturer's standard enamel finish.

2.4 BOXES, ENCLOSURES, AND CABINETS

- A. General Requirements for Boxes, Enclosures, and Cabinets: Boxes, enclosures, and cabinets installed in wet locations shall be listed for use in wet locations.
- B. Sheet Metal Outlet and Device Boxes: Comply with NEMA OS 1 and UL 514A.
- C. Cast-Metal Outlet and Device Boxes: Comply with NEMA FB 1, ferrous alloy, Type FD, with gasketed cover.
- D. Luminaire Outlet Boxes: Nonadjustable, designed for attachment of luminaire weighing 50 lb. Outlet boxes designed for attachment of luminaires weighing more than 50 lb shall be listed and marked for the maximum allowable weight.
- E. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.
- F. Box extensions used to accommodate new building finishes shall be of same material as recessed box.
- G. Device Box Dimensions: 4 inches square by 2-1/8 inches deep 4 inches by 2-1/8 inches by 2-1/8 inches deep (100 mm by 60 mm by 60 mm deep).
- H. Gangable boxes are prohibited.
- I. Hinged-Cover Enclosures: Comply with UL 50 and NEMA 250, Type 1 or Type 3R with continuous-hinge cover with flush latch unless otherwise indicated.
 1. Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
 2. Interior Panels: Steel; all sides finished with manufacturer's standard enamel.

PART 3 - EXECUTION

3.1 RACEWAY APPLICATION

- A. Outdoors: Apply raceway products as specified below unless otherwise indicated:
 1. Exposed Conduit: PVC-coated rigid steel conduit.
 2. Concealed Conduit, Aboveground: RNC, Type EPC-40-PVC.
 3. Underground Conduit: RNC, Type EPC-40-PVC, direct buried.

4. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): LFMC.
 5. Boxes and Enclosures, Aboveground: NEMA 250, Type 3R.
- B. Minimum Raceway Size: 3/4-inch trade size.
- C. Raceway Fittings: Compatible with raceways and suitable for use and location.
1. Rigid and Intermediate Steel Conduit: Use threaded rigid steel conduit fittings unless otherwise indicated. Comply with NEMA FB 2.10.
 2. PVC Externally Coated, Rigid Steel Conduits: Use only fittings listed for use with this type of conduit. Patch and seal all joints, nicks, and scrapes in PVC coating after installing conduits and fittings. Use sealant recommended by fitting manufacturer and apply in thickness and number of coats recommended by manufacturer.
 3. EMT: Use setscrew or compression, steel fittings. Comply with NEMA FB 2.10.
 4. Flexible Conduit: Use only fittings listed for use with flexible conduit. Comply with NEMA FB 2.20.
- D. Do not install nonmetallic conduit where ambient temperature exceeds 120 deg F.

3.2 INSTALLATION

- A. Comply with NECA 1 and NECA 101 for installation requirements except where requirements on Drawings or in this article are stricter. Comply with NECA 102 for aluminum conduits. Comply with NFPA 70 limitations for types of raceways allowed in specific occupancies and number of floors.
- B. Keep raceways at least 6 inches away from parallel runs of flues and steam or hot-water pipes. Install horizontal raceway runs above water and steam piping.
- C. Complete raceway installation before starting conductor installation.
- D. Comply with requirements in Section 26 05 29 "Hangers and Supports for Electrical Systems" for hangers and supports.
- E. Arrange stub-ups so curved portions of bends are not visible above finished slab.
- F. Install no more than the equivalent of three 90-degree bends in any conduit run except for control wiring conduits, for which fewer bends are allowed. Support within 12 inches of changes in direction.
- G. Conceal conduit and EMT within finished walls, ceilings, and floors unless otherwise indicated. Install conduits parallel or perpendicular to building lines.
- H. Support conduit within 12 inches of enclosures to which attached.
- I. Raceways Embedded in Slabs:
1. Run conduit larger than 1-inch trade size, parallel or at right angles to main reinforcement. Where at right angles to reinforcement, place conduit close to slab support. Secure raceways to reinforcement at maximum 10-foot intervals.

2. Arrange raceways to cross building expansion joints at right angles with expansion fittings.
 3. Arrange raceways to keep a minimum of 2 inches of concrete cover in all directions.
 4. Do not embed threadless fittings in concrete unless specifically approved by Contracting Officer for each specific location.
 5. Change from ENT to GRC or IMC before rising above floor.
- J. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound to threads of raceway and fittings before making up joints. Follow compound manufacturer's written instructions.
- K. Coat field-cut threads on PVC-coated raceway with a corrosion-preventing conductive compound prior to assembly.
- L. Raceway Terminations at Locations Subject to Moisture or Vibration: Use insulating bushings to protect conductors including conductors smaller than No. 4 AWG.
- M. Terminate threaded conduits into threaded hubs or with locknuts on inside and outside of boxes or cabinets. Install bushings on conduits up to 1-1/4-inch trade size and insulated throat metal bushings on 1-1/2-inch trade size and larger conduits terminated with locknuts. Install insulated throat metal grounding bushings on service conduits.
- N. Install raceways square to the enclosure and terminate at enclosures with locknuts. Install locknuts hand tight plus 1/4 turn more.
- O. Do not rely on locknuts to penetrate nonconductive coatings on enclosures. Remove coatings in the locknut area prior to assembling conduit to enclosure to assure a continuous ground path.
- P. Cut conduit perpendicular to the length. For conduits 2-inch trade size and larger, use roll cutter or a guide to make cut straight and perpendicular to the length.
- Q. Install pull wires in empty raceways. Use polypropylene or monofilament plastic line with not less than 200-lb tensile strength. Leave at least 12 inches of slack at each end of pull wire. Cap underground raceways designated as spare above grade alongside raceways in use.
- R. Install raceway sealing fittings at accessible locations according to NFPA 70 and fill them with listed sealing compound. For concealed raceways, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent plates or surfaces. Install raceway sealing fittings according to NFPA 70.
- S. Install devices to seal raceway interiors at accessible locations. Locate seals so no fittings or boxes are between the seal and the following changes of environments. Seal the interior of all raceways at the following points:
1. Where an underground service raceway enters a building or structure.
 2. Where otherwise required by NFPA 70.
- T. Comply with manufacturer's written instructions for solvent welding RNC and fittings.
- U. Expansion-Joint Fittings:
1. Install in each run of aboveground RNC that is located where environmental temperature change may exceed 30 deg F and that has straight-run length that exceeds 25 feet. Install

- in each run of aboveground RMC conduit that is located where environmental temperature change may exceed 100 deg F and that has straight-run length that exceeds 100 feet.
2. Install type and quantity of fittings that accommodate temperature change listed for each of the following locations:
 - a. Outdoor Locations Not Exposed to Direct Sunlight: 125 deg F temperature change.
 - b. Outdoor Locations Exposed to Direct Sunlight: 155 deg F temperature change.
 - c. Attics: 135 deg F temperature change.
 3. Install fitting(s) that provide expansion and contraction for at least 0.00041 inch per foot of length of straight run per deg F of temperature change for PVC conduits. Install fitting(s) that provide expansion and contraction for at least 0.000078 inch per foot of length of straight run per deg F of temperature change for metal conduits.
 4. Install expansion fittings at all locations where conduits cross building or structure expansion joints.
 5. Install each expansion-joint fitting with position, mounting, and piston setting selected according to manufacturer's written instructions for conditions at specific location at time of installation. Install conduit supports to allow for expansion movement.
- V. Flexible Conduit Connections: Comply with NEMA RV 3. Use a maximum of 72 inches of flexible conduit for equipment subject to vibration, noise transmission, or movement; and for transformers and motors.
1. Use LFMC in damp or wet locations subject to severe physical damage.
 2. Use LFMC or LFNC in damp or wet locations not subject to severe physical damage.
- W. Mount boxes at heights indicated on Drawings. If mounting heights of boxes are not individually indicated, give priority to ADA requirements. Install boxes with height measured to center of box unless otherwise indicated.
- X. Horizontally separate boxes mounted on opposite sides of walls so they are not in the same vertical channel.
- Y. Locate boxes so that cover or plate will not span different building finishes.
- Z. Support boxes of three gangs or more from more than one side by spanning two framing members or mounting on brackets specifically designed for the purpose.
- AA. Fasten junction and pull boxes to or support from building structure. Do not support boxes by conduits.
- 3.3 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS
- A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies. Comply with requirements in Section 26 05 44 "Sleeves and Sleeve Seals for Electrical Raceways and Cabling."
- 3.4 PROTECTION
- A. Protect coatings, finishes, and cabinets from damage and deterioration.

1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
2. Repair damage to PVC coatings or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION 26 05 33

SECTION 26 05 43 - UNDERGROUND DUCTS AND RACEWAYS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the following:
 - 1. Conduit, ducts, and duct accessories for direct-buried duct banks, and in single duct runs.
 - 2. Polymer Concrete In-Ground Pull Boxes.
- B. Related Requirements: Section 26 05 53 "Identification for Electrical Systems" for warning tape for underground ducts and raceways.

1.2 DEFINITION

- A. RNC: Rigid nonmetallic conduit.

1.3 ACTION SUBMITTALS

- A. Product Data: For the following:
 - 1. Duct-bank materials, including separators and miscellaneous components.
 - 2. Ducts and conduits and their accessories, including elbows, end bells, bends, fittings, and solvent cement.
 - 3. Handholes, boxes, and accessories.
- B. Shop Drawings for Precast or Factory-Fabricated Underground Utility Structures: Include plans, elevations, sections, details, attachments to other work, and accessories, including the following:
 - 1. Duct entry provisions, including locations and duct sizes.
 - 2. Reinforcement details.
 - 3. Frame and cover design.
- C. Shop Drawings for Factory-Fabricated Polymer Concrete Boxes: Include dimensioned plans, sections, and elevations, and fabrication and installation details, including the following:
 - 1. Duct entry provisions, including locations and duct sizes.
 - 2. Cover design.

1.4 QUALITY ASSURANCE

- A. Comply with ANSI C2.
- B. Comply with NFPA 70.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver ducts to Project site with ends capped. Store nonmetallic ducts with supports to prevent bending, warping, and deforming.

- B. Store factory-fabricated underground utility structures at Project site as recommended by manufacturer to prevent physical damage. Arrange so identification markings are visible.

1.6 COORDINATION

- A. Coordinate layout and installation of ducts and boxes with final arrangement of other utilities, site grading, and surface features as determined in the field.
- B. Coordinate elevations of duct entrances into boxes with final locations and profiles of ducts as determined by coordination with other utilities, underground obstructions, and surface features. Revise locations and elevations from those indicated as required to suit field conditions and to ensure that duct runs drain to handholes, and as approved by Contracting Officer.

PART 2 - PRODUCTS

2.1 CONDUIT

- A. Coated Steel Conduit: PVC-Coated rigid steel conduit.
 - 1. Comply with NEMA RN 1.
 - 2. Coating Thickness: 0.040 inch, minimum.
- B. Rigid Steel Conduit: Galvanized. Comply with ANSI C80.1.
- C. RNC: NEMA TC 2, Type EPC-40-PVC and Type EPC-80-PVC, UL 651, with matching fittings by same manufacturer as the conduit, complying with NEMA TC 3 and UL 514B.

2.2 NONMETALLIC DUCTS AND DUCT ACCESSORIES

- A. Duct Accessories:
 - 1. Duct Separators: Factory-fabricated rigid PVC interlocking spacers, sized for type and sizes of ducts with which used, and selected to provide minimum duct spacings indicated while supporting ducts during concreting or backfilling.

2.3 POLYMER CONCRETE IN-GROUND PULL BOXES

- A. Description: Comply with SCTE 77.
 - 1. Color: Gray, tan, or green as indicated on drawings.
 - 2. Configuration: Units shall be designed for flush burial and have open bottom, unless otherwise indicated.
 - 3. Cover: Weatherproof, secured by tamper-resistant locking devices and having structural load rating consistent with enclosure.
 - 4. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
 - 5. Cover Legend: Molded lettering, "ELECTRIC", "FIRE ALARM", "COMMUNICATIONS", or "CONTROL" as indicated.
 - 6. Handholes 12 inches wide by 24 inches long and larger shall have factory-installed inserts for cable racks and pulling-in irons.

- B. Polymer Concrete Handholes and Boxes with Polymer Concrete Cover: Molded of sand and aggregate, bound together with a polymer resin, and reinforced with steel or fiberglass or a combination of the two.

2.4 SOURCE QUALITY CONTROL

- A. Nonconcrete Handhole and Pull-Box Prototype Test: Test prototypes of manholes and boxes for compliance with SCTE 77. Strength tests shall be for specified tier ratings of products supplied.
 - 1. Strength tests of complete boxes and covers shall be by either an independent testing agency or the manufacturer. A qualified registered professional engineer shall certify tests by manufacturer.
 - 2. Testing machine pressure gages shall have current calibration certification complying with ISO 9000 and ISO 10012, and traceable to NIST standards.

PART 3 - EXECUTION

3.1 UNDERGROUND DUCT APPLICATION

- A. Ducts for Electrical Feeders 600 V and Less: RNC, NEMA Type EPC-40-PVC, in direct-buried duct bank, unless otherwise indicated.
- B. Ducts for Electrical Branch Circuits: RNC, NEMA Type EPC-40-PVC, in direct-buried duct bank, unless otherwise indicated.
- C. Underground Ducts Crossing Paved Paths, Walks and Driveways: RNC, NEMA Type EPC-40-PVC, encased in reinforced concrete.

3.2 UNDERGROUND ENCLOSURE APPLICATION

- A. Handholes and Boxes for 600 V and Less:
 - 1. Units in Driveway, Parking Lot, Off-Roadway Locations, Sidewalks, and all other locations Subject to Occasional, Nondeliberate Loading by Heavy Vehicles: Polymer concrete, SCTE 77, Tier 15 structural load rating.

3.3 EARTHWORK

- A. Excavation and Backfill: Comply with Division 31 Section "Earthwork," but do not use heavy-duty, hydraulic-operated, compaction equipment.
- B. Restore surface features at areas disturbed by excavation and reestablish original grades, unless otherwise indicated. Replace removed sod immediately after backfilling is completed.
- C. Restore areas disturbed by trenching, storing of dirt, cable laying, and other work.

3.4 DUCT INSTALLATION

- A. Slope: Pitch ducts a minimum slope of 1:300 down toward handholes and away from equipment. Slope ducts from a high point in runs between two handholes to drain in both directions.
- B. Curves and Bends: Use 5-degree angle couplings for small changes in direction. Use manufactured long sweep bends with a minimum radius of 48 inches, both horizontally and vertically, at other locations, unless otherwise indicated.
- C. Joints: Use solvent-cemented joints in ducts and fittings and make watertight according to manufacturer's written instructions. Stagger couplings so those of adjacent ducts do not lie in same plane.
- D. Duct Entrances to Polymer Concrete Handholes: Use end bells, spaced approximately 10 inches o.c. for 5-inch ducts, and vary proportionately for other duct sizes.
 - 1. Begin change from regular spacing to end-bell spacing 10 feet from the end bell without reducing duct line slope and without forming a trap in the line.
 - 2. Grout end bells into structure walls from both sides to provide watertight entrances.
- E. Sealing: Provide temporary closure at terminations of ducts that have cables pulled. Seal spare ducts at terminations. Use sealing compound and plugs to withstand at least 15-psig hydrostatic pressure.
- F. Pulling Cord: Install 100-lbf-test nylon cord in ducts, including spares.
- G. Direct-Buried Duct Banks:
 - 1. Support ducts on duct separators coordinated with duct size, duct spacing, and outdoor temperature.
 - 2. Space separators close enough to prevent sagging and deforming of ducts, with not less than 4 spacers per 20 feet of duct. Secure separators to earth and to ducts to prevent displacement during backfill and yet permit linear duct movement due to expansion and contraction as temperature changes. Stagger spacers approximately 6 inches between tiers.
 - 3. Excavate trench bottom to provide firm and uniform support for duct bank. Prepare trench bottoms as specified in Section 31 00 00 "Earthwork" for pipes less than 6 inches in nominal diameter.
 - 4. Install backfill as specified in Section 31 00 00 "Earthwork."
 - 5. Depth: Install top of duct bank at least 24 inches below finished grade, unless otherwise indicated.
 - 6. Set elevation of bottom of duct bank below the frost line.
 - 7. Install manufactured PVC Coated rigid steel conduit elbows for stub-ups at poles and equipment.
 - a. Couple steel conduits to ducts with adapters designed for this purpose, and encase coupling with 3 inches of concrete.

3.5 INSTALLATION OF POLYMER CONCRETE BOXES

- A. Install handholes and boxes level and plumb and with orientation and depth coordinated with connecting ducts to minimize bends and deflections required for proper entrances. Use box

extension if required to match depths of ducts, and seal joint between box and extension as recommended by the manufacturer.

- B. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1/2-inch sieve to No. 4 sieve and compacted to same density as adjacent undisturbed earth.
- C. Elevation: In paved areas and trafficways, set so cover surface will be flush with finished grade. Set covers of other handholes 1 inch above finished grade.
- D. Install handholes and boxes with bottom below the frost line.
- E. Install removable hardware, including pulling eyes, cable stanchions, cable arms, and insulators, as required for installation and support of cables and conductors and as indicated. Select arm lengths to be long enough to provide spare space for future cables, but short enough to preserve adequate working clearances in the enclosure.
- F. Field-cut openings for ducts and conduits according to enclosure manufacturer's written instructions. Cut wall of enclosure with a tool designed for material to be cut. Size holes for terminating fittings to be used, and seal around penetrations after fittings are installed.
- G. For enclosures installed in asphalt paving and subject to occasional, non-deliberate, heavy-vehicle loading, form and pour a concrete ring encircling, and in contact with, enclosure and with top surface screeded to top of box cover frame. Bottom of ring shall rest on compacted earth.
 - 1. Concrete: 3000 psi, 28-day strength with a troweled finish.
 - 2. Dimensions: 10 inches wide by 12 inches deep.

3.6 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections and prepare test reports:
 - 1. Demonstrate capability and compliance with requirements on completion of installation of underground ducts and utility structures.
 - 2. Pull aluminum or wood test mandrel through duct to prove joint integrity and test for out-of-round duct. Provide mandrel equal to 80 percent fill of duct. If obstructions are indicated, remove obstructions and retest.
- B. Correct deficiencies and retest as specified above to demonstrate compliance.

3.7 CLEANING

- A. Pull leather-washer-type duct cleaner, with graduated washer sizes, through full length of ducts. Follow with rubber duct swab for final cleaning and to assist in spreading lubricant throughout ducts.
- B. Clean internal surfaces of manholes, including sump. Remove foreign material.

END OF SECTION 26 05 43

SECTION 26 05 44 - SLEEVES AND SLEEVE SEALS FOR ELECTRICAL RACEWAYS AND CABLING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Sleeves for raceway and cable penetration of non-fire-rated construction walls and floors.
 - 2. Sleeve-seal systems.
 - 3. Sleeve-seal fittings.
 - 4. Grout.
 - 5. Silicone sealants.

PART 2 - PRODUCTS

- A. Wall Sleeves:
 - 1. Steel Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, zinc coated, plain ends.
 - 2. Cast-Iron Pipe Sleeves: Cast or fabricated "wall pipe," equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop unless otherwise indicated.
- B. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies: Galvanized-steel sheet; 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint, with tabs for screw-fastening the sleeve to the board.
- C. PVC-Pipe Sleeves: ASTM D 1785, Schedule 40.
- D. Molded-PVC Sleeves: With nailing flange for attaching to wooden forms.
- E. Molded-PE or -PP Sleeves: Removable, tapered-cup shaped, and smooth outer surface with nailing flange for attaching to wooden forms.
- F. Sleeves for Rectangular Openings:
 - 1. Material: Galvanized sheet steel.
 - 2. Minimum Metal Thickness:
 - a. For sleeve cross-section rectangle perimeter less than 50 inches and with no side larger than 16 inches, thickness shall be 0.052 inch.
 - b. For sleeve cross-section rectangle perimeter 50 inches or more and one or more sides larger than 16 inches, thickness shall be 0.138 inch.

2.2 SLEEVE-SEAL SYSTEMS

- A. Description: Modular sealing device, designed for field assembly, to fill annular space between sleeve and raceway or cable.
 - 1. Sealing Elements: EPDM rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.

2. Pressure Plates: Carbon steel.
3. Connecting Bolts and Nuts: Carbon steel, with corrosion-resistant coating, of length required to secure pressure plates to sealing elements.

2.3 SLEEVE-SEAL FITTINGS

- A. Description: Manufactured plastic, sleeve-type, water stop assembly made for embedding in concrete slab or wall. Unit shall have plastic or rubber water stop collar with center opening to match piping OD.

2.4 GROUT

- A. Description: Nonshrink; recommended for interior and exterior sealing openings in non-fire-rated walls or floors.
- B. Standard: ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.
- C. Design Mix: 5000-psi, 28-day compressive strength.
- D. Packaging: Premixed and factory packaged.

2.5 SILICONE SEALANTS

- A. Silicone Sealants: Single-component, silicone-based, neutral-curing elastomeric sealants of grade indicated below.
 1. Grade: Pourable (self-leveling) formulation for openings in floors and other horizontal surfaces that are not fire rated.
- B. Silicone Foams: Multicomponent, silicone-based liquid elastomers that, when mixed, expand and cure in place to produce a flexible, nonshrinking foam.

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION FOR NON-FIRE-RATED ELECTRICAL PENETRATIONS

- A. Comply with NECA 1.
- B. Sleeves for Conduits Penetrating Above-Grade Non-Fire-Rated Concrete and Masonry-Unit Floors and Walls:
 1. Interior Penetrations of Non-Fire-Rated Walls and Floors:
 - a. Seal annular space between sleeve and raceway or cable, using joint sealant appropriate for size, depth, and location of joint. Comply with requirements in Section 07 92 00 "Joint Sealants."

- b. Seal space outside of sleeves with mortar or grout. Pack sealing material solidly between sleeve and wall so no voids remain. Tool exposed surfaces smooth; protect material while curing.
 - 2. Use pipe sleeves unless penetration arrangement requires rectangular sleeved opening.
 - 3. Size pipe sleeves to provide 1/4-inch annular clear space between sleeve and raceway or cable unless sleeve seal is to be installed.
 - 4. Install sleeves for wall penetrations unless core-drilled holes or formed openings are used. Install sleeves during erection of walls. Cut sleeves to length for mounting flush with both surfaces of walls. Deburr after cutting.
 - 5. Install sleeves for floor penetrations. Extend sleeves installed in floors 2 inches above finished floor level. Install sleeves during erection of floors.
- C. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies:
- 1. Use circular metal sleeves unless penetration arrangement requires rectangular sleeved opening.
 - 2. Seal space outside of sleeves with approved joint compound for gypsum board assemblies.
- D. Roof-Penetration Sleeves: Seal penetration of individual raceways and cables with flexible boot-type flashing units applied in coordination with roofing work.
- E. Aboveground, Exterior-Wall Penetrations: Seal penetrations using cast-iron pipe sleeves and mechanical sleeve seals. Select sleeve size to allow for 1-inch annular clear space between pipe and sleeve for installing mechanical sleeve seals.
- F. Underground, Exterior-Wall and Floor Penetrations: Install cast-iron pipe sleeves. Size sleeves to allow for 1-inch annular clear space between raceway or cable and sleeve for installing sleeve-seal system.

3.2 SLEEVE-SEAL-SYSTEM INSTALLATION

- A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at raceway entries into building.
- B. Install type and number of sealing elements recommended by manufacturer for raceway or cable material and size. Position raceway or cable in center of sleeve. Assemble mechanical sleeve seals and install in annular space between raceway or cable and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.

3.3 SLEEVE-SEAL-FITTING INSTALLATION

- A. Install sleeve-seal fittings in new walls and slabs as they are constructed.
- B. Assemble fitting components of length to be flush with both surfaces of concrete slabs and walls. Position water stop flange to be centered in concrete slab or wall.
- C. Secure nailing flanges to concrete forms.

D. Using grout, seal the space around outside of sleeve-seal fittings.

END OF SECTION 26 05 44

SECTION 26 24 16 - PANELBOARDS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Lighting and appliance branch-circuit panelboards.

B. Related Requirements:

1. Section 26 43 13 "Surge Protection for Low-Voltage Electrical Power Circuits" for field-installed Surge Protective Device(s) for panelboards.

1.2 DEFINITIONS

- A. SVR: Suppressed voltage rating.
- B. SPD: Surge Protective Device.
- C. TVSS: Transient voltage surge suppressor.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of panelboard, switching and overcurrent protective device, transient voltage suppression device, accessory, and component indicated. Include dimensions and manufacturers' technical data on features, performance, electrical characteristics, ratings, and finishes.
- B. Shop Drawings: For each panelboard and related equipment.
1. Include dimensioned plans, elevations, sections, and details. Show tabulations of installed devices, equipment features, and ratings.
 2. Detail enclosure types and details for types other than NEMA 250, Type 1.
 3. Detail bus configuration, current, and voltage ratings.
 4. Short-circuit current rating of panelboards and overcurrent protective devices.
 5. Detail features, characteristics, ratings, and factory settings of individual overcurrent protective devices and auxiliary components.
 6. Include wiring diagrams for power, signal, and control wiring.

1.4 INFORMATIONAL SUBMITTALS

- A. Field Quality-Control Reports:
1. Test procedures used.
 2. Test results that comply with requirements.

3. Results of failed tests and corrective action taken to achieve test results that comply with requirements.
- B. Panelboard Schedules: For installation in panelboards. Submit final versions after load balancing.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data: For panelboards and components to include in emergency, operation, and maintenance manuals. In addition to items specified in Section 01 78 23 "Operation and Maintenance Data," include the following:
 1. Manufacturer's written instructions for testing and adjusting overcurrent protective devices.

1.6 MAINTENANCE MATERIAL SUBMITTALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 1. Keys: Six spares for each type of panelboard cabinet lock.

1.7 QUALITY ASSURANCE

- A. Source Limitations: Obtain panelboards, overcurrent protective devices, components, and accessories from single source from single manufacturer.
- B. Product Selection for Restricted Space: Drawings indicate maximum dimensions for panelboards including clearances between panelboards and adjacent surfaces and other items. Comply with indicated maximum dimensions.
- C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- D. Comply with NEMA PB 1.
- E. Comply with NFPA 70.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Remove loose packing and flammable materials from inside panelboards; install temporary electric heating (250 W per panelboard) to prevent condensation.
- B. Handle and prepare panelboards for installation according to NEMA PB 1.

1.9 PROJECT CONDITIONS

A. Environmental Limitations:

1. Rate equipment for continuous operation under the following conditions unless otherwise indicated:
 - a. Ambient Temperature: Not exceeding 23 deg F to plus 104 deg F.

B. Service Conditions: NEMA PB 1, usual service conditions.

C. Interruption of Existing Electric Service: Do not interrupt electric service to facilities occupied by Government or others unless permitted under the following conditions and then only after arranging to provide temporary electric service according to requirements indicated:

1. Notify Contracting Officer no fewer than seven days in advance of proposed interruption of electric service.
2. Do not proceed with interruption of electric service without Contracting Officer's written permission.
3. Comply with NFPA 70E.

1.10 COORDINATION

- ### A.
- Coordinate layout and installation of panelboards and components with other construction that penetrates walls or is supported by them, including electrical and other types of equipment, raceways, piping, encumbrances to workspace clearance requirements, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

1.11 WARRANTY

- ### A.
- Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace transient voltage suppression devices that fail in materials or workmanship within specified warranty period.

1. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR PANELBOARDS

A. Enclosures: Surface-mounted cabinets.

1. Rated for environmental conditions at installed location.
 - a. Outdoor Locations: NEMA 250, Type 4X.
2. Hinged Front Cover: Entire front trim hinged to box and with standard door within hinged trim cover.
3. Finishes:

- a. Panels and Trim: Steel and galvanized steel, factory finished immediately after cleaning and pre-treating with manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat.
 - b. Back Boxes: Same finish as panels and trim.
- 4. Directory Card: Inside panelboard door, mounted in transparent card holder.
- B. Incoming Mains Location: Top and bottom.
- C. Phase, Neutral, and Ground Buses:
 - 1. Material: Hard-drawn copper, 98 percent conductivity.
 - 2. Equipment Ground Bus: Adequate for feeder and branch-circuit equipment grounding conductors; bonded to box.
- D. Conductor Connectors: Suitable for use with conductor material and sizes.
 - 1. Main and Neutral Lugs: Mechanical type.
 - 2. Ground Lugs and Bus-Configured Terminators: Mechanical type.
- E. Future Devices: Mounting brackets, bus connections, filler plates, and necessary appurtenances required for future installation of devices.
- F. Panelboard Short-Circuit Current Rating: Fully rated to interrupt symmetrical short-circuit current available at terminals.

2.2 LIGHTING AND APPLIANCE BRANCH-CIRCUIT PANELBOARDS

- A. Panelboards: NEMA PB 1, lighting and appliance branch-circuit type.
- B. Mains: Circuit breaker.
- C. Branch Overcurrent Protective Devices: Bolt-on circuit breakers, replaceable without disturbing adjacent units.
- D. Doors: Concealed hinges; secured with flush latch with tumbler lock; keyed alike.

2.3 DISCONNECTING AND OVERCURRENT PROTECTIVE DEVICES

- A. Molded-Case Circuit Breaker (MCCB): Comply with UL 489, with interrupting capacity to meet available fault currents.
 - 1. Thermal-Magnetic Circuit Breakers: Inverse time-current element for low-level overloads, and instantaneous magnetic trip element for short circuits. Adjustable magnetic trip setting for circuit-breaker frame sizes 250 A and larger.
 - 2. Current-Limiting Circuit Breakers: Frame sizes 400 A and smaller; let-through ratings less than NEMA FU 1, RK-5.
 - 3. GFCI Circuit Breakers: Single- and two-pole configurations with Class A ground-fault protection (6-mA trip).

4. Ground-Fault Equipment Protection (GFEP) Circuit Breakers: Class B ground-fault protection (30-mA trip).
5. Molded-Case Circuit-Breaker (MCCB) Features and Accessories:
 - a. Standard frame sizes, trip ratings, and number of poles.
 - b. Lugs: Mechanical style, suitable for number, size, trip ratings, and conductor materials.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Receive, inspect, handle, and store panelboards according to NEMA PB 1.1.
- B. Examine panelboards before installation. Reject panelboards that are damaged or rusted or have been subjected to water saturation.
- C. Examine elements and surfaces to receive panelboards for compliance with installation tolerances and other conditions affecting performance of the Work.
- D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. Install panelboards and accessories according to NEMA PB 1.1.
- B. Mount top of trim 90 inches above finished floor unless otherwise indicated.
- C. Mount panelboard cabinet plumb and rigid without distortion of box. Mount recessed panelboards with fronts uniformly flush with wall finish and mating with back box.
- D. Install overcurrent protective devices and controllers not already factory installed.
- E. Install filler plates in unused spaces.
- F. Arrange conductors in gutters into groups and bundle and wrap with wire ties after completing load balancing.
- G. Comply with NECA 1.

3.3 IDENTIFICATION

- A. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs complying with Section 26 05 53 "Identification for Electrical Systems."
- B. Create a directory to indicate installed circuit loads after balancing panelboard loads; incorporate Government's final equipment designations. Obtain approval before installing. Use a computer or typewriter to create directory; handwritten directories are not acceptable.

- C. Panelboard Nameplates: Label each panelboard with a nameplate complying with requirements for identification specified in 26 05 53 "Identification for Electrical Systems."
- D. Device Nameplates: Label each new branch circuit device in existing distribution panelboard with a nameplate complying with requirements for identification specified in Section 26 05 53 "Identification for Electrical Systems."

3.4 FIELD QUALITY CONTROL

- A. Perform tests and inspections.
- B. Acceptance Testing Preparation:
 - 1. Test insulation resistance for each panelboard bus, component, connecting supply, feeder, and control circuit.
 - 2. Test continuity of each circuit.
- C. Tests and Inspections:
 - 1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
 - 2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
- D. Panelboards will be considered defective if they do not pass tests and inspections.
- E. Prepare test and inspection reports, including a certified report that identifies panelboards included and that describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

3.5 ADJUSTING

- A. Adjust moving parts and operable component to function smoothly, and lubricate as recommended by manufacturer.
- B. Load Balancing: After Substantial Completion, but not more than 60 days after Final Acceptance, measure load balancing and make circuit changes.
 - 1. Measure as directed during period of normal system loading.
 - 2. Perform load-balancing circuit changes outside normal occupancy/working schedule of the facility and at time directed. Avoid disrupting critical 24-hour services such as fax machines and on-line data processing, computing, transmitting, and receiving equipment.
 - 3. After circuit changes, recheck loads during normal load period. Record all load readings before and after changes and submit test records.
 - 4. Tolerance: Difference exceeding 20 percent between phase loads, within a panelboard, is not acceptable. Rebalance and recheck as necessary to meet this minimum requirement.

3.6 PROTECTION

- A. Temporary Heating: Prior to energizing panelboards, apply temporary heat to maintain temperature according to manufacturer's written instructions.

END OF SECTION 26 24 16

SECTION 26 05 53 - IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Identification for raceways.
2. Identification of power and control cables.
3. Identification for conductors.
4. Warning labels and signs.
5. Instruction signs.
6. Equipment identification labels.
7. Miscellaneous identification products.

B. Requirements:

1. Provide warning and tracer tape for all underground piping and conduit.
2. Provide valve boxes with square caps to encapsulate and protect tracer wire ends and to provide location of, and access to tracer wire. Refer to site details on Civil drawings.

C. Related Requirements:

1. Section 33 40 20 "Warning and Tracer Tape" for warning tape for underground ducts and raceways.

1.2 QUALITY ASSURANCE

- A. Comply with ANSI A13.1.
- B. Comply with NFPA 70.
- C. Comply with 29 CFR 1910.144 and 29 CFR 1910.145.
- D. Comply with ANSI Z535.4 for safety signs and labels.

1.3 COORDINATION

- A. Coordinate identification names, abbreviations, colors, and other features with requirements in other Sections requiring identification applications, Drawings, Shop Drawings, manufacturer's wiring diagrams, and the Operation and Maintenance Manual; and with those required by codes, standards, and 29 CFR 1910.145. Use consistent designations throughout Project.
- B. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.
- C. Coordinate installation of identifying devices with location of access panels and doors.

- D. Install identifying devices before installing acoustical ceilings and similar concealment.

PART 2 - PRODUCTS

2.1 POWER AND CONTROL CABLE IDENTIFICATION MATERIALS

- A. Comply with ANSI A13.1 for minimum size of letters for legend and for minimum length of color field for each raceway and cable size.
- B. Metal Tags: Brass or aluminum, 2 by 2 by 0.05 inch (50 by 50 by 1.3 mm), with stamped legend, punched for use with self-locking cable tie fastener.
- C. Write-On Tags: Polyester tag, 0.010-inch thick, with corrosion-resistant grommet and cable tie for attachment to conductor or cable.
 - 1. Marker for Tags: Permanent, waterproof, black ink marker recommended by tag manufacturer.

2.2 UNDERGROUND-LINE WARNING TAPE

- A. Tape:
 - 1. Recommended by manufacturer for the method of installation and suitable to identify and locate underground electrical and communications utility lines.
 - 2. Printing on tape shall be permanent and shall not be damaged by burial operations.
 - 3. Tape material and ink shall be chemically inert, and not subject to degrading when exposed to acids, alkalis, and other destructive substances commonly found in soils.
- B. Color and Printing:
 - 1. Comply with ANSI Z535.1 through ANSI Z535.5.
 - 2. Inscriptions for Red-Colored Tapes: ELECTRIC LINE, HIGH VOLTAGE.
 - 3. Inscriptions for Orange-Colored Tapes: TELEPHONE CABLE, COMMUNICATIONS CABLE, OPTICAL FIBER CABLE.
- C. Conductive Type:
 - 1. Detectable three-layer laminate, consisting of a printed pigmented polyolefin film, a solid aluminum-foil core, and a clear protective film that allows inspection of the continuity of the conductive core, bright-colored, compounded for direct-burial service.
 - 2. Overall Thickness: 5 mils.
 - 3. Foil Core Thickness: 0.35 mil.
 - 4. Weight: 28 lb/1000 sq. ft.
 - 5. 3-Inch Tensile According to ASTM D 882: 70 lbf, and 4600 psi.

2.3 WARNING LABELS AND SIGNS

- A. Comply with NFPA 70 and 29 CFR 1910.145.
- B. Baked-Enamel Warning Signs:

1. Preprinted aluminum signs, punched or drilled for fasteners, with colors, legend, and size required for application.
2. 1/4-inch grommets in corners for mounting.
3. Nominal size, 7 by 10 inches.

C. Metal-Backed, Butyrate Warning Signs:

1. Weather-resistant, nonfading, preprinted, cellulose-acetate butyrate signs with 0.0396-inch galvanized-steel backing; and with colors, legend, and size required for application.
2. 1/4-inch (6.4-mm) grommets in corners for mounting.
3. Nominal size, 10 by 14 inches.

D. Warning label and sign shall include, but are not limited to, the following legends:

1. Workspace Clearance Warning: "WARNING - OSHA REGULATION - AREA IN FRONT OF ELECTRICAL EQUIPMENT MUST BE KEPT CLEAR FOR 36 INCHES."
2. Arc Flash Hazard Warning: "WARNING – POTENTIAL ARC FLASH HAZARD – APPROPRIATE PPE AND TOOLS REQUIRED WHEN WORKING ON THIS EQUIPMENT."

2.4 INSTRUCTION SIGNS

- A. Engraved, laminated acrylic or melamine plastic, minimum 1/16-inch thick for signs up to 20 sq. inches and 1/8-inch thick for larger sizes.
1. Engraved legend with black letters on white face.
 2. Punched or drilled for mechanical fasteners.
 3. Framed with mitered acrylic molding and arranged for attachment at applicable equipment.

2.5 EQUIPMENT IDENTIFICATION LABELS

- A. Engraved, Laminated Acrylic or Melamine Label: Punched or drilled for screw mounting. White letters on a dark-gray background. Minimum letter height shall be 3/8 inch.

2.6 CABLE TIES

- A. General-Purpose Cable Ties: Fungus inert, self-extinguishing, one-piece, self-locking, Type 6/6 nylon.
1. Minimum Width: 3/16 inch.
 2. Tensile Strength at 73 deg F, According to ASTM D 638: 12,000 psi.
 3. Temperature Range: Minus 40 to plus 185 deg F.
 4. Color: Black except where used for color-coding.

2.7 MISCELLANEOUS IDENTIFICATION PRODUCTS

- A. Paint: Comply with requirements in painting Sections for paint materials and application requirements. Select paint system applicable for surface material and location (exterior or interior).
- B. Fasteners for Labels and Signs: Self-tapping, stainless-steel screws or stainless-steel machine screws with nuts and flat and lock washers.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Verify identity of each item before installing identification products.
- B. Location: Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment.
- C. Apply identification devices to surfaces that require finish after completing finish work.
- D. Attach signs and plastic labels that are not self-adhesive type with mechanical fasteners appropriate to the location and substrate.
- E. Cable Ties: For attaching tags. Use general-purpose type, except as listed below:
 - 1. Outdoors: UV-stabilized nylon.
- F. Warning and Tracer Tape: Provide warning and tracer tape for all underground piping and conduit.
 - 1. Encapsulate and protect wire ends with valve boxes with square caps.
 - 2. Install valve box at each end of underground conduit. Terminate tracer wire at valve box.
 - 3. Do not install tracer wire valve boxes in paved areas.
 - 4. Form and pour a concrete ring encircling, and in contact with, the valve box and with top surface screeded to top of box cover frame. Bottom of ring shall rest on compacted gravel.
 - a. Concrete: 4000 psi, 28-day strength with a troweled finish.
 - b. Dimensions: 10 inches wide by 12 inches deep.
 - 5. Refer to "Typical Tracer Wire Detail" on Civil drawings for additional requirements.
- G. Painted Identification: Comply with requirements in painting Sections for surface preparation and paint application.

3.2 IDENTIFICATION SCHEDULE

- A. Power-Circuit Conductor Identification, 600 V or Less: For conductors in pull boxes, junction boxes, and handholes, use color-coding conductor tape to identify the phase.
 - 1. Color-Coding for Phase and Voltage Level Identification, 600 V or Less: Use colors listed below for ungrounded feeder and branch-circuit conductors.

2. Color shall be factory applied.
 3. Colors for 240/120-V Circuits:
 - a. Phase A: Black.
 - b. Phase B: Red.
- B. Auxiliary Electrical Systems Conductor Identification: Identify field-installed alarm, control, and signal connections.
1. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, and pull points. Identify by system and circuit designation.
 2. Use system of marker tape designations that is uniform and consistent with system used by manufacturer for factory-installed connections.
 3. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and the Operation and Maintenance Manual.
- C. Locations of Underground Lines: Identify with underground-line warning tape and tracer wire for power and control wiring.
- D. Warning Labels for Indoor Cabinets, Boxes, and Enclosures for Power and Lighting:
1. Comply with 29 CFR 1910.145.
 2. Identify system voltage with black letters on an orange background.
 3. Apply to exterior of door, cover, or other access.
- E. Operating Instruction Signs: Install instruction signs to facilitate proper operation and maintenance of electrical systems and items to which they connect. Install instruction signs with approved legend where instructions are needed for system or equipment operation.
- F. Equipment Identification Labels: On each unit of equipment, install unique designation label that is consistent with wiring diagrams, schedules, and the Operation and Maintenance Manual. Apply labels to disconnect switches and protection equipment, central or master units, control panels, control stations, terminal cabinets, and racks of each system. Systems include power, lighting, control, communication, signal, monitoring, and alarm systems unless equipment is provided with its own identification.
1. Labeling Instructions:
 - a. Indoor Equipment: Engraved, laminated acrylic or melamine label. Unless otherwise indicated, provide a single line of text with 1/2-inch-high letters on 1-1/2-inch-high label; where two lines of text are required, use labels 2 inches high.
 - b. Outdoor Equipment: Engraved, laminated acrylic or melamine label.
 - c. Elevated Components: Increase sizes of labels and letters to those appropriate for viewing from the floor.
 - d. Fasten labels with appropriate mechanical fasteners that do not change the NEMA or NRTL rating of the enclosure.
 2. Equipment to Be Labeled:
 - a. Panelboards: Typewritten directory of circuits in the location provided by panelboard manufacturer. Panelboard identification shall be engraved, laminated acrylic or melamine label.

- b. Enclosures and electrical cabinets.
- c. Enclosed switches.
- d. Enclosed circuit breakers.
- e. Enclosed controllers.
- f. Push-button stations.
- g. Contactors.
- h. Monitoring and control equipment.

END OF SECTION 26 05 53

SECTION 26 43 13 - SURGE PROTECTION FOR LOW-VOLTAGE ELECTRICAL POWER CIRCUITS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes field-mounted SPDs for low-voltage (120 to 600 V) power distribution and control equipment.
- B. Related Requirements:

1.2 DEFINITIONS

- A. Inominal: Nominal discharge current.
- B. MCOV: Maximum continuous operating voltage.
- C. Mode(s), also Modes of Protection: The pair of electrical connections where the VPR applies.
- D. MOV: Metal-oxide varistor; an electronic component with a significant non-ohmic current-voltage characteristic.
- E. OCPD: Overcurrent protective device.
- F. SCCR: Short-circuit current rating.
- G. SPD: Surge protective device.
- H. VPR: Voltage protection rating.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
 - 1. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.
 - 2. Copy of UL Category Code VZCA certification, as a minimum, listing the tested values for VPRs, Inominal ratings, MCOVs, type designations, OCPD requirements, model numbers, system voltages, and modes of protection.

1.4 INFORMATIONAL SUBMITTALS

- A. Field quality-control reports.
- B. Sample Warranty: For manufacturer's special warranty.

1.5 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For SPDs to include in maintenance manuals.

1.6 WARRANTY

- A. Manufacturer's Warranty: Manufacturer agrees to replace or replace SPDs that fail in materials or workmanship within specified warranty period.
 - 1. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 GENERAL SPD REQUIREMENTS

- A. SPD with Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. Comply with NFPA 70.
- C. Comply with UL 1449.
- D. MCOV of the SPD shall be the nominal system voltage.

2.2 SERVICE ENTRANCE SUPPRESSOR

- A. SPDs: Comply with UL 1449, Type 1.
 - 1. SPDs with the following features and accessories:
 - a. Integral disconnect switch.
 - b. Internal thermal protection that disconnects the SPD before damaging internal suppressor components.
 - c. Indicator light display for protection status.
 - d. Form-C dry contacts.
 - e. Surge counter.
- B. Peak Surge Current Rating: The minimum single-pulse surge current withstand rating per phase shall not be less than 200 kA. The peak surge current rating shall be the arithmetic sum of the ratings of the individual MOVs in a given mode.
- C. Protection modes and UL 1449 VPR for grounded wye circuits with 208Y/120V, three-phase, four-wire circuits shall not exceed the following:
 - 1. Line to Neutral: 700 V.
 - 2. Line to Ground: 1200 V.
 - 3. Line to Line: 1000 V.
- D. SCCR: Equal or exceed 200 kA.
- E. Inominal Rating: 20 kA.

2.3 PANEL SUPPRESSORS

- A. SPDs: Comply with UL 1449, Type 2.
 - 1. Include LED indicator lights for power and protection status.
 - 2. Internal thermal protection that disconnects the SPD before damaging internal suppressor components.
 - 3. Include Form-C contacts.
- B. Peak Surge Current Rating: The minimum single-pulse surge current withstand rating per phase shall not be less than 100 kA. The peak surge current rating shall be the arithmetic sum of the ratings of the individual MOVs in a given mode.
- C. Comply with UL 1283.
- D. Protection modes and UL 1449 VPR for grounded wye circuits with 208Y/120 V, three-phase, four-wire circuits shall not exceed the following:
 - 1. Line to Neutral: 700 V for 208Y/120 V.
 - 2. Line to Ground: 700 V for 208Y/120 V.
 - 3. Neutral to Ground: 700 V for 208Y/120 V.
 - 4. Line to Line: 1200 V for 208Y/120 V
- E. SCCR: Equal or exceed 200 kA.
- F. Inominal Rating: 20 kA.

2.4 ENCLOSURES

- A. Enclosures: NEMA 250, Type 3R.

2.5 CONDUCTORS AND CABLES

- A. Power Wiring: Same size as SPD leads, complying with Section 26 05 19 "Low-Voltage Electrical Power Conductors and Cables."

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Comply with NECA 1.
- B. Install an OCPD or disconnect as required to comply with the UL listing of the SPD.
- C. Install SPDs with conductors between suppressor and points of attachment as short and straight as possible, and adjust circuit-breaker positions to achieve shortest and straightest leads. Do not splice and extend SPD leads unless specifically permitted by manufacturer. Do not exceed manufacturer's recommended lead length. Do not bond neutral and ground.

- D. Wiring:
 - 1. Power Wiring: Comply with wiring methods in Section 26 05 19 "Low-Voltage Electrical Power Conductors and Cables."
 - 2. Controls: Comply with wiring methods in Section 26 05 19 "Low-Voltage Electrical Power Conductors and Cables."

3.2 FIELD QUALITY CONTROL

- A. Perform the following tests and inspections with the assistance of a factory-authorized service representative.
 - 1. Compare equipment nameplate data for compliance with Drawings and Specifications.
 - 2. Inspect anchorage, alignment, grounding, and clearances.
 - 3. Verify that electrical wiring installation complies with manufacturer's written installation requirements.
- B. An SPD will be considered defective if it does not pass tests and inspections.

3.3 STARTUP SERVICE

- A. Complete startup checks according to manufacturer's written instructions.
- B. Do not perform insulation-resistance tests of the distribution wiring equipment with SPDs installed. Disconnect SPDs before conducting insulation-resistance tests, and reconnect them immediately after the testing is over.
- C. Energize SPDs after power system has been energized, stabilized, and tested.

3.4 DEMONSTRATION

- A. Train Owner's maintenance personnel to operate and maintain SPDs.

END OF SECTION 26 43 13

SECTION 311000 - SITE CLEARING

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 DESCRIPTION OF WORK

- A. Work Included: Provide labor, materials, and equipment necessary to complete the work of this Section, including but not limited to the following:
 - 1. Protecting existing trees and vegetation to remain, including temporary fencing for trees in close proximity to construction operations.
 - 2. Clearing and grubbing.
 - 3. Stripping topsoil.
- B. Alternates: N/A
- C. Items to Be Installed Only: Not Applicable.

1.3 RELATED WORK

- A. Related Work: The following items are not included in this Section and will be performed under the designated Sections:
 - 1. Section 31 2000 – EARTH MOVING for soil materials, excavating, backfilling, and site grading and removal of site utilities.
 - 2. Section 31 2500 – EROSION AND SEDIMENTATION CONTROLS for required erosion and sedimentation control measures.

1.4 DEFINITIONS

- A. Topsoil: Natural or cultivated surface-soil layer containing organic matter and sand, silt, and clay particles; friable, pervious, and black or a darker shade of brown, gray, or red than underlying subsoil; reasonably free of subsoil, clay lumps, gravel, and other objects more than 2 inches in diameter; and free of subsoil and weeds, roots, toxic materials, or other nonsoil materials.

- B. Tree Protection Zone: Area surrounding individual trees or groups of trees to be protected during construction, and defined by the drip line of individual trees or the perimeter drip line of groups of trees, unless otherwise indicated.

1.5 MATERIAL OWNERSHIP

- A. Except for materials indicated to remain the Owner's property, cleared materials shall become Contractor's property and shall be removed from the Project site and disposed of legally offsite.

1.6 SUBMITTALS

- A. Photographs sufficiently detailed, of existing conditions of trees and plantings, adjoining construction, and site improvements that might be misconstrued as damage caused by site clearing.
- B. Record drawings, according to Section 01 7000 - PROJECT CLOSEOUT identifying and accurately locating capped utilities and other subsurface structural, electrical, and mechanical conditions.

1.7 PROJECT CONDITIONS

- A. Traffic: Minimize interference with adjoining roads, streets, walks, and other adjacent occupied or used facilities during site-clearing operations.
 - 1. Do not close or obstruct streets, walks, or other adjacent occupied or used facilities without permission from the Owner and authorities having jurisdiction.
 - 2. Provide alternate routes around closed or obstructed traffic ways if required by authorities having jurisdiction.
- B. Salvable Improvements: Carefully remove items indicated to be salvaged and store on Owner's premises where indicated.
- C. Utility Locator Service: Notify utility locator service for area where Project is located before site clearing.
- D. Do not commence site clearing operations until erosion and sedimentation control measures are in place.
- E. Protection of Existing Improvements: Provide protection necessary to prevent damage to existing improvements indicated to remain in place or outside of the limit of work. Protect improvements on adjoining properties and on Owner's property.
 - 1. Restore improvements damaged by Contractor's clearing activities to their original condition, at no additional expense to the Owner.

PART 2 – PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 PREPARATION

- A. Protect and maintain benchmarks and survey control points from disturbance during construction.
- B. Locate and clearly flag trees and vegetation to remain or to be relocated.
- C. Protect existing site improvements to remain from damage during construction.
 - 1. Restore damaged improvements to their original condition, as acceptable to the Owner.

3.2 TREE PROTECTION

- A. Erect and maintain temporary fencing around tree protection zones before starting site clearing. Remove fence when construction is complete.
 - 1. Do not store construction materials, debris, or excavated material within fenced area.
 - 2. Do not permit vehicles, equipment, or foot traffic within fenced area.
 - 3. Maintain fenced area free of weeds and trash.
 - 4. Except as otherwise directed, cutting and trimming of existing trees will not be permitted.
- B. Do not excavate within tree protection zones, unless otherwise indicated.
- C. Where excavation for new construction is required within tree protection zones, hand clear and excavate to minimize damage to root systems. Use narrow-tine spading forks, comb soil to expose roots, and cleanly cut roots as close to excavation as possible.
 - 1. Cover exposed roots with burlap and water regularly.
 - 2. Temporarily support and protect roots from damage until they are permanently redirected and covered with soil.
 - 3. Coat cut faces of roots more than 1-1/2 inches in diameter with an emulsified asphalt or other approved coating formulated for use on damaged plant tissues.
 - 4. Backfill with soil as soon as possible.

- D. Repair or replace trees and vegetation indicated to remain that are damaged by construction operations, in a manner approved by the Engineer.
 - 1. Employ an arborist, licensed in jurisdiction where Project is located, to submit details of proposed repairs and to repair damage to trees and shrubs.
 - 2. Replace trees that cannot be repaired and restored to full-growth status, as determined by the Engineer.

3.3 CLEARING AND GRUBBING

- A. Remove obstructions, trees, shrubs, grass, and other vegetation to permit installation of new construction.
 - 1. Cut minor roots and branches of trees indicated to remain in a clean and careful manner where such roots and branches obstruct installation of new construction.
 - 2. Grind stumps and remove roots, obstructions, and debris extending to a depth of 18 inches below exposed subgrade in landscaped areas. Completely remove stumps and roots under pavement, sidewalks and building footprint.
 - 3. Use only hand methods for grubbing within tree protection zone.
 - 4. Chip removed tree branches and dispose of off-site.
- B. Fill depressions caused by clearing and grubbing operations with satisfactory soil material, or as otherwise noted, unless further excavation or earthwork is indicated.
 - 1. Place fill material in horizontal layers not exceeding a loose depth of 8 inches and compact each layer to a density equal to adjacent original ground.

3.4 TOPSOIL STRIPPING

- A. Remove sod and grass before stripping topsoil.
- B. Strip topsoil to whatever depths are encountered in a manner to prevent intermingling with underlying subsoil or other waste materials. Remove subsoil and nonsoil materials from topsoil, including trash, debris, weeds, roots, and other waste materials.
- C. Stockpile topsoil materials away from edge of excavations without intermixing with subsoil. Grade and shape stockpiles to drain surface water. Cover to prevent saturation, windblown dust or contamination by air-borne weed seed.
 - 1. Limit height of topsoil stockpiles to 72 inches.
 - 2. Do not stockpile topsoil within tree protection zones.
 - 3. Surround stockpiles with silt fence.

3.5 EXCESS SOIL

- A. Excess soil that is not needed after the completion of all final spreading and grassing shall become the property of the Contractor and shall be removed and disposed of offsite. Contractor shall test and legally dispose of all excess soil that is to be removed from the site.

3.6 SITE IMPROVEMENTS

- A. Remove existing above- and below-grade improvements as indicated and as necessary to facilitate new construction.

3.7 DISPOSAL

- A. Disposal: Remove surplus soil material, boulders, unsuitable topsoil, obstructions, demolished materials, and waste materials including trash and debris, and legally dispose of them off the Owner's property.
 - 1. Burning and burying on site is prohibited.
 - 2. Separate recyclable materials produced during site clearing from other nonrecyclable materials. Store or stockpile without intermixing with other materials and transport them to recycling facilities.

END OF SECTION 311000

SECTION 312000 - EARTH MOVING

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 DESCRIPTION OF WORK

- A. Work Included: Provide labor, materials, and equipment necessary to complete the work of this Section, including but not limited to the following:
 - 1. Preparing subgrades for structures and landscaping.
 - 2. Excavating and backfilling for pavements and structures.
 - 3. Subbase course for concrete pavements.
 - 4. Excavating and backfilling for utility trenches.
 - 5. Remove and replace unsuitable existing fill material.

1.3 Related Work:

- A. The following items are not included in this Section and will be performed under the designated Sections:
 - 1. Section 31 1000 - SITE CLEARING for site stripping, grubbing, stripping and stockpiling topsoil, and removal of above- and below-grade improvements and utilities.
 - 2. Section 31 2500 – EROSION AND SEDIMENTATION CONTROLS for temporary erosion and sedimentation control measures.

1.4 DEFINITIONS

- A. Backfill: Soil material or controlled low-strength material used to fill an excavation.
- B. Bedding Course: Course placed over the excavated subgrade in a trench before laying pipe.
- C. Borrow Soil: Satisfactory soil imported from off-site for use as fill or backfill.
- D. Excavation: Removal of material encountered above subgrade elevations and to lines and dimensions indicated.
 - 1. Authorized Additional Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions as directed by Engineer Authorized additional excavation and replacement material will be paid for according to Contract provisions for unit prices.
 - 2. Bulk Excavation: Excavation more than 10 feet in width and more than 30 feet in length.

3. Unauthorized Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions without direction by Engineer. Unauthorized excavation, as well as remedial work directed by Engineer, shall be without additional compensation.

E. Fill: Suitable soil materials used to raise existing grades.

F. Rock: Rock material in beds, ledges, unstratified masses, conglomerate deposits, and boulders of rock material that exceed 1 cu. yd. for bulk excavation or 3/4 cu. yd. for footing, trench, and pit excavation that cannot be removed by rock excavating equipment without systematic drilling, ram hammering, ripping, or blasting, when permitted.

G. Structures: Buildings, footings, foundations, retaining walls, slabs, tanks, curbs, mechanical and electrical appurtenances, or other man-made stationary features constructed above or below the ground surface.

H. Subbase Course: Course placed between the subgrade and base course for hot-mix asphalt pavement, or course placed between the subgrade and a cement concrete pavement or a cement concrete or hot-mix asphalt walk.

I. Subgrade: Surface or elevation remaining after completing excavation, or top surface of a fill or backfill immediately below subbase, drainage fill, or topsoil materials.

J. Utilities: On-site underground pipes, conduits, ducts, and cables, as well as underground services within buildings.

1.5 SITE INVESTIGATION

A. The Contractor shall satisfy themselves to the nature and location of the work, the general and local conditions, particularly those bearing upon transportation, disposal, handling and storage of materials, availability of labor, water, electric power, roads and uncertainties of weather, groundwater table or similar physical conditions at the site, the confirmation of subsurface materials to be encountered, the character of equipment and facilities needed prior to and during the prosecution of work and other matters which can affect the work or the cost thereof under this contract. Failure by the Contractor to acquaint themselves with all information concerning these conditions will not relieve them from responsibility for properly estimating the difficulty or cost of successfully performing the work.

1.6 SUBSURFACE DATA

A. Variations in existing ground or subsurface soil conditions from those indicated on the test pit logs shall not constitute grounds for changes in contract price or completion dates of this contract.

1.7 SUBMITTALS

- A. Product Data: For the following:
 - 1. Each type of plastic warning tape.
 - 2. Geotextile.
 - 3. Soil textural classification analysis for all imported fill materials.
- B. Material Test Reports: Contractor to hire a qualified testing agency indicating and interpreting test results for compliance of the following with requirements indicated:
 - 1. Classification according to ASTM D 2487 of each on-site and borrow soil material proposed for fill and backfill.
 - 2. Laboratory compaction curve according to ASTM D 1557 for each onsite and borrow soil material proposed for fill and backfill.
- C. Pre-excavation Photographs and Videotape: Show existing conditions of adjoining construction and site improvements, including finish surfaces that might be misconstrued as damage caused by earthwork operations. Submit before earthwork begins. Maintain catalog of up-to-date photographs at the site.

1.8 QUALITY CONTROL

- A. Compaction and materials testing results shall be submitted to the Engineer for review as outlined in the following sections.

PART 2 – PRODUCTS

2.1 SOIL MATERIALS

- A. General: Provide borrow soil materials when sufficient satisfactory soil materials are not available from excavations.
- B. Satisfactory Soils: ASTM D 2487 Soil Classification Groups GW, GP, GM, SW, SP, and SM or a combination of these groups; free of rock or gravel larger than 3 inches in any dimension, debris, waste, frozen materials, vegetation, and other deleterious matter.
- C. Common Fill: Imported Common Fill should consist of Satisfactory Soils having a maximum particle size of 6 inches and no more than 25 percent by weight passing the US No. 200 sieve.
- D. Unsatisfactory Soils: Soil Classification Groups GC, SC, CL, ML, OL, CH, MH, OH, and PT according to ASTM D 2487, or a combination of these groups.
 - 1. Unsatisfactory soils also include satisfactory soils not maintained within 2 percent of optimum moisture content at time of compaction.

- E. Base Course: Material meeting the minimum requirements for Dense-graded Crushed Stone, as defined by the MassDOT Standard Specifications for Highways and Bridges (Section M2.01.7). The gradation requirements for Dense-graded Crushed Stone are as follows:

Percent Passing	
Sieve Size	By Weight
2 in.	100
1½ in.	70-100
¾ in.	50-85
No. 4	30-55
No. 50	8-24
No. 200	3-10

- F. Bedding Course: Bedding course for utilities shall comply with the requirements of Screened Gravel listed below.

1. Screened Gravel: shall be uniformly graded with the maximum size of a particle between 3/8 inch and 3/4 inch. Screened Gravel shall consist of clean, hard and durable particles free from an excess of soft, elongated and disintegrated pieces or other objectionable material meeting MassDOT Standard Specifications for Highways and Bridges Section M.2.01.5. The gradation requirements for Screened Gravel are as follows:

Sieve Size	Percent Passing
5/8 inch	100
½ inch	85-100
3/8 inch	15-45
No 4.	0-15
No 8.	0-5

2.2 GEOTEXTILES

- A. Geotextile Filter Fabric: Nonwoven needle-punched geotextile, manufactured for subsurface drainage applications, made from polyolefins or polyesters; with elongation greater than 50 percent; complying with AASHTO M 288 and the following, measured per test methods referenced:

1. Survivability: Class 2; AASHTO M 288.
2. Grab Tensile Strength: 157 lbf; ASTM D 4632.
3. Sewn Seam Strength: 142 lbf; ASTM D 4632.
4. Tear Strength: 56 lbf; ASTM D 4533.
5. Puncture Strength: 56 lbf; ASTM D 4833.
6. Apparent Opening Size: No. 40 sieve, maximum; ASTM D 4751.
7. UV Stability: 50 percent after 500 hours' exposure; ASTM D 4355.

2.3 ACCESSORIES

- A. Detectable Warning Tape: Acid- and alkali-resistant polyethylene film warning tape manufactured for marking and identifying underground utilities, a minimum of 6 inches wide and 4 mils thick, continuously inscribed with a description of the utility, with metallic core encased in a protective jacket for corrosion protection, detectable by metal detector when tape is buried up to 30 inches deep; colored as follows:
1. Red: Electric.
 2. Yellow: Gas, oil, steam, and dangerous materials.
 3. Orange: Telephone and other communications.
 4. Blue: Water systems.
 5. Green: Sewer systems.

PART 3 – EXECUTION

3.1 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.
- B. Preparation of subgrade for earthwork operations including removal of vegetation, topsoil, debris, obstructions, and deleterious materials from ground surface is specified in Section 31 1000 - SITE CLEARING.
- C. Protect and maintain erosion and sedimentation controls, which are specified in Section 32 2500 - EROSION, during earthwork operations.
- D. Provide protective insulating materials to protect subgrade soils against freezing temperatures or frost.

3.2 DEWATERING

- A. Prevent surface water and ground water from entering excavations, from ponding on prepared subgrades, and from flooding Project site and surrounding area. Dispose of contaminated water in accordance with regulations of authorities having jurisdiction.
- B. Protect subgrades from softening, undermining, washout, and damage by rain or water accumulation.
1. Reroute surface water runoff away from excavated areas. Do not allow water to accumulate in excavations. Do not use excavated trenches as temporary drainage ditches.
 2. Install a dewatering system to keep subgrades dry and convey ground water away from excavations. Maintain until dewatering is no longer required.

3.3 EXPLOSIVES

- A. Explosives: Do not use explosives.

3.4 EXCAVATION, GENERAL

- A. Unclassified Excavation: Excavate to subgrade elevations regardless of the character of surface and subsurface conditions encountered. Unclassified excavated materials may include rock, soil materials, and obstructions. No changes in the Contract Sum or the Contract Time will be authorized for rock excavation or removal of obstructions.
 - 1. If excavated materials intended for fill and backfill include unsatisfactory soil materials and rock, replace with satisfactory soil materials.

3.5 EXCAVATION FOR STRUCTURES

- A. Excavate to indicated elevations and dimensions within a tolerance of plus or minus 1 inch. If applicable, extend excavations a sufficient distance from structures for placing and removing concrete formwork, for installing services and other construction, and for inspections.
 - 1. Excavation for Underground Tanks, Basins, and Mechanical or Electrical Utility Structures: Excavate to elevations and dimensions indicated within a tolerance of plus or minus 1 inch. Do not disturb bottom of excavations intended as bearing surfaces.

3.6 EXCAVATION FOR WALKS AND PAVEMENTS

- A. Excavate surfaces under walks and pavements to indicated lines, cross sections, elevations, and subgrades.

3.7 EXCAVATION FOR UTILITY TRENCHES

- A. Excavate trenches to indicated gradients, lines, depths, and elevations.
- B. Excavate trenches to uniform widths to provide the following clearance on each side of pipe or conduit.
- C. Trench Bottoms: Excavate and shape trench bottoms to provide uniform bearing and support of pipes and conduit. Shape subgrade to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits. Remove projecting stones and sharp objects along trench subgrade.
 - 1. For pipes and conduit less than 6 inches in nominal diameter and flat-bottomed, multiple-duct conduit units, hand-excavate trench bottoms and support pipe and conduit on an undisturbed subgrade.
 - 2. For pipes and conduit 6 inches or larger in nominal diameter, shape bottom of trench to support bottom 90 degrees of pipe circumference. Fill depressions with tamped sand backfill.

3. Excavate trenches 6 inches deeper than elevation required in rock or other unyielding bearing material to allow for bedding course.

3.8 OVER-EXCAVATION OF UNSUITABLE SOILS

- A. When approved by the Engineer, the Contractor may be required to remove unsuitable soils, fill, or natural soil materials in areas where fills are to be placed when determined to be undesirable in their location or condition. The Contractor shall be required to remove the undesirable material and backfill with approved material properly compacted.
- B. At locations where unstable soil is identified, the removal and replacement of such soil shall be as directed by the Engineer.
- C. At locations where soil is wet of optimum moisture, the Contractor shall provide a “good faith” effort in drying and discing these areas prior to completing over-excavation as approved by the Engineer.
- D. Where over-excavations are required adjacent or beneath the location of the proposed structure, undercut and backfill shall be done over a sufficient distance adjacent to the installation to prevent future operations from disturbing the completed drainage structure.
- E. All material removed in the work of over-excavation will be classified by the Engineer and Owner as either suitable for other use without excessive manipulation and utilized by the Contractor elsewhere in the work, or unsuitable for future use and disposed of by the Contractor as directed by the Engineer.
- F. The Contractor shall conduct over-excavation operations in such a way that the necessary measurements can be taken before any backfill is placed.
- G. Backfill in over-excavation areas shall be placed as a continuous operation along with the over-excavation operation. Backfill materials shall be consistent with the intended use. No backfill material shall be placed in water unless otherwise permitted by the Engineer.

3.9 SUBGRADE INSPECTION

- A. Notify Engineer when excavations have reached required subgrade.
- B. If the Engineer determines that unsatisfactory soil is present, continue excavation and replace with compacted backfill or fill material as directed and specified herein.
- C. The Contractor shall be responsible for maintaining stable soil subgrades. Fine-grained subgrade soils exposed during construction are anticipated to be easily disturbed by construction traffic and are likely to become unstable when above the optimum moisture content. The Contractor shall be responsible for managing construction traffic, stockpiling of materials, and providing routine maintenance to protect subgrades from disturbance. Where subgrades are damaged by freezing temperatures, frost, rain, accumulated water, or construction activities, they shall be reconstructed as directed by the Engineer, without additional compensation.

3.10 UNAUTHORIZED EXCAVATION

- A. Fill unauthorized excavation by extending bottom elevation without altering top elevation. Alternatively, the unauthorized excavation may be backfilled to design elevation using appropriate soil for the intended use. Lean concrete fill, with 28-day compressive strength of 2500 psi may also be used when approved by the Engineer.
 - 1. Fill unauthorized excavations under other construction or utility pipe as directed by the Engineer.

3.11 STORAGE OF SOIL MATERIALS

- A. Stockpile borrow soil materials and excavated satisfactory soil materials without intermixing. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
 - 1. Stockpile soil materials away from edge of excavations. Do not store within drip line of remaining trees, if applicable.

3.12 BACKFILL

- A. Place and compact backfill in excavations promptly, but not before completing the following:
 - 1. Construction below finish grade including, where applicable, dampproofing, waterproofing, and perimeter insulation.
 - 2. Observing and accepting subgrade.
 - 3. Surveying locations of underground utilities for Record Documents.
 - 4. Testing and inspecting underground utilities.
 - 5. Removing concrete formwork.
 - 6. Removing trash and debris.
 - 7. Removing temporary shoring and bracing, and sheeting.
 - 8. Installing permanent or temporary horizontal bracing on horizontally supported walls.
- B. Place backfill on subgrades free of mud, frost, snow, or ice.

3.13 UTILITY TRENCH BACKFILL

- A. Place backfill on subgrades free of mud, frost, snow, or ice.
- B. Place and compact bedding course on trench bottoms and where indicated. Shape bedding course to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits.

- C. Place and compact initial backfill of subbase material free of particles larger than 1 inch in any dimension, to a height of 12 inches over the utility pipe or conduit.
 - 1. Carefully compact initial backfill under pipe haunches and compact evenly up on both sides and along the full length of utility piping or conduit to avoid damage or displacement of piping or conduit. Coordinate backfilling with utilities testing.
- D. Backfill voids with satisfactory soil while installing and removing shoring and bracing.
- E. Place and compact final backfill of satisfactory soil to final subgrade elevation.
- F. Install warning tape directly above utilities, 12 inches below finished grade, except 6 inches below subgrade under pavements and slabs.

3.14 SOIL FILL

- A. Plow, scarify, bench, or break up sloped surfaces steeper than 1 vertical to 4 horizontal so fill material will bond with existing material.
- B. Place and compact fill material in layers to required elevations as shown on the contract drawings.
- C. Place soil fill on subgrades free of mud, frost, snow, or ice.
- D. All soils to be compacted to a minimum of 95% of its maximum density at optimum moisture content or as otherwise specified.

3.15 SOIL MOISTURE CONTROL

- A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill soil layer before compaction to within 2 percent of optimum moisture content.
 - 1. Do not place backfill or fill soil material on surfaces that are muddy, frozen, or contain frost or ice.
 - 2. Remove and replace, or scarify and air dry otherwise satisfactory soil material that exceeds optimum moisture content by 2 percent and is too wet to compact to specified dry unit weight.

3.16 COMPACTION OF SOIL BACKFILLS AND FILLS

- A. Place backfill and fill soil materials in layers not more than 8 inches in loose depth for material compacted by heavy compaction equipment, and not more than 4 inches in loose depth for material compacted by hand-operated tampers.
- B. Place backfill and fill soil materials evenly on all sides of structures to required elevations, and uniformly along the full length of each structure.

- C. Compact soil materials to not less than the following percentages of maximum dry unit weight according to ASTM D 1557:
 - 1. Under structures and pavements, scarify and recompact top 12 inches of existing subgrade and each layer of backfill or fill soil material at 95 percent; and areas within 10 feet of structures, building slabs, steps, and pavements at 92 percent.
 - 2. Under lawn or unpaved areas, scarify and recompact top 6 inches below subgrade and compact each layer of backfill or fill soil material at 90 percent.
 - 3. For utility trenches, compact each layer of initial and final backfill soil material at 95 percent.

3.17 GRADING

- A. General: Uniformly grade areas to a smooth surface, free of irregular surface changes. Comply with compaction requirements and grade to cross sections, lines, and elevations indicated.
 - 1. Provide a smooth transition between adjacent existing grades and new grades.
 - 2. Cut out soft spots, fill low spots, and trim high spots to comply with required surface tolerances.
- B. Site Grading: Slope grades to direct water away from buildings and to prevent ponding. Finish subgrades to required elevations within the following tolerances:
 - 1. Lawn or Unpaved Areas: Plus or minus 1 inch.
 - 2. Walks: Plus or minus 1/2 inch. Tolerance will not alleviate the contractor's responsibility to meet required slopes in Accessible areas.
 - 3. Pavements: Plus or minus 1/4 inch.

3.18 SUBBASE AND BASE COURSES

- A. Place subbase and base course on subgrades free of mud, frost, snow, or ice.
- B. On prepared subgrade, place subbase and base course under pavements and walks as follows:
 - 1. Install separation geotextile on prepared subgrade according to manufacturer's written instructions, overlapping sides and ends.
 - 2. Place subbase and base course 6 inches or less in compacted thickness in a single layer.
 - 3. Place subbase and base course that exceeds 6 inches in compacted thickness in layers of equal thickness, with no compacted layer more than 6 inches thick or less than 3 inches thick.
 - 4. Compact subbase and base course at optimum moisture content to required grades, lines, cross sections, and thickness to not less than 95 percent of maximum dry unit weight according to ASTM D 1557.

3.20 FIELD QUALITY CONTROL

- A. Testing Agency: Contractor to hire a qualified independent materials testing agency to perform field quality-control testing.
- B. Allow testing agency to inspect and test subgrades and each fill or backfill layer. Proceed with subsequent earthwork only after test results for previously completed work comply with requirements.
- C. Testing agency will test compaction of soils in place according to ASTM D 1556, ASTM D 2167, ASTM D 2922, and ASTM D 2937, as applicable. Tests will be performed at the following locations and frequencies:
 - 1. Paved Slab Areas: At subgrade and at each compacted fill and backfill layer, at least 1 test for every 2000 sq. ft. or less of paved area, but in no case fewer than 3 tests.
 - 2. Trench Backfill: At each compacted initial and final backfill layer, at least 1 test for each 150 feet or less of trench length, but no fewer than 2 tests.
- D. When testing agency reports that subgrades, fills, or backfills have not achieved degree of compaction specified, scarify and moisten or aerate, or remove and replace soil to depth required; recompact and retest until specified compaction is obtained.

3.21 PROTECTION

- A. Protecting Graded Areas: Protect newly graded areas from traffic, freezing, and erosion. Keep free of trash and debris.
- B. Repair and reestablish grades to specified tolerances where completed or partially completed surfaces become eroded, rutted, settled, or where they lose compaction due to subsequent construction operations or weather conditions.
 - 1. Scarify or remove and replace soil material to depth as directed by Engineer; reshape and recompact.
- C. Where settling occurs before Project correction period elapses, remove finished surfacing, backfill with additional soil material, compact, and reconstruct surfacing.
 - 1. Restore appearance, quality, and condition of finished surfacing to match adjacent work, and eliminate evidence of restoration to greatest extent possible.

3.22 DISPOSAL OF SURPLUS AND WASTE MATERIALS

- A. Disposal: Remove surplus satisfactory soil and waste material, including unsatisfactory soil, trash, and debris, and legally dispose of it off the property.

END OF SECTION 312000

SECTION 312500 - EROSION AND SEDIMENTATION CONTROLS

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 DESCRIPTION OF WORK

- A. Work Included: Provide labor, materials and equipment necessary to complete the work of this Section, including but not limited to the following:
 - 1. Control measures to prevent all erosion, siltation and sedimentation of wetlands, waterways, construction areas, adjacent areas and off-site areas.
 - 2. Control measures shall be accomplished adjacent to or in the following work areas:
 - a. Soil stockpiles and on-site storage and staging areas.
 - b. Cut and fill slopes and other stripped and graded areas.
 - c. Constructed and existing swales and ditches.
 - d. Protection of drainage structure inlets.
 - e. At edge of wetlands areas, if applicable, as shown on Drawings.
 - f. Protection of stockpile areas.
 - 3. Additional means of protection shall be provided by the Contractor as required for continued or unforeseen erosion problems, at no additional cost to the Owner
 - 4. Periodic maintenance of all sediment control structures shall be provided to ensure intended purpose is accomplished. Sediment control measures shall be in working condition at the end of each day.
 - 5. On a weekly basis and after any significant rainfall, sediment control structures shall be inspected for integrity. Any damaged device shall be corrected immediately.
- B. Alternates: Not Applicable.
- C. Items To Be Installed Only: Not Applicable.
- D. Items To Be Furnished Only: Not Applicable.

- E. Related Work: The following items are not included in this Section and will be performed under the designated Sections:
 - 1. Section 31 1000 – SITE CLEARING for protection of existing trees and other vegetation to remain.
 - 2. Section 31 2000 – EARTH MOVING for soil materials, excavating, backfilling, and site grading and removal of site utilities.

1.3 QUALITY ASSURANCE

- A. The Contractor shall develop, submit, and comply with the requirements of a Stormwater Pollution Prevention Plan (SWPPP) consistent with all applicable federal, state, and local regulations.
 - 1. If the total land disturbance for this project equals or exceeds one acre, or is part of a larger common plan of development or sale, the Contractor shall obtain coverage under the EPA National Pollutant Discharge Elimination System (NPDES) Construction General Permit (CGP) by filing a Notice of Intent (NOI) and preparing a site-specific SWPPP in accordance with EPA requirements.
 - 2. If the total land disturbance is less than one acre, the Contractor shall prepare and implement an Erosion and Sediment Control Plan (ESCP) that meets the performance standards of EPA 832/R-92-005 and all local authorities having jurisdiction.
 - 3. The SWPPP or ESCP shall be submitted to the Contracting Officer for review prior to the start of land-disturbing activities.
 - 4. Provide and maintain all temporary erosion and sedimentation control measures necessary to prevent discharge of soil, sediment, or pollutants to adjacent properties, wetlands, or waterways.
 - 5. The Contractor shall be the sole permittee of record and operator for purposes of any required NPDES Construction General Permit. The Government shall not be listed as a co-permittee.
- B. Erosion control measures shall be established at the beginning of construction and maintained during the entire period of construction. On-site areas which are subject to severe erosion, and off-site areas which are especially vulnerable to damage from erosion and/or sedimentation, are to be identified and receive special attention.
- C. All land-disturbing activities are to be planned and conducted to minimize the size of the area to be exposed at any one time, and the length of time of exposure.
- D. Surface water runoff originating upgradient of exposed areas should be controlled to reduce erosion and sediment loss during the period of exposure.
- E. When the increase in the peak rates and velocity of storm water runoff resulting from a land-disturbing activity is sufficient to cause accelerated erosion of the receiving stream bed, provide measures to control both the velocity and rate of release so as to minimize accelerated erosion and increased sedimentation of the stream.

- F. All land-disturbing activities are to be planned and conducted so as to minimize off-site sedimentation damage.
- G. The Contractor shall leave all disturbed areas stabilized at the completion of construction to prevent erosion or sediment from leaving the project site, including sand from washing onto paved surfaces. Stabilization may include temporary measures such as mulch, erosion matting, silt fencing, or other approved means. Permanent plantings are optional and may be included only if exercised under a separate bid option.
- H. The Contractor is responsible for cleaning out and disposing of all sediment once the storage capacity of the sediment facility is reduced by one-half.
- I. Inspect, repair, and maintain erosion and sedimentation control measures during construction until permanent vegetation has been established.
- J. Remove erosion and sedimentation controls and restore and stabilize areas disturbed during removal.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Siltation Fence: Fabricated or prefabricated unit consisting of the following filter fabric properties:

1) Grab Tensile Strength	90	ASTM D1682
2) Elongation at Failure (%)	50	ASTM D1682
3) Mullen Burst Strength (PSI)	190	ASTM D3786
4) Puncture Strength (lbs)	70	ASTMD751(modified)
5) Slurry Flow Rate (gal/min/sf)	0.5	Virginia DOT VTM-51
6) Equivalent Opening Size	40-80 US	Std Sieve CW-02215
7) Ultraviolet Radiation Stability (%)	90	ASTM G26
- B. Fencing: Steel posts shall be standard 6 foot long metal stamped drive stakes commonly used to support snow fences. Fencing shall be new four foot height wood lath snow fencing. Provide suitable steel staples or heavy nylon cord for securing filter cloth to support system.
- C. Silt Socks: The silt socks for construction of erosion control devices shall be 12" in diameter. In areas of slope greater than 2:1(horizontal: vertical), silt sock must be secured in place by stakes. Silt socks shall be either lapped or butted at the ends to create a continuous line.
- D. Stakes: Stakes for silt socks shall be one of the following materials: Wood stakes of sound hardwood 2 by 2 inches in size or steel reinforcing bars of at least No. 4 size. Lengths shall be approximately three feet.

- E. Protective Measures: As temporary coverings on ground areas subject to erosion, provide one of the following protective measures, and as directed by the contracting officer:
 - 1. Hay or straw temporary mulch, 100 pounds per 1,000 square feet.
 - 2. Wood fiber cellulose temporary mulch, 35 pounds per 1,000 square feet.
 - 3. Tackafier for anchoring mulch or straw shall be a non-petroleum based liquid bonding agent specifically made for anchoring hay or straw.
 - 4. Provide natural (jute, wood excelsior) or man-made (glass fiber) covering with suitable staples or anchors to secure to ground surface. Note that wire staples and non-biodegradable coverings shall not be used for any area that will be mown turf.
 - 5. Temporary vegetative cover for graded areas shall be undamaged, air dry threshed straw or hay free of undesirable weed seed.
 - 6. Provide temporary settling basins as shown on the contract drawings and described in the specifications.
- F. Stone for Construction Entrance: Shall be ASTM designation C-33, size No. 3 crushed stone.

PART 3 - EXECUTION

3.1 STRAW BALE BARRIERS

- A. Excavation shall be to the width of the bale and the length of the proposed barrier to a minimum depth of 4 inches.
- B. Bales shall be placed in a single row, lengthwise on proposed line, with ends of adjacent bales tightly abutting one another. In swales and ditches the barrier shall extend to such a length that the bottoms of the end bales are higher in elevation than the top of the lowest middle bale.
- C. Staking shall be accomplished to securely anchor bales by driving at least two stakes or rebars through each bale to a minimum depth of 18 inches.
- D. The gaps between bales shall be filled by wedging straw in the gaps to prevent water from escaping between the bales.
- E. The excavated soil shall be backfilled against the barrier. Backfill shall conform to ground level on the downhill side and shall be built up to 4 inches on the uphill side. Loose straw shall then be scattered over the area immediately uphill from a straw barrier.
- F. Inspection shall be frequent and repair or replacement shall be made promptly as needed.

3.2 STABILIZED CONSTRUCTION ENTRANCE AND STONE BERMS

- A. Stone as specified above.
- B. Length: 25 feet minimum.
- C. Thickness: Not less than twelve inches.
- D. Width: Not less than 12 feet.
- E. Washing: When necessary, wheels shall be cleaned to remove sediment prior to entrance onto public right-of-way. When washing is required, it shall be done on an area stabilized with crushed stone which drains into an approved sediment trap or sediment basin. All sediment shall be prevented from entering any storm drain, ditch, or watercourse through the use of sand bags, gravel boards or other approved methods.
- F. Maintenance: The entrance shall be maintained in a condition which will prevent tracking or flowing of sediment onto public rights-of-way. This may require periodic top dressing with additional stone as conditions demand and repair and/or cleanout of any measures used to trap sediment. All sediment spilled, dropped, washed or tracked onto public rights-of-way must be removed immediately.
- G. Place crushed stone berms in locations required and as directed. Berms shall have side slopes of 1:3 or less.
- H. Inspect stone berms periodically and replace and/or regrade crushed stone as required.

3.3 SILT FENCING

- A. Excavate a 6 inch trench along the upstream side of the desired fence location.
- B. Drive fence posts a minimum of 12" into the ground. Install fence, well-staked at maximum eight foot intervals in locations as shown on Drawings. Secure fabric to fence and bury fabric end within the six inch deep trench cut.
- C. Lay lower 12 inches of silt fence into the trench, 6 inches deep and 6 inches wide. Backfill trench and compact.
- D. Overlap joints in fabric at post to prevent leakage of silt at seam.

3.4 VEGETATIVE STABILIZATION / TEMPORARY SEEDING

- A. Grassing shall be applied only if used for temporary stabilization. Permanent plantings are optional and are not required under the base bid.

3.5 INLET PROTECTION

- A. Install silt fence, silt sacks, or straw bales around inlet as specified herein.

3.6 DUST CONTROL

- A. Throughout the construction period the Contractor shall carry on an active program for the control of fugitive dust within all site construction zones, or areas disturbed as a result of construction. Control methods shall include the following: Apply calcium chloride at a uniform rate of one and one-half (1 ½) pounds per square yard in areas subject to blowing. For emergency control of dust apply water to affected areas. The source of supply and the method of application for water are the responsibility of the contractor.
- B. The frequency and methods of application for fugitive dust control shall be as directed by the contracting officer.

3.7 TEMPORARY PROTECTIVE COVERINGS (AFTER GROWING SEASON)

- A. Place temporary coverings for erosion and sedimentation control on all areas that have been graded and left exposed. Temporary stabilization is required under the base bid to prevent sediment or sand from leaving the project site, including paved roads and parking areas. Contractor shall use one or more of the methods described herein (mulch, mesh/blanket matting, tackifiers) as approved by the Contracting Officer.
- B. Hay or straw shall be anchored in-place by one of the following methods and as approved by the government: Mechanical “crimping” with a tractor drawn device specifically devised to cut mulch into top two inches of soil surface or application of non-petroleum based liquid tackifier, applied at a rate and in accordance with manufacturer’s instructions for specific mulch material utilized.
- C. Placement of mesh or blanket matting and anchoring in place shall be in accordance with manufacturer’s printed instructions.
- D. Inspect protective coverings periodically and reset or replace materials as required.

3.8 SILT SOCKS

- A. Silt socks shall be constructed and installed in accordance with the approved SWPPP/ESCP and any applicable Order of Conditions prior to the start of work.

END OF SECTION 312500

SECTION 315000 - EXCAVATION SUPPORT AND PROTECTION

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 DESCRIPTION OF WORK

- A. Work Included: Provide labor, materials and equipment necessary to complete the work of this Section, including but not limited to the following:
 - 1. Temporary excavation support and protection systems.
- B. Alternates: Not Applicable.
- C. Items To Be Installed Only: Not Applicable.
- D. Items To Be Furnished Only: Not Applicable.
- E. Related Work: The following items are not included in this Section and will be performed under the designated Sections:
 - 1. Section 015000 - TEMPORARY FACILITIES for temporary utilities and support facilities.

1.3 PERFORMANCE REQUIREMENTS

- A. Design, furnish, install, monitor, and maintain excavation support and protection system capable of supporting excavation sidewalls and of resisting soil and hydrostatic pressure and superimposed and construction loads.
 - 1. Provide professional engineering services needed to assume engineering responsibility, including preparation of Shop Drawings and a comprehensive engineering analysis by a qualified professional engineer.
 - 2. Prevent surface water from entering excavations by grading, dikes, or other means.
 - 3. Install excavation support and protection systems without damaging existing buildings, pavements, and other improvements adjacent to excavation.
 - 4. Provide vibration monitory to prevent impacts on adjacent structures and utilities.

1.4 SUBMITTALS

- A. Shop Drawings: Prepared by or under the supervision of a qualified professional engineer for excavation support and protection systems.
 - 1. Include Shop Drawings signed and sealed by the qualified professional engineer responsible for their preparation.
- B. Qualification Data: For Installer and professional engineer.
- C. Photographs or videotape, sufficiently detailed, of existing conditions of adjoining construction and site improvements that might be misconstrued as damage caused by the absence of, the installation of, or the performance of excavation support and protection systems.

1.5 QUALITY ASSURANCE

- A. Preinstallation Conference: Conduct conference at Project site to comply with requirements in Division 01.
 - 1. Review methods and procedures related to excavation support and protection system including, but not limited to, the following:
 - a. Existing utilities and subsurface conditions.
 - b. Proposed excavations.
 - c. Proposed equipment.
 - d. Monitoring of excavation support and protection system.
 - e. Working area location and stability.
 - f. Coordination with waterproofing.
 - g. Abandonment or removal of excavation support and protection system.

1.6 PROJECT CONDITIONS

- A. Interruption of Existing Utilities: Do not interrupt any utility serving facilities occupied facilities unless permitted under the following conditions and then only after arranging to provide temporary utility according to requirements indicated:
 - 1. Notify Engineer no fewer than two days in advance of proposed interruption of utility.
 - 2. Do not proceed with interruption of utility without Engineer's written permission.

- B. Project-Site Information: The opinions expressed in this report are those of geotechnical engineer and represent interpretations of subsoil conditions, tests, and results of analyses conducted by geotechnical engineer. Owner will not be responsible for interpretations or conclusions drawn from the data.
 - 1. Make additional test borings and conduct other exploratory operations necessary for excavation support and protection.
 - 2. The geotechnical report is referenced elsewhere in the Project Manual.
- C. Survey Work: Engage a qualified land surveyor or professional engineer to survey adjacent existing buildings, structures, and site improvements; establish exact elevations at fixed points to act as benchmarks. Clearly identify benchmarks and record existing elevations.
 - 1. During installation of excavation support and protection systems, regularly resurvey benchmarks, maintaining an accurate log of surveyed elevations and positions for comparison with original elevations and positions. Promptly notify Architect if changes in elevations or positions occur or if cracks, sags, or other damage is evident in adjacent construction.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. General: Provide materials that are either new or in serviceable condition.
- B. Structural Steel: ASTM A 36/A 36M, ASTM A 690/A 690M, or ASTM A 992/A 992M.
- C. Steel Sheet Piling: ASTM A 328/A 328M, ASTM A 572/A 572M, or ASTM A 690/A 690M; with continuous interlocks.
- D. Cast-in-Place Concrete: ACI 301, of compressive strength required for application.
- E. Reinforcing Bars: ASTM A 615/A 615M, Grade 60, deformed.
- F. Timber Piling: ASTM D 25, species listed in AWPAC3, pressure-treated in accordance with AWPAC3.
- G. Grout: Suitable for service, minimum 4,000 psi.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards that could develop during excavation support and protection system operations.
 - 1. Shore, support, and protect utilities encountered.
- B. Install excavation support and protection systems to ensure minimum interference with roads, streets, walks, and other adjacent occupied and used facilities.
 - 1. Do not close or obstruct streets, walks, or other adjacent occupied or used facilities without permission from Owner and authorities having jurisdiction. Provide alternate routes around closed or obstructed traffic ways if required by authorities having jurisdiction.
- C. Locate excavation support and protection systems clear of permanent construction so that forming and finishing of concrete surfaces are not impeded.
- D. Monitor excavation support and protection systems daily during excavation progress and for as long as excavation remains open. Promptly correct bulges, breakage, or other evidence of movement to ensure that excavation support and protection systems remain stable.
- E. Promptly repair damages to adjacent facilities caused by installing excavation support and protection systems.

3.2 SOLDIER BEAMS AND LAGGING

- A. Install steel soldier beams before starting excavation. Space soldier beams at regular intervals not to exceed allowable flexural strength of wood lagging. Accurately align exposed faces of flanges to vary not more than 2 inches from a horizontal line and not more than 1:120 out of vertical alignment.
- B. Install wales horizontally at spacings indicated on the approved shop drawings and secure to soldier beams.

3.3 SHEET PILING

- A. Before starting excavation, install one-piece sheet piling lengths and tightly interlock to form a continuous barrier. Limit vertical offset of adjacent sheet piling to 60 inches. Accurately align exposed faces of sheet piling to vary not more than 2 inches from a horizontal line and not more than 1:120 out of vertical alignment. Cut tops of sheet piling to uniform elevation at top of excavation.

3.4 TIEBACKS

- A. Tiebacks: Drill for, install, grout, and tension tiebacks into position. Test load-carrying capacity of each tieback and replace and retest deficient tiebacks.
 - 1. Test loading shall be observed by a qualified professional engineer responsible for design of excavation support and protection system.
 - 2. Maintain tiebacks in place until permanent construction is able to withstand lateral earth and hydrostatic pressures.
 - 3. Inspect tiebacks periodically to confirm anchors exhibit no movement.

3.5 BRACING

- A. Bracing: Locate bracing to clear columns, floor framing construction, and other permanent work. If necessary to move brace, install new bracing before removing original brace.
 - 1. Do not place bracing where it will be cast into or included in permanent concrete work, unless otherwise approved by Engineer.
 - 2. Install internal bracing, if required, to prevent spreading or distortion of braced frames.
 - 3. Maintain bracing until structural elements are supported by other bracing or until permanent construction is able to withstand lateral earth and hydrostatic pressures.

3.6 REMOVAL AND REPAIRS

- A. Remove excavation support and protection systems when construction has progressed sufficiently to support excavation and bear soil and hydrostatic pressures. Remove in stages to avoid disturbing underlying soils or damaging structures, pavements, facilities, and utilities.
 - 1. Remove excavation support and protection systems to a minimum depth of 48 inches below overlaying construction and abandon remainder.
 - 2. Repair or replace, as approved by Architect, adjacent work damaged or displaced by removing excavation support and protection systems.

END OF SECTION 315000

SECTION 329300 - PLANTS

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.
- B. Examine all other Sections of the Specifications for requirements which affect work of this Section, whether or not such work is specifically mentioned in this Section.
- C. Coordinate work with trades affecting, or affected by, work of this Section. Cooperate with such trades to assure the steady progress of all work under the Contract.

1.2 WORK INCLUDED

- A. Refer to the Drawings for the extent and details of the work.
- B. The work of this Section consists of all seeding and related work, as shown on the Drawings or required herein and includes, but is not limited to, the following:
 - 1. Planting trees, shrubs, and perennials.
 - 2. Maintenance and guarantee.

1.3 SUBMITTALS

- A. Materials List: Submit a complete list of all materials proposed for use in this work, demonstrating complete conformance with the requirements specified.
- B. Submit soil analysis results for review by the Landscape Architect.
- C. Submit a list of plant nurseries to the Landscape Architect for approval of selected plant material species.
- D. Submit a manufacturer's certificate of compliance for controlled release fertilizer issued by the manufacturer only, showing the quantity of material ordered and the specific job application.
- E. For all plant materials, submit a complete list showing each type, each source of materials, and the measured size of each.

- F. Submit a physical sample of the mulch to be used in all planting areas from the company the mulch will be purchased from.

1.4 QUALITY ASSURANCE

- A. All plant materials shall be true to name according to "Standardized Plant Names", published by the American Joint Committee on Horticulture Nomenclature, 1942 edition. Each plant or bundle shall be tagged with the name and size of the plants, in accordance with the standards of the American Association of Nurserymen (ANN). In all cases, botanical names shall take precedence over common names.
- B. Quality and size shall conform to the current edition of "Horticultural Standards", for a number one grade nursery stock, as adopted by the ANN.
- C. All plants and plant materials shall comply with all federal, state and local laws and regulations requiring inspection for plant disease and insect control.

1.5 PRODUCT HANDLING

- A. Delivery and Storage:
 - 1. Deliver all items to the job site in their original containers, with all labels intact and legible at time of the Landscape Architect's inspection.
 - 2. Immediately remove from the site all plants which do not comply with the specified requirements.
 - 3. Use all means necessary to protect plant materials before, during and after installation and to protect the work and materials of all other trades.
 - 4. Replacements: In the event of damage, immediately make all repairs and replacements necessary, to the approval of the Landscape Architect and at no additional cost to the Owner.

1.6 JOB CONDITIONS

- A. Utilities: Determine the location of underground utilities and perform work in a manner which will avoid possible damage. Hand excavate, as required. Maintain grade stakes set by others until removal is mutually agreed upon by all parties concerned.
- B. Excavation: When conditions detrimental to plant growth are encountered, such as rubble fill, adverse drainage conditions, or obstructions, notify the Landscape Architect before spreading topsoil.

- C. Planting Restrictions: Plant during one of the following periods. Coordinate planting periods with maintenance periods to provide required maintenance from date of Substantial Completion.
 - 1. Spring Planting: April 1 through June 1.
 - 2. Fall Planting: September 1 through November 1.
- D. Weather Limitations: Proceed with planting only when existing and forecasted weather conditions permit planting to be performed when beneficial and optimum results may be obtained. Apply products during favorable weather conditions according to manufacturer's written instructions and warranty requirements.

PART 2 - PRODUCTS

2.1 GENERAL

- A. Loam - friable, typical of local cultivated topsoil, containing 5% (min.) decayed organic matter (humus), no toxic materials, from well drained, arable site, reasonably free of subsoil, stones, earth, clods, sticks, roots or debris.
 - 1. Test for acidity, fertility, and general texture by a recognized commercial or government agency. Report findings and recommendations to the Landscape Architect. Add soil conditioners, as per the report findings and recommendations by the Landscape Architect.
 - 2. Deliver no topsoil in frozen or muddy condition.
- B. Superphosphate - finely ground phosphate rock, eighteen percent (18%) minimum available phosphoric acid.
- C. Bone meal - commercial raw bone meal, finely ground, 1% nitrogen and 18% phosphorus acid (min.).
- D. Manure - well rotted, unleached, cattle manure, reasonably free of wood shavings, sawdust, or other litter and no chemicals or other ingredients harmful to plants. Dehydrated manure (Bovung) is acceptable.
- E. Fertilizer - All plants will be fertilized with a controlled release 16-8-16 analysis fertilizer, contained in polyethylene perforated bags with micropore holes. The bag shall contain 4 ounces minimum of water soluble fertilizer, to be effective for 8 years. Pills, spikes, tablets and injections are not considered controlled released packets.
 - 1. Deliver as specified, in standard containers, showing weight, analysis and manufacturer. Store in a weather-proof place.

- F. Peat - domestic or imported, of partially decomposed vegetable matter of natural occurrence, brown, clean, low in content of mineral and woody material; mildly acid, granulated or shredded, free from weedy grasses, sedges or rush.
- G. Lime - ground, dolomite limestone, 95% passing through a 100 mesh screen.
- H. Planting mixture - topsoil thoroughly incorporated with well rotted manure or equivalent dehydrated manure or bone meal and peat, proportioned 1 c.y. to 7 c.y. topsoil.
- I. Mulch - aged pine bark, consisting of the outer bark of pine trees, with minimum hardwood bark. Bark shall be thoroughly mixed and aged in stock piles a minimum of 6 months, partially decomposed, dark brown in color, and generally free of chunks of wood thicker than 1/4". Aged pine bark containing an excess of fine particles will not be acceptable.
- J. Water - The Contractor will furnish hose, hose connections and other implements required for watering all plant materials, until the completion of the project.

2.2 PLANT MATERIALS

- A. Furnish all plant materials consistent with the kinds of sizes indicated in these Specifications.
- B. All plants are to be nursery grown, unless authorized to be collected from other sources.
- C. Plants - in accordance with USDA Standard for Nursery Stock, Latest Edition, hardy under climatic conditions similar to the locality of the project, typical of species or variety, normal habit of growth, sound, healthy, vigorous, well-branched, densely foliated when in leaf, free of disease, insect pests, eggs or larvae, with well developed root systems.
- D. If plants of a specified kind or size are not available, substitutions may be made upon request and if approved by Landscape Architect.
- E. Plant Dimensions - conform to USDA Standard for Nursery Stock, Latest Edition, as specified. Exceptions as follows:
 - 1. Plants larger than specified may be used, if approved by the Landscape Architect and at no increase in the contract price. Increase spread of roots or earth ball in proportion to the size of the plant.
 - 2. Undersized plants (10% max.) in any one variety of grade may be used, if approved by the Landscape Architect. Provide sufficient plants above size to make average equal to or above specified grade. Undersized plants shall be larger than average size of next smaller grade.

- F. Balled and Burlapped (B&B) Plants - dig with firm natural earth roots. Made balls are unacceptable.
- G. Container Grown Plants - grow in container long enough for root system to have developed sufficiently to hold its soil together firm and whole. Plants loose in containers will not be acceptable.
- H. Protect B&B plants not planted immediately upon delivery with soil, wet moss or other acceptable material. Prevent voids among roots with careful filling. Bind no plants with wire or rope so as to not damage bark or break branches.
- I. Plants are subject to inspection and approval at the place of growth for conformity to Specifications as to quality, size and variety. The cost of expenses incurred by the Landscape Architect for such inspections shall be borne by the Contractor. The Landscape Architect reserves right of inspection upon delivery at the site or during progress of work or right of rejection due to damage suffered in handling or transportation. Remove defective plants immediately from site. Plants to be accompanied by a State Nursery inspection certification, if required.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Examine the areas and conditions under which the work of this Section will be performed. Correct conditions detrimental to the proper and timely completion of the work. Do not proceed until unsatisfactory conditions have been corrected.

3.2 PLANT OPERATIONS

- A. Plant nursery stock immediately upon delivery to the site and approval of the Landscape Architect. Layout individual tree and shrub locations and areas for multiple plantings. Stake locations, outline plant beds and obtain the Landscape Architect's approval before proceeding with planting work.
- B. Planting may be done whenever weather and soil conditions are favorable or as otherwise authorized by the Landscape Architect. If this is not feasible, heel-in material with damp soil or mulch to protect it from sun and wind.
- C. Notify the Landscape Architect at least one week prior to beginning planting operations. Stake locations of all plant material with surveyors bean poles 48 hours prior to planting for approval by the Landscape Architect.
- D. Excavate tree pits as show on the plans.

- E. Locate the pits prepared and backfilled with planting mixture to grade, prior to planting by staking and recording on plants for location when planting proceeds.
- F. Set plants in the center of the pits, plumb and straight, with the crown of the plants 1" higher, after settlement, than the surrounding finished grade.
- G. When B&B trees are set, cut and remove the entire burlap and strings from around the root ball and compact the topsoil mixture around the bases of the balls to fill all voids.
- H. Container grown plants with well developed root systems shall be roots cut, vertically along the sides of the root ball, with a utility knife to encourage outward root growth.
- I. Thoroughly compact planting mixture around the roots or balls and water immediately after plant pit is backfilled. Form a shallow basin, slightly larger than the pit, with a ridge of sod to facilitate and contain watering. Cultivate the soil in the shrub beds, rake smooth, and neatly outline after planting. Provide a 12" (min.) of loam between all ground covers.
- J. Distribute controlled release fertilizer packets within the planting pit, adjacent to the root ball, but not in direct contact with the roots. Placement depth shall be 6 to 8 inches. Packets shall not be cut, ripped or damaged.

1. Application rates as follows:

Plant Item	Size	No. of Packets
Deciduous Trees:	2 ½" – 3 ½" Caliper	3
	4-6" Caliper	4
Shrubs:	2-3'	2
	Over 3'	3
Evergreen Trees:	6-10'	4
	Over 10'	5
Groundcover:	1 packet per four plants	

- K. Prune each plant in accordance with ANN Standards to preserve natural character. Remove all dead wood, suckers and/or broken or badly bruised branches. Prune with clean, sharp tools. Paint cuts over 1" in diameter with approved tree paint; cover all exposed cambium and living tissue.
- L. Cover all tree and shrub pits immediately after planting with a 3" (min.) layer of specified mulch. Limit of mulch for trees shall be the area of the pit; for shrubs in beds, the entire area of the shrub bed.
- M. If rocks of underground obstructions are encountered in plant pit excavation, remove the obstructions to 3' (min.) below grade and 12" (min.) below the bottom of the ball or the roots.

- N. The Contractor shall furnish plans showing locations of underground utilities, as required.
- O. Remove all tape, string, rope and tags that are not part of the natural plant.

3.3 PLANT MAINTENANCE

A. Planting Maintenance

1. Maintenance shall begin immediately after each plant is planted and shall continue, for a minimum of 90 days, following the completion of all planting installations and until the final acceptance of all planting work.
2. Maintenance shall consist of keeping the plants in a healthy, growing condition and shall include, but is not limited to, watering, weeding, cultivating, re-mulching, tightening and repairing of guys, removal of dead material, resetting plants to proper grades or upright position, and maintaining the planting saucer.
 - a. Plants shall be inspected for watering needs at least twice each week and watered, as necessary, to promote plant growth and vitality.
 - b. Stakes shall be kept plumb and neat in appearance. Guys, wires, and anchoring cables shall be tightened and repaired weekly.
 - c. Planting beds and individual plant pits shall be kept free of weeds, and mulch shall be replaced, as required, to maintain the specified layer of mulch. Beds and individual pits shall be neat in appearance and maintained to the designed layout.
 - d. Plants that die during the maintenance period shall be removed and replaced at once, unless designated otherwise by the Landscape Architect.
 - e. Spraying for both insect pests and diseases shall be included during the maintenance period, as required and as directed.
3. During the maintenance period, any decline in the condition of plantings shall require the Contractor to take immediate action to identify potential problems and undertake corrective measures. If required, the Contractor shall engage professional arborists and/or horticulturalists to inspect the plant materials and to identify problems and recommend corrective procedures.

3.4 ACCEPTANCE STANDARDS FOR PLANTING

- A. Following the minimum required maintenance periods of planting, the Contractor shall request the Landscape Architect, in writing, for a formal inspection of the completed work. If the plant materials and workmanship for the site are acceptable, written notice will be given to the Contractor stating that the work has received acceptance and that the guarantee period has commenced from the date of acceptance.

- B. If a number of plants are sickly or dead at the time of inspection, or if, in the Landscape Architect's opinion, the workmanship is unacceptable, written notice will be given by the Landscape Architect to the Contractor in the form of a punch list, which itemizes necessary planting replacements and/or other deficiencies to be remedied. The Contractor's responsibility for maintenance of all of the plants shall be extended until replacements are made or other deficiencies are corrected. All dead and unsatisfactory plants shall be removed promptly from the project. Replacements shall conform in all respects to the Specifications for new plants and shall be planted in the same manner.

3.5 GUARANTEE FOR PLANT MATERIALS

- A. Plants shall be guaranteed for a period of one (1) year after written notification of acceptance and shall be alive and in satisfactory growth at the end of the guarantee period.
- B. At the end of the guarantee period, a final inspection will be held to determine whether any plant material replacements are required. Each plant shall show at least 75% healthy growth and shall have the natural character of its species, as determined by the Landscape Architect. Plants found unacceptable shall be removed promptly from the site and replaced during the normal planting season, until the plants live through one year. A final inspection for acceptance will be made after the replacement plantings have lived through one (1) year.

END OF SECTION 339300

SECTION 333000 - SEWERAGE SYSTEM

PART 1 - GENERAL

1.1 GENERAL PROVISIONS

- A. Attention is directed to all Sections within DIVISION 01 – GENERAL REQUIREMENTS, which are hereby made a part of this Section of the Specifications.

1.2 GENERAL SCOPE OF WORK

- A. This Section specifies requirements for the sewerage system.
- B. Definition(s): Regulatory Contractor retained Engineer/Sanitarian – licensed PE or Sanitarian per MassDEP permit. Project Engineer – Contractor’s engineer responsible for coordination and submittals.
- C. The work includes furnishing and installing gravity sewer pipe, sewer pump stations including precast tanks, pumps, floats, isolation valves, sewer force main pipe, and other structures and appurtenances required for the proposed system to convey sewerage. The work also includes installing new septic tanks and leaching field for on-site wastewater disposal. There are two pump stations in the project. Pump Station 2 is an existing pump station that services the comfort center and is denoted a CLIN 0002, Option 1. The work associated with Pump Station 2 includes replacement of the existing pumps with chopper pumps, replacing the existing pump station accessories, including floats, rails, check valves, and ladders and providing new electrical service. The existing structure for Pump Station 2 is to remain. Pump Station 1 is a newly proposed Pump Station. Pump Station 1 feeds two distribution boxes, that convey sewage flow to the proposed leaching field and is provided under CLIN 0001, base bid. CLIN 0007, Optional Exterior Force Main Valve Work, an optional contract line item (CLIN 0007), is included for exterior excavation at the Pump Station 2 force main for the purpose of locating, removing, and replacing an existing check valve and valve box, if directed by the contracting officer. This work will only be performed if the option is exercised. The contractor shall not include this work in the Base Bid or Option 1.
- D. The contractor shall provide soils and groundwater information to all manhole and tank manufactures prior to fabrication.
- E. Existing System Condition and Contractor Responsibility
 1. The existing Pump Station 2 system is known to be non-functional. The inability of the system to convey flow may be due to blockage, valve failure, obstruction, or other unknown conditions within the existing force main or appurtenances.
 2. The Contractor is advised that replacement of pumps and controls alone may not restore system functionality.
 3. The Contractor shall be responsible for verifying that the complete system, including Pump Station 2, force main, and downstream components, is fully operational at the completion of the Work.

4. Claims for additional compensation or time extensions based on the condition, obstruction, or performance of the existing force main, check valves, or downstream piping will not be considered, as these conditions are reasonably inferable from the Contract Documents.

5. The Contractor shall include in their bid all labor, equipment, and coordination necessary to achieve a fully functioning system, regardless of the condition of the existing infrastructure, except where work is explicitly identified as an optional CLIN.

6. Optional CLINs for force main cleaning (CLIN 0005) and CCTV inspection (CLIN 0006) are provided for the Government's use at its sole discretion and shall not relieve the Contractor of responsibility to deliver a functional system under the Base Bid and Option 1 if exercised.

7. Work associated with the exterior force main excavation and check valve replacement is specifically excluded from the Base Bid and Option 1 and is covered under CLIN 0007. The contractor shall have no responsibility to perform this work unless directed by the Contracting Officer.

1.3 RELATED SECTIONS

- A. Division 01 – General Requirements: All provisions of Division 01 apply to this section.
- B. Section 31 20 00 – Earth Moving: Excavation, bedding, backfill, and compaction for sewer piping and structures.
- C. Division 26 – Electrical: General electrical standards including wiring methods, grounding, bonding, identification, conduit, and general installation requirements. Unless otherwise specified in this section, all electrical work shall comply with applicable Division 26 requirements and the National Electrical Code (NEC).
- D. Control Panels, Motor Protection, and Generator Connections: Requirements for pump control panels, starters, overloads, alarm wiring, and temporary or permanent generator connections are specified in this section (333000). Div 26 does not provide motor protection or transfer switch requirements; the contractor shall comply with NEC Articles 445 and 702 for generator connections.
- E. All electrical work shall comply with National Electrical Code (NEC), National Electrical Manufacturers Association (NEMA) standards, Underwriters Laboratories (UL) listings, and applicable ANSI standards as referenced in this Section.
- F. Coordination between the pump control panels, motor protection, and generator connection shall be in accordance with manufacturer's instructions, this Section, and all related electrical standards listed above.

1.4 SUBMITTALS

- A. List of all materials proposed.

- B. Product information and Shop drawings for all material and structures prior to ordering materials.
- C. Manufacturer's warranty certificate for all pumps, alarm panels, and accessories.
- D. Manufacturer's one (1) year minimum warranty for all tanks to include water tightness and structural stability
- E. Shop drawings and wiring diagrams for emergency generator connection, including transfer switch and placard per paragraph 2.7.F.9.
- F. Shop Drawings for check valve and valve box associated with CLIN 0007, if exercised.

1.5 INSPECTION

- A. Inspection of the pipe may be made after delivery. The pipe will be subject to rejection at any time on account of failure to meet any of the specification requirements, even though pipe samples may have been accepted as satisfactory at the place of manufacture. Pipe rejected after delivery will be marked for identification and will be removed from the site at once.
- B. Imperfections in materials may be repaired, subject to approval of the Contracting Officer, after demonstration by the manufacturer that strong and permanent repairs result. Repairs will be carefully inspected before final approval.

1.6 DELIVERY, STORAGE & HANDLING

- A. All materials will be adequately protected from damage during transit. Pipes will not be dropped.
- B. All pipe and other appurtenances shall be inspected before placement in the work and any found to be defective from any cause, including damage caused by handling, and determined to be un-repairable, will be replaced at no cost to the government.
- C. Storage and handling of pipes, manholes and other sewer system appurtenances will be in accordance with the manufacturer's recommendations, subject to the approval of the contracting officer.

1.7 SEQUENCING AND DIAGNOSTIC REQUIREMENTS

A. Prior to installation of new pumps at Pump Station 2, the Contractor shall:

1. Verify that the discharge piping and force main are not obstructed to the extent that pump operation would be impaired.
2. Notify the CO/COR of any observed blockage, restriction, or abnormal condition.

B. The Contractor shall not assume that the Government will exercise CLIN 0005 (jetting) or CLIN 0006 (CCTV inspection).

C. If the Contractor determines that cleaning or inspection is necessary to achieve system functionality, the Contractor shall submit a written recommendation to the CO/COR.

D. No additional compensation shall be provided for delays or inefficiencies resulting from failure to evaluate existing system conditions prior to pump installation.

E. Installation of new pumps without verifying discharge conditions shall be at the Contractor's risk.

1.8 Existing Sewer System

- A. The Contractor is expected to make reasonable assumptions regarding the condition of existing sewer infrastructure, including the potential for partial or complete blockage, based on the age, performance, and known non-functionality of the system.

Such conditions shall not be considered unusual or materially different from those ordinarily encountered in similar work.

PART 2 - MATERIALS

2.1 POLYVINYL CHLORIDE PIPE (PVC) GRAVITY PIPE

- A. Pipe and Fittings: Polyvinyl chloride gravity pipe and fitting shall be Type PSM polyvinyl chloride (PVC) SDR 35, with full diameter dimensions conforming to the specifications for Type PSM Polyvinyl Chloride (PVC) Sewer Pipe and Fittings, ASTM Designation D3034.
- B. Joints: PVC pipe shall have an integral wall bell and spigot push-on joint with elastomeric gaskets secured in place in the bell of the pipe. The bell shall consist of an integral wall section, with a solid cross section elastomeric gasket, factory assembled, securely locked in place to prevent displacement during assembly. Elastomeric gaskets shall conform to ASTM D3212.

2.2 HIGH DENSITY POLYETHYLENE (HDPE) PRESSURE FORCE MAIN

- A. Pipe and Fittings: HDPE pressure pipe and fitting shall conform to ASTM D3035 for 160 psi pipe (SDR 11).
- B. Joints: The pipe shall be joined with butt, heat fusion joints as outlined in ASTM F2620. All joints shall be made in strict compliance with the manufacturer's recommendations. A factory qualified joining technician as designated by pipe manufacturer or experienced, trained technician shall perform all heat fusion joints.

2.3 DUCTILE IRON PIPE (FOR WET WELL AND VALVE PIT PIPING)

- A. Flanged Joint Pipe: Ductile iron pipe shall be flanged joint pipe, made in accordance ANSI A21.15/AWWA C115 and manufactured in accordance with ANSI A21.51/AWWA C151. Ductile iron pipe shall be Thickness Class 53.
- B. Lining: Pipes shall be cement-mortar lined, in accordance with ANSI A21.4/AWWA C104, except that the cement lining shall be double thickness.
- C. Coating: The exterior of all pipes shall be factory coated, with a double coat of asphaltic material conforming to ANSI A21.51/AWWA C151. The interior of all pipes shall have a seal coat of asphaltic material applied over the cement lining, in accordance with ANSI A21.4/AWWA C104.
- D. Fittings: Fittings shall be ductile or cast-iron conforming to ANSI Specification A21.10, 150 pounds per square inch minimum pressure rating.

2.4 CHECK VALVES (FORCE MAIN)

- A. Check valves shall be full port, flanged, swing check type designed for wastewater service.
- B. Valve body shall be ductile iron conforming to ASTM A536.
- C. Valve shall be rated for minimum 150psi working pressure.
- D. Valve seat shall be resilient (EDPM or Buna-N) suitable for sewerage service.
- E. Valve shall be non-clogging type suitable for solids handling wastewater applications.
- F. Provide external lever and weight or spring-assisted closure where recommended by the manufacturer.
- G. Acceptable Manufacturers: Val-Matic, Cla-Vel, APCO, or approved equal.

2.5 VALVE BOX WITH GRADE ACCESS

- A. Valve box shall be cast iron, adjustable, suitable for H-20 loading.
- B. Provide locking or blotted cover labeled "SEWER".
- C. Valve box shall be sized to allow full operation and maintenance of valve without removal of the surrounding structures.
- D. Provide extension sections as required to match finished grade.
- E. Interior clearance shall be sufficient for removal of valve components.
- F. Provide corrosion resistance coatings.

2.6 LEACHING SYSTEM

- A. Leaching system units shall consist of a 12-inch diameter, corrugated, perforated HDPE pipe with internal skimmer tabs. Pipe sections shall be 10 feet in length. Leaching system units shall be wrapped in fiber mat and geo-textile fabric. Basis of design is Presby Enviro-Septic or approved equal that meets the same hydraulic and treatment capacity.
- B. Leaching system units shall be surrounded by a bed of "System Sand" which facilitates wastewater treatment through a passive biological process. System Sand specification provided on detail sheet CU504.
- C. Couplings and fittings shall snap-lock and have offset adapters to connect pipe sections.
- D. Leaching field shall be vented as shown on the plans.

2.7 MANHOLES AND STRUCTURES

- A. Precast Units:
 - 1. Structure: Four (4) foot minimum inside diameter precast units (4,000 psi minimum compressive strength), with eccentric cone section tapering to twenty-four (24) inch diameter (minimum), or flat top, and one pour monolithic base section conforming to ASTM C478. All

- units shall be designed for HS-20 loading. Shall be sealed with two (2) coats of bituminous sealer
2. Precast Unit Joint Seals: Butyl rubber O-ring type seal conforming to ASTM C990 to be water tight.
 3. The date of manufacture, trademark and the name of the manufacturer shall be clearly marked on the inside of each precast section.
- B. Manhole Frame and Cover: Grey iron casting conforming to ASTM A48, heavy duty, with the word "SEWER" embossed on cover. Letter size shall be three inches. Any structure installed within a wetland resource area or at low points of topography where water intrusion could occur should have a water tight gasketed cover.
- C. Pipe Connections: Flexible sleeve or rubber gaskets cast into structure
- D. Bitumastic Coating: The entire exterior surface of all manholes, tanks, and chambers shall be coated with two coats of an approved bitumastic material to produce a dry film thickness of 0.07 inches (7 mils) per coat.
- E. Rungs: Manhole rungs shall be steel reinforced copolymer polypropylene plastic. Rungs shall be 14 inches wide. Copolymer polypropylene shall be Type II, Grade 16906, meeting ASTM specifications D2146. Steel reinforcing shall be 3/8-inch diameter, Grade 60, conforming to ASTM Specification A615 and shall be continuous throughout the rung. The portion of the legs to be embedded in the precast section shall have fins and be tapered to ensure a secure bond.
- F. Distribution Box: Shall be made of reinforced concrete. Sizing, inlets, and outlets as shown on plans.

2.8 LIFT STATION WET WELL AND VALVE PIT

- A. The wet well and valve pit structures shall conform to the latest ASTM Specification C478. Concrete shall have a minimum compressive strength of 5,000 pounds per square inch at 28 days. Steel reinforcing shall be in accordance with ASTM A615-75, Grade 60, 2-inch minimum cover. The structure shall be designed to withstand vertical and horizontal soil loads. Internal bracing may be used if it does not prevent or prohibit internal flow of sewerage. The structure shall have a minimum wall thickness of 6 inches and designed for H-20 Loading. The structure shall be cast in a maximum of two (2) pieces comprising of a base section and top section. Tunnel tanks will not be accepted. The shop drawing for the wet well shall be stamped by a Professional Structural Engineer licensed in Massachusetts. Pump Station 1 shall be vented, which shall be fitted with a charcoal filter.
- B. Wet well and valve pit shall be designed to be non-buoyant. The contractor shall provide buoyancy calculations stamped by a Massachusetts licensed professional engineer with shop drawings. Contractor shall provide adequate tie-downs, thicker structure floor/walls or foundation to prevent buoyancy as necessary.
- C. Precast units shall be designed for the following loads and possible combinations thereof:
1. H-20 loading, manhole riser with frame and cones, plus the weight of soil above.
 2. Weight of precast concrete structure.
 3. Initial handling and erection loadings.

- D. Joints between concrete structures shall be sealed with Kent Seal No. 2 butyl rubber joint sealant conforming to ASTM C990 or neoprene gaskets designed by manufacturer to be watertight, installed in accordance with manufacturer's instructions. All joints and openings are to be watertight.
- E. Wet well and valve pit are to be provided with co-polymer polypropylene steel reinforced manhole steps, conforming to ASTM 2146-68. Steps shall be installed 12 inches on center.
- F. Access
 - 1. Hatches
 - a. Exterior access hatches shall be of heavy-duty aluminum construction. Unit shall be supplied with hinged safety grate to provide protection against fall through. Hatch frame shall be designed to minimize water entering the wet well and valve vault. Angle type frames will not be accepted. Leaves shall be made from diamond patterned, aluminum plates and designed to withstand H2O loading. Leaves shall be equipped with heavy-duty, corrosion resistant hinges; spring, piston or torsion bar operators for easy operation; and an automatic hold open arm with release handle. Aluminum shall have a mill finish with bituminous coating applied to the exterior of the frame. All hardware shall be stainless steel. For security, the access hatches shall be equipped with a slam lock, removable key wrench and removable plug and a staple for a padlock. The wet well hatch frame shall be equipped to accept pump guides, cable holder and all other components required for a submersible lift rail pumping station.
 - b. Access hatches shall be cast into the wet well and valve vault covers and shall be installed in accordance with the manufacturer's instructions.
 - c. The access hatch dimensions shall be as shown on the plans.
 - d. Hatches shall be as manufactured by East Jordan Iron Works, Bilco, Campbell Foundry, US Foundry and Manufacturing, or equal.
- G. Manhole riser section, manhole steps, frames and covers shall be as specified elsewhere in Part 2 of this Specification.
- H. Alarm floats shall be provided within the wet well. Floats shall be Conery 2900 non-mercury series or approved equal. Floats shall be installed as shown on the contract drawings.

2.9 PUMPS & RELATED EQUIPMENT

- A. Sewer ejector pumps shall be Barnes Sithe or approved equal, shall be Explosion-Proof, shall be Solids Handling Wastewater pumps with a chopper blade, with a three-phase 208V.
 - 1. Pump Station 1 (newly proposed pump station). Each of the two pumps shall be capable of pumping wastewater at 200 gallons per minute against 20 feet of total dynamic head through a 4" forcemain.
 - 2. Pump Station 2 (existing pump station). Each of the two pumps shall be capable of pumping wastewater at 200 gallons per minute against 30 feet of total dynamic head through a 4" forcemain.
- B. Pump floats shall be installed per the Sewer Pump Station Detail. The "Lead pump on" float will start the discharge cycle which will run until the "Pump off" float is activated. The "Alarm Float" will sound the visible and audible alarm if activated. The "Lag Pump On Float" will activate the lag pump.

- C. Pumps shall have recessed cast iron impellers and serrated wear plate for shredding of solids or other method, approved by the contracting, which will prevent clogging.
- D. Two (2) pumps shall be supplied for both Pump Station 1 and Pump Station 2; pumps shall be on alternating cycles.
- E. Pumps shall be installed on stainless steel guide rails per manufacturer's specification.
- F. Control panels shall be supplied per pump manufacturer's specifications. At minimum the following standards shall be met:
 - 1. All controls shall be intrinsically safe.
 - 2. All control panels, motor starter enclosures, alarm panels, receptacle panels, and manual transfer equipment located in areas accessible to the public shall be installed in lockable, tamper-resistant NEMA 4X enclosures. Each enclosure shall include factory-provided padlock provisions. Locking hardware shall prevent unauthorized access to energized electrical components and comply with NEC, UL, NEMA, and ANSI requirements for equipment security.
 - 3. Control panel enclosures shall be stainless steel and shall have a NEMA 4X rating. The panel shall include a pump circuit breaker, control circuit breaker, alarm circuit fuse, control circuit fuse, I.E.C. rated motor starter with 3-pole ambient compensated bimetal overload relay, pump hand-off-auto switch (momentary in the hand position), alarm test switch, pump run lights, seal leak lights, alternator relay (solid state), override relay, control transformer with primary fusing, terminal blocks, ground lugs, auxiliary contacts for alarms, and all necessary wiring and brackets. Control panel shall conform to pump manufacturer specifications and shall be adequate for indoor and outdoor use.
 - 4. Stainless steel enclosure shall be of adequate size to accommodate interlocking breakers and individual breakers for each pump and control panel.
 - 5. The control panel shall be fitted with an amber colored polycarbonate alarm light. The light shall be approximately 3" high by 3-1/2" diameter. The bulb shall be 40 watt minimum. The alarm shall have a bright glow and shall flash during high water conditions. The alarm shall turn off when high water level drops.
 - 6. The control panel shall be fitted with an audible horn alarm that sounds during high water conditions. The horn shall turn off when high water level drops. A silence switch shall be provided for the audible alarm.
 - 7. The contractor shall coordinate mounting the control panels and powering the pumps. Coordination includes verification of mounting location, electrical supply, and final integration testing.
 - 8. Manual Transfer Equipment and Generator Connection:
 - a. Provide a NEMA 4X-rated receptacle for connection to a temporary generator. Receptacle voltage shall be 208V, 3-phase; the contractor shall coordinate required amperage with the sum of full-load amps (FLA) of all pumps and control equipment.
 - b. Provide a manual transfer switch (MTS) or load-break interlock rated for transfer under load, NEMA 4X, UL-listed, mechanically interlocked, and conforming to ANSI standards. The MTS shall:
 - 1). Be sized for the maximum combined full-load current of all connected equipment and coordinated with upstream protective devices.

- 2). Include verified short-circuit current rating (SCCR) and full-load rating, with documentation demonstrating compliance with UL 1008, NEC Articles 445 and 702, NEMA, and ANSI standards.
 - 3). Include a lighted jewel or visual indicator to show switch position (“Utility” / “Generator”) and power availability.
 - 4). Include durable engraved labels or placards stating: *“WARNING – Generator connection. Normal power must be disconnected before generator use. Back feed hazard.”*
 - 5). Prevent simultaneous connection of utility and generator sources under all operating conditions.
 - 6). Be suitable for outdoor installation, with environmental rating for temperature, UV exposure, and wet conditions.
 - 7). Be demonstrated to operate properly during system testing, including load testing under normal and emergency conditions.
 - c. Motor protection, transfer switch operation, and generator connections shall be coordinated with pump manufacturer recommendations and comply with UL, NEMA, ANSI, and NEC requirements.
 - d. Shop drawings, wiring diagrams, and calculations demonstrating compliance with all applicable standards shall be submitted to the Contracting Officer for review and approval prior to procurement and installation.
 9. Provide an elapsed time meter for each pump to record actual running time (maximum reading 99,999.99 hours).
 10. Provide an auto-dialer capable of programming and calling at least three phone numbers to report high water or alarm conditions.
- G. The contractor shall provide an elapsed time meter for each pump installed. Meter shall record actual running time of each pump. The meter shall have a maximum reading of 99,999.99 hours.
- H. The contractor shall provide an auto dialer capable of programing and calling at least three (3) phone numbers.

PART 3 - EXECUTION

3.1 EXCAVATION AND BACKFILLING

- A. The type of materials to be used in bedding and backfilling and the method of placement shall conform to the requirements of Section 31200, Earth Moving, and details shown on the Drawings.

3.2 PIPE INSTALLATION

- A. All sewer piping shall be laid accurately to the lines and grades shown in the Drawings and in conformance with pipe manufacturer's recommended procedures.
- B. Laying Pipe:
 1. Each length of pipe shall be laid with firm, full and even bearing, throughout its entire length, in a prepared trench. Pipe shall be laid with bells upgrated, unless otherwise approved by the Project Engineer and the Regulatory Engineer/Sanitarian. Every length of pipe shall be inspected and cleaned of all dirt and debris before being laid. The interior of the

pipe and the jointing seal shall be free from sand, dirt and trash. Extreme care shall be taken to keep the bells of the pipe free from dirt and rocks so that joints may be properly lubricated and assembled.

2. No length of pipe shall be laid until the proceeding lengths of pipe have been thoroughly embedded in place, to prevent movement or disturbance of the pipe alignment.
- C. Pipe Extension: Where an existing pipe is to be extended, the same type of pipe shall be used, unless otherwise approved by the Project Engineer in coordination with the Regulatory Engineer/Sanitarian.
- D. Full Lengths of Pipe: Only full lengths of pipe shall be used in the installation, with the exception that partial lengths of pipe may be used at the entrance to structures and to accommodate the required locations of service connection fittings.
- E. Pipe Entrances to Structures: All pipes entering structures shall be cut flush with the inside face of the structure, and the cut ends of the pipe surface within the structure shall be properly rounded and finished so that there will be no protrusion, ragged edges or imperfections that will impede or affect the hydraulic characteristics or the sewage flow. The method of cutting and finishing shall be subject to the approval of the Project Engineer.
- F. Protection During Construction: The Contractor shall protect the installation at all times during construction and the movement of construction equipment, vehicles and loads over and adjacent to any pipe shall be performed at the Contractor's risk and in accordance with all applicable federal, state and local safety regulations.
1. At all times when pipe laying is not in progress, all open ends of pipes shall be closed by approved, temporary, water-tight plugs. If water is in the trench when work is resumed, the plug shall not be removed until the trench has been properly dewatered and all danger of water entering the pipe eliminated. The Contractor is responsible for proper dewatering to ensure a stable pipe foundation. Proper dewatering is to one or two (2) feet (minimum) below the pipe invert to ensure joining of the pipe in a dry condition.
- G. Water Pipe -Sewer Pipe Separation: When a sewer pipe crosses above or below a water pipe, the following procedures shall be utilized. The Contractor shall comply with these following procedures.
1. Relation to Water Mains
 - a. Horizontal Separation: Whenever possible, sewers shall be laid at a minimum of ten (10) feet, horizontally, from any existing or proposed water main.
 - b. Vertical Separation: Whenever sewers must cross under water mains, the sewer must be at least eighteen (18) inches below the bottom of the water main, measured from the invert of the water main to the crown of the sewer line. When the elevation of the sewer cannot be varied to meet the above requirements, the water main shall be relocated to provide this separation or reconstructed with mechanical-joint pipe for a distance of ten (10) feet on each side of the sewer. One full length of water main should be centered over the sewer so that both joints will be as far from the sewer as possible.
 - c. When it is impossible to obtain horizontal and/or vertical separation as stipulated above, both the water main and sewer shall be constructed of mechanical-jointing cement lined ductile iron pipe or other equivalent, based on water tightness a structural soundness. Both pipes shall be pressure tested by an approved method to achieve water tightness and both pipes shall be encased in concrete.

3.3 LAYING DUCTILE IRON PIPE AND FITTINGS

- A. Ductile iron pipe fittings shall be installed in accordance with the requirements of ANSI/AWWA C600.
- B. Each length of pipe shall be laid with firm, full and even bearing throughout its entire length, in a trench prepared and maintained in accordance with Section 312000, Earthwork. The type of materials to be used in bedding and backfilling and method of placement shall conform to the requirements of Section 312000.
- C. All pipes shall be clean before laying. When laying is stopped for any reason, the open ends of the pipe shall be closed by watertight plugs or other approved means. If water is in the trench when work is resumed, the plug shall not be removed until the trench has been properly dewatered and all danger of water entering the pipe has been eliminated.
 - 1. Fittings, in addition to those shown on the Drawings, shall be provided, if required, to avoid utility conflicts.
- D. When cutting of pipe is required, the cutting shall be done by machine, leaving a smooth cut at right angles to the axis of the pipe. Cut ends of pipe to be used with a push-on bell shall be beveled to conform to the manufactured spigot end. Cement lining shall be undamaged.
- E. Maximum allowable deflection for pipe laid without fittings shall not exceed the allowable amount established by the pipe manufacturer and shall not exceed those shown in AWWA C600.
- F. The pipe shall be laid to the lines and grades as shown on the Drawings.

3.4 PVC PIPE JOINTS (Gravity Pipe)

- A. All joints shall be made watertight and shall be push on rubber gasket connections. Glued Joints will not be accepted.
- B. Pipe shall be jointed in strict accordance with the pipe manufacturer's instruction. Jointing of all pipes shall be done entirely in the trench.
- C. PVC Pipe
 - 1. Lubricant for jointing of PVC pipe shall be applied as specified by the pipe manufacturer. Use only lubricant supplied by the pipe manufacturer.
 - 2. PVC pipe shall be pushed home by hand or with use of a bar and block. The use of power equipment, such as a backhoe bucket, shall only be used at the direction of the manufacturer.
 - 3. Field-cut pipe ends shall be cut square and the pipe surface beveled to the size and shape of a factory-finished beveled end. All sharp edges shall be rounded off.

3.5 MANHOLES

- A. General Requirements: All manholes shall be built in accordance with the Details and in the locations shown on the Drawings.
 - 1. Structures shall be constructed of brick masonry, precast solid concrete block, cast-in-place concrete or precast concrete.

2. Personnel experienced and skilled in this work shall install all masonry. Any person not deemed to be such, by the Contracting Officer, shall be removed and replaced by a person so qualified.
 3. Manholes shall be constructed as soon as the pipe laying reaches the location of the manhole.
 4. The Contractor shall accurately locate each manhole and set accurate templates to conform to the required line and grade. Any manhole that is incorrectly located or oriented improperly shall be removed and rebuilt in its proper location, alignment and orientation, at no additional cost to the government.
- B. Foundations: All manholes shall be constructed on an eight (8) inch layer of compacted bedding material. The excavation shall be dewatered to provide a dry condition while placing bedding material and setting the base.
- C. Inverts: Brick invert channels shall be constructed in all manholes to provide a smooth channel for sewage flow through the structure and shall correspond in shape to the lower half of the pipe and extend vertically to the pipe crown. At changes in directions, the inverts shall be laid out in curves of the longest possible radii tangent to the centerline of the sewer pipes at the manhole side. Shelves shall be constructed to the elevation of the highest pipe crown and sloped to drain toward the flow channel at one inch per foot (1 "/foot).
1. Special care shall be taken in laying brick inverts. Joints shall not exceed three sixteenths of an inch in thickness and each brick shall be carefully laid in full cement mortar joint on bottom, side and end in one operation. No grouting or working in of mortar after laying of the brick will be permitted. Bricks forming the shaped inverts in manholes shall be laid on edge.
 2. Invert channels shall be built for future extensions, where shown on the Drawings and where directed by the Regulatory Engineer/Sanitarian.
- D. Steps: Steps shall be installed in all manholes, spaced twelve inches on center vertically and set securely in place during the construction of the wall.
- E. Precast Manholes: Precast manholes shall be installed only after Shop Drawings have been approved. The top grade of the precast concrete cone section shall be set sufficiently below finished grade to permit, a maximum of five and a minimum of two, courses of eight inch brick to be used as risers to adjust the grade of the manhole frame. Manhole frames shall be set on a grout pad to make a watertight fit.

3.6 VALVE PIT AND WET WELL STRUCTURES

- A. General Requirements: All structures shall be built and installed in accordance with the manufacturer's recommendations and as shown on the Drawings.
- B. The precast base section shall be set on twelve (12) inches of compacted crushed stone placed on compacted stable subgrade, or as otherwise shown on the Drawings. Joints of field assembled precast sections shall be waterproof. Manhole riser sections, frames and covers shall be installed, as required.

3.7 EXTERIOR FORCE MAIN VALVE EXCAVATION (CLIN 0007, OPTION 6)

- A. This work shall only be performed if CLIN 0007 is exercised by the contracting officer.
- B. Work includes:
 - 1. Excavation in constrained areas near existing structures.
 - 2. Protection of adjacent improvements and maintenance of pedestrian access.
 - 3. Dewatering as required.
 - 4. Removal and disposal of existing valve box and check valve.
 - 5. Installation of new check valve and valve box with grade access.
 - 6. Backfill, compaction, and surface restoration to match existing conditions.
- C. The contractor shall take all necessary precautions to avoid damage to nearby structures and utilities.
- D. Trench support systems shall be used as required to protect workers and adjacent infrastructure.
- E. Surface features including walkways, fencing, and disturbed areas shall be restored in kind.
- F. The contractor shall verify valve location prior to excavation.
- G. All work shall comply with applicable specification provisions.

3.8 CONNECTIONS TO EXISTING FACILITIES

- A. General Requirements: The Contractor shall make all required connections of the proposed sewer into existing sewer system, where and as shown on the Drawings and as required by the contracting officer.
- B. Permits and Coordination: The Contractor shall secure all required permits and coordinate all work with the appropriate authorities. Connections shall be scheduled to minimize disruption to existing service and shall comply with all applicable codes and standards. The CO/COR shall be notified at least 24 hours prior to beginning connection work.

3.9 PIPE CONNECTIONS TO NEW STRUCTURES

- A. Pipe connections for precast structures may be accomplished by any method described below. The Contractor shall make sure that the outside diameter of the pipe is compatible with the particular pipe connection used.
 - 1. The LOCK JOINT Flexible Manhole Sleeve cast in the wall of the manhole base. The stainless-steel strap and exposed sleeve shall be protected from corrosion with a bitumastic coating.
 - 2. KOR-N-SEAL neoprene boot cast into the manhole wall. The stainless-steel clamp shall be protected from corrosion with a bitumastic coating.
- B. Sewer manholes shall be constructed with drop connections when the proposed invert of the connection is at least two feet nine inches (2'-9") above the manhole invert. Drop connections for differences of less than two feet nine inches (2'-9") shall also be provided if approved by the Regulatory Engineer/Sanitarian.
- C. Each manhole pipe connection shall begin with a five (5) foot stub, prior to laying a full section of pipe. Pipe stubs for future connections shall be installed in the locations shown on the

drawings and the stub ends shall be sealed with a watertight plug.

3.10 LEAKAGE TEST (GRAVITY SEWER)

- A. General Requirements: The Contractor shall test the completed gravity sewer system, including manholes, septic tanks, pump station structure, and service connections, for leakage by infiltration, vacuum, exfiltration or low-pressure air exfiltration test. The test shall be conducted, as approved by the Regulatory Engineer/Sanitarian and Provincetown Health Department. The Contractor shall furnish all necessary equipment, materials and labor for performing the test.
 - 1. The Contractor shall notify the government at least forty-eight (48) hours prior to the start of testing. Testing shall only be performed in the presence of the Regulatory Engineer/Sanitarian.
 - 2. Each manhole and structure shall be tested immediately after assembly, including the connection of pipes, and prior to backfilling. All lift holes shall be plugged with non-shrink grout and all pipes entering the manhole shall be plugged and braced to prevent the plug from being drained into the manhole. The Contractor shall test the first section of pipe as soon as it is installed to demonstrate that the work conforms to these specifications.
 - 3. Sections of pipe tested for infiltration and exfiltration, prior to completion of the Contract, shall be subject to additional leakage test, if warranted, in the opinion of the Regulatory Engineer/Sanitarian, prior to acceptance of the Work.
- B. Infiltration and Exfiltration Testing: Where sewer pipe is laid on steep grades, the length to be tested by exfiltration at any one time shall be limited by the maximum allowable internal pressure on the pipe and joints at the lower end of the line. The maximum internal pressure at the lowest end shall not exceed 25 feet of water or 10.8 psi.
 - 1. The test period, wherein the measurements are taken, shall not be less than four hours in either type of test.
 - 2. Depending on field conditions, the following tests for leakage shall be employed:
 - a. Infiltration Test: The test may be used only when ground water levels are at least five feet above the top of the pipe, for the entire length of the section to be tested and during the entire period of the test. Ground water levels may be measured in a properly protected open trench or in standpipes installed in the backfill for ground water measurement, the lower ends shall be satisfactorily embedded in a mass of crushed stone or gravel to maintain free percolation and drainage. Infiltration through joints shall be measured by using a watertight weir, or any other approved device for volumetric measurement installed, at the lower end of the section under test.
 - b. Exfiltration Test: This test consists of filling the pipe with water to provide a head of at least five feet above the top of the pipe or five feet above ground water, whichever is higher, at the highest point of the pipe section under test, and the measuring of the loss of water from the line by the amount, which must be added for at least twenty-four hours prior to the taking of measurements. Exfiltration shall be measured by the drop of water level in a closed-end standpipe or in one of the sewer manholes available for convenient measuring.
 - i) When a standpipe and plug arrangement is used in the upper manhole of a section under test, a positive method of releasing entrapped air in the sewer shall be installed prior to taking measurements.

- ii) When independently testing manholes, all pipes and other openings into the manhole shall be suitably plugged and the plugs braced to prevent blowout. The manhole shall then be filled with water to the top of the cone section. If the excavation has not been backfilled and observation indicates no visible leakage, that is, no water is visibly moving down the surface of the manhole, the manhole may be considered to be satisfactorily watertight. If the test, as described above, is unsatisfactory, as determined by the Regulatory Engineer/Sanitarian, or if the manhole excavation has been backfilled, the test shall be continued. A period of time, up to twenty-four (24) hours, may be permitted if the Contractor so wishes to allow for absorption. At the end of this period, the manhole shall be refilled to the top of the cone, if necessary, and the measuring time of at least twenty-four hours begun. At the end of the test period, the manhole shall be refilled to the top of the cone, measuring the volume of water added.
- c. Leakage Requirements: The total leakage of any section tested shall not exceed the rate of fifty (50) gallons per day, per inch of nominal pipe diameter, per mile of pipe. For purposes of determining the maximum allowable leakage, manholes shall be considered as sections of forty-eight-inch (48") diameter pipe, with the length equal to the height of the manhole. The leakage for each manhole shall not exceed one (1) gallon per vertical foot for a twenty-four (24) hour period. If the manhole fails this test, but the leakage does not exceed three (3) gallons per vertical foot, per day, repairs by approved methods may be made, as directed by the Regulatory Engineer/Sanitarian, to bring the leakage with the allowable rate of one (1) gallon per foot per day. Leakage due to a defective section or joint or exceeding the three (3) gallon per vertical foot, per day, shall be cause for the rejection of the manhole. It shall be the Contractor's responsibility to uncover the manhole, as necessary, and to disassemble, reconstruct or replace it. The manhole shall be retested and, if satisfactory, interior joints shall be filled and pointed.

C. Low-Pressure Air Exfiltration Testing

- 1. The sewer pipes, service pipes and manholes shall be tested for leakage by the use of low-pressure air or vacuum, as approved by the Regulatory Engineer/Sanitarian. Air test procedures may be dangerous and the Contractor shall take all necessary precautions to prevent blowouts.
- 2. Pneumatic plugs shall have a sealing length equal to or greater than the diameter of the pipe to be tested.
 - a. Pneumatic plugs shall resist internal test pressures without requiring external bracing or blocking.
 - b. All air used shall pass through a single control panel.
 - c. Three individual hoses shall be used for the following connections:
 - 1). From control panel to pneumatic plugs for installation.
 - 2). From control panel to sealed line for introducing the low pressure air.

- 3). From sealed line to control panel for continually monitoring the air pressure rise in the sealed line.

3. The following testing procedures shall be explicitly followed:

- a. All pneumatic plugs shall be seal tested before being used in the actual test installation. One length of pipe shall be laid on the ground and sealed at both ends with the pneumatic plugs to be checked. Air shall be introduced into the plugs to 25 psig. The sealed pipe shall be pressurized to 5 psig. The plugs shall hold against this pressure without bracing and without movement of the plugs out of the pipe.
- b. After the pipe has been backfilled and cleaned, pneumatic plugs shall be placed in the line at each manhole and inflated to 25 psi. Low-pressure air shall be introduced into this sealed line until the internal air pressure reaches 4 psi greater than the average back pressure of any ground water that may be over the pipe. At least two minutes shall be allowed for the air pressure to stabilize. After the stabilization period (3.5 psi minimum pressure in the pipe), the portion of pipe tested shall be acceptable if the time required in minutes for the pressure to decrease from 3.5 to 3.0 psi (greater than the average back pressure of any ground water that may be over the pipe) is not less than the time indicated in the following table:

Pipe Size (in.)	Time (sec.)
4	190L
6	0.427L
8	0.760L
10	1.187L
12	1.709L
15	2.671L

Where L = length of pipe being tested.

- D. Correction of Defective Work: If leakage exceeds the specified amount, the Contractor shall make the necessary repairs or replacements required to permanently reduce the leakage to within the specified limit and the tests shall be repeated until the leakage requirement is met.
- E. The generator connection shall be coordinated for all live testing or temporary shutdowns. The contractor shall provide all labor, material and equipment required to demonstrate proper operation of the generator connection.
- F. Compliance with Agency Requirements: In the event of conflict between the leakage test requirements specified herein with the leakage test requirements of agencies having jurisdiction over all or any portion of the sewer system installed under this Contract, the more restrictive requirements shall govern.

3.11 PRESSURE TESTING OF FORCE MAIN

- A. Hydrostatic and leakage test shall be conducted in accordance with AWWA Standard C600, and

as directed by the Contracting Officer. Testing shall be conducted by a certified independent water testing company.

- B. Conduct pipe tests after concrete thrust blocks have cured to the required 3000 psi strength. Fill pipe 24 hours prior to testing and apply test pressure to stabilize system. Use only potable water.
- C. Prior to pressure testing, the entire pipe section shall be flushed to remove any rocks or debris, which may have inadvertently entered the pipe during construction. Once the pipe section has been filled at normal pressure and all entrapped air removed, the Contractor shall raise the pressure to 150 psi or two times the operation pressure (whichever is greater) by a special pressure pump, taking water from a small tank of proper dimensions for satisfactorily measuring the rate of pumpage into the pipe. This pressure shall be maintained for a minimum of 2 hours, during which time the line shall be checked for leaks. Measured rate of water leakage shall not exceed the allowable leakage listed below.

Allowable leakage in gallons per hour, per 1,000 feet of exterior pipeline:

Test Pressure		Nominal Pipe Diameter [inches]					
		4	6	8	10	12	16
150	psi	0.36	0.55	0.74	0.92	1.10	1.47

Interior piping in vaults, building, etc., shall have zero leakage. Should leakage exceed this rate, the Contractor shall immediately locate the leak or leaks and repair them. Pipe will be accepted only when leakage is zero, or less than the allowable amount. Approval does not absolve the Contractor from responsibility if leaks develop later within the period warranty.

3.12 CLEANING AND REPAIR

- A. The Contractor shall clean the entire sewer system of all debris and obstructions. This shall include the removal of all form work from structures, concrete and mortar droppings, construction debris and dirt. The system shall be thoroughly flushed clean, and the Contractor shall furnish all necessary hose, pumps, pipe and other equipment that may be required for this purpose. No debris shall be flushed into existing sewers, storm drains and/or streams.
- B. All the work of cleaning and repair shall be performed at no additional cost to the government.

3.12 DEFLECTION TESTING

- A. Allowable Deflection Test for Gravity Pipe
 - 1. Pipe deflection, measured not less than ninety days (90) after the backfill has been completed, as specified, shall not exceed five (5.0) percent. Deflection shall be computed by multiplying the amount of deflection (nominal diameter less minimum diameter when measured) by 100 and dividing by the nominal diameter of the pipe.
 - 2. Deflection shall be measured with a rigid mandrel (Go/No-Go) device, cylindrical in shape and constructed with a minimum of nine or ten evenly spaced arms or prongs. Drawings of the mandrel, with complete dimensions, shall be submitted to the Regulatory Engineer/Sanitarian for each diameter of pipe to be tested. The mandrel shall be hand

pulled by the Contractor through all sewer lines.

3. Any section of sewer not passing the mandrel shall be uncovered at the Contractor's expense and the bedding and backfill replaced to prevent excessive deflection. Repaired pipe shall be retested.

3.13 FINAL INSPECTION

- A. Upon completion of the work and before final acceptance by the government, the entire sewer system shall be subjected to a final inspection in the presence of the government and Regulatory Engineer/Sanitarian as required by MassDEP permit. The work shall not be considered as complete until all requirements for line, grade, cleanliness, leakage tests and other requirements have been met.

3.14 SYSTEM PERFORMANCE REQUIREMENTS

- A. The completed sewerage system, including Pump Station 1, Pump Station 2 (if Option 1 is exercised), force main, and disposal system, shall be demonstrated to operate as a complete and functional system.
- B. The Contractor shall demonstrate that Pump Station 2 is capable of conveying flow through the force main without alarm conditions, backup, or failure.
- C. Functional testing shall include:
 1. Wet well fill and pump cycling
 2. Verification of flow through the force main.
 3. Observation of normal pump operation without excessive cycling, cavitation, or blockage
- D. Failure of the system to perform as intended shall require the Contractor to identify and correct deficiencies at no additional cost to the Government, except for work explicitly covered under unexercised optional CLINs.
- E. Partial system functionality or isolated component operation shall not be considered acceptable.

END OF SECTION 333000



Commonwealth of Massachusetts
Executive Office of Energy & Environmental Affairs

Department of Environmental Protection

Southeast Regional Office • 20 Riverside Drive, Lakeville MA 02347 • 508-946-2700

Maura T. Healey
Governor

Kimberley Driscoll
Lieutenant Governor

Rebecca L. Tepper
Secretary

Bonnie Heiple
Commissioner

May 15, 2025

Cape Cod National Seashore
National Park Service
Attn: John Cullen
Property Manager
171 Racepoint Road
Provincetown, MA 02657

RE: **PROVINCETOWN - BWR**
Province Lands Visitor Center
171 Racepoint Road
25-WP63-0002-APP
PLAN APPROVAL

Dear John Cullen:

The Southeast Regional Office of the Massachusetts Department of Environmental Protection ("MassDEP") has completed its Technical Review of the application for the above-referenced authorization and determined that, based on information presently in the record, the application is technically complete. This application is for installation of a Title 5 wastewater treatment system located at the Province Lands Visitor Center on Cape Cod National Seashore owned property; therefore, this application is for federally owned facilities. The facility is located at 171 Racepoint Road, Provincetown. **MassDEP hereby approves the plans for the proposed upgrade.**

As part of the application submitted by Langan Engineering and Environmental Services, Inc., MassDEP received:

One (1) set of engineering plans, dated January 25, 2024, consisting of 17 sheets titled:

*Septic System for Province
Lands Visitor Center
CACO 299311
Cape Cod National Seashore
Historic District, 171 Race Point Road
Provincetown, MA 02657*

One (1) set of soil evaluation form documents, dated September 23, 2024, consisting of 14 sheets titled:

Soil Suitability Assessment for On-Site Sewage Disposal

One (1) set of specifications consisting of 81 sheets titled:

SECTION 013543 – ENVIROMENTAL PROTECTION

One (1) completed BRP WP 63 application, signed and dated by John Cullen.

The project consists of one system, described below:

The proposed work is to abandon the existing leaching facilities in accordance with 310 CMR 15.354. The existing wet well for the pump station serving the one (1) story building on site will be repurposed with new duplex pumps, floats, rails, ladders, controls and control panel being installed. The building sewer will be altered with the addition of two (2) manholes to direct flow into a 14,000-gallon settling tank, followed by a 7,000-gallon settling tank. Each of the setting tanks will be equipped with a Zable filter on the effluent end. Effluent from the settling tanks will be directed via gravity into a 7,000-gallon dosing chamber. The dosing chamber will contain two (2) pumps which each discharge into a designated soil absorption system. Each soil absorption system will have a minimum of one (1) inspection port.

MassDEP hereby approves the plan for the proposed upgrade subject to the following:

1. Construction shall be in strict conformance with the submitted application materials, engineering plans cited above, proposed flow volume, and provisions of this approval. No changes shall be made without approval from the Department. All system components shall be marked with magnetic tape or equivalent.
2. The applicant or engineer shall notify MassDEP at this office of the following construction events to allow the opportunity for a MassDEP construction inspection:
 - a. At bottom of bed excavation milestone to verify the elevation of the leaching field and minimum of six (6) inch crushed stone base for all septic tanks and distribution boxes.
 - b. After installation of the leaching units for the soil absorption system.
 - c. Before final back fill of distribution box and tanks.
3. MassDEP shall issue a Certificate of Compliance prior to operation of the system. An inspection and operational test of the entire system must be performed prior to the system being put on-line and prior to backfilling. The inspection and operational test shall be scheduled at least seven (21) days in advance so that Department can be present.
4. The facility served by the system and the system itself shall be open to inspection by MassDEP and the Provincetown Board of Health.
5. A written certification that the system was constructed in accordance with the submitted designs cited above shall be submitted by a Professional Engineer registered in the Commonwealth of Massachusetts.

6. The owner/operator of the system shall properly operate and maintain the system at all times in accordance with the approved plan.
7. Withing 30 days from the date of the clear water test, the design engineer shall submit signed and stamped as-built plans to the Department.
8. Copies of the approved plans cited above must be kept on-site and used for construction purposes.

The issuance of this approval does not relieve the involved parties from complying with any applicable Massachusetts and local laws and regulations.

If you should have any questions on any information provided with this letter, please contact Aaron Raposo at Aaron.Raposo@mass.gov.

Sincerely,

A handwritten signature in dark ink, appearing to read "Andrew Osei", is written over a horizontal line.

Andrew Osei, PE, Chief
Wastewater Management Program
Bureau of Water Resources

ECC:

Provincetown Board of Health
Lezli Rowell, Health Agent - lrowell@provincetown-ma.gov

Cape Cod National Sea Shore
John Cullen, Property Manager - John_cullen@nps.gov

Langan Engineering and Environmental Services, Inc.
Robert Phillips, PE, PhD, consultant - rphillips@langan.com

Submittal List with Review Estimate Template

National Park Service (NPS) - Denver Service Center (DSC) | 1-27-21

SUBMITTAL LIST										ACTION SUBMITTAL REVIEW ESTIMATE							
Park Acronym/Project Management Information System (PMIS) Number:						CACO				(Complete this portion for Architect / Engineer (A/E) construction service task order negotiation.)							
Project Title:						Replace Septic PLVC											
SUBMITTAL			REQUIREMENTS (Indicate with X)							ARCHITECT / ENGINEER REVIEW TIME (Indicate in whole or partial hours.)							
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	INFORMATIONAL				ACTION				ARCHITECT	CIVIL ENGINEER	STRUCTURAL ENGINEER	MECHANICAL ENGINEER	ELECTRICAL ENGINEER	PROJECT MANAGER COORDINATION	LANDSCAPE ARCHITECT
			CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER							
11100	1.9	Construction Schedule and Schedule of Values				X											
11100	1.9	List of Subcontractors and Suppliers.				X											
11100	1.9	Excavation protection plan, trenching plan				X											
11100	1.9	Permitting documentation, inspection reports, and coordination with the Registered Sanitarian or Engineer				X											
11100	1.9	Testing and start-up procedures for the combined septic system and electrical controls.				X											
11100	1.9	Record Drawings and O&M manuals at project closeout.				X											
11100	1.9	Third part testing firm, resumes and reports				X											
013100	1.4	Project Management Submittals				X											
013233	1.2	Construction Images				X											
013523	1.3	LOTO Acknowledgement				X											
014000	1.4	Quality Control Plan			X			X									
016799	1.3	Affirmative Procurement															
017340	1.2	Certificates and Receipts				X											
017700	1.3	Closeout Submittal				X											
017823	1.2	Operation and Maintenance Data				X											
017900	1.3	Instruction Program				X											
017419	1.4	Waste Management Plan				X											
032000	1.6	Shop Drawings				X											
033000	1.6	Mix Design, certificates				X											
260519	1.3	Electrical Power Conductors and Cables		X	X												
260523	1.3	Electrical Power Cables		X	X												
260526	1.2	Grounding and Bonding Materials		X	X												

[illegible]